THE SPHERE PACKING PROBLEM IN DIMENSION 4 AND THE TWENTY-FOUR-CELL CONJECTURE

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Abstract. We prove that every Voronoi cell of a unit-ball packing in $\mathbb{R}^4$ has at least the volume of the regular twenty-four-cell, with equality only for D_4, unless 24 to 27 other centres lie within $\sqrt{6}$ of its centre, not all at distance 2. Hence the twenty-four-cell conjecture holds for every contact configuration, D_4 is a strict local minimum of the cell volume, and no periodic packing avoiding the exception is denser than D_4; the remaining cells reduce to two explicit inequalities about at most twenty-seven centres. The cells alone bound the density of every packing of unit balls in $\mathbb{R}^4$ by 0.63477, below the three-point bound.

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1. Introduction

The sphere packing problem in dimension four asks for the supremum Δ_4 of the densities of packings of unit balls in $\mathbb{R}^4$. The D_4 root lattice, or checkerboard lattice, scaled so that nearest centres lie at distance 2, packs the balls at density $\pi^2/16 = 0.61685\ldots$, and by a theorem of Korkine and Zolotareff [KZ1872] no lattice packing of $\mathbb{R}^4$ is denser. Whether a packing that is not a lattice can be denser is open.[1]

Let $\mathcal{C} \subset \mathbb{R}^4$ be the centre set of a packing of unit balls, so that $|c - c'| \geq 2$ for distinct $c, c' \in \mathcal{C}$, and let $c \in \mathcal{C}$. The Voronoi cell

$$V_c = \{z \in \mathbb{R}^4 : |z - c| \leq |z - c'| \text{ for all } c' \in \mathcal{C}\}$$

of a point of the D_4 lattice is a regular 24-cell circumscribed about the unit ball, of volume 8. The *twenty-four-cell conjecture* asserts that every Voronoi cell of every unit-ball packing of $\mathbb{R}^4$ has volume at least 8. It implies $\Delta_4 = \pi^2/16$, because the cells tile space and each contains one ball (Section 3).[2] In dimension three the corresponding statement, with the rhombic dodecahedron in place of the 24-cell, is false (Section 1.4). In dimension four it is open; Musin [Mus18] discusses it and proposes an approach through the contact graph on S^3.

This paper proves the conjecture at every centre whose neighbours within $2\sqrt{2}$ touch it, at every centre with at most twenty-three or at least twenty-eight other centres within $\sqrt{6}$, and in an explicit neighbourhood of the D_4 configuration, and it reduces the remaining centres to two explicit statements about at most twenty-seven neighbours; the cases left open are those of Conjecture 1.3 below.

1.1. Results.

Two radii govern which neighbours matter. A centre at distance d from c cuts V_c only outside the ball of radius $d/2$ about c, so centres at distance $2\sqrt{2}$ or more never cut the 24-cell that a D_4 configuration about c would give (Lemma 2.7); and the centres whose half-spaces meet the ball $B(c, \sqrt{3/2})$ are exactly those within $\sqrt{6}$ of c (Figure 4). A *contact* of c is a centre at distance exactly 2, and R denotes the D_4 root system, normalised so that its 24 roots have length $\sqrt{2}$.

Theorem 1.1 (General local bound)**.** *Let V_c be the Voronoi cell of a centre c of a unit-ball packing of $\mathbb{R}^4$, and let $d_1 \leq d_2 \leq \ldots$ be the distances from c of the other centres closer than $2\sqrt{2}$. If every d_i equals 2, or if $\Phi(d_1, d_2, \ldots) > 8$ for the function Φ of Proposition 7.1, then $\operatorname{vol}(V_c) \geq 8$, with equality if and only if the centres within $2\sqrt{2}$ of c form a copy of $\sqrt{2}\,\mathsf{R}$ about c.*

When every neighbour within $2\sqrt{2}$ is a contact, the theorem is the twenty-four-cell conjecture for contact configurations, and its proof divides by the number m of contacts, which is at most 24 by Musin's theorem [Mus08]. For $m \leq 22$ a covering estimate on S^3 gives $\operatorname{vol}(V_c) \geq 8.0463\ldots$ whatever the directions (Theorem 1.7). For $m = 24$ the contacts form

[1] The lattice problem is a problem about positive definite quadratic forms: the densest lattice packing in dimension n corresponds to the form of given minimum with the least determinant, and Korkine and Zolotareff found it for $n = 4$ and $n = 5$ [KZ1872]. Blichfeldt settled $n = 6, 7, 8$ in the 1920s and 1930s, and Cohn and Kumar $n = 24$ [CK09]. Among all packings, periodic or not, the answer is known in dimensions 1, 2, 3 [Thu10, FT42, Hal05], 8 [Via17] and 24 [CKMRV17] only.

[2] In a periodic packing with period lattice L and N centres in a fundamental domain, the cells of one period tile a fundamental domain, so their volumes add up to $\det L$ and the density is $N \operatorname{vol}(B^4)/\det L$, with $\operatorname{vol}(B^4) = \pi^2/2$. A lower bound v for every cell therefore bounds the density by $\operatorname{vol}(B^4)/v$, and $v = 8$ gives $\pi^2/16$. A general packing can be enlarged to a saturated one, in which every cell lies within distance 2 of its centre, and the same count over a large ball gives the same bound (Lemma 3.1). The implication runs one way only: a density bound says nothing about an individual cell.

a copy of $\sqrt{2}\,\mathsf{R}$ and V_c is the 24-cell; this is the classification of twenty-four-point kissing configurations, due to de Laat, Leijenhorst and de Muinck Keizer [LLM24] and proved here from their positive kernel, whose properties we verify in exact arithmetic (Theorems 5.25 and 5.30).[3] The case $m = 23$ is neither small enough for a covering estimate nor rigid enough for a classification. We prove it by an identity: inside the ball of radius $\sqrt{3/2}$ no three contact caps meet, so the volume of the cell there is an exact function of the pair angles, and the bound becomes an inequality between pair angles that a three-point semidefinite certificate proves (Theorems 5.40, 5.69 and 5.72).

Theorem 1.2 (Further cases of the local bound)**.** *Let c be a centre of a unit-ball packing of $\mathbb{R}^4$ and Y the set of other centres within $\sqrt{6}$ of c. The conclusion of Theorem 1.1 holds at c in each of the following cases:*

(a) $|Y| \le 23$*, at any distances;*
(b) $|Y| \ge 31$*, and then* $\mathrm{vol}(V_c) > 8.0179$*;*
(c) $28 \le |Y| \le 30$*;*
(d) $|Y| = 24$*, and Y lies on the rays from c through the roots of a copy of* R*;*
(e) *the contact directions of c contain all but at most two of the normalised roots of a copy of* R*, or more generally one of the three orthogonal pairs of normalised roots at each vertex of its 24-cell;*
(f) *no centre lies at a distance from c strictly between $2+\varepsilon_0$ and $\sqrt{6}$, where $\varepsilon_0 = 4\cdot 10^{-26}$.*

These are Theorems 7.8, 7.9, 7.25, 7.32 and 7.35, Propositions 7.37 and 7.38, and Corollaries 7.5 and 7.41, case (c) at $|Y| = 28$ with Proposition 7.34 and Theorem 5.78. Case (f) makes the D_4 configuration a strict local minimum of the cell volume among all packings, in an explicit neighbourhood; the constant ε_0 comes from a robust form of the classification (Theorem 5.75). Cases (a), (b) and (c) rest on semidefinite certificates whose kernels carry the distance of each centre from c as a label, and their verification is exact or in interval arithmetic.

What Theorems 1.1 and 1.2 leave is the following.

Conjecture 1.3 (Crowded near-contacts)**.** *The conclusion of Theorem 1.1 holds at every centre of every unit-ball packing of $\mathbb{R}^4$, also when some centre lies at a distance strictly between 2 and $2\sqrt{2}$ from it and $\Phi(d_1, d_2, \dots) \le 8$.*

[3] The classification states that any 24 unit vectors in $\mathbb{R}^4$ with pairwise inner products at most $\frac{1}{2}$ form a copy of the D_4 root system. Its proof has two halves. The first is a semidefinite bound: the second level of the hierarchy for spherical codes in the form of de Laat and Vallentin [dLV15] bounds the size of a code by the value $K(\emptyset,\emptyset)$ of any positive kernel K on the subsets of at most two points of the sphere that satisfies a family of linear inequalities. The authors of [LLM24] found such a kernel with value exactly 24, at truncation degrees 14 and 16, reduced by the symmetry of the sphere and solved to forty digits, and rounded it to an exact rational point by the procedure of [CdLL24]; that point is public [LLM24d], under a licence that permits redistribution. Because the bound is exactly 24 and a code of size 24 exists, the equality case forces every inner product of such a code into the zero set of the two-point polynomial of the kernel, which is $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$. Both the bound and its equality case are proved in the text (Lemma 5.27), the positivity of the kernel is proved from the form in which it is built (Lemma 5.28), and every property of the rational point that these lemmas use is verified in exact arithmetic (Proposition 5.29): the format of the data, the positive definiteness of all 127 blocks, the construction of the zonal matrices, the four polynomial identities, the value 24 and the zeros of the two-point polynomial, the last also in Lean. The second half is classical. Scaled by $\sqrt{2}$ the vectors have norm 2 and integer inner products, so they lie among the norm-2 vectors of an integral lattice of rank at most 4; these form a simply laced root system, and among the root systems of rank at most 4 only D_4 has 24 roots (Lemma 5.26). The earlier bounds on the same question are the three-point bound of Bachoc and Vallentin [BV08], whose value in dimension four, 24.5797 in [BV08], has been lowered to 24.0569 at degree 16 [MV10, MO18] and to 24.0472 at degree 20 [Lei25], and Musin's theorem [Mus08], which bounds the size of the code but does not identify it.

The twenty-four-cell conjecture is Theorem 1.1 together with Conjecture 1.3. By Theorem 1.2 and Theorem 5.78, a centre at which Conjecture 1.3 fails has between 24 and 27 other centres within $\sqrt{6}$, one of them at a distance between $2+\varepsilon_0$ and $\sqrt{6}$, and at most 24 within 2.0161. For a set Y of centres within $\sqrt{6}$ of c let $T(Y)$ be the volume of the part of $B(c,\sqrt{3/2})$ that no half-space of Y cuts off, and $K(Y)$ the convex hull of c and the images of the centres of Y under inversion in the sphere of radius 2 about c. Proposition 7.23 shows that Conjecture 1.3 follows from two statements:

(G) if $|Y|=24$ and $T(Y)\le 8$, then the cell $V(Y)$ that Y alone cuts out satisfies $\mathrm{vol}(V(Y)\cap K(Y))\ge 8$, with equality only for a copy of $\sqrt{2}\,\mathsf{R}$;
(C) if $|Y|\ge 25$, then $T(Y)>8$.

We prove (C) for $|Y|\ge 29$ (Theorems 7.25, 7.32 and 7.35), at thirty and twenty-nine with a kernel on triples of directions that carry the range of their distance from c as a type; for $25\le|Y|\le 28$ we prove lower bounds on the number of centres within 2.05, 2.1, 2.15 and 2.2 (Proposition 7.34), and with the same kernel we prove (C) for $|Y|=28$ when at most 14 centres lie within 2.0161, when 15 do and at most 3 lie beyond 2.35, and when 16 do and none lies beyond 2.35 (Proposition 7.37), and with a kernel of higher degree, combined from kernels found for parts of the case, for every Y with $|Y|=28$ (Proposition 7.38); so (C) holds for $|Y|\ge 28$. We prove (G) along every push-out of the root system and in a neighbourhood of it (Theorems 7.9 and 7.40). Statement (G) in general, and (C) for $25\le|Y|\le 27$, are open, and Section 7.4 sets out what each needs; Figure 1 shows the cases. The density bound does not require them cell by cell: Lemma 3.3 shows that it suffices to move volume between neighbouring cells, antisymmetrically, so that every centre ends with at least 8, and the theorems above say that no transfer is needed at the centres they cover. Whether such a transfer exists is open as well.

Theorem 1.4 (Density bound)**.** *Let $\mathcal{C}$ be the centre set of a unit-ball packing of $\mathbb{R}^4$ that is periodic or saturated, and suppose that every centre of $\mathcal{C}$ satisfies the hypothesis of Theorem 1.1 or one of the conditions of Theorem 1.2. Then the density of the packing is at most $\pi^2/16$, with equality for the D_4 lattice at nearest-neighbour distance 2.*

Without the hypothesis on the centres the same cell bounds still control the density. At a centre with 24 to 27 others within $\sqrt{6}$ the cell is not shown to reach 8, but it is shown to exceed 7.7742, and this gives the following bound for every packing (Section 8).

Theorem 1.5 (Density from the cells)**.** *Every packing of unit balls in $\mathbb{R}^4$ has density at most* 0.63477.

This is below the three-point bound 0.63611 of Cohn, de Laat and Salmon [CdLS22] and the linear programming bound 0.64774 of Cohn and Elkies [CE03], and it uses nothing but the volume of each cell inside the ball of radius $\sqrt{3/2}$ about its centre.

1.2. **Methods.** The single-deviation case has a proof that uses no certificate at all, and it is the model for the others. Call the centres within $2\sqrt{2}$ of c its *active neighbours*, and write $V_c^{\mathrm{act}}\supseteq V_c$ for the cell that they alone cut out.

Theorem 1.6 (Local cell bound, single deviation)**.** *Let c be a centre of a unit-ball packing of $\mathbb{R}^4$ whose active neighbours lie in root directions of a copy of R except for at most one, and, if there is one, suppose that some root direction of R is taken by no active neighbour. Then $\mathrm{vol}(V_c^{\mathrm{act}})\ge 8$, with equality if and only if the active neighbours of c form a copy of $\sqrt{2}\,\mathsf{R}$ centred at c. Nothing is assumed about the deviating direction or about the size of its deviation.*

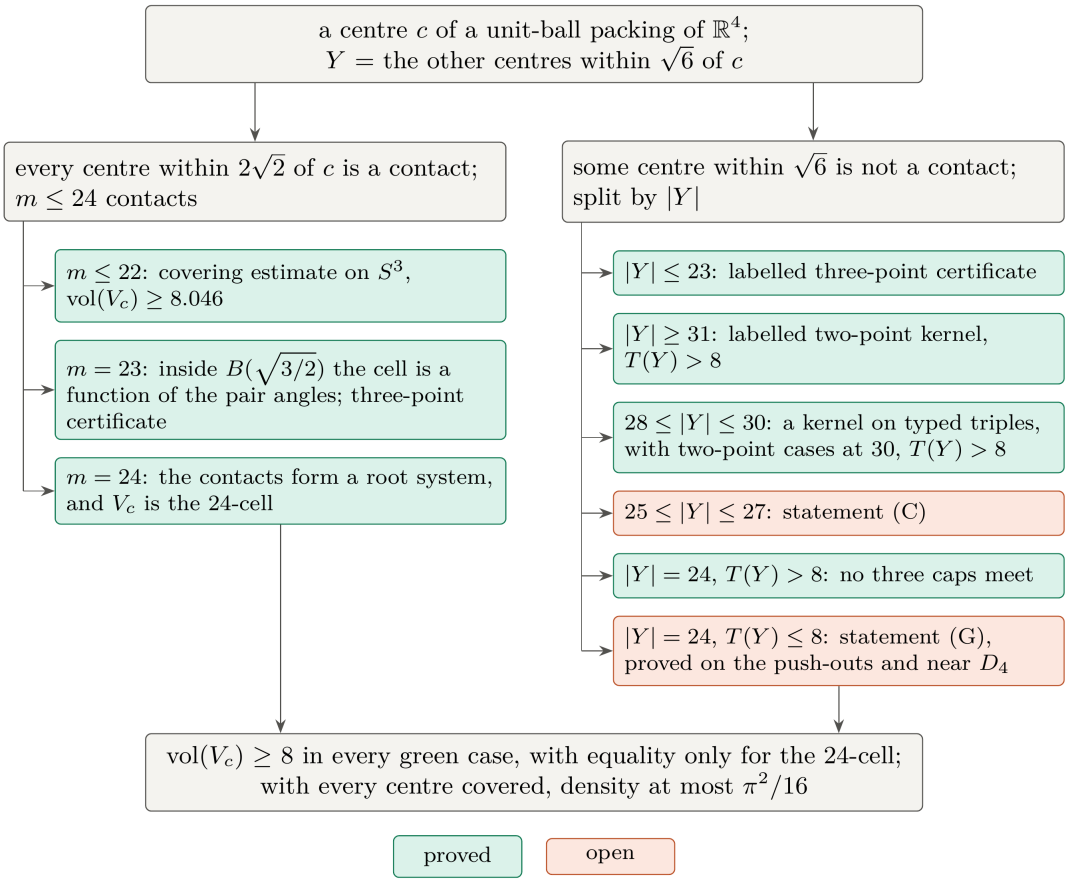

FIGURE 1. The cases of the local bound $\mathrm{vol}(V_c) \ge 8$. Left, contact configurations, by the number m of contacts: Corollary 5.17 and Theorems 5.25 and 5.40. Right, a centre within $\sqrt{6}$ of c that is not a contact, by the number $|Y|$ of centres within $\sqrt{6}$: Theorems 7.8, 7.25, 7.32 and 7.35, Propositions 7.37 and 7.38, and Lemma 7.6, statement (C) of Proposition 7.23 with the counts of Proposition 7.34, and statement (G) with Theorems 7.9 and 7.40. Green cases are proved and orange cases are open.

Fix the deviating contact direction and let Q be the intersection of the half-spaces of the twenty-three contact directions that remain at roots. The cell at the all-contact corner is Q cut by one further half-space at unit distance from the centre, and $\mathrm{vol}(Q) = \frac{25}{3}$; so the cell has volume at least 8 exactly when that half-space cuts at most $\frac{1}{3}$ off Q (Proposition 6.2). The sixteen roots at 60° and 120° to the deviating root cut out the cross-polytope

$$B = \{x : |x_0| + |x_1| + |x_2| + |x_3| \le 2\}$$

in the frame of four pairwise orthogonal roots (Lemma 6.3), so $Q \subseteq B$, and the enclosure loses nothing in the one direction that matters, since $\{x_0 \ge 1\}$ cuts exactly $\frac{1}{3}$ off both. What remains is a question about the cross-polytope alone, answered by Theorem 6.6: among the half-spaces at unit distance from the centre, those normal to a vertex direction cut off the most. Its proof, by hand, writes the cap volume as a divided difference of $a(2\sqrt{a}-1)_+^4$, that is, as an average of the third derivative of this function over a simplex, and uses that the third derivative is nondecreasing and, beyond $\frac{1}{4}$, concave (Figure 25).

The covering estimate measures the cell radially. With $\delta(\theta)$ the angular distance from $\theta \in S^3$ to the nearest contact direction, the cell at the all-contact corner has volume $\frac{1}{4}\int_{S^3} \sec^4 \delta$ (Lemma 5.15). The contacts are 60° apart, so caps of radius r about them leave at least $2\pi^2 - m\pi(2r - \sin 2r)$ of the sphere uncovered, and integrating gives $\mathrm{vol}(V_c) \geq \frac{\pi m}{3} \tan^3 r_m$ with $2r_m - \sin 2r_m = 2\pi/m$.

Theorem 1.7 (Local cell bound, few contacts)**.** *Let c be a centre of a unit-ball packing of $\mathbb{R}^4$ every active neighbour of which is a contact, and suppose there are at most 22 of them. Then $\mathrm{vol}(V_c) \geq 8.0463\ldots > 8$, whatever their directions. With no bound on their number the same argument gives $\mathrm{vol}(V_c) \geq 7.7989\ldots$.*

With several deviating directions and twenty-four contacts, an exact free-volume identity (Theorem 5.1) turns the bound into a comparison of two unions of pieces of the cell.

Theorem 1.8 (Multi-direction positivity)**.** *Let the configuration be at the all-contact corner with a full active set, and let D be any set of $m \geq 2$ roots whose directions are deviated, with arbitrary tilts and arbitrary deviation directions. Then, in the notation of Theorem 5.1,*

$$\mathrm{vol}\Big(\bigcup_{j\in D} E_j\Big) \;\geq\; \mathrm{vol}\Big(\bigcup_{j\in D} C_j\Big).$$

It follows from the classification, since twenty-four contacts admit no deviation (Corollary 5.31). The corresponding statement with sums of volumes in place of volumes of unions is a different assertion, which neither implies it nor follows from it (Remark 5.2); an additivity argument of this kind in the preprint [BBB26] of the first, second and fourth authors is not correct as stated there and is not used here.

Beyond contacts, the cell is measured inside $B(c, \sqrt{3/2})$. There no three half-spaces of the centres within $\sqrt{6}$ cut a common point (Lemma 7.6), so $\mathrm{vol}(V_c) \geq T(Y)$, and $T(Y)$ is $\frac{9}{8}\pi^2$ less the caps of the centres plus their pairwise overlaps, an explicit function of the distances $|y|$ and of the angles between the directions. A two-point or three-point kernel that carries the distance from c as a label, continuous or by ranges, then bounds $T(Y)$ from below over all configurations at once (Section 4). That is how Theorems 7.8, 7.25, 7.32 and 7.35 and Propositions 7.37 and 7.38 are proved; the same kernels without the label count the centres near c (Theorems 5.78 and 5.79). For twenty-four centres T can fall below 8, and the cell has to be bounded through the hull $K(Y)$ (Proposition 7.4); this is where (G) enters.

1.3. **Organisation.** Section 2 fixes the root system, the 24-cell and three reductions: shell localisation, the root-aligned case, and monotonicity in the contact radii. Section 3 passes from cells to density. Section 4 proves the positivity theorems on which every certificate of the paper rests, the general form in which a positive kernel bounds a sum over a configuration, and the lemmas that make a finite check of a certificate into a proof; it also lists the certificates. Section 5 proves the free-volume identity and the covering bound, settles twenty-four contacts by the classification, and proves the case of twenty-three contacts. Section 6 proves Theorem 1.6. Section 7 treats neighbours that do not touch: the distance criterion, the cases of Theorem 1.2, the second order of the volume at the root system, and the statements (G) and (C). Section 8 proves Theorem 1.5. Section 9 compares the method with others, Section 10 summarises what is proved and what is open, and Section 11 describes how the computations are checked. Two appendices follow the cell further than the theorems need: Appendix B computes the defect of a single deviation region by region, and Appendix C studies several deviating contacts to second order and along exact paths. No proof in the main text uses them. Figure 2 shows the objects of the argument in the

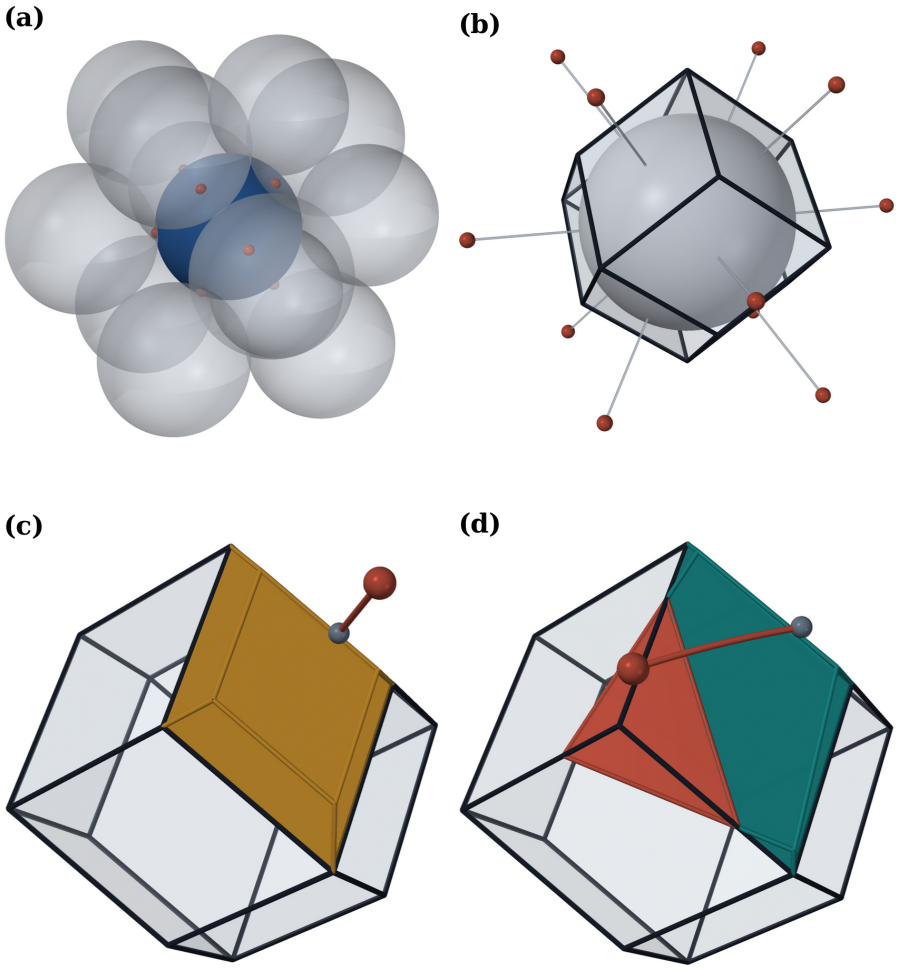

FIGURE 2. The three-dimensional analogue. (a) The all-contact corner: the unit ball about the centre (blue), its twelve contact points (clay), and the twelve unit balls that touch it, at distance 2 in the contact directions of the face-centred cubic lattice. (b) The Voronoi cell of the centre, the rhombic dodecahedron, with one facet at distance 1 for each contact and volume $4\sqrt{2} = 5.6569$. (c) Monotonicity in the radii: one neighbour moves from distance 2 out to 2.6, its half-space recedes, and the cell grows from 5.6569 to 5.9666; the ochre solid is the volume gained. (d) A deviation: one contact direction is tilted by 25° at distance 2; the cell gains the teal volume and loses the clay volume, and rises from 5.6569 to 5.7470. The volumes are computed from the half-space intersections.

three-dimensional analogue, where they can be drawn: the face-centred cubic lattice, the rhombic dodecahedron and its twelve contacts.

1.4. **Background.** In the plane, Thue [Thu10] announced the hexagonal bound and Fejes Tóth [FT42] gave the first complete proof, $\Delta_2 = \pi/\sqrt{12}$. In three dimensions Kepler's assertion that the face-centred cubic arrangement is optimal was proved by Hales, through the programme begun in [Hal97] and completed in [Hal05], and then in a machine-checked form [Hal17]. There the local statement that plays the part of the twenty-four-cell conjecture is

false: a Voronoi cell of a packing of $\mathbb{R}^3$ can be smaller than the rhombic dodecahedron, and the least is the regular dodecahedron, as Fejes Tóth conjectured and Hales and McLaughlin proved [HM10].[4] The lattice case is older and easier. Korkine and Zolotareff [KZ1872] showed that the quaternary form of D_4 has the least determinant among positive quaternary forms of given minimum; Voronoi's reduction theory [Vor08], whose polyhedral language we use, placed such results on a systematic footing, and Martinet [Mar03] gives the modern account. Upper bounds for all packings begin with Blichfeldt's method [Bli29], which gives $\Delta_4 \le \frac{3}{4}$, and were sharpened by the simplex bound of Rogers [Rog64]. Gruber and Lekkerkerker [Gru87] survey the classical theory, Conway and Sloane [CS88] is the standard source for D_4 and its Voronoi cell, and Bezdek [Bez10] collects the local and combinatorial side of the subject.

Cohn and Elkies [CE03] recast the upper bound as a linear programme over radial Schwartz functions whose values and Fourier transforms obey sign conditions; equality requires a single function with a prescribed sequence of zeros. Viazovska [Via17] constructed that function in dimension 8 from quasimodular forms, and the Leech case followed [CKMRV17]. In dimension 4 the programme cannot be sharp: Li [Li25] bounds its optimum from below by a finite-dimensional reduction, and in dimension 4 that lower bound is a centre density of 0.12914461 against the 0.125 of D_4.[5] The three-point bound of Cohn, de Laat and Salmon [CdLS22], the best upper bound known in dimension four, and the refinements of Cohn and Zhao [CZ14], which couple the packing bound to spherical-code bounds, also stay above $\pi^2/16$.

The kissing problem, which asks only how many unit balls can touch one, was settled in dimension four by Musin [Mus08], who showed that the answer is 24 by adding a polynomial constraint to the linear programming bound; the three-dimensional statement is the theorem of Schütte and van der Waerden [SW52]. A kissing bound constrains the contacts at one centre and says nothing about volume, so it does not imply the packing bound; it is the first ingredient of Theorem 1.1. The route taken here is local. It uses the symmetry of D_4, the Weyl group of order 192 and the self-duality of the 24-cell, with the polyhedral facts of [Zie95, Gru03] and the root-system conventions of [Hum90], and no Fourier analysis. Semidefinite programming enters only as a search for certificates, which are then checked in exact rational arithmetic, in interval arithmetic and, where the text says so, in the Lean proof assistant.

2. The D_4 root system and its Voronoi cell

2.1. **Lattice and roots.** Write $D_4 = \{m \in \mathbb{Z}^4 : m_1 + m_2 + m_3 + m_4 \equiv 0 \pmod 2\}$ for the standard integral D_4 lattice, and set $\Lambda = \sqrt{2}\, D_4$, which has nearest-neighbour distance 2, the natural normalisation for a unit-ball packing. The associated root system is

$$\mathsf{R} \;=\; \{\pm e_i \pm e_j : 1 \le i < j \le 4\}, \qquad |\mathsf{R}| = 24, \qquad |\alpha|^2 = 2 \;\text{ for every } \alpha \in \mathsf{R}.$$

The Weyl group $W(D_4)$ acts on R by signed permutations of the coordinates with an even number of sign changes; it has order 192 and acts transitively on R.

[4] The regular dodecahedron circumscribed about a unit ball has volume $5.5503\ldots$, against $4\sqrt{2} = 5.6568\ldots$ for the rhombic dodecahedron; it occurs as a Voronoi cell when twelve balls touch the central one at the vertices of an icosahedron, a configuration that does not extend to a dense packing. This is why Hales's proof uses a hybrid of Voronoi cells and Delaunay simplices rather than Voronoi cells alone.

[5] The centre density is the density divided by $\mathrm{vol}(B^4) = \pi^2/2$, so the centre density of D_4 at distance 2 is $\frac{1}{8}$. Cohn and Elkies [CE03] report an upper bound of 0.13126 for it; the gap between 0.12914 and 0.125 is a property of the programme, not of the functions tried.

Lemma 2.1 (Gram values on R)**.** *For distinct $\alpha, \beta \in \mathsf{R}$, the normalised inner product $s_{\alpha\beta} = \langle \alpha, \beta \rangle / 2$ takes one of the four values $-1, -\frac{1}{2}, 0, \frac{1}{2}$, and $s_{\alpha\beta} = -1$ occurs only for $\beta = -\alpha$. The three remaining values correspond respectively to $\alpha - \beta \in \mathsf{R}$ (for $s = \frac{1}{2}$), $\alpha \perp \beta$ (for $s = 0$), and $\alpha + \beta \in \mathsf{R}$ (for $s = -\frac{1}{2}$).*

Proof. Since $\alpha, \beta \in \mathbb{Z}^4$, $\langle \alpha, \beta \rangle \in \mathbb{Z}$, and Cauchy–Schwarz gives $|\langle \alpha, \beta \rangle| \le |\alpha||\beta| = 2$ with equality forcing $\beta = \pm\alpha$. For the remaining integer values $\langle \alpha, \beta \rangle \in \{-1, 0, 1\}$, expand $|\alpha \mp \beta|^2 = |\alpha|^2 + |\beta|^2 \mp 2\langle \alpha, \beta \rangle = 4 \mp 2\langle \alpha, \beta \rangle \in \{2, 4, 6\}$. An integer vector of squared length 2 has two entries ± 1 and the others 0, so it lies in R; hence $\alpha - \beta \in \mathsf{R}$ when $\langle \alpha, \beta \rangle = 1$ and $\alpha + \beta \in \mathsf{R}$ when $\langle \alpha, \beta \rangle = -1$. ∎

The four values -1 (antipodal pairs), $\frac{1}{2}$ (acute pairs), 0 (orthogonal pairs) and $-\frac{1}{2}$ (obtuse pairs) enter every computation at the all-contact corner below; the Hessian of Proposition 7.13 and the polytope Q of Section 6 are built from them.

2.2. **The reference cell.**

Definition 2.2. The *reference cell* is

$$\mathcal{V} \;=\; \{ z \in \mathbb{R}^4 : \langle z, \alpha \rangle \le \sqrt{2} \text{ for all } \alpha \in \mathsf{R} \}.$$

Lemma 2.3. *$\mathcal{V}$ has exactly 24 vertices, all of norm $\sqrt{2}$:*

$$\operatorname{vert}(\mathcal{V}) \;=\; \{ \pm\sqrt{2}\, e_i : 1 \le i \le 4 \} \;\cup\; \Big\{ \tfrac{1}{\sqrt{2}} (\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4) : \epsilon_i \in \{\pm 1\} \Big\},$$

an 8-point axis family and a 16-point sign family. $\operatorname{vol}(\mathcal{V}) = 8$*, and the symmetry group of $\mathcal{V}$, the Weyl group $W(F_4)$ of order 1152, contains $W(D_4)$ as a normal subgroup of index 6, the quotient being the group S_3 of automorphisms of the Dynkin diagram of D_4.*

Proof. Since $W(D_4)$ is transitive on R, it suffices to describe the facet $F = \mathcal{V} \cap \{ z_1 + z_2 = \sqrt{2} \}$. On F the constraint $|z_1| + |z_2| \le \sqrt{2}$ forces $z_1, z_2 \ge 0$; write $z_1 = \frac{1+a}{\sqrt{2}}$, $z_2 = \frac{1-a}{\sqrt{2}}$ and $z_j = \frac{b_j}{\sqrt{2}}$ for $j = 3, 4$. The constraints $|z_i| + |z_j| \le \sqrt{2}$ with $i \in \{1, 2\}$, $j \in \{3, 4\}$ become $|a| + |b_j| \le 1$, and the constraint for $\{3, 4\}$ follows from these. So F is the square bipyramid $|a| + \max(|b_3|, |b_4|) \le 1$, whose vertices $(a, b_3, b_4) = (\pm 1, 0, 0)$ and $(0, \pm 1, \pm 1)$ are the points $\sqrt{2}\, e_1$, $\sqrt{2}\, e_2$ and $\frac{1}{\sqrt{2}}(1, 1, \pm 1, \pm 1)$. A unit step in a has length 1 in $\mathbb{R}^4$ and a unit step in b_j has length $\frac{1}{\sqrt{2}}$, so every edge of F has length $\sqrt{2}$: F is a regular octahedron of edge $\sqrt{2}$ and volume $\frac{4}{3}$. Every vertex of $\mathcal{V}$ is a vertex of a facet, which gives the 24 listed points. The facet hyperplanes lie at distance 1 from 0, so $\mathcal{V}$ is the union of 24 pyramids of height 1 over its facets, each of volume $\frac{1}{4} \cdot \frac{4}{3} = \frac{1}{3}$, and $\operatorname{vol}(\mathcal{V}) = 8$. The symmetry group of the regular 24-cell is $W(F_4)$, of order 1152 [Cox73, §8.2]. It permutes the three families of eight vertices (the axis family and the two halves of the sign family split by the parity of the number of minus signs) through all of S_3, and the subgroup fixing each family is $W(D_4)$, of order 192. ∎

The 24-cell is the Voronoi cell of D_4 [Cox73, Ch. VIII], [CS88, Ch. 4].[6]

[6] Since the 24-cell is the one object every later section perturbs, its arithmetic is collected here. Take R to be the 24 integer vectors $\pm e_i \pm e_j$, $i < j$, each of squared length 2; the lattice D_4 they generate is $\{ x \in \mathbb{Z}^4 : \sum x_i \text{ even} \}$, of covolume 2, with minimum $\sqrt{2}$ and kissing number 24. Scaling by $\sqrt{2}$ puts the nearest centres at distance 2, as a unit-ball packing requires, and the covolume of $\Lambda = \sqrt{2}\, D_4$ is $4 \cdot 2 = 8$, which is the volume of one Voronoi cell and the number 8 that every bound in this paper is measured against. The cell itself, $\mathcal{V} = \{ x : \langle x, \alpha \rangle \le 1 \text{ for all } \alpha \in \mathsf{R}/\sqrt{2} \}$, is the regular 24-cell: 24 vertices, 96 edges, 96 triangular faces, 24 octahedral facets, Schläfli symbol $\{3, 4, 3\}$, self-dual, and with each facet centred on

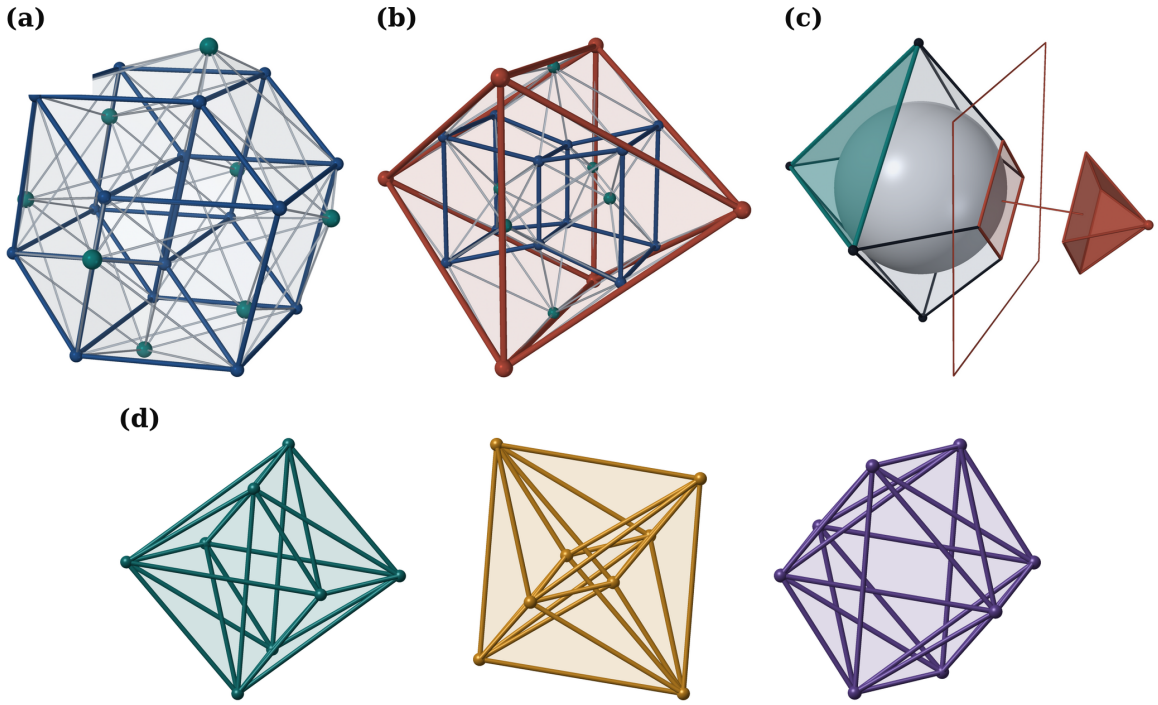

FIGURE 3. (a) The reference cell, the regular 24-cell $\mathcal{V}$, in an orthogonal projection to three dimensions: the 8 vertices of the axis family (teal), the 16 of the sign family (blue), the 32 edges inside the sign family and the 64 joining the two families (grey). (b) The Schlegel diagram of the same cell from one octahedral facet (clay): the remaining 18 vertices and 84 edges fill the interior of that facet with the other 23 octahedra. (c) The cap inequality one dimension down. The octahedron of inradius 1 carries its inscribed ball; the half-space at unit distance normal to a vertex cuts off the clay pyramid, drawn displaced along its normal, and the one normal to a facet touches that facet (teal) and cuts off nothing. (d) The same 24 vertices read as three 16-cells, which triality permutes, drawn side by side in one orthogonal projection, each with its own 24 edges and the hull of its image: the axis family (teal) and the sign family split by the parity of the number of minus signs (ochre, violet).

Figure 3 draws the polytope from its coordinates. Panel (a) is an orthogonal projection to $\mathbb{R}^3$, in which the axis family and the sign family are told apart by colour, (b) is the Schlegel diagram from one octahedral facet, (c) is the three-dimensional form of the cap inequality of Section 6, and (d) is the decomposition into the three 16-cells that triality permutes.

Definition 2.4 (Support function and angular gap)**.** For $u \in S^3$, the support function of $\mathcal{V}$ is $h(u) = \max_{z\in\mathcal{V}}\langle z, u\rangle$, and the *angular gap* is $\gamma(u) = h(u) - 1$.

one of the 24 contact directions. Its vertices split under $W(D_4)$ into the 8 points $\pm\sqrt{2}\,e_i$ and the 16 points $(\pm1, \pm1, \pm1, \pm1)/\sqrt{2}$, which is the axis and sign division used throughout. Two numbers recur: the support function of $\mathcal{V}$ equals 1 at every root direction, which is Lemma 2.8 in one line; and the circumradius is $\sqrt{2}$, so the inradius-to-circumradius ratio is $1/\sqrt{2}$, and a centre at distance $2\sqrt{2}$ or more cannot cut the cell (Lemma 2.7). The density $\pi^2/16$ is $\mathrm{vol}(B^4)/8 = (\pi^2/2)/8$.

Lemma 2.5. *For every $u \in S^3$, $h(u) \geq 1$, with equality exactly on the 24 unit root directions $\alpha/\sqrt{2}$, $\alpha \in \mathsf{R}$. The maximum $h(u) = \sqrt{2}$ is attained exactly at the 24 vertex directions $v/\sqrt{2}$, v a vertex of $\mathcal{V}$, so $\gamma(u) \in [0, \sqrt{2}-1]$ throughout, vanishing exactly on root directions.*

Proof. For $u \in S^3$ and $\alpha \in \mathsf{R}$, $\langle u, \alpha\rangle \leq |\alpha| = \sqrt{2}$, so $u \in \mathcal{V}$ and $h(u) \geq \langle u, u\rangle = 1$. If $h(u) = 1$, the point u maximises $\langle z, u\rangle$ over $\mathcal{V}$ and lies on its boundary, so $\langle u, \alpha\rangle = \sqrt{2}$ for some α, and equality in the Cauchy–Schwarz inequality gives $u = \alpha/\sqrt{2}$. Conversely $h(\alpha/\sqrt{2}) \leq 1$ by the definition of $\mathcal{V}$. For the upper bound, $h(u) = \max_v \langle v, u\rangle \leq \max_v |v| = \sqrt{2}$ over the vertices v of Lemma 2.3, with equality exactly when $u = v/|v|$ for a vertex v. ∎

2.3. **Conventions used throughout.**

We fix, once and for all, the following conventions.

Notation 2.6. A *packing centre* is a point c in the centre set $\mathcal{C}$ of a unit-ball packing; its *active neighbours* are those centres $y \in \mathcal{C}$ with $|y-c| < 2\sqrt{2}$ (Section 2.4 shows this is the only range that can matter). An active neighbour's *contact direction* is $u = (y-c)/|y-c| \in S^3$. The *all-contact corner* is the configuration in which every active radius equals the minimum packing radius 2 exactly. The normalised roots are $u_i = \alpha_i/\sqrt{2}$. A deviating contact direction is written $u_1 = \cos\varphi\, u_0 + \sin\varphi\, \eta$, where u_0 is the root it deviates from, φ the tilt angle and η a unit vector orthogonal to u_0, the deviation direction; the volume defect $\mathrm{vol}(V_c) - 8$ of such a configuration is written $\Delta(\varphi, \eta)$.

Appendix A collects every piece of notation used more than once in this paper, symbol by symbol, for reference.

2.4. **Shell localisation.**

Lemma 2.7 (Shell localisation)**.** *Let c be a centre of a unit-ball packing and $R > 0$. A centre y with $|y-c| \geq 2R$ does not cut the ball $B(c, R)$: its half-space $\{z : |z-c| \leq |z-y|\}$ contains $B(c, R)$. In particular the centres at distance at least $2\sqrt{2}$ from c contain $B(c, \sqrt{2})$, and with it the reference cell $c + \mathcal{V}$, in their half-spaces; and if no other centre lies within $2\sqrt{2}$ of c, then $V_c \supseteq c + \mathcal{V}$ and $\mathrm{vol}(V_c) \geq 8$.*

Proof. The half-space of y is $\{z : \langle z-c, u\rangle \leq |y-c|/2\}$, with u the unit vector from c towards y, and $|y-c|/2 \geq R$. The reference cell has circumradius $\sqrt{2}$ (Lemma 2.3). ∎

The lemma does not say that only the centres at distance in $[2, 2\sqrt{2})$ can affect $\mathrm{vol}(V_c)$. That is true of the part of V_c inside $B(c, \sqrt{2})$ and false outside it. A cell that reaches beyond $\sqrt{2}$, as the cell of the deletion of a root does, its circumradius being 2, can be cut there by a centre farther than $2\sqrt{2}$: with the twenty-three roots of a deletion at distance 2 and one more centre at distance 2.9 on the axis of the deleted root, which is a packing, the cell loses $\frac{1}{3}(2-1.45)^4$ and has volume $\frac{1328453}{160000} = 8.3028\ldots$ instead of $\frac{25}{3}$. Every argument below that bounds the cell of a packing either stays inside $B(c, \sqrt{2})$ or says what it assumes about the centres beyond $2\sqrt{2}$.

2.5. **The root-aligned case.**

Lemma 2.8 (Root-aligned configurations)**.** *If every active neighbour direction u (in the sense of Notation 2.6) is an exact root direction $\alpha/\sqrt{2}$, $\alpha \in \mathsf{R}$, then $\mathrm{vol}(V_c) \geq 8$.*

Proof. Take $c = 0$. An active neighbour at distance $d \geq 2$ in the root direction $\alpha/\sqrt{2}$ contributes the half-space $\{z : \langle z, \alpha/\sqrt{2}\rangle \leq d/2\}$, which contains $\{z : \langle z, \alpha/\sqrt{2}\rangle \leq 1\}$ and hence $\mathcal{V}$, since $h(\alpha/\sqrt{2}) = 1$ by Lemma 2.5. The other centres lie at distance at least $2\sqrt{2}$ and contain $\mathcal{V}$ in their half-spaces by Lemma 2.7. So $V_c \supseteq \mathcal{V}$ and $\mathrm{vol}(V_c) \geq \mathrm{vol}(\mathcal{V}) = 8$.

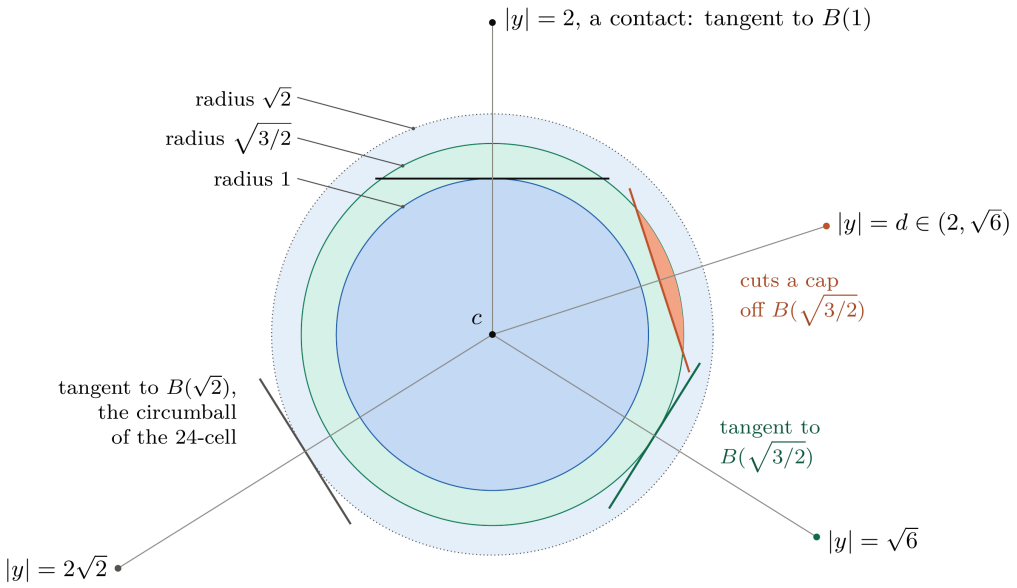

FIGURE 4. The radii that decide which centres cut V_c, in a plane through c, to scale. A centre y cuts V_c along the bisector $\langle x, y/|y| \rangle = |y|/2$. A contact, $|y| = 2$, gives a facet tangent to the unit ball; a centre with $2 < |y| < \sqrt{6}$ cuts a cap off the ball $B(\sqrt{3/2})$, which is where Lemma 7.6 measures the cell; at $|y| = \sqrt{6}$ the bisector is tangent to $B(\sqrt{3/2})$, and at $|y| = 2\sqrt{2}$ to $B(\sqrt{2})$, the ball circumscribed about the 24-cell (Lemma 2.7).

Equality forces $V_c = \mathcal{V}$, so each of the 24 facet hyperplanes of $\mathcal{V}$ bounds V_c, and each is then the bisector of a contact in its root direction. ■

This is the base case that every other configuration, deviating by a tilt angle away from a root, or having a contact radius above the minimum, is compared against; Section 2.6 handles the radius freedom, and Section 6 a deviation from a root.

2.6. **Radial reduction.**

Lemma 2.9 (Monotonicity in the contact radii)**.** *Let $u_1, \dots, u_n$ be the directions from c of its active neighbours and $d_1, \dots, d_n$ their distances. For fixed directions, the volume of $\bigcap_i \{z : \langle z - c, u_i \rangle \le d_i/2\}$ is non-decreasing in each d_i, so it is at least the volume of the same intersection with every d_i replaced by 2. When every d_i equals 2 the directions are pairwise at least 60° apart and form a contact configuration (Definition 5.7); when some d_i exceeds 2 they need not be, and the intersection with every d_i set to 2 is then a lower bound that belongs to no packing.*

Proof. Moving a bounding hyperplane away from c can only enlarge the region it helps bound. Two centres at distance 2 from c and at least 2 from each other make an angle of at least 60° at c. ■

The lemma does not say that the minimum over packing-valid radii is attained at the all-contact corner by a contact configuration. The corner is a lower bound but not a packing. Two centres at distance 2.4 from c and 2 from each other are 49.25° apart as seen from c, and pulled in to distance 2 they overlap (Figure 27(a)). Forty-eight centres at distance

2.6131 in the directions of the binary octahedral group, the vertices of two 24-cells in dual position, make a packing of 49 balls whose central cell has volume 19.216; at the corner the 48 directions have inner products up to 0.7071, and the cell has volume 6.594. Hence the reduction to contact configurations holds when every active neighbour touches the centre, and Section 7 deals with the neighbours that do not.

For centres whose active neighbours all touch, the lemma reduces the bound to contact configurations, which Section 5 treats by the number of contacts and Section 6 by the deviation from a root.

3. From the local bound to the density bound

For a packing with centre set $\mathcal{C}$, periodic or not, the density is the upper limit

$$\delta(\mathcal{C}) = \limsup_{R\to\infty} \frac{\operatorname{vol}(B^4)\,\#\{c\in\mathcal{C} : |c|\le R\}}{\operatorname{vol}(B_R)},$$

where B_R is the ball of radius R about 0 and $\operatorname{vol}(B^4) = \pi^2/2$. A packing is *saturated* if no further unit ball can be added to it without overlap. The local bound passes to density in two settings.[7]

Lemma 3.1 (Cells and density)**.** *Let $v > 0$, and let $\mathcal{C}$ be the centre set of a unit-ball packing of $\mathbb{R}^4$ with $\operatorname{vol}(V_c) \ge v$ for every $c \in \mathcal{C}$. If $\mathcal{C}$ is periodic, or if every cell V_c lies in the ball of radius 2 about c, then $\delta(\mathcal{C}) \le \operatorname{vol}(B^4)/v$; for $v = 8$ this is $\pi^2/16$. Every cell of a saturated packing lies in the ball of radius 2 about its centre.*

Proof. Let $\mathcal{C}$ be periodic, with period lattice L and N centres in a fundamental domain. The cells are permuted by L, they tile $\mathbb{R}^4$, and the cells of one period tile a fundamental domain up to translations by L; so their volumes add up to $\det L$, which is therefore at least vN, and $\delta(\mathcal{C}) = N\operatorname{vol}(B^4)/\det L \le \operatorname{vol}(B^4)/v$. If every cell lies within 2 of its centre, the cells of the centres in B_R have disjoint interiors and lie in B_{R+2}, so $v\,\#\{c : |c| \le R\} \le \operatorname{vol}(B_{R+2})$; divide by $\operatorname{vol}(B_R)$ and let $R \to \infty$. Finally, if $z \in V_c$ and $|z-c| \ge 2$, then $|z-c'| \ge |z-c| \ge 2$ for every centre c', and a unit ball about z can be added; so in a saturated packing $|z-c| < 2$ on V_c. ∎

Every packing is contained in a saturated one: the packings containing $\mathcal{C}$, ordered by inclusion, have the union of every chain as an upper bound, since the packing condition concerns two centres at a time, and Zorn's lemma gives a maximal element. Adding centres does not lower the density.

Corollary 3.2 (The conjecture gives the density)**.** *If every Voronoi cell of every saturated unit-ball packing of $\mathbb{R}^4$ has volume at least 8, then $\Delta_4 = \pi^2/16$.*

Proof. Enlarge a given packing to a saturated one and apply Lemma 3.1; the D_4 lattice attains $\pi^2/16$ (Lemma 2.3). ∎

Proof of Theorem 1.4. Every centre satisfies a hypothesis of Theorem 1.1 or of Theorem 1.2, so every cell has volume at least 8, and Lemma 3.1 applies to a periodic or saturated packing. The D_4 lattice has every cell a 24-cell of volume 8 and density $\pi^2/16$. ∎

[7] This is the classical passage from cells to density, on which the bounds of Blichfeldt and Rogers also rest [Bli29, Rog64]; see [Gru87] for its general form. It runs in one direction only, since an upper bound for the density bounds no single cell. The conjecture is therefore the stronger statement, and Lemma 3.3 below asks for less.

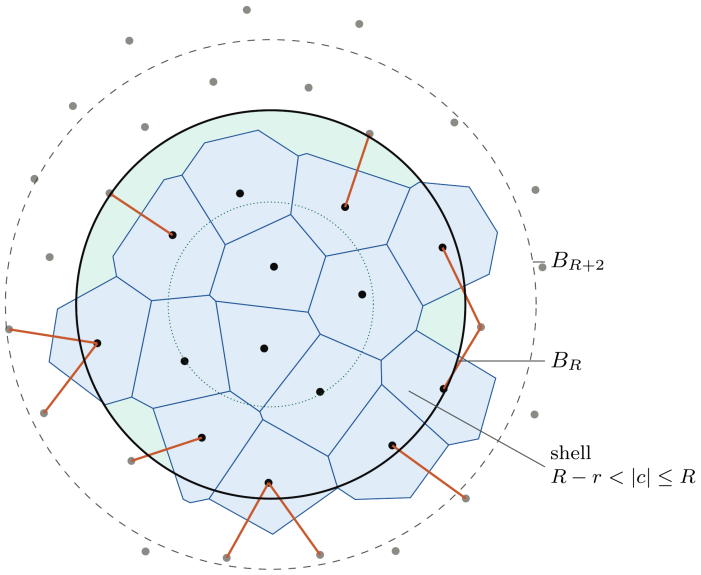

FIGURE 5. The counting of Lemmas 3.1 and 3.3, drawn in the plane for a saturated packing of unit discs, to scale. Every cell lies within 2 of its centre, so the cells of the centres in B_R (blue) lie in B_{R+2}. In the sum of the transfers over the centres in B_R, every pair inside B_R cancels; what remains are the pairs drawn in orange, from a centre in the shell $R - r < |c| \le R$ (green) to a centre outside B_R, and their number grows like the area of the shell, one power of R below the number of centres.

The bound need not hold cell by cell for the density to follow. Volume can be moved between neighbouring cells, provided that what one cell receives another gives up.

Lemma 3.3 (Redistribution)**.** *Let $r > 0$ and $M \ge 0$. Suppose that to every ordered pair of distinct centres c, c' with $|c - c'| < r$ of every saturated unit-ball packing of $\mathbb{R}^4$ a number $\tau(c, c')$ is assigned, with $\tau(c', c) = -\tau(c, c')$ and $|\tau(c, c')| \le M$, such that*

$$\sigma(c) = \operatorname{vol}(V_c) + \sum_{c' \colon 0 < |c' - c| < r} \tau(c, c') \;\ge\; 8$$

at every centre c. Then $\Delta_4 = \pi^2/16$.

Proof. Let $\mathcal{C}$ be saturated and $N(R) = \#\{c \in \mathcal{C} : |c| \le R\}$. By Lemma 3.1 the cells of the centres in B_R lie in B_{R+2}, so $\sum_{|c| \le R} \operatorname{vol}(V_c) \le \operatorname{vol}(B_{R+2})$. In $\sum_{|c| \le R} \sum_{c'} \tau(c, c')$ the terms with $|c'| \le R$ cancel in pairs, and every other term has $R - r < |c| \le R$. The unit balls about the centres in that shell are disjoint and lie in $B_{R+1} \setminus B_{R-r-1}$, so there are at most $\operatorname{vol}(B_{R+1} \setminus B_{R-r-1}) / \operatorname{vol}(B^4) = O(R^3)$ of them, and each has at most $\operatorname{vol}(B_{r+1}) / \operatorname{vol}(B^4)$ centres within r. Hence $8N(R) \le \sum_{|c| \le R} \sigma(c) \le \operatorname{vol}(B_{R+2}) + O(R^3)$, and $\delta(\mathcal{C}) \le \pi^2/16$ follows as in Lemma 3.1 (Figure 5). Every packing lies in a saturated one. ∎

With $\tau = 0$ the hypothesis is the twenty-four-cell conjecture, so Lemma 3.3 asks for less: a centre at which $\operatorname{vol}(V_c) < 8$ is allowed, provided that its neighbours have volume to spare and pass it on. The theorems of this paper give $\sigma(c) = \operatorname{vol}(V_c) \ge 8$ with $\tau = 0$ at every centre that Theorems 1.1 and 1.2 cover, so a transfer is needed only towards the centres of Conjecture 1.3 (Section 10).

A hypothesis on the centres of a packing need not survive the enlargement to a saturated packing, since the added centres change the neighbourhoods of the old ones; this is why Theorem 1.4 is stated for packings that are periodic or already saturated, while Corollary 3.2 needs the local bound at every centre of every saturated packing, which is the twenty-four-cell conjecture.

4. Positive kernels and certificates

Several theorems of this paper bound a quantity over every finite configuration of points that keep a minimal distance from one another. Some bound the number of points of S^3 whose pairwise inner products are at most a given t (Theorems 5.25, 5.78 and 5.79). The others bound a sum, over the points and their pairs, of explicit functions of distances and angles; the main example is the volume $U(Y)$ of a union of caps, which decides the cell volume once no three caps meet (Theorems 5.72, 7.8, 7.25, 7.32 and 7.35 and Propositions 7.37, 7.38 and 8.1). All of these are proved in the same way, and each proof has two halves of different kinds. The first half is analysis. A kernel that is positive by a theorem turns the statement about all configurations into an inequality between explicit functions of one, two or three points, and that inequality has to hold on a compact domain of a few real variables. The second half decides the inequality, which no longer mentions configurations, by finitely many evaluations, each one either exact or enclosed with outward rounding. This section proves the first half once, in the generality in which the later sections use it, says what the second half has to establish and why finitely many evaluations establish it, and lists the certificates of the paper in Table 1. The proofs that follow cite it, and none of them depends on the way in which a certificate was found.

4.1. **Positivity on the three-sphere.** For $k \geq 0$ let U_k be the Chebyshev polynomial of the second kind and $G_k = U_k/(k+1)$, so that $G_k(\cos\theta) = \sin((k+1)\theta)/((k+1)\sin\theta)$ and $G_k(1) = 1$. It is the Gegenbauer polynomial of S^3: the functions on S^3 that are harmonic polynomials of degree k and invariant under the rotations fixing a point e are the multiples of $G_k(\langle\cdot, e\rangle)$.

Lemma 4.1 (Zonal positivity)**.** *Let $W \subset S^3$ be finite and $a\colon W \to \mathbb{R}$. Then $\sum_{x,x'\in W} a(x)\,a(x')\,G_k(\langle x, x'\rangle) \geq 0$ for every $k \geq 0$.*

Proof. Let $\mathcal{H}_k$ be the space of restrictions to S^3 of the harmonic homogeneous polynomials of degree k on $\mathbb{R}^4$, which has dimension $(k+1)^2$, and let (Y_m) be a basis of it that is orthonormal for the surface measure σ, of total mass $2\pi^2$. The kernel $Z(x,x') = \sum_m Y_m(x)Y_m(x')$ does not depend on the choice of the basis, so it is invariant under the orthogonal group, and it is therefore a function $g(\langle x, x'\rangle)$ of the inner product alone, a polynomial of degree at most k. For fixed $x' = e$ it lies in $\mathcal{H}_k$, so it is orthogonal on S^3 to every polynomial of lower degree, and in particular to $h(\langle x, e\rangle)$ whenever $\deg h < k$. The measure σ carries $u = \langle x, e\rangle$ to a multiple of $(1-u^2)^{1/2}\,du$ on $[-1,1]$, so g is orthogonal to the polynomials of lower degree for that weight, and the orthogonal polynomials of that weight are the U_k; hence g is a multiple of G_k. On the diagonal, $Z(x,x) = g(1)$ is a constant whose integral over S^3 is $\dim \mathcal{H}_k = (k+1)^2$, so $g(1) = (k+1)^2/2\pi^2$ and

$$\sum_m Y_m(x)\,Y_m(x') = \frac{(k+1)^2}{2\pi^2}\,G_k(\langle x, x'\rangle),$$

which is the addition theorem. The sum in the lemma is therefore

$$\frac{2\pi^2}{(k+1)^2}\sum_m \Big(\sum_{x\in W} a(x)\,Y_m(x)\Big)^2 \;\geq\; 0. \qquad \blacksquare$$

This is the case of S^3 of Schoenberg's theorem [Sch42], and with $a = 1$ it is the linear programming bound of Delsarte, Goethals and Seidel [DGS77] in its primal form. The kernels of this paper also carry, at each point, a vector of numbers attached to it, most often the values of a few polynomials at the distance of a centre from c, and the same argument applies to them.

Lemma 4.2 (Kernels with labels)**.** *Let W be a finite set, let $\xi\colon W \to S^3$ and $p\colon W \to \mathbb{R}^{r+1}$ be any maps, and let A be a positive semidefinite matrix of order $r+1$. Then*

$$\sum_{x,x' \in W} G_k(\langle \xi(x), \xi(x') \rangle)\; p(x)^{\mathsf{T}} A\, p(x') \;\geq\; 0$$

for every $k \geq 0$, and for $k = 0$ the sum is $v^{\mathsf{T}} A v$ with $v = \sum_{x \in W} p(x)$.

Proof. Write $A = \sum_j \lambda_j e_j e_j^{\mathsf{T}}$ with $\lambda_j \geq 0$. The sum is

$$\sum_j \lambda_j \sum_{x,x' \in W} a_j(x)\, a_j(x')\, G_k(\langle \xi(x), \xi(x') \rangle), \qquad a_j(x) = e_j \cdot p(x),$$

and each inner sum is nonnegative by Lemma 4.1, applied to the image of ξ with the weights of points that ξ sends to the same direction added together. For $k = 0$ the claim is $G_0 = 1$. ∎

Triples of points are handled by the matrices of Bachoc and Vallentin [BV08]. The three-point matrices are built from the Legendre polynomials P_k, which play the same role on S^2. Fix a degree d and, for $0 \leq k \leq d$, let $Y_k(u,v,t)$ be the square matrix of order $d-k+1$ with entries

$$\begin{aligned} Y_k(u,v,t)_{ij} = T_i(u)\, T_j(v)\, \big((1-u^2)(1-v^2)\big)^{k/2} \\ \times\, P_k\Big(\frac{t-uv}{\sqrt{(1-u^2)(1-v^2)}}\Big), \end{aligned} \tag{4.1}$$

for $0 \leq i, j \leq d-k$, where T_i is the Chebyshev polynomial of the first kind; since P_k has the parity of k, each entry is a polynomial in u, v, t. Bachoc and Vallentin write $u^i v^j$ in place of $T_i(u)T_j(v)$; the two families differ by a fixed congruence of the matrices and are interchangeable in everything below, and the Chebyshev basis is the better conditioned one on $[-1, \frac{1}{2}]$. Let S_k be the average of Y_k over the six permutations of (u,v,t).

Lemma 4.3 (Positivity of the three-point matrices)**.** *For every finite $W \subset S^3$ and every k, the matrix $\sum_{(x,y,z) \in W^3} S_k(\langle x,y \rangle, \langle x,z \rangle, \langle y,z \rangle)$ is positive semidefinite.*

Proof. Fix $e \in S^3$ and write $x^\perp = x - \langle e,x \rangle e$ for $x \in S^3$. With $u = \langle e,x \rangle$, $v = \langle e,y \rangle$, $t = \langle x,y \rangle$ one has $|x^\perp|^2 = 1-u^2$ and $\langle x^\perp, y^\perp \rangle = t - uv$, so the scalar factor of (4.1) is $Q_k(x,y) = |x^\perp|^k |y^\perp|^k P_k(\langle x^\perp, y^\perp \rangle / (|x^\perp||y^\perp|))$, read as 0 for $k \geq 1$ and as 1 for $k = 0$ when one of the projections vanishes. The kernel $P_k(\langle \xi, \eta \rangle)$ is positive definite on the unit sphere of $e^\perp$, a copy of S^2, by [Sch42], and multiplying a positive definite kernel by $a(x)a(y)$ keeps it positive definite; so Q_k is positive definite on S^3, and for any functions $a_0, \dots, a_{d-k}$ the matrix with entries $\sum_{x,y \in W} a_i(x) a_j(y) Q_k(x,y)$ is positive semidefinite, because $c^{\mathsf{T}} M c = \sum_{x,y} b(x) Q_k(x,y) b(y)$ with $b = \sum_i c_i a_i$. Taking $a_i(x) = T_i(\langle e,x \rangle)$ gives $\sum_{x,y \in W} Y_k(\langle e,x \rangle, \langle e,y \rangle, \langle x,y \rangle) \succeq 0$ for every e; summing over $e \in W$ gives the same for the sum over W^3 of Y_k, and that sum is unchanged by permuting the three arguments, so it equals the sum of S_k. ∎

In Section 7 each point also carries a type, the range in which the distance of its centre from c lies, and the three-point matrices are then taken block by block over the types.

Lemma 4.4 (Triples of typed directions)**.** *Let $W \subset S^3$ be finite, and let each $x \in W$ carry a type $\tau(x)$ from a finite set $\mathcal{T}$. For $0 \le k \le d$ and $e, x, y \in S^3$ let $Y_k^\tau(e; x, y)$ be the square matrix indexed by the pairs (s, i) with $s \in \mathcal{T}$ and $0 \le i \le d - k$ whose entry in row (s, i) and column (s', j) is $Y_k(\langle e, x\rangle, \langle e, y\rangle, \langle x, y\rangle)_{ij}$, with Y_k as in (4.1), if $\tau(x) = s$ and $\tau(y) = s'$, and 0 otherwise. Let $f_1, \dots, f_d$ be positive semidefinite matrices indexed by $\mathcal{T}$, and for each $s \in \mathcal{T}$ let $F_0^s, \dots, F_d^s$ be positive semidefinite matrices of the orders of $Y_0^\tau, \dots, Y_d^\tau$. Then*

$$\sum_{x,y \in W} \sum_{k=1}^{d} G_k(\langle x, y\rangle)\, (f_k)_{\tau(x)\tau(y)} + \sum_{e,x,y \in W} \sum_{k=0}^{d} \langle F_k^{\tau(e)},\, Y_k^\tau(e; x, y)\rangle \;\ge\; 0. \tag{4.2}$$

Proof. Write $f_k = \sum_j v_j v_j^{\mathsf T}$. Then the first sum is $\sum_k \sum_j \sum_{x,y} v_j(\tau(x))\, G_k(\langle x, y\rangle)\, v_j(\tau(y))$, and each inner sum is nonnegative by Lemma 4.1. Fix $e \in W$ and k. The proof of Lemma 4.3, run with the functions $a_{(s,i)}(x) = T_i(\langle e, x\rangle)$ for $\tau(x) = s$ and $a_{(s,i)}(x) = 0$ otherwise, shows that $\sum_{x,y \in W} Y_k^\tau(e; x, y)$ is positive semidefinite, so its inner product with $F_k^{\tau(e)}$ is nonnegative. Summing over e and k gives the second sum. ∎

Split the left side of (4.2) according to which of the points coincide. The terms with $x = y$ in the first sum and $e = x = y$ in the second add up to $\sum_{x \in W} q_{\tau(x)}$, with one constant q_s for each type. The terms in which exactly two points of W occur, $x \ne y$ in the first sum and $\{e, x, y\} = \{x', y'\}$ in the second, add up for each unordered pair $\{x', y'\}$ to a polynomial $Q_{\tau(x')\tau(y')}(\langle x', y'\rangle)$, and the terms with e, x, y distinct add up for each set of three points to a polynomial $R_{\tau\tau'\tau''}$ of their three inner products.[8] Hence every finite set of typed points of S^3 satisfies

$$0 \;\le\; \sum_{x} q_{\tau(x)} + \sum_{\{x,y\}} Q_{\tau(x)\tau(y)}(\langle x, y\rangle) + \sum_{\{x,y,z\}} R_{\tau(x)\tau(y)\tau(z)}(\langle x, y\rangle, \langle x, z\rangle, \langle y, z\rangle). \tag{4.3}$$

4.2. **From a positive kernel to an inequality.** The next theorem is the step that every bound on a union of caps in this paper takes. It holds for any numbers s_y and $\pi_{yy'}$; the geometry enters only when they are chosen.

Theorem 4.5 (Two-point certificates)**.** *Let $\mathcal{D} \subset \mathbb{R}^4 \setminus \{0\}$ and $p\colon \mathcal{D} \to \mathbb{R}^{r+1}$, let $A_0, \dots, A_D$ be symmetric matrices of order $r + 1$ with $A_1, \dots, A_D$ positive semidefinite, and put*

$$K(y, y') = \sum_{k=0}^{D} G_k(\langle \hat y, \hat y'\rangle)\; p(y)^{\mathsf T} A_k\, p(y'), \qquad \hat y = y/|y|.$$

[8] Write $\mathbf{t}_m(u) = (T_0(u), \dots, T_m(u))$ and $F_k^s[s', s'']$ for the block of F_k^s in the rows of type s' and the columns of type s''. Then $q_s = \sum_{k=1}^{d} (f_k)_{ss} + \mathbf{t}_d(1)^{\mathsf T} F_0^s[s, s]\, \mathbf{t}_d(1)$ and

$$\begin{aligned} Q_{st}(u) = {} & 2 \textstyle\sum_{k=1}^{d} G_k(u) (f_k)_{st} + 2\, \mathbf{t}_d(1)^{\mathsf T} \big(F_0^s[s, t] + F_0^t[t, s]\big) \mathbf{t}_d(u) \\ & + \textstyle\sum_{k=0}^{d} (1 - u^2)^k\, \mathbf{t}_{d-k}(u)^{\mathsf T} \big(F_k^s[t, t] + F_k^t[s, s]\big) \mathbf{t}_{d-k}(u). \end{aligned}$$

The second term collects the triples in which the pole e is one of the two points and x, y are the two points in either order; only $k = 0$ survives there, because the scalar factor of (4.1) vanishes at $\langle e, x\rangle = 1$ for $k \ge 1$. The third collects the triples with $x = y \ne e$, where that factor is $(1 - u^2)^k$. Finally R is the sum, over the three choices of the pole among the three points, of $2 \sum_k \langle F_k^{\tau(e)}, Y_k^\tau(e; x, y)\rangle$ with x, y the other two; it does not change when two points of the same type are exchanged together with their inner products.

Suppose that for some $z \in \mathbb{R}^{r+1}$ and $t \in \mathbb{R}$ the matrix $\begin{pmatrix} A_0 & z \\ z^{\mathsf{T}} & t \end{pmatrix}$ is positive semidefinite. Then for every finite $Y \subset \mathcal{D}$ and all real numbers s_y, for $y \in Y$, and $\pi_{yy'}$, for $\{y, y'\} \subset Y$,

$$\sum_{y \in Y} s_y - \sum_{\{y,y'\} \subset Y} \pi_{yy'} \;\le\; \sum_{y \in Y} \Big(s_y + \tfrac{1}{2} K(y,y) - z \cdot p(y) \Big) + \frac{t}{2} + \sum_{\{y,y'\} \subset Y} \big(K(y,y') - \pi_{yy'} \big). \tag{4.4}$$

In particular, if $K(y,y') \le \pi_{yy'}$ for every pair and $s_y + \frac{1}{2} K(y,y) - z \cdot p(y) \le m$ for every y, then the left side is at most $|Y|\, m + \frac{t}{2}$.

Proof. Put $v = \sum_{y \in Y} p(y)$. By Lemma 4.2, with $\xi(y) = \hat{y}$, the terms of degree $k \ge 1$ add a nonnegative amount to $\sum_{y,y' \in Y} K(y,y')$, and the term of degree 0 adds $v^{\mathsf{T}} A_0 v$. Hence

$$2 \sum_{\{y,y'\}} K(y,y') = \sum_{y,y' \in Y} K(y,y') - \sum_{y \in Y} K(y,y) \;\ge\; v^{\mathsf{T}} A_0 v - \sum_{y \in Y} K(y,y).$$

The quadratic form of the positive semidefinite matrix, evaluated at the vector $(v, -1)$, gives $v^{\mathsf{T}} A_0 v - 2 z \cdot v + t \ge 0$. The two together give $-\sum_{\{y,y'\}} K(y,y') \le \frac{1}{2} \sum_y K(y,y) - z \cdot v + \frac{t}{2}$, and adding $\sum_y s_y + \sum_{\{y,y'\}} (K(y,y') - \pi_{yy'})$ to both sides gives (4.4), since $z \cdot v = \sum_y z \cdot p(y)$. ∎

In Section 7 the theorem is applied with $s_y = S(|y|)$, the volume of the cap that the centre y cuts from $B(c, \sqrt{3/2})$, and $\pi_{yy'}$ the volume of the overlap of two caps, so that the left side is the volume $U(Y)$ of their union; and $p(y)$ is the vector of Chebyshev polynomials of a rescaled distance $|y|$, which is how the kernel carries the distance as a label. The hypothesis $K \le \pi$ on the pairs is then an inequality between two functions of three real variables, the two distances and the inner product of the directions, on the region that the packing condition $|y - y'| \ge 2$ allows, and the bound on $s_y + \frac{1}{2} K(y,y) - z \cdot p(y)$ is an inequality in one variable. Three further forms of the same argument occur.

Triples with types. Adding (4.3) to (4.4) changes the pair terms to $K - \pi + Q$ and adds a sum of the polynomials R over the triples. Once each of these is bounded above by a constant for each combination of types, the right side depends on Y only through the number of its centres in each of finitely many ranges of distance, and the bound becomes a maximisation over finitely many integer vectors ((7.9) in the proof of Theorem 7.32).

Counting points. With constants in place of s_y and π, the positivity of Lemmas 4.1 and 4.3, summed over all ordered pairs and triples of a code and separated by the points that coincide, bounds the number of its points; this is the form of the proofs of Theorems 5.78 and 5.79, and with N fixed and a weight on the pairs it is Lemma 5.71.

Subsets of two points. The classification of twenty-four contacts uses a kernel on the subsets of at most two points of S^3, positive in the sense of Section 5.5.1; Lemma 5.27 turns it into a bound on the size of a code, and the equality case of that bound identifies the code.

4.3. **What a certificate must satisfy, and how it is checked.** A certificate is a finite list of rational numbers: the coefficients of p, the entries of the matrices A_k, of z and of t, and, where triples enter, the entries of the matrices of Lemmas 4.3 and 4.4. Two kinds of statement have to be proved about it. The first is that certain symmetric rational matrices are positive semidefinite. The second is that certain explicit functions of at most three real variables, built from the certificate and from the geometric functions of the problem, stay below stated constants on stated compact domains. The geometric functions are the volume S of a cap and the volume Π of the overlap of two caps (Section 7), and the weight ω of the pair angles (Section 5.12). Both are decided by finitely many operations, and the next two lemmas say why that is enough.

Lemma 4.6 (Exact positivity). *Let A be a symmetric rational matrix. If $A = LDL^{\mathsf{T}}$ with L unit lower triangular and D diagonal with nonnegative entries, then A is positive semidefinite, and positive definite if the entries of D are positive. If A is positive definite and $t \geq z^{\mathsf{T}} A^{-1} z$, then $\begin{pmatrix} A & z \\ z^{\mathsf{T}} & t \end{pmatrix}$ is positive semidefinite.*

Proof. For the first statement, $x^{\mathsf{T}} A x = \sum_i d_i \big((L^{\mathsf{T}} x)_i\big)^2 \geq 0$, and when every d_i is positive this vanishes only if $L^{\mathsf{T}} x = 0$, that is if $x = 0$, since L is invertible. For the second,

$$\begin{pmatrix} A & z \\ z^{\mathsf{T}} & t \end{pmatrix} = \begin{pmatrix} I & 0 \\ z^{\mathsf{T}} A^{-1} & 1 \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & t - z^{\mathsf{T}} A^{-1} z \end{pmatrix} \begin{pmatrix} I & A^{-1} z \\ 0 & 1 \end{pmatrix},$$

a congruence with a block diagonal matrix whose blocks are positive semidefinite. ∎

The factorisation is carried out over $\mathbb{Q}$, so the test involves no rounding. It is applied to every matrix of the certificates of Theorems 5.72, 5.78, 5.79, 7.25, 7.32 and 7.35, and the number t of Theorem 4.5 is taken to be $z^{\mathsf{T}} A_0^{-1} z$ rounded up, so that the second statement gives the positivity of the block matrix exactly.

Lemma 4.7 (Enclosures on a box). *Let $R = \prod_{i=1}^{n} [a_i, b_i]$ be a box with centre c and half-widths $h_i = \frac{1}{2}(b_i - a_i)$.*

(i) *Let P be a polynomial of degree at most δ_i in its i-th variable, written $P = \sum_\beta c_\beta B_\beta$ in the tensor Bernstein basis $B_\beta(x) = \prod_i \binom{\delta_i}{\beta_i} (x_i - a_i)^{\beta_i} (b_i - x_i)^{\delta_i - \beta_i} / (b_i - a_i)^{\delta_i}$. Then $\min_\beta c_\beta \leq P \leq \max_\beta c_\beta$ on R.*

(ii) *Let f be twice continuously differentiable near R, with $|\partial_i \partial_j f| \leq M_{ij}$ on R. Then on R*

$$f \;\leq\; f(c) + \sum_i |\partial_i f(c)|\, h_i + \tfrac{1}{2} \sum_{i,j} M_{ij} h_i h_j,$$

and the same bound with the signs of the last two terms reversed holds from below.

(iii) *Let f be nondecreasing in some of its variables and nonincreasing in the others. Then its largest value on R is taken at the corner with the upper ends of the first variables and the lower ends of the others, and its least value at the opposite corner.*

Proof. (i) The B_β are nonnegative on R and add up to 1 there, by the binomial theorem in each variable, so P is a convex combination of its coefficients at every point of R. (ii) By Taylor's theorem with the remainder in Lagrange form along the segment from c to x, $f(x) = f(c) + \nabla f(c) \cdot (x - c) + \frac{1}{2}(x - c)^{\mathsf{T}} \nabla^2 f(\zeta)(x - c)$ for some $\zeta \in R$, and $|x_i - c_i| \leq h_i$. (iii) Move the variables one at a time to the end of their intervals at which f does not decrease. ∎

Every enclosure is evaluated either in exact rational arithmetic or in interval or ball arithmetic with outward rounding [MKC09, Ch. 5],[Joh17], so that the number computed is at least the true upper bound. The domain of an inequality is a box, or a semialgebraic part of one: the pairs allowed by $|y - y'| \geq 2$, which in the distances and the inner product is the condition $\langle \hat{y}, \hat{y}' \rangle \leq a(|y|, |y'|)$ of (7.2), or the triples of inner products whose Gram matrix is positive semidefinite. It is covered by finitely many boxes, and on each box either an exact test or an enclosure of the defining condition shows that the box misses the domain, or one of the enclosures of Lemma 4.7 shows that the function stays below its constant there. Since every point of the domain lies in a box of the second kind, the inequality holds on the whole domain. The boxes are produced by repeated bisection (Appendix D), and that their union is the box one started from is checked on their boundaries in exact arithmetic; the counts quoted are those of the final covers. Polynomials built from the certificate are

bounded by (i) or (ii). The functions S, Π and ω are not polynomials. They enter through (iii), by the monotonicity of S and of Π (Lemma 7.24), with the corner values computed from their closed forms in ball arithmetic, or through (ii) with tables of rigorous bounds for ω and its first two derivatives. Floating point is used only to find a certificate; once its entries are fixed as the rationals they denote, nothing that follows depends on how they were obtained, and a numerical optimum that has not been made into a certificate in this way is never used as a bound.

4.4. **The certificates of this paper.** Table 1 lists the certificates on which proofs rest, the kernel each one uses, its degree, the size of the finite check of its inequalities, and the value it proves against the value it needs. The theorems are stated and proved in the sections named there, which also say how each certificate was found; Remarks 5.80, 7.26 and 7.36 say where the certificates of Theorems 5.78, 5.79, 7.25, 7.32 and 7.35 are tight and what each of them needs.

Result	Kernel	Degree	Finite check	Proved / needed
Theorem 5.30	second level [LLM24]	14, 16	127 blocks, exact	24 / 24
Theorem 5.72	triples, contacts	8	313 780 boxes	0.0929000 / 0.0928556
Theorem 7.8	triples, distances	8	36 533 boxes	≥ 0 / ≥ 0
Theorem 5.78	triples, counting	10	2.7 million boxes	23.9613 / 23.96
Theorem 5.79	triples, counting	10	1.33 million boxes	24.8260 / 24.8
Theorem 7.25	pairs, labelled	12, 4	191 740 boxes	3.08537 / 3.10330
Theorem 7.32	pairs; typed triples	12, 4; 8	193 million boxes	3.0992 / 3.10330
Theorem 7.35	pairs; typed triples	12, 4; 8	16.3 million boxes	3.10147 / 3.10330
Proposition 7.37	pairs; typed triples	12, 4; 8	30.0 million boxes	3.10247 / 3.10330
Proposition 7.38	pairs; typed triples	12, 4; 10	1234.1 million boxes	3.10328 / 3.10330
Proposition 8.1, $M = 24$	pairs; typed triples	6 (triples)	104 028 boxes	3.25388 / 3.3291
Proposition 8.1, $M = 26$	pairs; typed triples	8 (triples)	7.5 million boxes	3.32906 / 3.3291
Proposition 7.34	pairs, labelled	12, 4	one split per entry	3.10276 / 3.10330

TABLE 1. The certificates on which proofs rest. The degree of a kernel on pairs with a distance label is given in the angle and in the distance; a semicolon separates the kernel on pairs from the kernel on typed triples. The last column gives the value that the certificate proves and the value that the theorem needs. In the first row the bound on the size of a code is exactly 24, as the classification needs. In the next four rows the certificate proves a lower bound, which has to exceed the number on the right; for Theorem 7.8 it is the sign of every term of its proof. In the remaining rows it proves an upper bound on $U(Y)$, which has to lie below the number on the right. For Theorem 7.32 and Proposition 7.34 the value is the largest over the cases of the proof.

5. The general local problem

5.1. **The free-volume identity in general.** Throughout this subsection the configuration is at the all-contact corner with a full active set: the active directions are $w_1, \dots, w_{24}$, one for each root, with $w_i = u_i$ except on a subset D of deviating indices, where $w_j = \cos\varphi_j\, u_j + \sin\varphi_j\, \eta_j$. A configuration with a proper active subset is covered a fortiori, since

dropping a constraint only enlarges the cell. Write

$$V_c = \bigcap_{i=1}^{24} \{\langle x, w_i\rangle \le 1\}, \qquad C_j = \mathcal{V} \cap \{\langle x, w_j\rangle \ge 1\}, \qquad E_j = V_c \cap \{\langle x, u_j\rangle \ge 1\},$$

for $j \in D$. Thus C_j is the part of the reference cell that the j-th half-space cuts off as it tilts, and E_j is the region the same motion frees beyond the j-th facet.

Theorem 5.1 (Free-volume identity)**.** *For every D and every choice of tilts and deviation directions,*

$$\operatorname{vol}(V_c) \;=\; \operatorname{vol}(\mathcal{V}) \;-\; \operatorname{vol}\Big(\bigcup_{j\in D} C_j\Big) \;+\; \operatorname{vol}\Big(\bigcup_{j\in D} E_j\Big). \tag{5.1}$$

In particular $\operatorname{vol}(V_c) \ge 8$ *if and only if* $\operatorname{vol}\big(\bigcup_j E_j\big) \ge \operatorname{vol}\big(\bigcup_j C_j\big)$.

Proof. V_c and $\mathcal{V}$ are intersections of half-spaces that agree in the $24-|D|$ constraints indexed outside D. A point of $\mathcal{V}$ fails to lie in V_c exactly when $\langle x, w_j\rangle > 1$ for some $j \in D$, so $\mathcal{V} \setminus V_c = \bigcup_{j\in D} C_j$ up to a null set; a point of V_c fails to lie in $\mathcal{V}$ exactly when $\langle x, u_j\rangle > 1$ for some $j \in D$, so $V_c \setminus \mathcal{V} = \bigcup_{j\in D} E_j$. Both sets are finite unions of polytopes, hence measurable, and $\operatorname{vol}(V_c) = \operatorname{vol}(\mathcal{V}) - \operatorname{vol}(\mathcal{V} \setminus V_c) + \operatorname{vol}(V_c \setminus \mathcal{V})$. ∎

For $|D| \le 1$ the two unions are single sets and (5.1) reduces to $\operatorname{vol}(V_c) = 8 - \operatorname{vol}(C) + \operatorname{vol}(E)$, which is the form used, through the cap form of Proposition 6.2, in Section 6. For $|D| \ge 2$ the unions matter, as follows.

Remark 5.2 (The sums are not the unions)*.* Neither family is disjoint once $|D| \ge 2$. The sets E_j overlap whenever two deviating roots are adjacent, that is when $\langle u_j, u_k\rangle = \frac{1}{2}$: the pyramid Pyr_j of Lemma 6.1 has one of its eight lateral facets on the hyperplane $\{\langle x, u_k\rangle = 1\}$, so removing the constraint at u_k opens E_j out past that facet and into the region where $\langle x, u_k\rangle > 1$, which is where E_k lives. At an adjacent pair with both tilts $30°$ the overlap has volume $0.0137\ldots$, against a defect of $0.1845\ldots$, so it is not a small correction. The sets C_j can also overlap, though only at large tilts and only for adjacent pairs; over random packing-valid configurations of two to four deviating directions the largest overlap is 1.6×10^{-3}.

Consequently the quantity $\sum_j \operatorname{vol}(E_j) - \sum_j \operatorname{vol}(C_j)$ is *not* equal to $\operatorname{vol}(V_c) - 8$, and its sign is neither implied by nor implies the cell bound: in samples it runs both above the defect (for instance 0.1798 against 0.1570) and below it (for instance 0.20945 against 0.20971). Only the union form (5.1) is an identity, and it is the union form that Theorem 1.8 asserts.

Remark 5.3 (Why the single-deviation theorem does not reduce the general case)*.* Splitting the comparison of Theorem 1.8 into m separate ones, each settled by Theorem 6.8, fails for two structural reasons.

The first is the overlapping just described. On its own it can be controlled: Bonferroni's inequality[9] gives

$$\operatorname{vol}\Big(\bigcup_j E_j\Big) \;\ge\; \sum_j \operatorname{vol}(E_j) \;-\; \sum_{j<k} \operatorname{vol}(E_j \cap E_k),$$

and together with $\operatorname{vol}(\bigcup_j C_j) \le \sum_j \operatorname{vol}(C_j)$ this reduces the inequality to

$$\sum_j \big(\operatorname{vol}(E_j) - \operatorname{vol}(C_j)\big) \;\ge\; \sum_{j<k} \operatorname{vol}(E_j \cap E_k). \tag{5.2}$$

[9]For any measure and any finitely many sets, a point that lies in exactly $n \ge 1$ of them is counted $n - \binom{n}{2} \le 1$ times by the right side of the inequality and once by the left side, so the inequality holds pointwise and therefore after integration.

The overlaps on the right are second order in the tilts, the same order as each term on the left, so the order of magnitude does not rule out (5.2).

The second obstruction closes the route. The E_j appearing in (5.2) are not the sets the single-deviation theorem controls. With one root deviating, E_j is cut out by the other twenty-three root half-spaces, and Lemma 6.1 identifies it inside the pyramid Pyr_j. With m roots deviating it is cut out by the $24-m$ fixed half-spaces together with the $m-1$ other tilted ones. The eight lateral facets of Pyr_j lie on the hyperplanes of the eight roots at 60° to α_j, so for an adjacent k the pyramid meets $\{\langle x, u_k\rangle = 1\}$ in a facet and not transversally: dropping that constraint opens E_j out, and reimposing it in the tilted position $\{\langle x, w_k\rangle \le 1\}$ slices back in along a whole face. The two effects do not cancel, and neither dominates: over random packing-valid configurations with $2 \le m \le 6$ and tilts up to 0.9, $\mathrm{vol}(E_j)$ both exceeds its single-deviation value and falls short of it, in the latter case by more than 0.1, against defects of order 0.3. Therefore Theorem 6.8 gives no bound on $\mathrm{vol}(E_j) - \mathrm{vol}(C_j)$ once $m \ge 2$, and (5.2) cannot be reached term by term.

What would suffice is a replacement for Pyr_j: a body, depending on the whole configuration rather than on one root, that contains each E_j, is cut by the tilted half-spaces in a controlled way, and for which the analogue of Theorem 6.6 can be proved. No such body is known.

Corollary 5.4. *In the setting above with $|D| = m \ge 2$, $\mathrm{vol}(V_c) \ge 8$ if and only if*

$$\mathrm{vol}\Big(\bigcup_{j\in D} E_j\Big) \;\ge\; \mathrm{vol}\Big(\bigcup_{j\in D} C_j\Big).$$

Figure 6 draws the three terms of (5.1).

5.2. **A fixed-body form of the multi-direction problem.** For $m \ge 2$ the cap reformulation of Section 6 still gives one extremal problem about one fixed polytope, as in the case $m = 1$, and not a sum of m problems.

Let D be the set of deviating roots, $|D| = m$, and put

$$Q_D = \bigcap_{i\notin D} \{x \in \mathbb{R}^4 : \langle x, u_i\rangle \le 1\}, \qquad \mathrm{Cap}_{Q_D}(w) = \{x \in Q_D : \langle x, w\rangle \ge 1\}.$$

For $m = 1$ this is the polytope Q of Section 6.1.

Proposition 5.5 (Multi-cap form)**.** *Suppose Q_D is bounded. Then, for every choice of tilts and deviation directions,*

$$\mathrm{vol}(V_c) \;=\; \mathrm{vol}(Q_D) \;-\; \mathrm{vol}\Big(\bigcup_{j\in D} \mathrm{Cap}_{Q_D}(w_j)\Big), \tag{5.3}$$

and therefore $\mathrm{vol}(V_c) \ge 8$ if and only if

$$\mathrm{vol}\Big(\bigcup_{j\in D} \mathrm{Cap}_{Q_D}(w_j)\Big) \;\le\; \mathrm{vol}(Q_D) - 8. \tag{5.4}$$

The right-hand side depends only on D, not on the tilts, and equals $\mathrm{vol}(\bigcup_{j\in D} \mathrm{Cap}_{Q_D}(u_j))$, so (5.4) holds with equality at the undeviated configuration. It equals $m/3$ exactly when no two roots of D are adjacent, and exceeds $m/3$ otherwise.

Proof. $V_c = Q_D \cap \bigcap_{j\in D}\{\langle x, w_j\rangle \le 1\}$ by definition, so $Q_D \setminus V_c = \bigcup_{j\in D} \mathrm{Cap}_{Q_D}(w_j)$ up to a null set, which is (5.3). Taking $w_j = u_j$ makes $V_c = \mathcal{V}$ and gives $\mathrm{vol}(\bigcup_j \mathrm{Cap}_{Q_D}(u_j)) = \mathrm{vol}(Q_D) - 8$. If no two roots of D are adjacent then $\mathrm{Cap}_{Q_D}(u_j) = \mathrm{Pyr}_j$ for each j, since by the proof of Lemma 6.1 only the eight roots at 60° to α_j are active on $\{\langle x, u_j\rangle \ge 1\}$ and

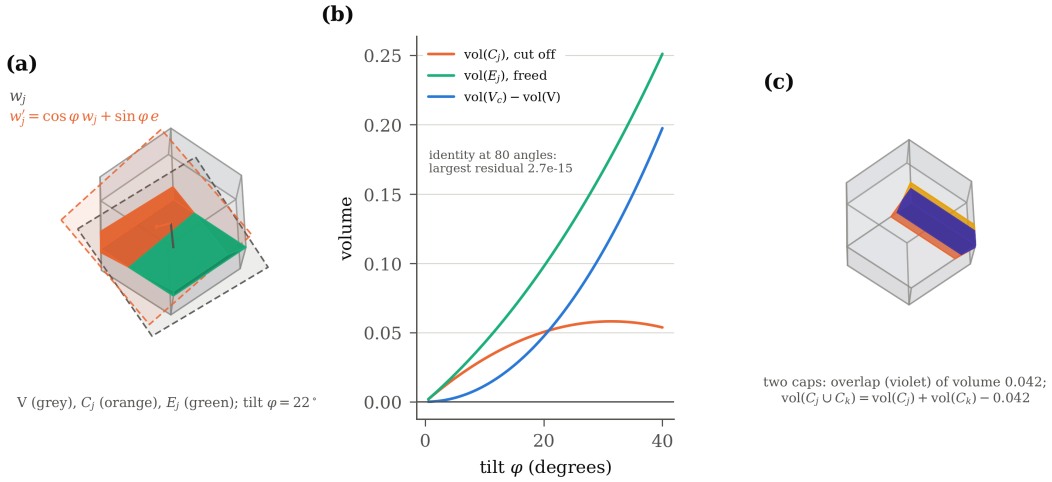

FIGURE 6. The free-volume identity of Theorem 5.1, drawn on real geometry. Because the identity holds in every dimension, it is drawn in three, where its pieces can be seen: the reference cell is the Voronoi cell of the face-centred cubic contacts, the rhombic dodecahedron $\{x : \langle x, w_i\rangle \le 1\}$ over the twelve directions $(\pm 1, \pm 1, 0)/\sqrt{2}$ and permutations. (a) One direction w_j tilted by 22° to w_j'; the cap C_j is the part of the cell the tilted half-space cuts off and E_j the region the motion frees beyond the old facet, both computed as half-space intersections and drawn to scale. (b) The two volumes and their difference against the tilt, each piece computed as a polytope of its own; the identity holds at every angle to the precision of the arithmetic. (c) Two directions tilted towards each other: the caps overlap, so for $m \ge 2$ the corrections are volumes of unions and not sums of volumes (Remark 5.2).

none of them lies in D; those pyramids are pairwise disjoint, again by Lemma 6.1, so the union has volume $m/3$. If two roots of D are adjacent, the corresponding $\mathrm{Cap}_{Q_D}(u_j)$ strictly contains Pyr_j, since the constraint carrying one of its lateral facets has been removed. ■

Hence the multi-direction problem is again an extremal problem for caps of one fixed polytope, with all the parameters in the cutting half-spaces; what has changed from $m = 1$ is that the quantity to bound is the volume of a *union* of m caps rather than of one cap, and Remark 5.3 shows the union cannot be traded for a sum.

Remark 5.6 (The multi-cap inequality). The inequality (5.4) holds for every D with Q_D bounded and every packing-valid choice of w_j, with equality only at $w_j = u_j$, by Corollary 5.31 and therefore by the classification of Theorem 5.25. A proof of it that does not go through the classification would have to bound the union of the m caps directly; the rest of this section shows why the standard ways of doing so fail.

Q_D is bounded precisely when the $24 - m$ undeviated root directions positively span $\mathbb{R}^4$, and an exhaustive enumeration over all packing-valid sets D gives the following: for $m \le 5$ it is bounded in every case, all 24, 276, 2024, 10626 and 42504 of them respectively; at $m = 6$ it fails for exactly 24 of the 134596 packing-valid sets, the first in lexicographic order being the six roots $e_1 \pm e_2$, $e_1 \pm e_3$, $e_1 \pm e_4$, whose complement lies in the closed half-space $\{x_1 \le 0\}$ and therefore leaves Q_D unbounded in the direction e_1. Beyond $m = 6$ unbounded cases become common, and for those a direct argument needs a bounded substitute for Q_D first; (5.4) as stated does not reach them.

5.3. **Polar and boundary forms of the same question.** The multi-cap form keeps a fixed body and moves the cutting half-spaces. A third form of the same question discards the reference cell and states the problem in terms of the contact configuration alone. It makes the relation to the 24-cell conjecture explicit and shows what a proof of Theorem 1.8 cannot look like.

Definition 5.7. A *contact configuration* is a finite set $W \subset S^3$ with $\langle w, w' \rangle \leq \frac{1}{2}$ for all distinct $w, w' \in W$. Its *cell* is

$$V_c(W) = \{x \in \mathbb{R}^4 : \langle x, w \rangle \leq 1 \text{ for all } w \in W\}.$$

The condition is exactly packing validity.[10] Every local configuration considered in this paper, deviated or not, is a contact configuration, and V_c is its cell.

Proposition 5.8 (Polar form)**.** *For every contact configuration W,*

$$V_c(W) = \mathrm{conv}(W)^\circ, \tag{5.5}$$

the polar dual of the convex hull of W. Consequently

(i) $\mathrm{vol}(V_c(W))$ *depends on W only through* $\mathrm{conv}(W)$*;*
(ii) $W \subseteq W'$ *implies* $V_c(W') \subseteq V_c(W)$*, so enlarging a contact configuration never increases the volume of its cell;*
(iii) $|W| \leq 24$*, by Musin's kissing bound* [Mus08]*.*

Theorem 1.8 is equivalent to the assertion $\mathrm{vol}(\mathrm{conv}(W)^\circ) \geq 8$ *for every contact configuration whose cell is bounded, with equality only when W is the set of* 24 *root directions.*

Proof. The polar of a compact convex set K containing the origin is $K^\circ = \{x : \langle x, p \rangle \leq 1 \text{ for all } p \in K\}$, and a linear functional attains its maximum over $\mathrm{conv}(W)$ at a point of W, so $\mathrm{conv}(W)^\circ$ is cut out by the constraints indexed by W alone, which is (5.5). Item (i) is immediate from (5.5) and (ii) from the definition. Item (iii) is the kissing-number theorem in dimension four, since a contact configuration is precisely a set of directions to mutually non-overlapping unit balls touching a fixed one. The final sentence is Theorem 5.1 together with the observation that a packing configuration at the all-contact corner, every active neighbour at distance exactly 2, is a contact configuration of exactly the shape in Definition 5.7: two centres at distance 2 from c and at least 2 apart make an angle of at least 60° at c. A configuration obtained by pulling neighbours that do not touch in to distance 2 need not be one (Section 7). ∎

Corollary 5.9 (A worst configuration exists, and may be taken maximal)**.** *The infimum of* $\mathrm{vol}(V_c(W))$ *over all contact configurations with bounded cell is attained, and it is attained at a configuration to which no further direction can be added.*

Proof. By Proposition 5.8(iii) a contact configuration has at most 24 elements, so for each $m \leq 24$ the set $\mathcal{C}_m = \{(w_1, \ldots, w_m) \in (S^3)^m : \langle w_i, w_j \rangle \leq \frac{1}{2} \ (i \neq j)\}$ is a closed subset of a compact space, hence compact, and its points have pairwise distinct entries. The volume functional $f(W) = \mathrm{vol}(V_c(W)) \in (0, \infty]$ is lower semicontinuous on $\mathcal{C}_m$: if S is a compact subset of the interior of $V_c(W)$ then $\langle x, w \rangle < 1$ on $S \times W$, hence the same holds for every configuration near W, so $S \subseteq V_c(W')$ and $f(W') \geq \mathrm{vol}(S)$; taking the supremum over such

[10] If unit balls centred at $2w$ and $2w'$ are to have disjoint interiors then $|2w - 2w'| \geq 2$, that is $|w - w'| \geq 1$, that is $2 - 2\langle w, w' \rangle \geq 1$; so the pairwise bound $\frac{1}{2}$ is the whole of the packing constraint on the directions, and the bisector of the segment from 0 to $2w$ is the hyperplane $\langle x, w \rangle = 1$, which is the constraint defining $V_c(W)$. The directions of the D_4 roots, $u_i = \alpha_i/\sqrt{2}$, form a contact configuration with the pairwise inner products taking the four values $\pm\frac{1}{2}$, 0 and -1, so the bound is attained but never exceeded (Lemma 2.1).

S and using inner regularity gives $\liminf_{W'\to W} f(W') \geq f(W)$. A lower semicontinuous function on a compact set attains its infimum, so f attains its infimum on each $\mathcal{C}_m$, and therefore on the union of the finitely many $\mathcal{C}_m$ with $m \leq 24$. If a minimiser W^* admits an additional admissible direction, extend it, repeatedly if necessary; the process stops after at most $24 - |W^*|$ steps and each step leaves the volume no larger by Proposition 5.8(ii), so the resulting maximal configuration is again a minimiser. ∎

So the inequality concerns maximal contact configurations only, and an extremal configuration exists. The remaining question, whether the maximal configurations of 24 directions are the root system alone, is the rigidity half of the kissing problem in dimension four; it is settled in Theorem 5.25 below by the semidefinite certificate of Theorem 5.30.

The second form replaces the volume by the boundary measure.

Lemma 5.10 (Boundary form)**.** *Let W be a contact configuration with bounded cell, and let $F_w = V_c(W) \cap \{\langle x, w\rangle = 1\}$ for $w \in W$. Then*

$$\operatorname{vol}(V_c(W)) = \tfrac{1}{4} \sum_{w\in W} \operatorname{vol}_3(F_w) = \tfrac{1}{4} \operatorname{area}(\partial V_c(W)), \tag{5.6}$$

so $\operatorname{vol}(V_c(W)) \geq 8$ if and only if the boundary 3-volume of the cell is at least 32.

Proof. The origin is interior to $V_c(W)$, so the cell is the union over its facets of the pyramids $\operatorname{conv}(\{0\} \cup F)$, with pairwise disjoint interiors. Each facet lies on a hyperplane $\langle x, w\rangle = 1$ with $|w| = 1$, at distance exactly 1 from the origin, so the corresponding pyramid has 4-volume $\frac{1}{4}\operatorname{vol}_3(F)$. Summing gives (5.6); the facets of $V_c(W)$ are among the F_w, and any F_w that is not a facet is lower-dimensional and contributes nothing. ∎

At the root configuration all 24 sets F_w are facets, each a regular octahedron of 3-volume $\frac{4}{3}$, and (5.6) reads $8 = \frac{1}{4} \cdot 32$. The next lemma shows why every facet is an octahedron of exactly that size, and the same computation marks the limit of facet-by-facet arguments.

Lemma 5.11 (Facet shape and inball)**.** *Let W be a contact configuration, $w \in W$, and write $g_{ww'} = \langle w, w'\rangle$ and $\widehat{w'} = (w' - g_{ww'}w)/\sqrt{1 - g_{ww'}^2}$ for the unit vector along the orthogonal projection of w' onto $w^\perp$, defined whenever $g_{ww'} \neq \pm 1$. Then, inside the 3-space $w^\perp$,*

$$F_w - w = \Big\{ y \in w^\perp : \langle y, \widehat{w'}\rangle \leq \rho(g_{ww'}) \ \text{ for all } w' \in W \setminus \{w\} \Big\}, \qquad \rho(g) = \sqrt{\frac{1-g}{1+g}}, \tag{5.7}$$

with the convention that $w' = -w$ imposes no constraint. The function ρ is strictly decreasing on $(-1, 1]$ and $\rho(\frac{1}{2}) = 1/\sqrt{3}$, so $F_w - w$ contains the 3-ball of radius $1/\sqrt{3}$ about the origin of $w^\perp$ and

$$\operatorname{vol}_3(F_w) \ \geq \ \frac{4\pi}{3}\Big(\frac{1}{\sqrt{3}}\Big)^3 = \frac{4\pi}{9\sqrt{3}} = 0.806133\ldots \tag{5.8}$$

Proof. Write $x = w + y$ with $y \in w^\perp$. Then $\langle x, w\rangle = 1$ automatically, and for $w' \neq w$ the constraint $\langle x, w'\rangle \leq 1$ reads $g_{ww'} + \langle y, w'\rangle \leq 1$, that is $\langle y, w' - g_{ww'}w\rangle \leq 1 - g_{ww'}$, since $y \perp w$. Dividing by $|w' - g_{ww'}w| = \sqrt{1 - g_{ww'}^2}$ turns this into $\langle y, \widehat{w'}\rangle \leq (1-g)/\sqrt{1-g^2} = \rho(g)$ at $g = g_{ww'}$, which is (5.7). On $(-1, 1]$ the quotient $(1-g)/(1+g)$ is strictly decreasing, hence so is ρ, and $\rho(\frac{1}{2}) = \sqrt{(1/2)/(3/2)} = 1/\sqrt{3}$. Since $g_{ww'} \leq \frac{1}{2}$ for every $w' \neq w$, every constraint in (5.7) sits at distance at least $1/\sqrt{3}$ from the origin of $w^\perp$, so the ball of that radius is contained in $F_w - w$, and (5.8) is its volume. ∎

At the root configuration the neighbours with $g = \frac{1}{2}$ are the eight roots at 60° to α_w; their projections into $w^\perp$ have pairwise inner products in $\{-1, -\frac{1}{3}, \frac{1}{3}\}$, so they are the eight vertex directions of a cube, and the eight constraints $\langle y, \widehat{w'}\rangle \le 1/\sqrt{3}$ cut out exactly the regular octahedron $\{y : |y_1|+|y_2|+|y_3| \le 1\}$ of 3-volume $\frac{4}{3}$, the remaining sixteen constraints being inactive. The bound (5.8) is therefore not attained there: the octahedron circumscribes the ball of radius $1/\sqrt{3}$ and has $1.654\ldots$ times its volume.

Remark 5.12 (Why no facet-by-facet argument can succeed). Combining (5.6) and (5.8) with $|W| \le 24$ gives, for every contact configuration,

$$\mathrm{vol}(V_c(W)) \;\ge\; \frac{|W|}{4} \cdot \frac{4\pi}{9\sqrt{3}} \;=\; \frac{|W|\pi}{9\sqrt{3}},$$

which at the largest possible value $|W| = 24$ is $8\pi/(3\sqrt{3}) = 4.836798\ldots$, short of 8 by more than 3. The inequality $V_c(W) \supseteq B^4$, which follows from $|w| = 1$ for every w and is the same computation run in one step instead of facet by facet, gives slightly more, $\pi^2/2 = 4.934802\ldots$.

Both bounds belong to a class that cannot reach 8, as one example shows. An argument that uses only the pairwise condition $g_{ww'} \le \frac{1}{2}$, and nothing about how the rest of the configuration is arranged, can produce only a bound of the form $\mathrm{vol}_3(F_w) \ge \beta$ with β a universal constant; the largest such constant is $\beta^* = \inf \mathrm{vol}_3(F_w)$, the infimum being over all contact configurations and all w. The root configuration gives $\beta^* \le \frac{4}{3}$, and a square antiprism gives less. Keep all eight tight neighbours but replace the cube by a square antiprism: in an orthonormal frame with $w = e_0$ and $w^\perp = \langle e_1, e_2, e_3\rangle$, take

$$w'_j = \tfrac{1}{2}\, e_0 + \tfrac{\sqrt{3}}{2}\, \widehat{n}_j, \qquad \widehat{n}_j \in \tfrac{1}{\sqrt{3}}\{(\pm\sqrt{2}, 0, 1),\ (0, \pm\sqrt{2}, 1),\ (\pm 1, \pm 1, -1)\}. \tag{5.9}$$

Each $\widehat{n}_j$ is a unit vector, so $\langle w, w'_j\rangle = \frac{1}{2}$; and the $\widehat{n}_j$ have pairwise inner products in $\{\frac{1}{3}, -\frac{1}{3}, \frac{\sqrt{2}-1}{3}, -\frac{\sqrt{2}+1}{3}\}$, so $\langle w'_i, w'_j\rangle = \frac{1}{4} + \frac{3}{4}\langle \widehat{n}_i, \widehat{n}_j\rangle \le \frac{1}{2}$ and the configuration is admissible. By (5.7) the resulting facet is $\{y : \langle y, \sqrt{3}\,\widehat{n}_j\rangle \le 1\}$, and integrating its height over the square $|y_1|, |y_2|$ gives its volume in closed form:

$$\mathrm{vol}_3(F_w) = 16\sqrt{2} - \frac{64}{3} = 1.2940836\ldots \;<\; \frac{4}{3}.$$

Fed into (5.6) together with $|W| \le 24$, the constant β^* gives $\mathrm{vol}(V_c(W)) \ge \frac{|W|}{4}\beta^*$, and this lower bound is at most

$$6\beta^* \;\le\; 96\sqrt{2} - 128 \;=\; 7.764502\ldots$$

Hence no bound of this class reaches 8, and its ceiling lies below the covering estimate of Theorem 5.16. For a configuration with $|W| \le 23$ the same computation returns at most $7.441\ldots$, so the method fails even where the inequality has the most room.

What is missing is the interaction between facets: the direction w' that makes F_w small is itself a direction whose own facet $F_{w'}$ is then constrained, and (5.6) becomes usable only if that trade is accounted for across the configuration. This is the obstruction of Remark 5.3, seen from the boundary.

Remark 5.13 (Two global routes that do not reach the equality case). Two standard tools apply to (5.5), and both lose at the configuration where the inequality is tight, so neither gives the sharp constant.

The polar volume has the integral form $\mathrm{vol}(K^\circ) = \frac{1}{4}\int_{S^3} h_K(\theta)^{-4}\, d\theta$, where h_K is the support function and $d\theta$ is the surface measure of total mass $2\pi^2$. Applying Jensen's inequality to the convex function $t \mapsto t^{-4}$ and the normalised measure gives $\mathrm{vol}(K^\circ) \ge (\pi^2/2)\,\bar{h}^{-4}$

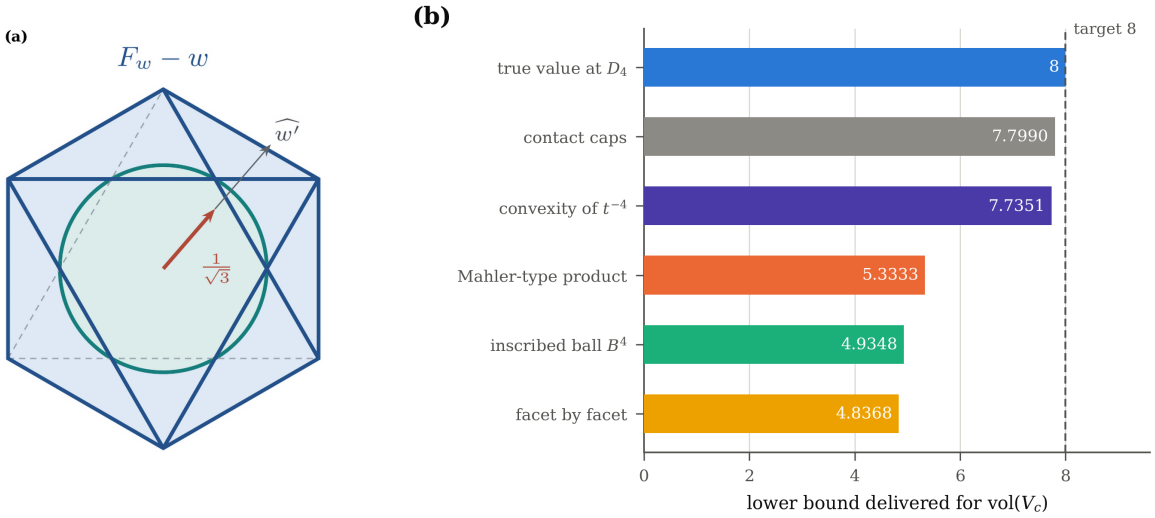

FIGURE 7. (a) A single facet of the reference cell, drawn inside the 3-space orthogonal to its own contact direction. At the root configuration the eight tight neighbours cut out the regular octahedron (Lemma 5.11); the pairwise contact condition on its own guarantees no more than the inscribed ball, and the gap between the two has to be absorbed by the boundary form (5.6). (b) The bounds that the boundary form and the other global routes deliver, all of them evaluated at the D_4 configuration where the inequality is tight (Remarks 5.12 and 5.13).

with $\bar{h}$ the mean of h_K over the sphere, and the target 8 would follow from $\bar{h} \le (\pi^2/16)^{1/4} = 0.886226\ldots$. For $K = \mathrm{conv}(W)$ at the root configuration, $h_K(\theta)$ is $2^{-1/2}$ times the sum of the two largest absolute coordinates of θ, and $\bar{h} = 0.89372\ldots$, evaluated numerically at four million sample directions with a standard error of 2×10^{-5}, so the criterion fails by a small margin. No refinement of the estimate repairs this: Jensen's inequality is an equality only when h_K is constant, that is only for a ball, so at any polytope the bound is strictly below the true volume, and at the root configuration it returns $7.7351\ldots$ against the true 8. No inequality of this shape is sharp at the root configuration.

The Mahler-type route fares worse. Mahler's conjectured lower bound $\mathrm{vol}(K)\,\mathrm{vol}(K^\circ) \ge 4^n/n!$, itself open in dimension four, would give $\mathrm{vol}(K)\,\mathrm{vol}(K^\circ) \ge \frac{32}{3}$; but at the root configuration $\mathrm{conv}(W)$ is the 24-cell of circumradius 1, of volume 2, and the product is 16, so the inequality carries a factor $\frac{3}{2}$ of slack at the one configuration where the bound must be exact, and dividing by $\mathrm{vol}(K) \le 2$ returns only $\frac{16}{3}$. Using $\mathrm{vol}(K) \le \mathrm{vol}(B^4) = \pi^2/2$ instead returns $64/(3\pi^2) = 2.16\ldots$.

Both evaluations, together with the facet-local bound of Remark 5.12, the boundary identity (5.6) and the facet description (5.7), are checked numerically and collected in Figure 7.

Remark 5.14 (What the multi-direction case is). Proposition 5.8 identifies Theorem 1.8 with a named problem. The assertion that every Voronoi cell of a unit-ball packing of $\mathbb{R}^4$ has volume at least that of the regular 24-cell, with equality only at the D_4 configuration, is the 24-cell conjecture. For contact configurations, where every neighbour that can cut the cell touches the centre, Theorem 1.6 settles the part of it in which at most one contact direction leaves a root, and Theorem 1.8 is the remainder, proved in Corollary 5.31 by

the classification of Theorem 5.25 and by nothing else in this paper; what the conjecture asks beyond contact configurations is Conjecture 1.3. The reformulations of this subsection do not make a direct proof easier. They fix the exact shape of the obstacle and exclude three families of approach, and Remark 5.3 excludes a fourth. Partial results towards the conjecture, by a different route, are in [Mus18].

5.4. **A bound from the contact caps.** The boundary form measures the cell from outside, facet by facet. Measuring from the centre outward instead gives a bound for every configuration at once, however many of its contact directions deviate.

Lemma 5.15 (Radial form)**.** *Let W be a contact configuration and let*

$$\delta(\theta) = \min_{w \in W} \arccos\langle \theta, w \rangle, \qquad \theta \in S^3,$$

be the angular distance from θ to the nearest contact direction. Then $V_c(W)$ is bounded if and only if $\delta < \pi/2$ everywhere, and in that case

$$\operatorname{vol}(V_c(W)) \;=\; \frac{1}{4} \int_{S^3} \sec^4(\delta(\theta))\, d\theta, \tag{5.10}$$

where $d\theta$ is the surface measure on S^3, of total mass $2\pi^2$.

Proof. In polar coordinates the volume of a star body with radial function ρ is $\frac{1}{4}\int_{S^3} \rho^4$. The radial function of a polar dual is the reciprocal of the support function of the body, so by Proposition 5.8, $\rho_{V_c(W)}(\theta) = 1/h_{\mathrm{conv}(W)}(\theta)$ with $h_{\mathrm{conv}(W)}(\theta) = \max_w \langle \theta, w \rangle$. Since every w is a unit vector, that maximum is $\max_w \cos \angle(\theta, w) = \cos \delta(\theta)$, which is positive exactly when $\delta(\theta) < \pi/2$; where it fails the cell contains the ray through θ and is unbounded. Substituting $\rho = 1/\cos\delta$ gives (5.10). ∎

So the problem is to show that a set of $m \le 24$ points on S^3 with pairwise angular distance at least 60° cannot be so evenly spread that the mean of $\sec^4 \delta$ drops below $32/(2\pi^2)$. In this form a covering estimate on S^3 already gives most of the bound.

Theorem 5.16 (Covering bound)**.** *Let W be a contact configuration with $|W| = m$ and bounded cell, and let r_m be the unique solution in $(0, \pi/2)$ of*

$$2r - \sin 2r = \frac{2\pi}{m}. \tag{5.11}$$

Then

$$\operatorname{vol}(V_c(W)) \;\ge\; \frac{\pi m}{3} \tan^3 r_m. \tag{5.12}$$

Proof. A geodesic cap of angular radius r in S^3 has surface measure

$$C(r) = \int_0^r (4\pi \sin^2 t)\, dt = \pi\,(2r - \sin 2r),$$

the integrand being the area of the 2-sphere of radius $\sin t$ that the cap meets at angular distance t from its centre; in particular $C(\pi) = 2\pi^2$, the whole measure. The set $\{\delta \le r\}$ is the union of the m caps of radius r about the points of W, so

$$\sigma(\{\delta > r\}) \;\ge\; a(r) := 2\pi^2 - m\,\pi\,(2r - \sin 2r),$$

where σ denotes surface measure. The function $2r - \sin 2r$ has derivative $2 - 2\cos 2r \ge 0$, vanishing only at $r = 0$, and increases from 0 to π on $[0, \pi/2]$; since $m \ge 5$ forces $2\pi/m < \pi$, equation (5.11) has exactly one root r_m there, and $a > 0$ on $[0, r_m)$, $a(r_m) = 0$.

Now $\sec^4\delta \geq 1$, so by the layer-cake formula and the substitution $s = \sec^4 r$, under which $ds = 4\sec^4 r\,\tan r\,dr = d(\sec^4 r)$,

$$\int_{S^3} \sec^4\delta\,d\theta \;=\; 2\pi^2 + \int_1^\infty \sigma(\{\sec^4\delta > s\})\,ds \;=\; 2\pi^2 + \int_0^{\pi/2} \sigma(\{\delta > r\})\,d(\sec^4 r).$$

The integrand is nonnegative, so restricting the range to $[0, r_m]$ and replacing $\sigma(\{\delta > r\})$ by the smaller $a(r)$ only decreases the value. Integrating by parts, with $a(0) = 2\pi^2$, $a(r_m) = 0$ and $a'(r) = -4\pi m \sin^2 r$,

$$\int_0^{r_m} a(r)\,d(\sec^4 r) = \big[a(r)\sec^4 r\big]_0^{r_m} + 4\pi m \int_0^{r_m} \sin^2 r\,\sec^4 r\,dr = -2\pi^2 + 4\pi m \int_0^{r_m} \tan^2 r\,\sec^2 r\,dr,$$

and the last integral is $\frac{1}{3}\tan^3 r_m$. The two occurrences of $2\pi^2$ cancel, leaving $\int_{S^3}\sec^4\delta \geq \frac{4\pi m}{3}\tan^3 r_m$, which with (5.10) is (5.12). ∎

The right-hand side of (5.12) decreases as m grows, since a larger m both multiplies by more and shrinks r_m; the second effect wins. The values for the whole admissible range $5 \leq m \leq 24$, a bounded cell in $\mathbb{R}^4$ needing at least five facets and Proposition 5.8(iii) allowing at most 24, are these.

m	r_m (degrees)	bound	m	r_m (degrees)	bound
5	60.536972	29.042559	15	40.255989	9.536151
6	56.406799	21.440235	16	39.340721	9.227382
7	53.197354	17.503703	17	38.502427	8.962031
8	50.601922	15.119204	18	37.730497	8.731395
9	48.441431	13.525786	19	37.016289	8.528959
10	46.603050	12.387172	20	36.352679	8.349750
11	45.011524	11.533082	21	35.733735	8.189905
12	43.614387	10.868477	22	35.154469	**8.046376**
13	42.373749	10.336286	23	34.610650	7.916728
14	41.261436	9.900225	24	34.098663	7.798989

TABLE 2. The covering bound (5.12) over the whole admissible range of m. It exceeds 8 for every $m \leq 22$ and falls below 8 at $m = 23$. Each r_m is the root of (5.11), computed to the digits shown by bisection on an exactly evaluated monotone function. That the bound exceeds 8 for $5 \leq m \leq 22$ is also checked in Lean, in interval arithmetic: it says $2\rho - \sin 2\rho < 2\pi/m$ at $\rho = \arctan c$, $c^3 = 24/(\pi m)$.

Corollary 5.17 (The local bound for at most 22 contacts)**.** *Every contact configuration W with $|W| \leq 22$ and bounded cell satisfies* $\mathrm{vol}(V_c(W)) \geq 8.0463\ldots > 8$*, with no hypothesis on how many of its directions deviate from roots and no hypothesis on the size of those deviations. Consequently*

$$\mathrm{vol}(V_c) \;\geq\; \tfrac{\pi\cdot 24}{3}\tan^3 r_{24} = 7.798989\ldots$$

holds for every contact configuration, and the local bound $\mathrm{vol}(V_c) \geq 8$ *is left undecided only for configurations with exactly* 23 *or exactly* 24 *contacts.*

Proof. Immediate from Theorem 5.16 and Table 2, using $|W| \leq 24$ from Proposition 5.8(iii) and the monotonicity visible in the table, which over the finite range $5 \leq m \leq 24$ is a comparison of twenty numbers. ∎

Corollary 5.17 is Theorem 1.7 of the introduction, and it settles one point that the deviation picture does not reach. A configuration in which several of the twenty-four contacts are absent, not deviated, is not a perturbation of the root configuration, and the argument of Section 6 says nothing about it. Corollary 5.17 disposes of every such configuration with at most twenty-two contacts, whatever its directions.

Proposition 5.18 (The covering bound is the best area-based bound)**.** *Let* W *be a contact configuration with bounded cell and let* $\Omega_w = \{\theta \in S^3 : \langle \theta, w\rangle \geq \langle \theta, w'\rangle \text{ for all } w' \in W\}$ *be its spherical Voronoi cells, so that* $\delta(\theta)$ *is the distance to* w *on* Ω_w *and* $\sum_w \sigma(\Omega_w) = 2\pi^2$*. Write* ρ_w *for the radius of the cap of the same measure,* $C(\rho_w) = \sigma(\Omega_w)$*. Then*

$$\mathrm{vol}(V_c(W)) \;\geq\; \frac{\pi}{3} \sum_{w \in W} \tan^3 \rho_w, \tag{5.13}$$

and the function $\phi(s) = \tan^3(C^{-1}(s))$ *is strictly convex, with* $\phi'(s) = \frac{3}{4\pi} \sec^4(C^{-1}(s))$*. Consequently the right-hand side of* (5.13) *is smallest, over all ways of distributing the total measure* $2\pi^2$ *among* m *cells, exactly when the cells have equal measure, and its value there is* $\frac{\pi m}{3} \tan^3 r_m$*: the bound of Theorem 5.16.*

Proof. On Ω_w the integrand of (5.10) is $\sec^4 d(\theta, w)$, an increasing function of the distance to w, so its sublevel sets are the caps about w. By the bathtub principle, among all sets of measure $\sigma(\Omega_w)$ the integral is least on the sublevel set of that measure, which is the cap of radius ρ_w; and

$$\int_{\{d(\theta,w) \leq \rho\}} \sec^4 d(\theta, w)\, d\theta = \int_0^{\rho} \sec^4 t \cdot 4\pi \sin^2 t \, dt = 4\pi \int_0^{\rho} \tan^2 t \, \sec^2 t \, dt = \frac{4\pi}{3} \tan^3 \rho.$$

Summing over w and dividing by 4 gives (5.13). For the convexity, $C'(\rho) = 4\pi \sin^2 \rho$, so

$$\phi'(s) = \frac{3 \tan^2 \rho \, \sec^2 \rho}{4\pi \sin^2 \rho} = \frac{3 \sec^4 \rho}{4\pi}, \qquad \rho = C^{-1}(s),$$

which is strictly increasing in ρ and hence in s. Jensen's inequality for a convex function then gives $\sum_w \phi(\sigma(\Omega_w)) \geq m\, \phi(2\pi^2/m)$, with equality only when all the measures agree, and $\phi(2\pi^2/m) = \tan^3 r_m$ by the definition (5.11) of r_m. ∎

So Theorem 5.16 is the strongest conclusion that the areas of the spherical Voronoi cells can support, and at the root configuration those areas are all equal, so no redistribution argument can improve it there. What the estimate gives away is shape. It replaces each cell by the round set of the same measure, and a round set is exactly the worst case for an integrand that grows with the distance. At the root configuration each cell is a spherical polytope with eight facets and circumradius 45°, while the cap of the same measure has radius 34.0986…°; that difference is the whole of the remaining 0.2010. Any improvement must therefore begin by proving that a spherical Voronoi cell of a 60°-separated set cannot be round, which is a statement about the shape of the cells and not about their areas.

Remark 5.19 (The shape is not local to a cell). Working one cell at a time fails in both decompositions that the configuration provides.

On Ω_w the integrand of (5.10) is $\sec^4 d(\theta, w)$, so the Voronoi decomposition localises the identity exactly, and $\operatorname{vol}(V_c(W)) \geq 8$ would follow from

$$\int_{\Omega_w} \sec^4 d(\theta, w)\, d\theta \;\geq\; \frac{16}{\pi^2}\, \sigma(\Omega_w) \tag{5.14}$$

at every w, that is from the mean of $\sec^4$ over each cell being at least $16/\pi^2 = 1.621138\ldots$. At the root configuration every cell attains that value, so (5.14) would be sharp; but it does not hold for every configuration. A direction may carry nine contacts at $60°$: writing $w' = \frac{1}{2}w + \frac{\sqrt{3}}{2}n$ with n a unit vector orthogonal to w, the condition $\langle w', w''\rangle \leq \frac{1}{2}$ becomes $\langle n', n''\rangle \leq \frac{1}{3}$, and nine unit vectors of $\mathbb{R}^3$ with pairwise inner products at most $\frac{1}{3}$ exist, by the case of nine points in Schütte and van der Waerden's determination of the optimal spherical codes of small size [SW51]. The cell of w is then smaller than any cell of the root configuration, and the mean of $\sec^4$ over it is $1.5732\ldots$; moving the nine contacts out to $61°$, so that every inner product is strictly below $\frac{1}{2}$, still gives $1.5954\ldots$. Enlarging the configuration only shrinks Ω_w and lowers the mean, so every completion violates (5.14) there.

The Delaunay decomposition does not localise. Inside a Delaunay cell the nearest contact direction need not be a vertex of that cell, so $\int_T \sec^4 \delta$ is not a functional of T; on a 22-point configuration there are directions whose nearest contact lies outside the cell containing them and beats every vertex of that cell by more than 0.1 in cosine. Even the functional it would have to be, the distance to the cell's own vertices, fails the corresponding inequality. For the regular spherical simplex of edge $60°$, parametrised by the barycentric coordinates λ of the cone it spans, the identities $|x|^2 = \frac{1}{2}(1+q)$ with $q = \sum_i \lambda_i^2$ and $\langle x, v_i\rangle = \frac{1}{2}(1+\lambda_i)$ turn the mean of $\sec^4$ into

$$\frac{4\,\mathbb{E}\big[(1+M)^{-4}\big]}{\mathbb{E}\big[(1+q)^{-2}\big]} = \frac{4/5}{0.518254\ldots} = 1.543643\ldots, \qquad \mathbb{E}\big[(1+M)^{-4}\big] = \tfrac{1}{5}, \tag{5.15}$$

M being the largest coordinate and both expectations taken over a uniformly distributed λ. The value falls short of $16/\pi^2$ by $0.077495\ldots$, which is 4.78%. That simplex has circumradius $37.7612\ldots°$, so Proposition 5.38 below does not exclude it from a saturated configuration, and it does occur as a Delaunay cell of a contact configuration; at edge $62°$, where every inner product is strictly below $\frac{1}{2}$, the mean is still $1.5955\ldots$. It occurs nowhere in the root system, whose $60°$ graph has clique number three. Thus the locally densest arrangement is not one the extremal configuration contains.

So the gain over Proposition 5.18 that a proof needs is not additive over cells. Cells worse than round exist in both decompositions, and what makes the root configuration extremal is the way its cells fit together, not the shape of any one of them. All of this is checked numerically, with integrals over S^3 by Monte Carlo; the margins above are percentages, not last digits.

Remark 5.20 (What the estimate leaves). Two contact counts survive Corollary 5.17, and they are of different kinds. At $m = 24$ the configuration is a full contact shell and the question is the one Theorem 1.8 asks, whether two or more of the twenty-four directions can leave the roots; the estimate returns $7.7989\ldots$ there, a shortfall of 0.2010. At $m = 23$ the configuration need not be a deviation of the root system, so Theorem 1.8 does not reach it and Corollary 5.17 stops one short; the estimate returns $7.9167\ldots$, a shortfall of 0.0833. Section 5.5 disposes of the first count and reduces the second to the configurations with bounded cell, and Sections 5.11 and 5.12 prove the bound for those.

Proposition 5.21 (Every cell reaches out to $37.7612\ldots^\circ$)**.** *Let W be a contact configuration with bounded cell. Then every vertex of every spherical Voronoi cell Ω_w is at angular distance at least $R_0 = \arccos\sqrt{5/8} = 37.7612\ldots^\circ$ from w, and in particular the covering radius of W on S^3 is at least R_0.*

Proof. A vertex V of Ω_w is a point of S^3 at which at least four cells meet, so there are at least four points of W, one of them w, at the common distance $R = d(V, w)$ from V and none nearer. Let $u_1, \ldots, u_4$ be unit vectors in the tangent space at V pointing along the geodesics from V to four of them. Four unit vectors in $\mathbb{R}^3$ cannot have all six pairwise inner products below $-\frac{1}{3}$, since $0 \le \left|\sum_i u_i\right|^2 = 4 + 2\sum_{i<j}\langle u_i, u_j\rangle$ forces $\sum_{i<j}\langle u_i, u_j\rangle \ge -2$ and there are six terms. Take a pair with $\langle u_i, u_j\rangle \ge -\frac{1}{3}$. The two corresponding points of W lie at distance R from V in directions at angle α with $\cos\alpha \ge -\frac{1}{3}$, so by the spherical law of cosines their separation γ satisfies

$$\cos\gamma = \cos^2 R + \sin^2 R\,\cos\alpha \;\ge\; \cos^2 R - \tfrac{1}{3}\sin^2 R.$$

Packing validity gives $\gamma \ge 60^\circ$, hence $\cos\gamma \le \frac{1}{2}$, so $\cos^2 R - \frac{1}{3}(1-\cos^2 R) \le \frac{1}{2}$ and $\cos^2 R \le \frac{5}{8}$. The farthest point of S^3 from W is a vertex of the cell containing it, so the covering radius is at least R_0 as well. ∎

This improves on what Theorem 5.16 extracts along the way, since that argument stops at $r_{24} = 34.0986\ldots^\circ$, but it does not help the bound. Knowing that the survival function is positive out to R_0 bounds it below only by the measure of a cap of radius $R_0 - r$ about one deep hole, and feeding that into the integral changes the constant in the fifth decimal: 7.79899 becomes 7.79902, and even allowing one such hole per cell and treating them as disjoint gives only 7.79966. The missing 0.2003 is a statement about how much of the sphere lies far from W, not about how far the farthest point is.

The overlaps the estimate discards can be put back exactly, and the result shows how much room the method has left.

Lemma 5.22 (Three caps do not meet)**.** *Three directions of a contact configuration have circumradius at least $\arccos\sqrt{2/3} = 35.2644\ldots^\circ$ on their span. Consequently, if $r < \arccos\sqrt{2/3}$, no point of S^3 lies within r of three contact directions at once.*

Proof. Let w_1, w_2, w_3 be the three directions, and let $p \in S^3$ satisfy $\langle p, w_i\rangle \ge \cos R$ for each i, for some $R \le \frac{\pi}{2}$; the point of their span equidistant from all three, at angular distance R, is one such point, and so is any point within R of all three. Since $\langle w_i, w_j\rangle \le \frac{1}{2}$ for $i \ne j$, $|w_1 + w_2 + w_3|^2 \le 3 + 3 = 6$, and the Cauchy–Schwarz inequality gives

$$3\cos R \;\le\; \langle p, w_1 + w_2 + w_3\rangle \;\le\; \sqrt{6},$$

so $\cos R \le \sqrt{2/3}$. This proves both statements. ∎

Proposition 5.23 (Putting the overlaps back)**.** *Let W be a contact configuration with $|W| = m$ and bounded cell, write γ_{ij} for the angle between w_i and w_j, and let*

$$\Lambda(r, \gamma) = 4\pi \int_{\gamma/2}^{r} \sin^2 t \left(1 - \tan\tfrac{\gamma}{2}\,\cot t\right) dt \qquad (r \ge \tfrac{\gamma}{2}), \tag{5.16}$$

the measure of the intersection of two caps of radius r whose centres are γ apart, and $\Lambda(r, \gamma) = 0$ for $r < \gamma/2$. Then for every $R \le r_* := \arccos\sqrt{2/3} = 35.2644\ldots^\circ$,

$$\operatorname{vol}(V_c(W)) \;\ge\; \frac{1}{4}\Big[2\pi^2 + \int_0^R (2\pi^2 - mC(r))\, d(\sec^4 r)\Big] \;+\; \frac{1}{4}\sum_{i<j}\int_0^R \Lambda(r, \gamma_{ij})\, d(\sec^4 r). \tag{5.17}$$

At $R = r_m$ the first bracket is the covering bound $\frac{\pi m}{3}\tan^3 r_m$ of Theorem 5.16, and $r_m < r_*$ for every $m \geq 23$. Below r_* the right-hand side is not an estimate: by Lemma 5.22 the caps of radius r meet only in pairs there, so $\sigma(\{\delta \leq r\}) = mC(r) - \sum_{i<j}\Lambda(r,\gamma_{ij})$ exactly, and (5.17) is the true value of the integral truncated at R, that is the volume of $V_c(W)$ inside the ball of radius $\sec R$; the largest admissible R is r_*, where the ball has radius $\sqrt{3/2}$.

Proof. For Λ, split the lens by the hyperplane bisecting w_i and w_j; the half on the w_i side is the part of the cap about w_j lying beyond that hyperplane, which is at angular distance $\gamma/2$ from w_j. Parametrise the cap about w_j by the distance t and a direction on the 2-sphere it meets at distance t, of measure $4\pi\sin^2 t\,dt$; the coordinate of that direction along the bisector normal is uniform on $[-1,1]$, and the half-space condition reads $s \geq \cot t\,\tan(\gamma/2)$, which is vacuous for $t \leq \gamma/2$ and otherwise excludes a fraction $\frac{1}{2}(1+\cot t\tan\frac{\gamma}{2})$. Doubling gives (5.16). The rest is the layer-cake step in the proof of Theorem 5.16, run with the exact survival function on $[0,R]$ and with the integrand discarded above R: writing $\sec^4\min(\delta,R) = 1+\int_0^R \mathbf{1}[\delta>r]\,d(\sec^4 r)$ and integrating over S^3 gives the bracket, and the survival function is $2\pi^2 - mC(r) + \sum_{i<j}\Lambda(r,\gamma_{ij})$ on $[0,R]$ by Lemma 5.22. ∎

At the configuration obtained by deleting one root, where $m = 23$ and the 88 close pairs are all at exactly 60°, the right-hand side of (5.17) is

$$7.997885\ldots \text{ at } R = r_{23}, \qquad 8.034340\ldots \text{ at } R = r_*, \tag{5.18}$$

against $7.916728\ldots$ for the plain estimate and $\frac{25}{3}$ for the true volume. The first stops the integration where the covering estimate stops, at the radius $r_{23} = 34.6106\ldots°$ at which the caps would exactly exhaust the sphere if they were disjoint, and it falls short of 8 by $0.002115\ldots$. The second carries the same identity to its limit r_*, the smallest circumradius of three contact directions, and passes 8 with $0.034340\ldots$ to spare. So the pairwise estimate is not blocked at the deletion configuration; Proposition 5.41 works out what it asks of an arbitrary configuration of 23 directions, and Section 5.11 continues from there. At $m = 24$ the same computation gives $7.858738\ldots$ at $R = r_{24}$ and $7.906940\ldots$ at $R = r_*$, against the true value 8; there the truncation discards the deep holes of the root system, which sit at 45°.

Remark 5.24 (What the constant is and is not). Two facts limit what $7.798989\ldots$ says.

It is not a density bound. A lower bound v for every cell would give packing density at most $\mathrm{vol}(B^4)/v$, and $7.798989\ldots$ would give $0.632749\ldots$, below the three-point bound 0.63611 of [CdLS22]; but the constant holds only for cells whose active neighbours all touch, and a general packing has other cells. For every cell the paper proves 7.7742 (Theorem 8.2). Unlike a density bound, the covering estimate concerns one cell at a time, as Theorem 1.8 requires, and so it settles the range $m \leq 22$ outright.

The only fact about the configuration that the estimate uses is that m caps cannot cover more than m times the area of one cap. Proposition 5.23 shows how much of the loss pair data recover: almost all of it at $m = 23$ and little of it at $m = 24$. What neither version reaches is the tail: both stop at r_m, while the survival function stays positive out to the covering radius, 45° at the root configuration. Going further means a lower bound on the covering radius better than r_m, and Proposition 5.21 shows that this changes the bound only in the fifth decimal. The rest of the tail is a statement about how much of the sphere lies far from W, and it is the same statement as the rigidity of the kissing arrangement.

Everything in this subsection, the cap-area formula, the radial identity, the layer-cake step, the closed form, the whole of Table 2 including its monotonicity, and the absence of

any violation of (5.12) at the root configuration, at its subsets and at random configurations, is checked numerically, and so are Lemma 5.22, (5.16) and the two values above.

5.5. **Twenty-four contacts.** At the top of the range the problem changes, because a contact configuration of maximal size is rigid.

Theorem 5.25 (Twenty-four contacts)**.** *Let W be a contact configuration with $|W| = 24$. Then W is the image of the 24 normalised roots of D_4 under an orthogonal map, and consequently $V_c(W)$ is a regular 24-cell and $\mathrm{vol}(V_c(W)) = 8$.*

The theorem was first proved by de Laat, Leijenhorst and de Muinck Keizer [LLM24, Thm. 5.3], by a semidefinite computation. It is proved here in full. The proof has two halves. The first is an inequality: a contact configuration has at most 24 elements, and one with exactly 24 has all its pairwise inner products in $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$. That is Theorem 5.30 below, and it is a consequence of an explicit positive kernel on the two-element subsets of S^3. The kernel was found by the authors of [LLM24] and deposited with their paper [LLM24d]; what is proved here is that any kernel with its properties bounds the size of a contact configuration (Lemma 5.27), that a kernel built from positive definite blocks in the way theirs is built is positive (Lemma 5.28), that the deposited data have every property required, which is a finite computation carried out in Section 5.5.3, and that at the value 24 the bound forces the inner products. No statement of [LLM24] is used; what is taken from that work is the certificate itself, which is data, and the construction of the kernels in which it is expressed, which is a definition. The second half is classical, and is the following lemma.

Lemma 5.26 (From the inner products to the root system)**.** *Let W be a set of 24 unit vectors in $\mathbb{R}^4$ whose pairwise inner products lie in $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$. Then $\sqrt{2}\,W$ is the set of roots of a copy of the D_4 root lattice, and W is the image of the normalised roots of D_4 under an orthogonal map.*

Proof. The vectors of $\sqrt{2}\,W$ have norm 2 and integer inner products, so they generate an integral lattice $L \subset \mathbb{R}^4$ of rank at most 4, and they are among the vectors of norm 2 of L. The vectors of norm 2 of an integral lattice form a root system, simply laced, since two of them have inner product in $\{0, \pm 1, \pm 2\}$ and the reflection in one of them preserves the lattice [CS88, Ch. 4], and the lattice they generate is a root lattice, an orthogonal sum of lattices of the types A_n, D_n and E_n by the classification of simply laced root systems [Hum90, Ch. 2]; it has a basis of roots, its simple roots. Since $\sqrt{2}\,W$ generates L, the lattice L is such a root lattice, of rank $r \le 4$, and $|\sqrt{2}\,W| = 24$ is at most the number of roots of L. The root lattices of rank at most 4 and their numbers of roots are A_1, A_2, A_3, A_4 with 2, 6, 12, 20, D_4 with 24, and the orthogonal sums A_1^2, A_1A_2, A_1^3, A_1A_3, A_2^2, $A_1^2A_2$, A_1^4 with 4, 8, 6, 14, 12, 10, 8. Only D_4 reaches 24, so L is isometric to the D_4 lattice and $\sqrt{2}\,W$ is its full set of roots. The D_4 root system is unique up to isometry, which gives the last statement.

The finite content of this argument can also be checked without the classification. A lattice generated by norm-2 vectors has a basis of such vectors, whose Gram matrix has 2 on the diagonal and -1, 0 or 1 off it. Of order 1 to 4 there are 1, 3, 23 and 393 positive definite matrices of that shape. Counting the integer solutions of $x^{\mathsf{T}}Gx = 2$ for each, exactly and inside a box that provably contains them all, gives the largest counts 2, 6, 12 and 24. The count 24 occurs only at determinant 4, with each root having 8 neighbours at inner product 1, 6 at 0, 8 at -1 and one at -2; this is D_4. The next count at rank 4 is 20, at determinant 5, and belongs to A_4. The enumeration was carried out in exact integer arithmetic and repeated in Lean, whose kernel also exhibits, in every lattice with 24 roots,

four roots with the Cartan matrix of D_4 as Gram matrix, so that the lattice contains a copy of the D_4 lattice of its own determinant and is that lattice. ∎

5.5.1. *The bound of the second level, and its equality case.* For $t \geq 0$ let $\mathcal{I}_t$ be the set of subsets of S^3 of size at most t whose pairwise inner products are at most $\frac{1}{2}$; the empty set belongs to $\mathcal{I}_t$, and a contact configuration is an element of $\mathcal{I}_t$ for t large enough. Every subset of an element of $\mathcal{I}_t$ is again one. For a symmetric real function K on $\mathcal{I}_2 \times \mathcal{I}_2$ and a set $Q \in \mathcal{I}_4$ put

$$A_2K(Q) \;=\; \sum_{\substack{J_1,J_2\in\mathcal{I}_2\\ J_1\cup J_2=Q}} K(J_1,J_2), \tag{5.19}$$

the sum running over ordered pairs, so that $A_2K(\emptyset) = K(\emptyset,\emptyset)$. Call K *positive* if for every finite family $J_1,\dots,J_m$ of elements of $\mathcal{I}_2$ the matrix $\big(K(J_i,J_j)\big)_{i,j}$ is positive semidefinite.[11] This is the second level of the hierarchy that de Laat and Vallentin [dLV15] set up for packing problems in discrete geometry, and the bound it gives is the following; we include the short proof because the equality case, which the theorem needs, is read off from it.[12]

Lemma 5.27 (The bound of the second level)**.** *Let K be positive, and suppose that $A_2K(\{x\}) \leq -1$ for every $x \in S^3$ and that $A_2K(Q) \leq 0$ for every $Q \in \mathcal{I}_4$ with $|Q| \geq 2$. Then every contact configuration C satisfies*

$$|C| \;\leq\; K(\emptyset,\emptyset).$$

If $|C| = K(\emptyset,\emptyset)$, then $A_2K(\{x\}) = -1$ for every $x \in C$ and $A_2K(Q) = 0$ for every $Q \subseteq C$ with $2 \leq |Q| \leq 4$.

Proof. The set C is finite, since its points are at pairwise angular distance at least $\pi/3$ on a compact sphere, so we may argue with finite sums. Let $\mathcal{J}$ be the family of subsets of C of size at most 2; it lies in $\mathcal{I}_2$. Positivity of K on this family, with every coefficient equal to 1, gives

$$0 \;\leq\; \sum_{J_1,J_2\in\mathcal{J}} K(J_1,J_2) \;=\; \sum_{\substack{Q\subseteq C\\ |Q|\leq 4}} A_2K(Q),$$

the second expression being the first with the pairs grouped by their union Q, which is a subset of C of size at most 4 and hence an element of $\mathcal{I}_4$. The term $Q = \emptyset$ is $K(\emptyset,\emptyset)$. The $|C|$ terms with $|Q| = 1$ are each at most -1, and every term with $|Q| \geq 2$ is at most 0. Hence $0 \leq K(\emptyset,\emptyset) - |C|$, which is the bound. If $|C| = K(\emptyset,\emptyset)$ the chain

$$0 \;\leq\; \sum_{Q} A_2K(Q) \;=\; K(\emptyset,\emptyset) + \sum_{|Q|=1} A_2K(Q) + \sum_{|Q|\geq 2} A_2K(Q) \;\leq\; K(\emptyset,\emptyset) - |C| + 0 \;=\; 0$$

[11] The hierarchy of [dLV15] works with continuous positive kernels on the compact space $\mathcal{I}_2$, because it is stated for an optimisation problem whose feasible set has to be closed. For the bound on one given code only the finite Gram matrices above enter, so continuity is not needed and is not assumed here; the certificate kernel is in fact a polynomial in inner products and is continuous.

[12] The bound is the generalisation to four points of the Delsarte linear programming bound and of the three-point bound of Bachoc and Vallentin [BV08]: with $\mathcal{I}_1$ in place of $\mathcal{I}_2$ the same argument gives $|C| \leq K(\emptyset,\emptyset)$ for a positive kernel on the one-element subsets of S^3, which is the Delsarte bound in the form of [DGS77], and the equality argument is the one that identifies the extremal codes there, for instance in [CK07]. The only thing that changes with the level is the set of subsets over which the pairs are grouped.

collapses, so the two sums take the values $-|C|$ and 0 that bound them; since each singleton term is at most -1 and each remaining term at most 0, every singleton term equals -1 and every remaining term equals 0.[13] ∎

The kernels to which the lemma is applied have a particular form, and their positivity is a matter of that form alone.

Lemma 5.28 (Gram kernels are positive)**.** *Let Λ be a finite set, let H be a real or complex inner product space, and for each $\lambda \in \Lambda$ let $\psi_{\lambda,1}, \dots, \psi_{\lambda,n_\lambda}$ be maps from $\mathcal{I}_2$ to H. Put*

$$Z_\lambda(J_1, J_2)_{ab} \;=\; \langle \psi_{\lambda,a}(J_1), \psi_{\lambda,b}(J_2) \rangle,$$

the inner product being conjugate linear in its first argument when H is complex, and let X_λ be a real symmetric positive semidefinite matrix of order n_λ for each λ. Then the function

$$K(J_1, J_2) \;=\; \sum_{\lambda \in \Lambda} \sum_{a,b=1}^{n_\lambda} (X_\lambda)_{ab}\, Z_\lambda(J_1, J_2)_{ab}$$

satisfies $K(J_2, J_1) = \overline{K(J_1, J_2)}$, and its real part is a positive kernel on $\mathcal{I}_2$; when K is real valued, K itself is one.[14]

Proof. Write $X_\lambda = R_\lambda^{\mathsf{T}} R_\lambda$ with R_λ real, which is possible for a real symmetric positive semidefinite matrix. For $J_1, \dots, J_m \in \mathcal{I}_2$ and real numbers $c_1, \dots, c_m$,

$$\begin{aligned} \sum_{i,j} c_i c_j\, K(J_i, J_j) &= \sum_{\lambda} \sum_{c} \Big\langle \sum_{i,a} c_i (R_\lambda)_{ca} \psi_{\lambda,a}(J_i),\; \sum_{j,b} c_j (R_\lambda)_{cb} \psi_{\lambda,b}(J_j) \Big\rangle \\ &= \sum_{\lambda,c} \Big\| \sum_{i,a} c_i (R_\lambda)_{ca} \psi_{\lambda,a}(J_i) \Big\|^2 \;\geq\; 0, \end{aligned}$$

using $(X_\lambda)_{ab} = \sum_c (R_\lambda)_{ca} (R_\lambda)_{cb}$ and the sesquilinearity of the inner product. The symmetry $K(J_2, J_1) = \overline{K(J_1, J_2)}$ follows from $\langle v, w \rangle = \overline{\langle w, v \rangle}$ and the symmetry of X_λ. The displayed quantity is real and nonnegative, so it equals its real part, which is $\sum_{i,j} c_i c_j \operatorname{Re} K(J_i, J_j)$; that is positivity of $\operatorname{Re} K$, and of K when the two agree. ∎

5.5.2. *The certificate kernel.* The kernel that settles twenty-four points is of the form in Lemma 5.28, with the index set Λ the sixty signatures $\lambda = (\lambda_1, \lambda_2)$, $\lambda_1 \geq \lambda_2 \geq 0$, $\lambda_1 + \lambda_2 \leq 14$, of irreducible representations of $\mathrm{O}(4)$ other than $(1,1)$, $(3,3)$, $(5,5)$ and $(7,7)$,[15] and with the maps $\psi_{\lambda,a}$ the functions that [LLM24, §2–3] attach to a subset J of S^3 of size at most 2: an irreducible representation ρ_λ of $\mathrm{GL}(2, \mathbb{C})$ on a space W_λ carrying a Hermitian inner product, a fixed vector $w_k \in W_\lambda$ for each admissible index $a = (i, j, k)$, a section $s(J) \in \mathrm{O}(4)$ carrying a reference configuration ϵ onto J, a fixed 2×4 complex matrix ω,

[13] An exact bound identifies the configuration by this argument alone, which uses nothing about C beyond $|C| = K(\emptyset, \emptyset)$, nothing about the kernel beyond positivity and the two families of inequalities, and not its optimality. What it needs, and what a bound such as the three-point value 24.047 can never give, is that $K(\emptyset, \emptyset)$ is attained by a code; hence the certificate must have objective value 24, not merely a value below 25, and its rounding to an exact rational point with objective exactly 24 is the decisive step of its construction.

[14] Nothing about the maps $\psi_{\lambda,a}$ is used except that they take values in an inner product space, so the positivity of the certificate kernel does not depend on the representation theory that produces the zonal matrices, only on the zonal matrices being Gram matrices, which is their definition (5.21). The representation theory makes the entries of Z_λ polynomials in the inner products and makes every invariant positive kernel expressible in this form; the second of these properties, the completeness of the family, is needed by an optimiser and not here, since only one kernel is ever used and it is given.

[15] Of the sixty-four pairs with $\lambda_1 \geq \lambda_2 \geq 0$ and $\lambda_1 + \lambda_2 \leq 14$, the construction of [LLM24, §3] leaves out the four with $\lambda_1 = \lambda_2$ odd, and the deposited data carry a block for each of the other sixty.

and a scalar $\xi_{\lambda,a}(J)$ which is a polynomial in the inner product of the two points of J times powers of $\sqrt{2 \pm 2\langle x, y\rangle}$, and then

$$\psi_{\lambda,a}(J)\colon\ \gamma \ \longmapsto\ \xi_{\lambda,a}(J)\,\rho_\lambda(\omega\gamma\, s(J)\,\epsilon)\,w_k, \qquad \gamma \in \mathrm{O}(4), \tag{5.20}$$

an element of the space $H = L^2(\mathrm{O}(4), W_\lambda)$ of square integrable W_λ-valued functions on $\mathrm{O}(4)$ with the inner product $\langle f_1, f_2\rangle = \int_{\mathrm{O}(4)} \langle f_1(\gamma), f_2(\gamma)\rangle\, d\gamma$ against Haar probability measure. The *zonal matrices* are the Gram matrices

$$\begin{aligned} Z_\lambda(J_1, J_2)_{ab} &= \langle \psi_{\lambda,a}(J_1), \psi_{\lambda,b}(J_2)\rangle \\ &= \xi_{\lambda,a}(J_1)\,\xi_{\lambda,b}(J_2) \int_{\mathrm{O}(4)} \big\langle \rho_\lambda(\omega\gamma s(J_1)\epsilon)w_{k_1},\ \rho_\lambda(\omega\gamma s(J_2)\epsilon)w_{k_2}\big\rangle\, d\gamma, \end{aligned} \tag{5.21}$$

which are invariant under the diagonal action of $\mathrm{O}(4)$ on (J_1, J_2) because Haar measure is, and whose entries are polynomials in the inner products among the points of $J_1 \cup J_2$; the scalars ξ are chosen so that this is so, and the $\sqrt{2 \pm 2u}$ they carry absorb the square roots that the section $s(J)$ introduces. This is the construction of [LLM24], which extends the one of [dLV15] and of the equiangular lines computations that preceded it, and it is recomputed from scratch in Section 5.5.3. The *certificate kernel* is

$$K(J_1, J_2) \ = \sum_{\lambda \in \Lambda} \sum_{a,b} (X_\lambda)_{ab}\, Z_\lambda(J_1, J_2)_{ab}, \tag{5.22}$$

where the X_λ are the sixty rational matrices of the deposited data [LLM24d]. Its values are rational polynomials in the inner products, so it is real valued, and it is $\mathrm{O}(4)$-invariant. For $Q \in \mathcal{I}_4$ the quantity $A_2K(Q)$ of (5.19) is therefore a polynomial in the inner products among the points of Q; call it p_1, $p_2(u)$, $p_3(u_1, u_2, u_3)$ and $p_4(u_1, \dots, u_6)$ according to $|Q| = 1, 2, 3, 4$, the u being the inner products of the pairs of Q in a fixed order. Since a set of at most four points of $\mathbb{R}^4$ is determined up to an orthogonal map by its Gram matrix, these polynomials determine A_2K on all of $\mathcal{I}_4$.[16]

Proposition 5.29 (What the verification establishes)**.** *The following statements about the deposited data hold, each by an exact computation described in Section 5.5.3.*

(i) *Each of the sixty matrices X_λ, and each of the sixty-seven further matrices M that enter the sums of squares of* (iii)*, is a real symmetric positive definite matrix with rational entries.*

(ii) $K(\emptyset, \emptyset) = 24$.

(iii) *There are polynomials $\sigma_1, \dots, \sigma_4$, each a sum of terms $w \cdot \langle M,\, m\, m^{\mathsf{T}}\rangle$ with M one of the matrices of* (i)*, m a vector of monomials in the inner products of Q, and w a weight which is a nonnegative constant, or $(u+1)(\frac{1}{2}-u)$ for an inner product u of Q, or the determinant of the Gram matrix of Q, or an elementary symmetric function of the family of the $(u+1)(\frac{1}{2}-u)$ over the inner products of Q, or, for $|Q| = 4$, an elementary symmetric function of the four 3×3 principal minors of the Gram matrix of Q, such that the four identities*

$$p_1 + 1 + \sigma_1 = 0, \qquad p_2 + \sigma_2 = 0, \qquad p_3 + \sigma_3 = 0, \qquad p_4 + \sigma_4 = 0$$

hold identically as polynomials in the inner products.

[16] Four points of $\mathbb{R}^4$ with a given Gram matrix G are the columns of a 4×4 matrix P with $P^{\mathsf{T}}P = G$, and any two such matrices differ by an orthogonal factor on the left, since $P_2 = UP_1$ with U orthogonal whenever $P_1^{\mathsf{T}}P_1 = P_2^{\mathsf{T}}P_2$; for fewer than four points the same holds after adding zero columns. This is why the constraints of the second level are polynomial identities in one, three and six inner products and why the domain of each is described by the positivity of a Gram matrix. It is also why the level stops at four: five points of $\mathbb{R}^4$ have Gram matrices constrained by a determinant condition, and the third level would need it.

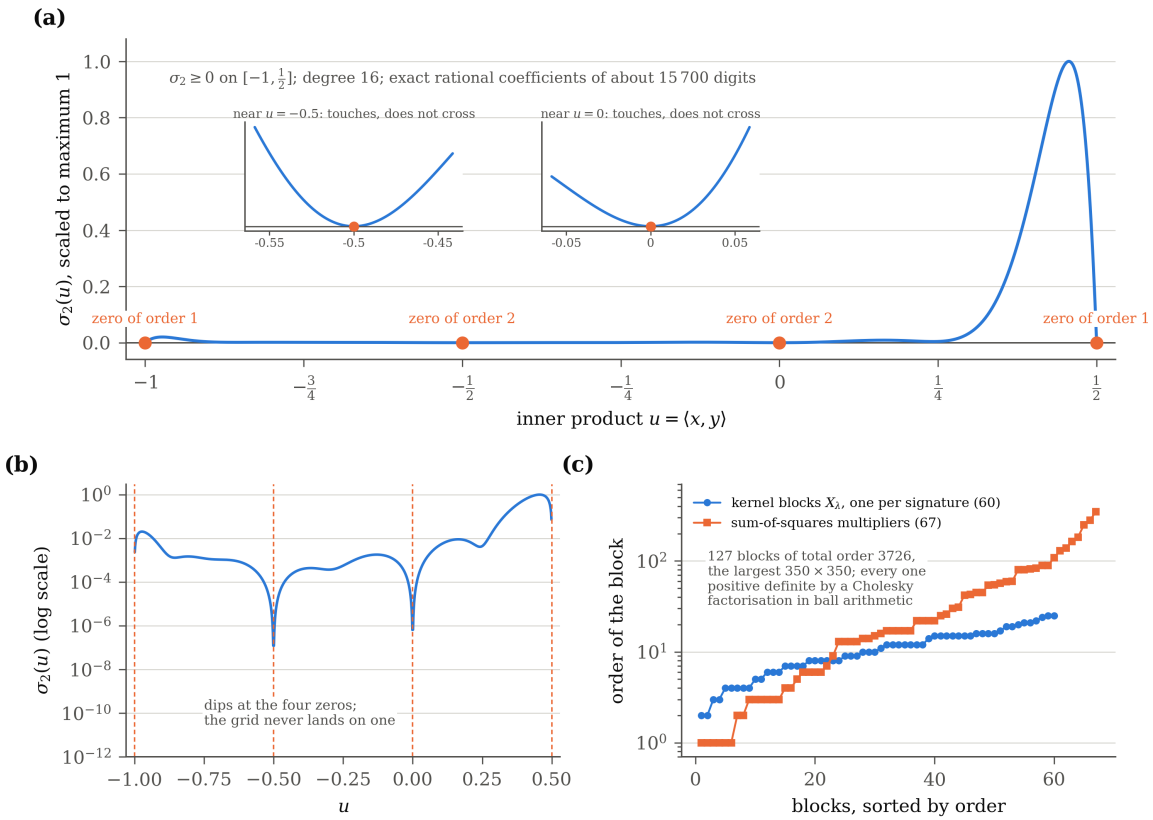

FIGURE 8. The certificate, as Proposition 5.29 reads it. (a) The two-point polynomial σ_2 on $[-1, \frac{1}{2}]$, evaluated exactly from the deposited rational coefficients and scaled by its maximum: it is nonnegative, it vanishes to first order at -1 and $\frac{1}{2}$ and to second order at $-\frac{1}{2}$ and 0, and nowhere else; the insets show the two double zeros, where the curve touches the axis without crossing it. (b) The same on a logarithmic scale. (c) The orders of the 127 positive definite blocks, the sixty kernel blocks X_λ and the sixty-seven sum-of-squares multipliers; the largest is 350×350.

(iv) *The polynomial σ_2 has degree* 16*, and its zeros in* $[-1, \frac{1}{2}]$ *are exactly* -1*,* $-\frac{1}{2}$*,* 0 *and* $\frac{1}{2}$*, of multiplicities* 1*,* 2*,* 2 *and* 1*.*

Every weight in (iii) is nonnegative at every $Q \in \mathcal{I}_4$: the inner products of Q lie in $[-1, \frac{1}{2}]$, so each $(u+1)(\frac{1}{2}-u)$ is nonnegative; the Gram matrix of Q is positive semidefinite, so its determinant and its principal minors are nonnegative; and an elementary symmetric function of nonnegative quantities is nonnegative. Each $\langle M, mm^{\mathsf{T}} \rangle = m^{\mathsf{T}} M m$ is nonnegative because M is positive definite. Hence $\sigma_k(Q) \geq 0$ on $\mathcal{I}_4$ for $k = 1, \ldots, 4$, and (iii) says

$$A_2K(\{x\}) = -1 - \sigma_1 \leq -1, \qquad A_2K(Q) = -\sigma_{|Q|}(Q) \leq 0 \quad (2 \leq |Q| \leq 4), \tag{5.23}$$

for every $x \in S^3$ and every $Q \in \mathcal{I}_4$ with $|Q| \geq 2$.

Theorem 5.30 (Twenty-four points)**.** *Let $C \subset S^3$ be a set whose pairwise inner products are at most $\frac{1}{2}$. Then $|C| \leq 24$. If $|C| = 24$, then every pairwise inner product of C lies in $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$.*

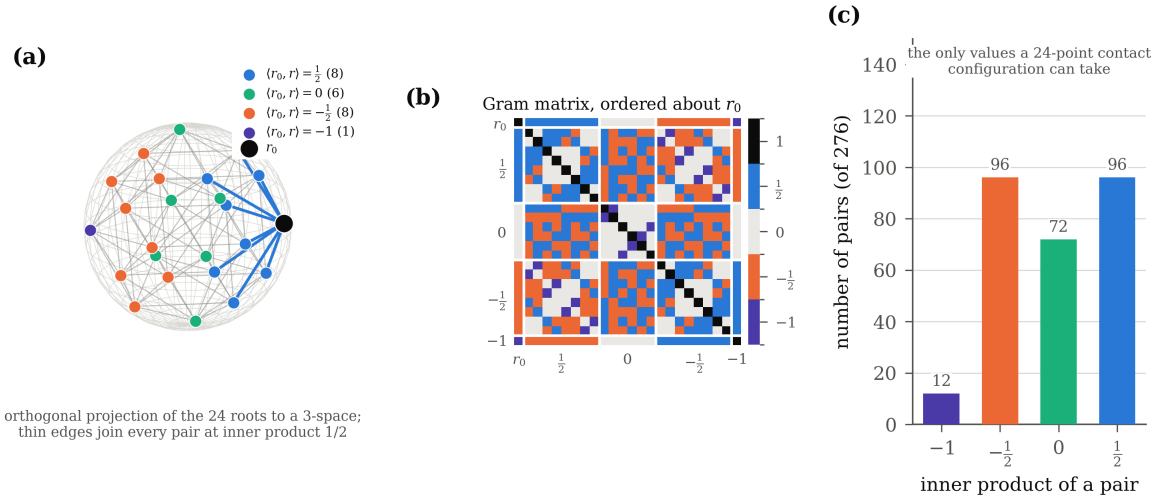

FIGURE 9. What Theorem 5.30 forces. (a) The twenty-four normalised roots of D_4, projected orthogonally to a generic 3-space so that no two coincide, with every pair at inner product $\frac{1}{2}$ joined; one root r_0 is singled out and the others are coloured by their inner product with it, eight at $\frac{1}{2}$, six at 0, eight at $-\frac{1}{2}$ and the antipode at -1. (b) The Gram matrix of the twenty-four directions, rows and columns ordered by these classes; it takes no value outside $\{1, \frac{1}{2}, 0, -\frac{1}{2}, -1\}$. (c) The 276 pairs by inner product: 96, 72, 96 and 12. The theorem says that every contact configuration of size 24 has a Gram matrix with the values of (b), and Lemma 5.26 that it then is (a) up to an orthogonal map.

Proof. By Proposition 5.29(i) and Lemma 5.28, the certificate kernel (5.22) is positive; it is real valued, so no real part has to be taken. By (5.23) it satisfies the hypotheses of Lemma 5.27, and by Proposition 5.29(ii) that lemma gives $|C| \le K(\emptyset, \emptyset) = 24$. If $|C| = 24$ the equality case of the lemma gives $A_2 K(\{x, y\}) = 0$ for every pair of distinct points $x, y \in C$, that is $\sigma_2(\langle x, y\rangle) = 0$ by (5.23), and Proposition 5.29(iv) places $\langle x, y\rangle$ in $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$. ∎

The first assertion is Musin's theorem [Mus08] that the kissing number in four dimensions is 24, obtained here a second time from a different certificate. The present paper needs the second assertion, the equality case. Neither the three-point bound of Bachoc and Vallentin [BV08], at the first level of the same hierarchy and with best computed value 24.047 [Lei25], nor Musin's argument identifies the configuration; the equality case is available here only because the bound equals 24 and is not merely below 25.

Proof of Theorem 5.25. A contact configuration is a subset of S^3 whose pairwise inner products are at most $\frac{1}{2}$, that is a spherical code of minimal angle at least $\pi/3$. By Theorem 5.30 the pairwise inner products of its 24 points lie in $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$, and Lemma 5.26 makes it a copy of the root system. The cell of the root configuration is the 24-cell of volume 8 by Lemma 2.3, and volume is invariant under orthogonal maps. ∎

5.5.3. *The certificate, checked.* What remains is Proposition 5.29, a statement about a finite set of rational numbers. The numbers are those of de Laat, Leijenhorst and de Muinck Keizer [LLM24]. They computed the second level of the hierarchy of [dLV15] for spherical codes

in S^3 with maximal inner product $\frac{1}{2}$, reduced by the symmetry of the problem to the form (5.22) with truncation degrees 14 and 16, solved it to forty digits with the clustered low-rank solver of [LdL24], rounded the numerical optimum to an exact rational feasible point with objective value exactly 24, and deposited that point at 4TU.ResearchData [LLM24d] under the MIT licence. The licence permits redistribution, and the supplementary archive of this paper [BBMB26] carries the data with the licence notice, so that every input to Proposition 5.29 travels with the paper.[17] Their paper is, at the time of writing, a preprint; nothing below depends on it.

We verified Proposition 5.29 from the deposited data independently of the authors' implementation, in exact rational arithmetic and in the rigorous ball arithmetic of Arb [Joh17]. The verification has seven steps.

1. The data are read and their format checked: 127 blocks of total dimension 3726, the largest of order 350; sixty of them are indexed by the signatures λ of $\mathrm{O}(4)$ with $|\lambda| \le 14$, each with exactly as many rows as the admissible index tuples of [LLM24, §3] prescribe, and the other sixty-seven are the multipliers of the sums of squares for the constraints on two, three and four points.
2. Every block is positive definite: a Cholesky factorisation in ball arithmetic at 256 bits has every pivot a ball inside the positive reals, the least about 1.4×10^{-15}, and the 81 blocks of order at most 16 are also factored as LDL^{T} exactly over $\mathbb{Q}$.
3. The zonal matrices Z_λ are computed from the definition (5.21), exactly, as polynomials in the six inner products of four points.
4. Each of the 125 prefactors of the sum-of-squares terms is a nonnegative rational multiple of one of the weights of Proposition 5.29(iii).
5. The four constraint polynomials $A_2K(Q)+\sigma_{|Q|}$, and $A_2K(\{x\})+1+\sigma_1$, are assembled from the zonal matrices and the blocks, and are identically zero.
6. The objective, the entry of the block $\lambda = (0,0)$ that is $K(\emptyset,\emptyset)$, equals 24.
7. The polynomial σ_2 has degree 16, and a Sturm sequence over $\mathbb{Q}$ shows that its zeros in $[-1, \frac{1}{2}]$ are exactly $-1, -\frac{1}{2}, 0, \frac{1}{2}$, of multiplicities $1, 2, 2, 1$. This step is also checked in Lean.

Step 3 is the substantial one.[18] What it needs, for each signature λ and each admissible pair of indices k_1, k_2, is the matrix

$$P(S)_{k_1,k_2} \;=\; \int_{\mathrm{O}(4)} \rho_{0,k_1}(\omega\gamma\varepsilon)\,\rho_{0,k_2}(\omega\gamma S)\,d\gamma,$$

and the integral is taken monomial by monomial in the sixteen entries of γ; on such integrals over the classical groups see [GL08]. The answer is homogeneous of degree $|\lambda| \le 14$ in the seven free entries of S, so it has at most $\binom{20}{6} = 38\,760$ coefficients for every λ. Each of the two factors of the integrand is a product of powers of single entries of a 2×2 matrix, so the

[17] The deposited archive has MD5 checksum `02acd5270f7b3fa799abdeb5291706fd` and 234 megabytes in 331 files. The checksum fixes which bytes the statements below refer to, and allows the copy in the supplementary archive, the deposited copy and any downloaded copy to be compared byte for byte. It is verified before the data are read, so the proof does not depend on the server that hosts the data.

[18] The construction as described in [LLM24] expands the product of the two representation polynomials first and holds the expansion in a table indexed by the exponent matrix of γ, which needs about 128 gigabytes; the certificate itself took about two weeks on eight cores at that memory. Taken in triples as described here, the whole of step 3 runs in exact rational arithmetic in about two hours on two cores and under 400 megabytes. Run one signature at a time, the expansion-first construction takes about a second for $(3,1)$ and $(4,0)$, half a minute for $(10,0)$ and four minutes for $(6,4)$ and $(8,2)$, and passes three gigabytes for $(10,2)$: its cost grows with λ_2 far faster than with $|\lambda|$.

monomials can be taken in triples and each triple added directly to the coefficient of $P(S)$ it contributes to, without expanding the product of the two representation polynomials first. Two facts keep the number of integrals small. The integral of a monomial $\prod \gamma_{ij}^{e_{ij}}$ over $\mathrm{O}(4)$ vanishes unless every row sum and every column sum of the exponent matrix (e_{ij}) is even, since $\gamma \mapsto D\gamma$ and $\gamma \mapsto \gamma D$ with D diagonal of entries ± 1 preserve Haar measure; this leaves 7.85×10^9 of the 2.39×10^{11} triples. And the integral is unchanged by permuting the rows or the columns of the exponent matrix, for the same reason with permutation matrices in place of D, so it depends only on a canonical form of (e_{ij}), of which at most 2 505 430 occur in any one of the 490 entries.

In step 5 the sum (5.19) is taken over ordered pairs, which gives it 3, 9, 12 and 6 terms for $|Q| = 1, 2, 3, 4$; an unordered pair $J_1 \neq J_2$ enters with the factor 2 that its two orders supply and a pair $J_1 = J_2$ once. All four identities hold coefficient by coefficient: the one-point identity, the two-point identity in one variable, the three-point identity in three variables and the four-point identity in six. In the last, the two sides are supported on the same 53 572 monomials, and on each of them the coefficient coming from the sum-of-squares blocks is the exact negative of the coefficient coming from the zonal matrices, both rationals of about 15 700 digits (Figure 10).

Three checks that do not depend on the outcome fix the construction of step 3. The integration reproduces the seven test values deposited with [LLM24d], and satisfies seventy further identities that follow from the rows and the columns of an orthogonal matrix being unit vectors. For the signatures $\lambda = (k, 0)$ and two single points at inner product u the matrix Z_λ must be a multiple of the zonal spherical harmonic kernel of degree k on S^3, and it is, exactly, with ratio $8^k/(k+1)^2$ for every $k \leq 14$, the denominator being the dimension of the space of harmonics of degree k on S^3. And Z_λ must be a positive semidefinite kernel on the subsets of size at most two of any set of points, which nothing in the computation forces; on random four-point sets its least eigenvalue is zero to the working precision for every signature, including those with $\lambda_2 > 0$, which the comparison with the Gegenbauer polynomials does not reach. The entries were also compared with those of the authors: for the thirteen signatures with $|\lambda| \leq 10$ and $\lambda_2 \leq 4$ all 88 entries agree coefficient by coefficient.

The seven steps establish Proposition 5.29 item by item: step 2 is (i), step 6 is (ii), steps 3, 4 and 5 are (iii), and step 7 is (iv). Lemmas 5.27 and 5.28, proved above, turn that list into Theorem 5.30. Nothing in the argument is taken from [LLM24] without being recomputed or proved here; the authors of the certificate proved the same bound and the same equality case in their Sections 2 to 5, and a reader who wants a second account of Lemma 5.27 will find it there.

Everything after Theorem 5.30, the passage from the four inner products to the root system, is Lemma 5.26, proved above by a classical argument and checked by an exact enumeration, twice, the second time in Lean. Every statement below that invokes Theorem 5.25, or the robust form of its certificate in Section 5.13, therefore rests on Proposition 5.29, that is on a finite exact computation reproducible from the supplementary archive, and on nothing outside this paper.

Corollary 5.31 (Multi-direction positivity, from the classification)**.** *Theorem 1.8 holds, with equality: a packing configuration at the all-contact corner with a full active set, twenty-four neighbours at distance exactly* 2*, is a contact configuration of size* 24*, so by Theorem 5.25 it is a copy of the root system, its cell is a regular* 24*-cell, and* $\mathrm{vol}(V_c) = 8$*. By Theorem 5.1 the two unions in Theorem 1.8 then have equal volume.*

5.6. **A volume identity exact at the root system.** The truncated estimates of Section 5.4 and Section 5.11 cut the cell at a radius below $\sqrt{2}$. At $m = 24$ no such truncation

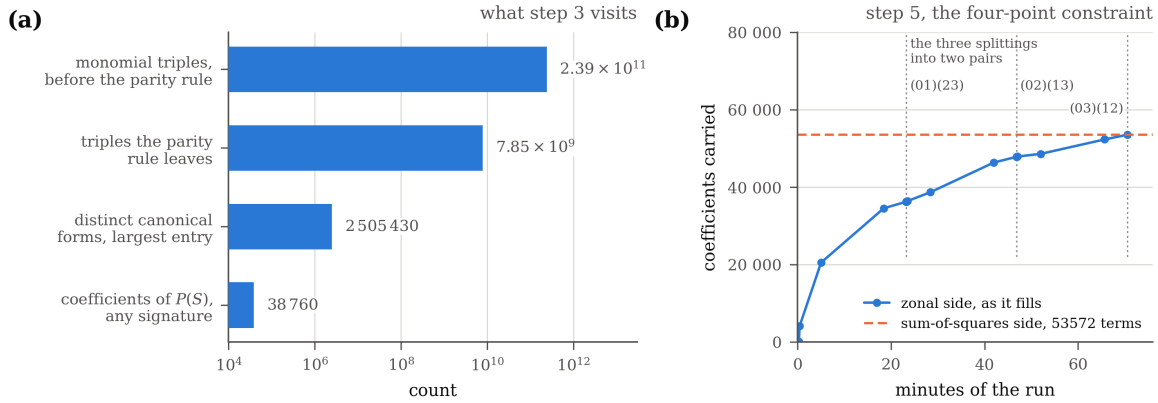

FIGURE 10. Steps 3 and 5 of the verification of the certificate. (a) The size of step 3. The integral over O(4) is taken one monomial triple at a time; the parity rule on the exponent matrix discards all but about a thirtieth of the triples, invariance under permuting rows and columns collapses the rest onto at most 2 505 430 canonical forms in the largest of the 490 entries, and the answer never has more than $\binom{20}{6} = 38\,760$ coefficients. (b) Step 5 for the four-point identity. The curve is the number of monomials of the zonal side as the signatures are added, through the three ways of splitting four points into two pairs; it ends at 53 572, the support of the sum-of-squares side as well. The two sides have the same support and opposite coefficients, so their sum is zero at each of the 53 572 monomials.

can be sharp: the 24-cell has circumradius $\sqrt{2}$ (Lemma 2.3), so a ball of smaller radius removes volume from it, and the lower bound falls strictly below the true value 8 at the root configuration. This subsection truncates at $\sqrt{2}$ itself and keeps three terms of inclusion and exclusion. The resulting lower bound is exact at the root configuration, and it converts the statement $\mathrm{vol}(V_c) \geq 8$ for 24 contacts into one inequality between the pair and triple angles of the contact directions, of the same form as the inequality proved for 23 contacts in Section 5.12. Remark 5.36 records how far the three-point relaxation of that inequality reaches.

Throughout, $B = \{x \in \mathbb{R}^4 : |x| \leq \sqrt{2}\}$, and for a unit vector w we write $K(w) = \{x \in B : \langle x, w\rangle > 1\}$. By orthogonal invariance the numbers

$$\begin{aligned} c &= \mathrm{vol}\, K(w), \\ \omega_2(u) &= \mathrm{vol}\big(K(w) \cap K(w')\big), \\ \omega_3(u,v,t) &= \mathrm{vol}\big(K(w) \cap K(w') \cap K(w'')\big) \end{aligned}$$

depend only on the inner products $u = \langle w, w'\rangle$ and, for ω_3, on

$$(u,v,t) = \big(\langle w, w'\rangle,\ \langle w, w''\rangle,\ \langle w', w''\rangle\big).$$

Lemma 5.32 (Third-order bound). *Let $w_1, \dots, w_m$ be unit vectors in $\mathbb{R}^4$, write $u_{ij} = \langle w_i, w_j\rangle$ and $V = \{x : \langle x, w_i\rangle \leq 1$ for all $i\}$. Then*

$$\mathrm{vol}(V) \;\geq\; 2\pi^2 - mc + \sum_{i<j} \omega_2(u_{ij}) - \sum_{i<j<k} \omega_3(u_{ij}, u_{ik}, u_{jk}), \tag{5.24}$$

with equality if and only if $V \setminus B$ and the set of points of B lying in at least four of the sets $K(w_i)$ both have measure zero.

Proof. First $\mathrm{vol}(V) \geq \mathrm{vol}(V \cap B) = \mathrm{vol}(B) - \mathrm{vol}\big(\bigcup_i K(w_i)\big)$, since $B \setminus V = \bigcup_i K(w_i)$, and $\mathrm{vol}(B) = \frac{1}{2}\pi^2(\sqrt{2})^4 = 2\pi^2$. For a point lying in exactly $n \geq 1$ of the sets, the identity $\sum_{j=0}^{3}(-1)^j\binom{n}{j} = -\binom{n-1}{3}$ gives

$$\binom{n}{1} - \binom{n}{2} + \binom{n}{3} \;=\; 1 + \binom{n-1}{3} \;\geq\; 1,$$

with equality exactly when $n \leq 3$. Hence the indicator of the union is bounded above, pointwise, by the sum of the indicators of the sets, less the sum over pairs, plus the sum over triples, with equality off the points lying in four or more of the sets. Integration over B gives (5.24), and the two inequalities used are equalities exactly under the stated conditions. ∎

Lemma 5.33 (The cap and the pair term)**.** *The cap volume is* $c = \frac{1}{2}\pi^2 - \frac{4}{3}\pi$. *For* $-1 \leq u \leq 0$ *one has* $\omega_2(u) = 0$, *and* $\omega_3(u,v,t) = 0$ *unless* u, v *and* t *are all positive. For* $0 < u < 1$, *with* $\gamma = \arccos u$,

$$\omega_2(u) = 2\pi \int_{\gamma/2}^{\pi/4} \Big(1 - \tfrac{1}{2}\sec^2\theta\Big)^2 d\theta = 2\pi\big(F(\tfrac{\pi}{4}) - F(\tfrac{\gamma}{2})\big), \tag{5.25}$$
$$F(\theta) = \theta - \tfrac{3}{4}\tan\theta + \tfrac{1}{12}\tan^3\theta.$$

In particular $\omega_2(\frac{1}{2}) = \pi(9\pi + 26\sqrt{3} - 72)/54 = 0.0760762770\ldots$.

Proof. Write $x = (y, z)$ with $y = \langle x, w\rangle$ and z in the orthogonal complement of w. Then $K(w)$ is the set $1 < y \leq \sqrt{2}$, $|z|^2 \leq 2 - y^2$, and

$$c = \int_1^{\sqrt{2}} \tfrac{4}{3}\pi(2-y^2)^{3/2}\,dy = \tfrac{16}{3}\pi \int_{\pi/4}^{\pi/2} \cos^4\phi\,d\phi$$
$$= \tfrac{16}{3}\pi\Big(\tfrac{3\pi}{32} - \tfrac{1}{4}\Big),$$

by the substitution $y = \sqrt{2}\sin\phi$. If $x \in K(w) \cap K(w')$ then $\langle x, w + w'\rangle > 2$ and $|w + w'|^2 = 2 + 2u$, so $|x|^2 > 2/(1+u)$; for $u \leq 0$ this exceeds 2 and the intersection is empty, and the same argument applied to any two of three directions gives the statement on ω_3. For $0 < u < 1$ decompose $x = y + z$ with y in the plane P spanned by w, w' and $z \perp P$. The conditions are $\langle y, w\rangle > 1$, $\langle y, w'\rangle > 1$ and $|z|^2 \leq 2 - |y|^2$, so $\omega_2(u)$ is the integral of $\pi(2 - |y|^2)_+$ over the region of P cut out by the first two. In polar coordinates on P with w at angle 0 and w' at angle γ, the ray at angle $\theta \in (\gamma - \frac{\pi}{2}, \frac{\pi}{2})$ meets that region in $\rho > \max(\sec\theta, \sec(\gamma - \theta))$, and the reflection $\theta \mapsto \gamma - \theta$ exchanges the two terms, so it suffices to take $\theta \geq \gamma/2$, where the maximum is $\sec\theta$, and to double. Since

$$\int_{\sec\theta}^{\sqrt{2}} \pi(2 - \rho^2)\rho\,d\rho = \pi\Big(1 - \tfrac{1}{2}\sec^2\theta\Big)^2$$

for $\sec\theta \leq \sqrt{2}$, while for $\theta > \pi/4$ the ray does not meet the region inside B, (5.25) follows, and $\gamma/2 < \pi/4$ because $u > 0$. Differentiation gives $F'(\theta) = 1 - \sec^2\theta + \frac{1}{4}\sec^4\theta = (1 - \frac{1}{2}\sec^2\theta)^2$. At $\gamma = \pi/3$, $F(\frac{\pi}{4}) = \frac{\pi}{4} - \frac{2}{3}$ and $F(\frac{\pi}{6}) = \frac{\pi}{6} - \frac{13}{18\sqrt{3}}$. ∎

Proposition 5.34 (Exactness at the root system)**.** *For the* 24 *normalised roots of* D_4, (5.24) *holds with equality, and*

$$8 \;=\; 32\pi - 10\pi^2 + 96\,\omega_2(\tfrac{1}{2}) - 96\,\omega_3(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}). \tag{5.26}$$

Consequently $\omega_3(\frac{1}{2},\frac{1}{2},\frac{1}{2}) = \frac{1}{16}\pi^2 - \pi + \frac{13\sqrt{3}}{27}\pi - \frac{1}{12} = 0.0118567031\ldots$.

Proof. The cell is the 24-cell, of volume 8, whose vertices all have norm $\sqrt{2}$ (Lemma 2.3); so it lies in B. By Lemma 5.33 a point of $K(w_i) \cap K(w_j)$ forces $u_{ij} > 0$, and the only positive inner product between distinct normalised roots is $\frac{1}{2}$. Suppose a point of B lies in four of the sets. The four corresponding roots, scaled to norm $\sqrt{2}$, then have Gram matrix $I + J$, with 2 on the diagonal and 1 off it, whose determinant is 5. They are therefore linearly independent and span a sublattice L of full rank of the D_4 lattice, whose determinant is 4; but $\det L = 4\,[D_4 : L]^2$, and 5 is not of that form. Thus every point of B lies in at most three of the sets, and Lemma 5.32 gives equality. In the sum over pairs only the pairs at inner product $\frac{1}{2}$ contribute; each root has eight such neighbours ($e_1 + e_2$ has $e_1 \pm e_3$, $e_1 \pm e_4$, $e_2 \pm e_3$, $e_2 \pm e_4$), giving 96 pairs. In the sum over triples only triangles of roots pairwise at $\frac{1}{2}$ contribute; each such pair lies in exactly three of them (for $e_1 + e_2$ and $e_1 + e_3$ the third root is $e_1 + e_4$, $e_1 - e_4$ or $e_2 + e_3$), giving $96 \cdot 3/3 = 96$ triangles. With $2\pi^2 - 24c = 32\pi - 10\pi^2$ from Lemma 5.33 this is (5.26), and the value of ω_3 follows from the value of $\omega_2(\frac{1}{2})$. ∎

The value of $\omega_3(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ obtained this way agrees with a direct quadrature of the triple intersection to 10^{-7}, and the cap, pair and triple volumes agree with Monte Carlo estimates in $\mathbb{R}^4$ within their standard errors.

Corollary 5.35 (The 24-contact case as an inequality between angles)**.** *For a contact configuration $W = \{w_1, \dots, w_{24}\}$ put*

$$E(W) = \sum_{i<j} \omega_2(u_{ij}) - \sum_{i<j<k} \omega_3(u_{ij}, u_{ik}, u_{jk}).$$

Then $\mathrm{vol}(V_c(W)) \geq 32\pi - 10\pi^2 + E(W)$*. The root configuration has $E = \tau$, where*

$$\tau = 8 - 32\pi + 10\pi^2 = 6.1650790960\dots,$$

and every W with $E(W) \geq \tau$ has $\mathrm{vol}(V_c(W)) \geq 8$*.*

Proof. Lemma 5.32 with $m = 24$, and Proposition 5.34. ∎

Remark 5.36 (The three-point relaxation of Corollary 5.35). The inequality $E(W) \geq \tau$ for all contact configurations of 24 directions would give the volume statement of Theorem 5.25 at the three-point level, without the four-point kernel of Theorem 5.30, and it has the form treated in Section 5.12. Splitting each pair over the 22 triples that contain it, $E(W) = \sum_{i<j<k} h(u_{ij}, u_{ik}, u_{jk})$ with $h(u,v,t) = \frac{1}{22}(\omega_2(u)+\omega_2(v)+\omega_2(t)) - \omega_3(u,v,t)$. In the notation of Section 5.12, a function $f = \sum_k f_k G_k$ with $f_k \geq 0$ for $k \geq 1$ and positive semidefinite matrices F_k satisfying

$$\begin{aligned} h(u,v,t) \;&\geq \tfrac{1}{22}\big(f(u) + f(v) + f(t)\big) + F(u,v,t) \\ &\quad + \tfrac{1}{22}\big(F(1,u,u) + F(1,v,v) + F(1,t,t)\big) \end{aligned}$$

on every admissible triple give $E(W) \geq \frac{1}{2}(576 f_0 - 24 f(1)) - 4F(1,1,1)$ for every W of size 24. Because the root configuration attains τ, a certificate reaching τ must be sharp, and exact rounding needs it to be attained at a finite degree.

The relaxation does not reach τ. Its dual is a probability measure p on admissible triples subject to the two-point conditions $1 + \frac{23}{3}\,\mathbb{E}_p[P_k(u) + P_k(v) + P_k(t)] \geq 0$ for $1 \leq k \leq d$ and, writing $\Sigma_k = S_k(1,u,u) + S_k(1,v,v) + S_k(1,t,t)$, the three-point conditions

$$24\, S_k(1,1,1) + 552\, \mathbb{E}_p[\Sigma_k] + 12144\, \mathbb{E}_p[S_k] \succeq 0$$

for $0 \leq k \leq d$, in the matrices S_k of [BV08]. Every such p satisfies $2024\,\mathbb{E}_p[h] \geq V_d$, where V_d is the value of the best degree-d certificate, so a single feasible p with $2024\,\mathbb{E}_p[h] < \tau$ rules

out every degree-d certificate at once. Restricting p to a finite set of triples only shrinks the dual feasible set, so such a p is a proof and not an estimate.

At degree 6 the computation gives such a p in floating point. We solved the dual on a grid of admissible triples with every constraint held with a margin $\varepsilon = 10^{-6}$, and rechecked the returned measure against the constraints without the margin: the measure deposited in the supplement has value 5.9424 on 2170 triples, and a rerun with a later solver gives 5.9851 on 522 triples, with least eigenvalue 10^{-6} in the moment matrices and least two-point slack 7.7×10^{-3}; both values lie below $\tau = 6.1650791$ by more than 0.18. So, up to floating point, no degree-6 certificate of this form exists. This is not a proof in the sense of Section 5.12: a proof would round the measure to rationals, check the moment matrices by an exact $LDL^{\top}$, and bound ω_3 rigorously in the value, and the supplement does not carry such a check. Nothing in this paper depends on the conclusion.

At degrees 8 through 20, on a grid of 1264 triples, the dual minimum is 6.0395, 6.0700, 6.0879, 6.1060, 6.1256, 6.1416 and 6.1516, short of τ by 0.126, 0.095, 0.077, 0.059, 0.039, 0.023 and 0.014. These are double-precision values, but the direction of the discretisation is favourable: restricting the measure to a grid only shrinks the dual feasible set and so only raises the minimum, and refining the grid by a factor of 2.6 at degrees 6 and 8 moves the value by less than 0.005. At degree 20 the shortfall is still 1.4×10^{-2}, where a certificate that is to be rounded exactly would need it to be zero.

Four further computations show that the shortfall is structural and not numerical. First, the same two-point and three-point conditions, read as conditions on a code of N points instead of 24, remain feasible up to $N = 26.61$ at degree 4, $N = 24.77$ at degree 6 and $N = 24.24$ at degree 8: the relaxation does not detect that 24 is the largest number of contacts. Since $E(W) \geq \tau$ holds with equality at the root configuration and, by Theorem 5.25, only there, a certificate for it is a certificate of uniqueness, and a relaxation that does not detect maximality cannot detect uniqueness. Second, the minimising p is not a perturbation of the root configuration: it carries about a third of its pair mass at $u = \frac{1}{2}$ and spreads the rest continuously over $(-1, 0)$, where the root configuration has atoms at -1, $-\frac{1}{2}$ and 0 alone. Requiring the two-point slacks at $k = 2$ and $k = 4$, which vanish exactly at the root configuration, to exceed ε raises the degree-8 minimum only from 6.0395 to 6.0489 at $\varepsilon = 0.1$, and is infeasible at $\varepsilon = 0.3$; so the shortfall is not sharpness at the root configuration, and treating a neighbourhood of it separately does not remove it. Third, the largest s for which the moment matrices admit $\succeq sI$ falls from 0.75 at degree 4 to 1.7×10^{-2} at degree 6 and to about 10^{-4} at degrees 8 and above: from degree 8 on the dual has no usable interior. That is a property of the feasible set, not of the arithmetic, and it makes interior-point solvers degrade there. Fourth, the root configuration never becomes a local minimum of the relaxation. Minimising the directional derivative of the objective over the directions the degree-d conditions leave open at the root configuration gives -1.5705×10^{-4} at degree 6 and -1.0951×10^{-4}, -8.7210×10^{-5}, -7.8417×10^{-5} and -7.6223×10^{-5} at degrees 8, 10, 12 and 14, each computed on the support of the direction with the closed form of ω_3 rather than a table, and each agreeing to four figures with the value the table gives. Fitting $L + Cq^d$ to the five returns $L = 7.25 \times 10^{-5}$ with $q = 0.654$ and residuals below 9×10^{-7}, and dropping any one of the five moves L only within $[7.03, 7.43] \times 10^{-5}$. The improvement available at the root configuration therefore does not tend to zero with the degree, where a certificate reaching τ would need it to be zero.

That the three-point level is the wrong level is not a conclusion of ours alone. Cohn, de Laat and Leijenhorst [CdLL24] settle a family of spherical codes by three-point bounds with exact rounding, and record that the one member of it they cannot settle is the 24 points of

R in $\mathbb{R}^4$, for which, in their words, three-point bounds do not seem to be sharp. Two of the three are authors of [LLM24], and they turned to the four-point computation there instead.

The three-point level therefore does not decide this question, and Theorem 5.25 is proved from the four-point kernel of Theorem 5.30, whose equality case does certify uniqueness. What Corollary 5.35 adds is an exact reduction of its volume consequence to a single inequality between pair and triple angles; what the computations above add is the location of the obstruction. Closing the 24-contact case needs a method that certifies the uniqueness of the root configuration, which a longer computation at higher precision does not supply; the second level of the hierarchy supplies one and the first does not.

Remark 5.37 (The second moment). The strongest elementary constraint on 24 contact directions is a second moment. For 24 unit vectors in $\mathbb{R}^4$ the Gram matrix $G = (u_{ij})$ is positive semidefinite of rank at most 4 with trace 24, so Cauchy–Schwarz applied to its nonzero eigenvalues gives $\|G\|_F^2 \geq 24^2/4 = 144$, that is

$$\sum_{i<j} u_{ij}^2 \;\geq\; 60, \tag{5.27}$$

with equality exactly when the configuration is a tight frame. The root system attains it, its frame operator being $6I$: the 96 pairs at $\frac{1}{2}$ contribute 24, and the 96 at $-\frac{1}{2}$ and the 12 at -1 contribute 36.

Write P for the set of pairs with $u_{ij} > 0$, the only ones on which ω_2 and ω_3 are supported. Since $u_{ij} \leq \frac{1}{2}$ there, each such pair contributes at most $\frac{1}{4}$ to the second moment, so a bound $\sum_P u_{ij}^2 \geq 24$ would force $|P| \geq 96$, exactly the count of the root system. That bound does not follow from (5.27) alone: it needs the pairs outside P to carry at most 36, which is the root system's own value and is not available in advance.

Even with that bound the route falls short. The ratio $\omega_2(u)/u^2$ increases on $(0, \frac{1}{2}]$, from 0.0487 at $u = 0.05$ to 0.3043 at $u = \frac{1}{2}$, so for a fixed second moment $\sum_P \omega_2(u_{ij})$ is smallest when the mass is spread over as many pairs as possible at as small a value as possible. A budget $\sum_P u_{ij}^2 = 24$ spread over all 276 pairs, at $u = 0.2949$, gives $\sum_P \omega_2 = 5.1357$, short of $\tau = 6.1651$ by 1.0294, with the triple term still to be subtracted; the same budget concentrated on 96 pairs at $\frac{1}{2}$ gives the root system's 7.3033. The second moment fixes how much inner-product mass a configuration carries but not where that mass sits, and the difference between the two placements exceeds the entire margin to be proved.

What is missing here and in Remark 5.36 is the same thing: a reason for the positive inner products to concentrate at $\frac{1}{2}$ instead of spreading. That reason is the uniqueness of the root system among 24-point configurations with $u_{ij} \leq \frac{1}{2}$, and the equality case of the second-level bound, Theorem 5.30, supplies it.

The extendability of a configuration is a statement about the reach of its own cell. This converts the remaining case into a question about one polytope.

Proposition 5.38 (Extendability and the circumradius)**.** *Let W be a contact configuration with bounded cell and write*

$$g(W) = \min_{\theta \in S^3} \max_{w \in W} \langle \theta, w \rangle = \cos(\textit{covering radius of } W).$$

The following are equivalent:

(i) *some $v \in S^3$ has $\langle v, w \rangle \leq \frac{1}{2}$ for all $w \in W$, that is $W \cup \{v\}$ is again a contact configuration;*
(ii) *$V_c(W)$ contains a point of norm 2, that is its circumradius is at least 2;*
(iii) *$g(W) \leq \frac{1}{2}$, that is the covering radius of W on S^3 is at least $60°$.*

We call W saturated *when these fail, so that $g(W) > \frac{1}{2}$.*

Proof. For a unit vector v, the condition $\langle v, w\rangle \le \frac{1}{2}$ for all w says exactly $\langle 2v, w\rangle \le 1$ for all w, that is $2v \in V_c(W)$; and $|2v| = 2$. Therefore (i) and (ii) are the same statement, since $V_c(W)$ is convex and contains the origin, so it meets the sphere of radius 2 if and only if its circumradius is at least 2. For (iii), the circumradius is $\max_\theta \rho(\theta) = \max_\theta 1/\cos\delta(\theta)$ by Lemma 5.15, which is $1/g(W)$; and $1/g(W) \ge 2$ is $g(W) \le \frac{1}{2}$. ∎

Corollary 5.39 (The reduction to twenty-three contacts)**.** *Every contact configuration with bounded cell satisfies* $\mathrm{vol}(V_c) \ge 8$*, with equality only for the root configuration, except possibly for a configuration of exactly* 23 *directions whose cell has circumradius less than* 2*.*

Proof. By Corollary 5.17 the bound holds when $|W| \le 22$, with strict inequality. By Theorem 5.25 it holds with equality when $|W| = 24$. Let $|W| = 23$. If the circumradius of $V_c(W)$ is at least 2 then by Proposition 5.38 there is a v with $W' = W \cup \{v\}$ a contact configuration of size 24, and $V_c(W') \subseteq V_c(W)$, so $\mathrm{vol}(V_c(W)) \ge \mathrm{vol}(V_c(W')) = 8$ by Theorem 5.25; the inequality is strict because $V_c(W)$ contains a point of norm 2 while the 24-cell has circumradius $\sqrt{2}$. The only case not covered is $|W| = 23$ with circumradius below 2. ∎

The remaining case is therefore a single, sharply posed statement, and it is no longer about deviations.

Theorem 5.40 (The 23-point case)**.** *Every contact configuration of* 23 *directions with bounded cell has* $\mathrm{vol}(V_c) > 8$*. In particular a contact configuration of* 23 *directions whose cell has circumradius less than* 2*, if one exists, has cell volume above* 8*.*

The proof occupies the rest of this section and is of a different kind from everything before it. Section 5.7 to Section 5.9 approach the statement through the configuration: what a saturated set of 23 directions would have to look like, how much of a root system it can contain, which codes inside the 600-cell qualify, and how the codes of 23 points behave as their minimal angle is pushed to 60°; that material locates the difficulty but is not used in the proof. Section 5.11 and Section 5.12 give the proof. There the statement is turned into an inequality between the pair angles of any 23 contact directions (Theorem 5.69), and the inequality is established by a semidefinite certificate that reads the triples of directions, verified in exact and interval arithmetic (Theorem 5.72). Two remarks first on the size of what is asked. The configuration obtained by deleting one root from D_4 has cell the 24-cell together with the pyramid of Lemma 6.1, of volume $\frac{25}{3} = 8.333\ldots$, and its circumradius is exactly 2: the apex $2u_0$ is the point that the deleted root would occupy. Hence that configuration sits exactly on the boundary of Proposition 5.38(ii), and is extendable by the single direction u_0. And the covering estimate of Theorem 5.16 gives $7.916728\ldots$ at $m = 23$, short of 8 by $0.083272\ldots$, against a true value of $\frac{25}{3}$ at the configuration just described; so the estimate has to be improved by about a fifth of its slack there, not by all of it, which is a different situation from $m = 24$, where the true value is 8 and no slack exists. Figure 11(a) plots the range, and Figure 11(b) records what settles each part of it.

5.7. **The saturated case.** Everything Theorem 5.40 asserts can be written down explicitly, since it concerns a single polytope with twenty-three facets and no deviation parameters. Fix a contact configuration $W = \{w_1, \ldots, w_{23}\}$ with bounded cell and circumradius below 2, and write $s_{ij} = \langle w_i, w_j\rangle$. The same statement takes three exact forms.

Radially. By Lemma 5.15 the quantity to bound is $\frac{1}{4}\int_{S^3} \sec^4\delta$, so the target is

$$\int_{S^3} \sec^4(\delta(\theta))\, d\theta \;\ge\; 32. \tag{5.28}$$

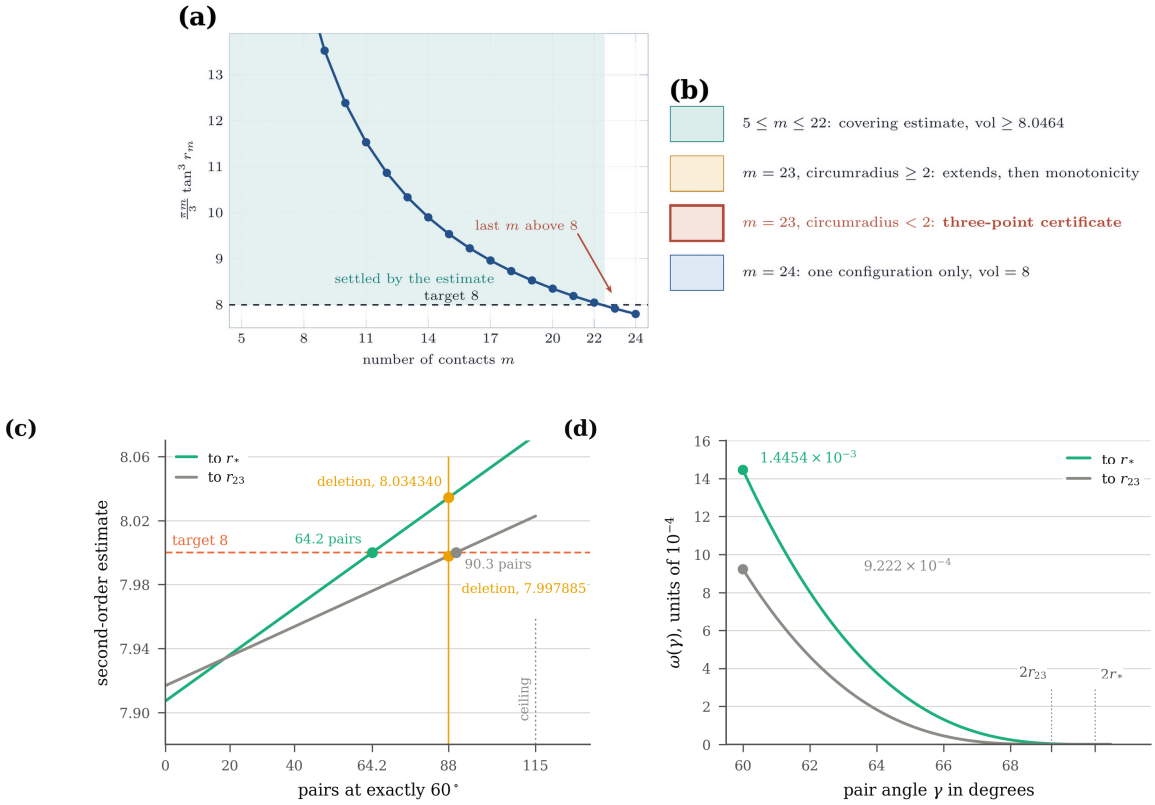

FIGURE 11. (a) The covering bound (5.12) at every admissible contact count m, against the target 8. It clears the target for all $m \le 22$ and falls below at $m = 23$. (b) How each part of the range of contact counts is settled: the estimate up to 22 (Corollary 5.17), extension followed by monotonicity at 23 when the circumradius reaches 2 (Proposition 5.38), and the classification at 24 (Theorem 5.25). The third row is settled by the pair inequality of Theorem 5.69 and its certificate, Theorem 5.72. (c) The pairwise estimate at $m = 23$, for the two integration limits r_{23} and r_*: the right-hand side of (5.17) as a function of the number of pairs at exactly 60°, with the bracket alone at 0 pairs, the deletion configuration at 88 on both lines, the crossings of the target at 90.3 and at 64.2 pairs, and the elementary ceiling of Proposition 5.41 at 115. (d) The per-pair weight $\omega(\gamma)$ for each integration limit: $0.000922\ldots$ and $0.001445\ldots$ at 60°, vanishing at $2r_{23} = 69.2213\ldots°$ and at $2r_* = 70.5288\ldots°$.

Saturation says $\delta < 60°$ everywhere, and Proposition 5.21 says $\max \delta \ge 37.7612\ldots°$, so the distance function is confined to a band of width about 22°.

On the boundary. By Lemma 5.10 the same target is $\sum_i \mathrm{vol}_3(F_i) \ge 32$, an average of $\frac{32}{23} = 1.391304\ldots$ per facet. That number is larger than the $\frac{4}{3}$ of a facet of the root configuration, and larger than the $16\sqrt{2} - \frac{64}{3} = 1.294084\ldots$ of the antiprism facet of Remark 5.12. The theorem therefore says that twenty-three facets, each containing the ball of radius $1/\sqrt{3}$ by Lemma 5.11, are on average larger than every facet shape that occurs at twenty-four contacts: the loss of a facet has to be paid for by the others, which is the sense in which the remaining case is not a perturbation of the root configuration.

At the vertices. If I indexes four active directions at a vertex z_I of the cell, then $\langle z_I, w_i\rangle = 1$ for $i \in I$, so writing $z_I = \sum_{i\in I} a_i w_i$ gives $G_I a = \mathbf{1}$ and

$$|z_I|^2 = \mathbf{1}^\top G_I^{-1}\mathbf{1}, \qquad G_I = (s_{ij})_{i,j\in I}. \tag{5.29}$$

Circumradius below 2 therefore says $\mathbf{1}^\top G_I^{-1}\mathbf{1} < 4$ at every vertex. Two cases of (5.29) are used below. Four pairwise orthogonal active directions give $G_I = \mathrm{Id}$ and $|z_I|^2 = 4$ exactly, so a saturated configuration has no vertex at which four active directions are mutually orthogonal. And four directions with all pairwise inner products equal to g give $|z_I|^2 = 4/(1+3g)$, which is at least 4 precisely when $g \le 0$; so at every vertex of a saturated configuration the active directions must be positively correlated on average.

Proposition 5.41 (What the pairwise estimate needs)**.** *For $m = 23$ write*

$$\omega(\gamma) = \frac{1}{4}\int_0^{r_*} \Lambda(r,\gamma)\, d(\sec^4 r)$$

for the contribution a single pair at angle γ makes to the right-hand side of (5.17) at $R = r_*$. Then ω is strictly decreasing on $[60^\circ, 2r_*]$ and vanishes from $2r_* = 70.5288\ldots^\circ$ on, with $\omega(60^\circ) = 0.001445408\ldots$, and a contact configuration of 23 directions with bounded cell satisfies $\mathrm{vol}(V_c) \ge 8$ as soon as

$$\sum_{i<j} \omega(\gamma_{ij}) \;\ge\; 0.092855570\ldots, \tag{5.30}$$

which holds in particular once it has 65 pairs at exactly 60°. The deletion configuration has 88, and no configuration of 23 directions has more than 115.

Proof. That ω is decreasing and eventually zero is immediate from (5.16), since $\Lambda(r,\gamma)$ is decreasing in γ and vanishes for $r \le \gamma/2$. Inequality (5.30) is (5.17) at $R = r_*$ with the value $7.907144430\ldots$ of the bracket subtracted from 8. Numerically $7.907144430 + 65 \times 0.001445408 = 8.001096\ldots$, while 64 pairs give $7.999651\ldots$, which is the sense in which 65 suffice and 64 do not; the crossing is at 64.24 pairs.

For the last assertion, fix $w \in W$ and let S be the set of w_i with $\langle w, w_i\rangle = \frac12$. Write $w_i = \frac12 w + \frac{\sqrt3}{2}u_i$ with u_i a unit vector of $w^\perp$. Then $\langle w_i, w_j\rangle = \frac14 + \frac34\langle u_i, u_j\rangle$, so the contact condition $\langle w_i, w_j\rangle \le \frac12$ reads $\langle u_i, u_j\rangle \le \frac13$: the u_i are unit vectors of $\mathbb{R}^3$ at pairwise angle at least $\arccos\frac13 = 70.5287\ldots^\circ$, and the caps of angular radius $35.2643\ldots^\circ$ about them are disjoint on S^2. Each such cap has area $2\pi(1-\cos 35.2643\ldots^\circ) = 1.152986\ldots$ against the 4π of S^2, so $|S| \le 10$. Every vertex of the graph of pairs at 60° therefore has degree at most 10, and a graph on 23 vertices of maximum degree 10 has at most $\lfloor 230/2\rfloor = 115$ edges. ∎

Remark 5.42 (The one route that is not blocked). With the integration stopped at r_{23}, as in (5.18), the estimate misses 8 at the deletion configuration by three pairs. Carried to r_* it does not miss: the deletion has 88 pairs against the 64.24 that (5.30) asks for, a surplus of 0.034, and the weight falls away quickly once a pair opens up, with $\omega(62^\circ)/\omega(60^\circ) = 0.55504\ldots$ and $\omega(66^\circ)/\omega(60^\circ) = 0.09077\ldots$. What would settle the local bound at twenty-three contacts, for every configuration and not only for the saturated ones, is a lower bound on the weighted number of pairs at or near 60° that every contact configuration of 23 directions satisfies. Remark 5.45 says how much of it the pair angles supply on their own, and Section 5.11 sets the statement out in full. Figure 11(c) and (d) plot these numbers for both integration limits.

Two further facts about the pair angles show how far a bound of this kind can be pushed and where it stops. The first is a lower bound on the total overlap of the contact caps that a saturated configuration is forced to have.

Proposition 5.43 (Covering multiplicity)**.** *Let W be a saturated contact configuration with $|W| = m$. Then*

$$\sum_{i<j} \Lambda\left(\tfrac{\pi}{3}, \gamma_{ij}\right) \;\geq\; \frac{1}{2}\left(\frac{\left(m\,C(\pi/3)\right)^2}{2\pi^2} - m\,C(\pi/3)\right), \tag{5.31}$$

where $C(r) = \pi(2r - \sin 2r)$ is the measure of a cap of radius r. At $m = 23$ the right-hand side is $155.172054\ldots$, against $162.566121\ldots$ at the deletion configuration, so the inequality is within five per cent of binding there.

Proof. Write $N(\theta)$ for the number of $w \in W$ with $\langle \theta, w\rangle > \frac{1}{2}$. Then $\int_{S^3} N = m\,C(\pi/3)$, and $\int_{S^3} N(N-1)$ counts ordered pairs of caps containing θ, so $\int_{S^3} N^2 = m\,C(\pi/3) + 2\sum_{i<j} \Lambda(\frac{\pi}{3}, \gamma_{ij})$. Cauchy–Schwarz gives $\int N^2 \geq (\int N)^2/2\pi^2$, and rearranging is (5.31). ∎

The second fact is negative. The estimate of Proposition 5.23 depends on a configuration only through the multiset of its pair angles, and the question is how far that multiset alone carries it, when one relaxes the multiset to an arbitrary measure subject to every inequality of the same character that a configuration satisfies.

Lemma 5.44 (Pair-angle constraints)**.** *The pair-angle multiset μ of a contact configuration of 23 directions, a measure on $[60°, 180°]$ of total mass $\binom{23}{2} = 253$, satisfies*

$$\int \Lambda(t, \gamma)\, d\mu \;\geq\; 23\,C(t) - 2\pi^2 \quad \text{for all } t, \qquad \int G_k(\cos\gamma)\, d\mu \;\geq\; -\tfrac{23}{2} \quad \text{for } k \geq 1,$$

where G_k are the Gegenbauer polynomials of S^3 normalised by $G_k(1) = 1$.

Proof. The first family is Bonferroni's inequality $\sigma(\bigcup_i C_i(t)) \geq \sum_i \sigma(C_i(t)) - \sum_{i<j} \Lambda(t, \gamma_{ij})$ together with $\sigma(\bigcup_i C_i(t)) \leq 2\pi^2$. The second is the positive definiteness of G_k on S^3, which gives $\sum_{i,j} G_k(\langle w_i, w_j\rangle) \geq 0$, that is $23 + 2\sum_{i<j} G_k(\cos\gamma_{ij}) \geq 0$. ∎

Remark 5.45 (What the pair angles alone can give). Minimise $\int \omega\, d\mu$, for the weight ω of Proposition 5.41, over all measures with the properties of Lemma 5.44. This is a linear programme, and solved on a grid of 1201 angles with 60 radii between r_{23} and 90° and Gegenbauer degrees up to 24 its optimum is $0.07380\ldots$, four fifths of the $0.092855\ldots$ that (5.30) requires; for the weight integrated only to r_{23} it is $0.04157\ldots$ against $0.083272\ldots$, one half. Adding radii or degrees does not raise either value, and the same measure attains both: 103.7 pairs at 62.3°, 90.0 at 103.0°, 39.2 at 122.6° and 20.1 at 154.1° (Figure 12).

So a bound that reads only the pair angles, through constraints of these two kinds, reaches four fifths of the way and no further. The minimising measure gives every direction nine neighbours at 62.3° and nearly eight at 103°, which is consistent at any one direction, and Bonferroni's inequality and positive definiteness cannot detect that twenty-three such neighbourhoods do not fit together on one sphere. Saturation is not expressible at that level either, since the covering radius is not a function of the pair angles, which is why Proposition 5.43 does not help. The proof of the bound at twenty-three contacts therefore uses the triples of directions: the three-point form of Section 5.11 and its certificate in Section 5.12.

One family of candidates can be ruled out completely, the symmetric one. A configuration of a prime number of directions with any symmetry transitive on them is a single orbit of a rotation of that order, and in dimension four such an orbit carries a single parameter.

Proposition 5.46 (No symmetric candidate)**.** *No contact configuration of 23 directions in $\mathbb{R}^4$ admits a group of symmetries transitive on its directions.*

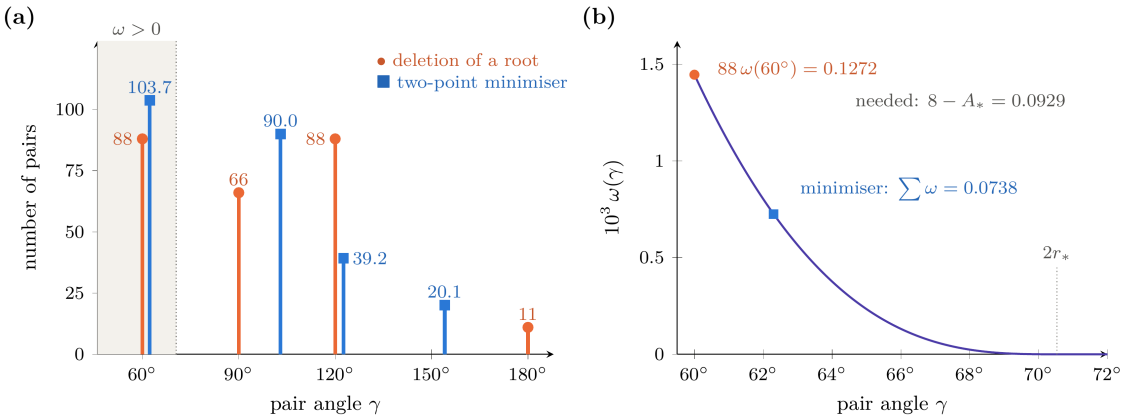

FIGURE 12. Pair angles at twenty-three contacts. (a) The 253 pair angles of the deletion of a root (clay) and the measure that minimises $\sum_{i<j}\omega(\gamma_{ij})$ over all pair distributions satisfying the Bonferroni inequality and positive definiteness (blue, Remark 5.45). Only pairs closer than $2r_* = 70.53°$ (shaded) carry weight. (b) The weight ω of Proposition 5.41. The deletion collects $88\,\omega(60°) = 0.1272$ from its 88 pairs at 60°; the minimiser moves its 103.7 close pairs out to 62.3°, where ω has halved, and collects 0.0738, short of the 0.0929 that (5.30) requires.

Proof. Let W be such a configuration and $G \le O(4)$ transitive on it. Then 23 divides $|G|$, so G has an element g of order 23 by Cauchy's theorem. Since $\det g$ is a 23rd root of unity in $\{\pm1\}$ it is 1, so $g \in SO(4)$ and g is a rotation through angles $t_1 = 2\pi a/23$ and $t_2 = 2\pi b/23$ in two orthogonal planes P_1, P_2, with $(a,b) \neq (0,0)$. The orbits of $\langle g\rangle$ on W have 1 or 23 elements. If one of them had a single element, the other 22 directions could not contain an orbit of 23, so g would fix every member of W; but the cell is bounded, so W spans $\mathbb{R}^4$, and g would be the identity. Hence the orbit of any $w_0 \in W$ has 23 elements and is all of W.

Write $w_0 = p + q$ with $p \in P_1$, $q \in P_2$ and $c = |p|^2$, so $|q|^2 = 1 - c$ and $c \in [0,1]$. Then $w_k = g^k w_0$ and

$$\langle w_0, w_k\rangle = \langle p, g^k p\rangle + \langle q, g^k q\rangle = c\cos(kt_1) + (1-c)\cos(kt_2),$$

which does not depend on the directions of p and q inside their planes and is affine in c. Transitivity makes every pairwise inner product one of these, so the contact condition is the system of 22 affine inequalities

$$c\cos(kt_1) + (1-c)\cos(kt_2) \;\le\; \tfrac{1}{2}, \qquad k = 1, \dots, 22,$$

in the single variable $c \in [0,1]$. Each of the 528 pairs (a,b) is therefore decided by a one-variable linear programme. None of the 528 is feasible. The quantity

$$\delta(a,b) = \min_{c\in[0,1]} \max_{1\le k\le 22} \Big(c\cos(kt_1) + (1-c)\cos(kt_2) - \tfrac{1}{2}\Big)$$

is the least violation available to the type (a,b), and it is positive exactly when that type is infeasible. Its minimum over the 528 types is $0.0740002839\ldots$, attained at $(a,b) = (8,17)$. The maximum of affine functions is convex, so each $\delta(a,b)$ is found exactly by evaluating at the endpoints and at the crossings of pairs of the 22 functions; the computation is carried out at 60 digits, and a margin of 0.074 in an inner product is not a rounding effect. ∎

Remark 5.47 (What that removes). Proposition 5.46 is not the proof of Theorem 5.40; its content is the following. A saturated configuration, if one exists, has no symmetry acting transitively on its directions: it is not a single orbit of anything. That removes every construction as a single orbit, and it is consistent with the rest of what is known about the case, since the only known contact configurations of 23 directions are the deletions, whose symmetry group is the stabiliser of a root in the Weyl group and has four orbits on the remaining directions, of sizes $1, 8, 6, 8$. A search looked for the rest: it minimised $\max_I |z_I|$ directly, from perturbed deletions, from random starts and from partially completed root systems, and over 202 starts it reached 12 contact configurations, all with the inner-product multiset of a deletion and all with g exactly $\frac{1}{2}$. The search is exploration and enters no proof.

The one configuration that comes close to the description is the deletion of a root, which lies on the wrong side of the circumradius condition and is moreover rigid, so a saturated configuration cannot be reached by bending it.

Proposition 5.48 (The deletion configurations are rigid)**.** *Let $W = \mathsf{R} \setminus \{u_0\}$ be the contact configuration obtained by deleting one root. Every first-order motion of W through contact configurations is an infinitesimal rotation.*

Proof. Scale the roots to a_i with $|a_i|^2 = 2$, so that the contact condition reads $\langle a_i, a_j \rangle \le 1$ and every quantity below is an integer. Write $i \sim j$ when $\langle a_i, a_j \rangle = 1$; deleting the root a_0 leaves 23 directions and a set T of 88 unordered pairs $\{i, j\}$ with $i \sim j$. A first-order motion is a tuple $d = (d_1, \dots, d_{23})$ with

$$\langle d_i, a_i \rangle = 0 \quad \text{for every } i, \qquad \langle d_i, a_j \rangle + \langle a_i, d_j \rangle \le 0 \quad \text{for every } \{i, j\} \in T, \tag{5.32}$$

the first condition keeping each direction on its sphere and the second keeping each tight pair from closing. The infinitesimal rotations $d_i = A a_i$ with A antisymmetric satisfy both with equality and span a space of dimension 6.

Let $c_i = \langle a_i, a_0 \rangle$ record the position of a_i relative to the deleted root; the values $-2, -1, 0, 1$ occur $1, 8, 6, 8$ times. Put

$$y_{ij} = \begin{cases} 1, & (c_i, c_j) \in \{(-1,-1),\ (-1,1)\}, \\ 2, & (c_i, c_j) \in \{(-2,-1),\ (-1,0),\ (0,1)\}, \\ 3, & (c_i, c_j) = (1,1), \end{cases} \qquad \mu_i = \begin{cases} -6, & c_i = -1, \\ -8, & \text{otherwise}, \end{cases}$$

the first for $\{i, j\} \in T$ with the pair written so that $c_i \le c_j$, the second for each i. Those six are the only types that occur among the pairs of T, and a direct integer computation gives the equilibrium relation

$$\sum_{j \,:\, j \sim i} y_{ij}\, a_j + \mu_i\, a_i = 0 \qquad \text{for every } i. \tag{5.33}$$

Pair (5.33) with d_i and sum over i. The diagonal terms contribute $\mu_i \langle d_i, a_i \rangle = 0$, and each unordered pair of T is met twice, contributing $y_{ij}(\langle d_i, a_j \rangle + \langle a_i, d_j \rangle)$, so

$$\sum_{\{i,j\} \in T} y_{ij} \big(\langle d_i, a_j \rangle + \langle a_i, d_j \rangle \big) = 0.$$

Every weight is positive and every bracket is nonpositive by (5.32), so every bracket vanishes: a first-order motion holds all 88 pairs at equality. The linear system consisting of those 88 equalities together with the 23 conditions $\langle d_i, a_i \rangle = 0$ has, over $\mathbb{Q}$, a solution space of dimension exactly 6, and the rotations already occupy 6 dimensions of it, so the two coincide. ∎

Since the deletion configuration is a tensegrity framework, 23 bars from the centre and 88 struts, infinitesimal rigidity is rigidity in the sense of Roth and Whiteley [RW81, Thm. 5.7]: no continuous deformation through contact configurations moves it except by rotation. A saturated 23-point configuration therefore cannot be a small perturbation of a deletion, which is the same conclusion the facet average reached above, now from the contact graph, not from the volume. The stress (5.33) and the dimension count are exact integer computations.

The same argument, with nothing deleted, places the root system itself.

Corollary 5.49 (The root system is rigid)**.** *Every first-order motion of* R *through contact configurations of* 24 *directions is an infinitesimal rotation, and* R *is rigid: a contact configuration of* 24 *directions sufficiently close to* R *is a rotation of it.*

The first half of this is not new. Cohn, Jiao, Kumar and Torquato [CJKT11, Prop. 3.3] prove that the D_4 root system is infinitesimally jammed, through the A_2 subsystems and not through a stress. We give the proof again because the stress, and not the jamming alone, is needed by Theorem 5.51, and because the same stress handles the deletion of Proposition 5.48. Neither their argument nor ours carries a radius as it stands; Theorem 5.51 supplies one.

Proof. The set T is now the set of the 96 pairs at $\langle a_i, a_j\rangle = 1$, the edges of the 24-cell, and every direction lies in 8 of them (Figure 13(b)). The eight roots at 60° from a root a_i sum to $4a_i$, so the constant stress, weight 1 on every pair of T and $\mu_i = -4$, satisfies (5.33); pairing it with a first-order motion as before forces all 96 brackets to vanish, and the system of those 96 equalities with the 24 conditions $\langle d_i, a_i\rangle = 0$ has, over $\mathbb{Q}$, a solution space of dimension 6, which the rotations fill. Rigidity follows from infinitesimal rigidity by [RW81, Thm. 5.7], the pairs at inner products 0, -1 and -2 staying strictly slack under a small motion. The count and the dimension are exact integer computations, and the equilibrium relation, the sum of the eight neighbours of every root being four times the root, is also checked in Lean. ∎

Corollary 5.49 is the local half of Theorem 5.25: the root system is isolated among contact configurations of 24 directions, and that much is elementary. What the semidefinite kernel of Theorem 5.30 supplies, and what no argument in a neighbourhood of R can, is the global half, that there is no such configuration at any distance from R; it is the one ingredient of the general theorems that rests on a certificate rather than on a closed argument, and no way of deriving it from anything local is known.

The rigidity of Corollary 5.49 comes with no radius. Roth and Whiteley produce a neighbourhood but not a measure of it, and a statement of isolation with no number in it cannot be combined with anything quantitative. It can be made explicit at little cost: the stress argument that proves Corollary 5.49 proves rather more if the quadratic terms are carried along instead of discarded, and what the carrying needs is one constant, the smallest nonzero singular value of the linear map that the argument linearises.

Write $\mathcal{T} = \{\tau \in (\mathbb{R}^4)^{24} : \langle \tau_i, a_i\rangle = 0\}$ for the tangent space, a subspace of dimension 72, and $\Lambda\colon \mathcal{T} \to \mathbb{R}^{96}$ for the *rigidity operator*

$$(\Lambda\tau)_{ij} = \tfrac{1}{\sqrt{2}}\big(\langle a_i, \tau_j\rangle + \langle a_j, \tau_i\rangle\big), \qquad \{i,j\} \in T, \tag{5.34}$$

where T is, as in the proof of Corollary 5.49, the set of the 96 pairs at $\langle a_i, a_j\rangle = 1$ and the factor normalises the roots. The first-order motions of Corollary 5.49 are the τ with $\Lambda\tau \le 0$, and that corollary says $\ker\Lambda$ is the 6-dimensional space of infinitesimal rotations.

Lemma 5.50 (Spectrum of the rigidity operator)**.** *The singular values of* Λ *are* $0,\ 1,\ \sqrt{5/2},\ \sqrt{3},\ 2$*, with multiplicities* $6,\ 29,\ 8,\ 21,\ 8$*. In particular the smallest nonzero singular value is exactly* 1*, so*

$$\|\Lambda\tau\|_2 \ \geq\ \|\tau\|_2 \qquad \text{for every } \tau \in \mathcal{T} \text{ orthogonal to } \ker\Lambda. \tag{5.35}$$

Moreover, the orthogonal projection Π *of* $\mathbb{R}^{96}$*, indexed by the pairs of* T*, onto* $\operatorname{im}\Lambda$ *has every diagonal entry equal to* $\frac{11}{16}$*, and consequently*

$$\|g\|_\infty \ \leq\ \tfrac{\sqrt{11}}{4}\,\|g\|_2, \qquad \|g\|_2 \ \leq\ \tfrac{\sqrt{11}}{4}\,\|g\|_1 \qquad \text{for every } g \in \operatorname{im}\Lambda. \tag{5.36}$$

Proof. The statement is a finite exact statement about one integral matrix. We set out what is computed and why it determines the spectrum; the computation is in integer arithmetic, from the integral roots, and is repeated in Lean.

Let $\Lambda' = \sqrt{2}\,\Lambda$, whose coefficients $\langle a_i, \cdot\rangle$ are integral, and let P be the orthogonal projection onto $\mathcal{T}$, with blocks $I - \frac{1}{2}a_i a_i^{\mathsf{T}}$, which is rational. Then $N = P\Lambda'^{\mathsf{T}}\Lambda' P$ has $4N$ integral, and the nonzero eigenvalues of N are twice the squares of the nonzero singular values of Λ. A direct computation in integer arithmetic gives the matrix identity

$$4N\,(4N - 8I)\,(4N - 20I)\,(4N - 24I)\,(4N - 32I) = 0,$$

so every eigenvalue of N lies in $\{0, 2, 5, 6, 8\}$, and the rank of $4N - cI$ over a field of positive characteristic, which bounds the rank over $\mathbb{Q}$ from below, is 66, 67, 88, 75, 88 for $c = 0, 8, 20, 24, 32$; the five multiplicities that result sum to 96, so each bound is met and the multiplicities are 30, 29, 8, 21, 8. The kernel of N contains the 24 radial directions, on which P vanishes, together with $\ker\Lambda$, and $30 = 24 + 6$ accounts for it exactly. Dividing by two and taking square roots gives the singular values. The identity and the ranks are computed by fraction-free elimination over $\mathbb{Z}$, with no floating point at any step; the 96 tight pairs the computation starts from, and the neighbour sum of Corollary 5.49, are checked again by Lean's kernel.

For the second statement, an orthogonal map R that permutes the roots, $Ra_i = a_{\pi(i)}$, acts on $\mathcal{T}$ by $(R\tau)_{\pi(i)} = R\tau_i$ and on $\mathbb{R}^{96}$ by permuting the pairs, and (5.34) gives $\Lambda(R\tau)_{\pi(i)\pi(j)} = (\Lambda\tau)_{ij}$. Therefore $\operatorname{im}\Lambda$ is invariant under these permutations and Π commutes with them. The group of such maps, the Weyl group of F_4, acts transitively on the 96 edges of the 24-cell, which are the pairs of T, so the diagonal entries of Π are all equal; they sum to $\operatorname{tr}\Pi = \operatorname{rank}\Lambda = 72 - 6 = 66$ by the first statement, so each is $\frac{66}{96} = \frac{11}{16}$. For $g \in \operatorname{im}\Lambda$ and a pair p, $|g_p| = |\langle g, \Pi e_p\rangle| \leq \|g\|_2\,\|\Pi e_p\|_2 = \frac{\sqrt{11}}{4}\|g\|_2$, and then $\|g\|_2^2 \leq \|g\|_\infty\|g\|_1 \leq \frac{\sqrt{11}}{4}\|g\|_2\|g\|_1$. The value $\frac{11}{16}$ of every diagonal entry is also checked directly, in integer arithmetic: with $2K = \Lambda'(2P)\Lambda'^{\mathsf{T}}$, which is integral and annihilated by $x(x-4)(x-10)(x-12)(x-16)$, the projection onto $\ker K$ is $(2K - 4I)(2K - 10I)(2K - 12I)(2K - 16I)/7680$, and each of its diagonal entries is $\frac{5}{16}$, in integer arithmetic and again in Lean. ∎

The second statement makes the passage from $\|\Lambda\tau\|_1$, which the stress argument below controls, to $\|\Lambda\tau\|_2$ inexpensive: the trivial $\|g\|_2 \leq \|g\|_1$ is sharp only for a vector concentrated on one pair, and the image of Λ contains no such vector.

Theorem 5.51 (A radius for the rigidity)**.** *Let* $W = \{w_1, \dots, w_{24}\} \subset S^3$ *be a contact configuration and put*

$$d(W) = \min_{R \in O(4)}\ \min_{\pi}\ \Big(\sum_{i=1}^{24} \big|w_{\pi(i)} - Ra_i/\sqrt{2}\big|^2\Big)^{1/2},$$

the minimum over orthogonal maps and over bijections π. Then either $d(W) = 0$, and W is an orthogonal image of the normalised root system, or

$$d(W) \;\geq\; \frac{2}{\sqrt{397}} \;=\; 0.1003771\ldots$$

Proof. Both minima are over compact sets and are attained; relabel and rotate so that they are attained at $\pi = \mathrm{id}$ and $R = I$, write $u_i = a_i/\sqrt{2}$, and put $\varepsilon_i = w_i - u_i$, $e_i = |\varepsilon_i|$ and $\|\varepsilon\|^2 = \sum_i e_i^2 = d(W)^2$.

Since $|w_i| = |u_i| = 1$, expanding $|u_i + \varepsilon_i|^2 = 1$ gives

$$\rho_i := \langle u_i, \varepsilon_i\rangle = -\tfrac{1}{2}e_i^2, \tag{5.37}$$

so the radial part of each displacement is quadratically small. Put $\tau_i = \varepsilon_i - \rho_i u_i$, so that $\tau \in \mathcal{T}$ and $|\tau_i|^2 = e_i^2 - \rho_i^2$.

Minimality in R fixes the gauge: differentiating $R \mapsto \sum_i |w_i - Ru_i|^2$ along $R = e^{sA}$ with A antisymmetric and setting $s = 0$ gives $\sum_i \langle \varepsilon_i, Au_i\rangle = 0$, and since $\langle u_i, Au_i\rangle = 0$ the same holds for τ. Hence τ is orthogonal to every infinitesimal rotation, hence to $\ker\Lambda$ by Corollary 5.49, and (5.35) applies.

Now the contact condition, kept to second order. For a pair $\{i,j\} \in T$, expanding $\langle w_i, w_j\rangle \leq \frac{1}{2}$ and using $\langle u_i, \varepsilon_j\rangle = \langle u_i, \tau_j\rangle + \frac{1}{2}\rho_j$,

$$g_{ij} := (\Lambda\tau)_{ij} \;\leq\; -\tfrac{1}{2}(\rho_i + \rho_j) - \langle \varepsilon_i, \varepsilon_j\rangle \;\leq\; \tfrac{1}{4}(e_i^2 + e_j^2) + e_i e_j$$

by (5.37) and Cauchy–Schwarz. Every index lies in exactly eight pairs of T, so $\sum_T (e_i^2 + e_j^2) = 8\|\varepsilon\|^2$ and $\sum_T e_i e_j \leq \frac{1}{2}\sum_T (e_i^2 + e_j^2) = 4\|\varepsilon\|^2$, and therefore $\sum_T g_{ij}^+ \leq 6\|\varepsilon\|^2$ for the positive parts. The constant stress of Corollary 5.49 gives $\sum_T g_{ij} = 0$ exactly, so the negative parts sum to as much as the positive ones and $\|\Lambda\tau\|_1 = 2\sum_T g_{ij}^+ \leq 12\|\varepsilon\|^2$. With (5.35) and (5.36),

$$\|\tau\|_2 \;\leq\; \|\Lambda\tau\|_2 \;\leq\; \tfrac{\sqrt{11}}{4}\,\|\Lambda\tau\|_1 \;\leq\; 3\sqrt{11}\,\|\varepsilon\|^2.$$

Finally $\sum_i \rho_i^2 = \frac{1}{4}\sum_i e_i^4 \leq \frac{1}{4}\|\varepsilon\|^4$ by (5.37), so

$$\|\varepsilon\|^2 = \|\tau\|_2^2 + \sum_i \rho_i^2 \;\leq\; 99\|\varepsilon\|^4 + \tfrac{1}{4}\|\varepsilon\|^4 \;=\; \tfrac{397}{4}\|\varepsilon\|^4,$$

and $\|\varepsilon\|$ is either 0 or at least $2/\sqrt{397}$. ∎

Corollary 5.52 (The twenty-four-contact case near the root system)**.** *A contact configuration W with $|W| = 24$ and $d(W) < 2/\sqrt{397}$ is an orthogonal image of the normalised roots, so $V_c(W)$ is a regular 24-cell and $\mathrm{vol}(V_c(W)) = 8$. Nothing in this uses a certificate.*

Proof. Theorem 5.51 gives $d(W) = 0$, and the volume is then Lemma 2.3 together with the invariance of volume under orthogonal maps. ∎

The line between the certificate and the closed arguments of this paper falls as follows. The global classification at 24 contacts is Theorem 5.30, proved from the verified kernel. What this section supplies is the quantitative statement that the classification by itself does not give: an explicit isolation radius about the normalised root system, $2/\sqrt{397}$ in the root-sum-of-squares displacement and about 0.02 per direction, within which Theorem 5.25 is proved directly and the kernel is not used. Beyond that radius only the kernel is used, on the rest of the 66-dimensional quotient of $(S^3)^{24}$ by $O(4)$, which is by far the larger part. The constant $397/4$ is not optimal either. What the argument uses is the least value ν of $\|\Lambda\tau\|_1$ over unit vectors $\tau \in \mathcal{T}$ orthogonal to $\ker\Lambda$, for which Lemma 5.50 gives $\nu \geq 4/\sqrt{11} = 1.206$, and the radius is $(144/\nu^2 + \frac{1}{4})^{-1/2}$. A displacement of a single direction,

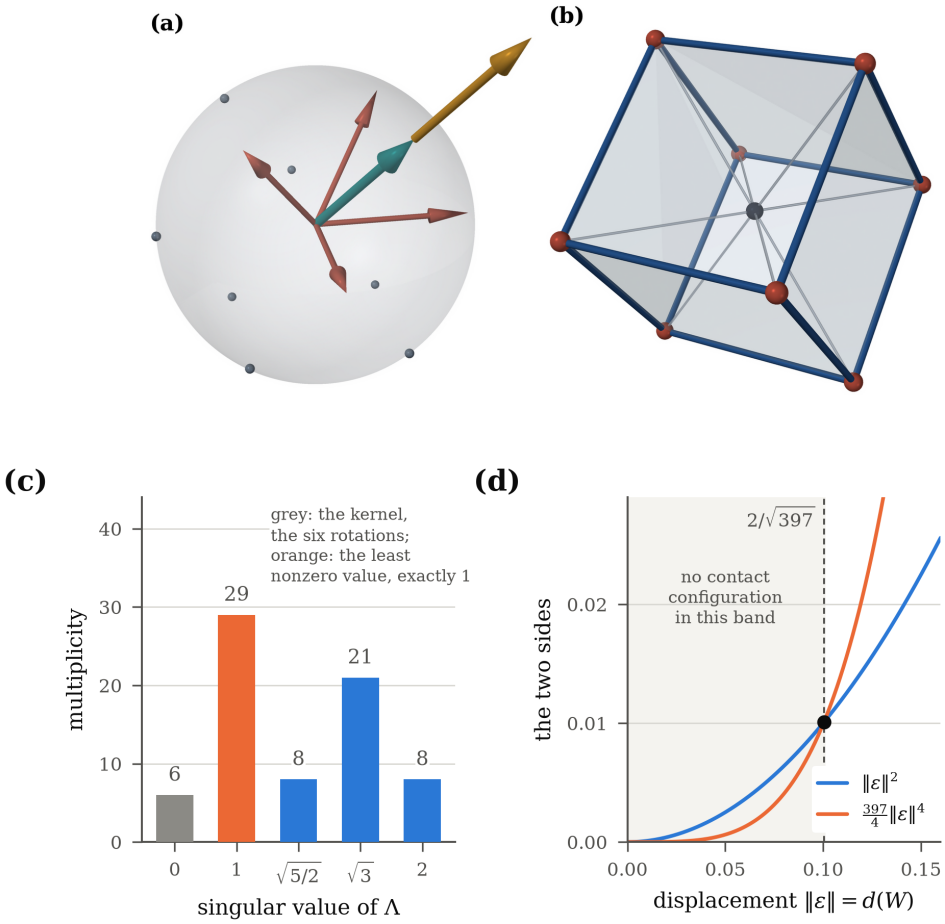

FIGURE 13. (a) The neighbour sum in dimension three, where it fits in one picture: a contact direction of the face-centred cubic lattice (teal), the four contact directions at 60° from it (clay), and their sum (ochre), which is twice the direction itself. (b) The same identity in dimension four, drawn where it lives. The eight roots at inner product 1 from a root α fill the hyperplane $\langle \alpha, x \rangle = 1$, a copy of $\mathbb{R}^3$, as the vertices of a cube centred at $\alpha/2$ (dark); eight times that centre is 4α. (c) The singular values of the rigidity operator Λ on the 72-dimensional tangent space, with multiplicities (Lemma 5.50). The kernel is the six infinitesimal rotations, and the smallest nonzero value is one of the two constants that turn the first-order argument into a radius; the other is the weight $\frac{11}{16}$ that every pair carries in the image of Λ. (d) The two sides of the inequality that closes Theorem 5.51. Feasibility forces $\|\varepsilon\|^2 \le \frac{397}{4}\|\varepsilon\|^4$, so the displacement is either zero or at least $2/\sqrt{397}$: the shaded band holds no contact configuration of 24 directions.

with its rotational part removed, has $\|\Lambda\tau\|_1/\|\tau\|_2 = \sqrt{96/11} = 2.954$, so $\nu \le 2.954$ and no sharpening of the constant can take this argument past 0.244. No argument of this kind reaches a configuration far from R, and the remaining difficulty lies with those configurations. Figure 13(a) and (b) draw the neighbour sum that makes the stress work, Figure 13(c) the spectrum that supplies the constant, and Figure 13(d) the two sides of the final inequality.

Beyond that radius only numerical exploration is available. Searches started from each of the twenty-four deletions, from greedy random packings and from annealed random starts produced no contact configuration of 23 directions with $g(W) > \frac{1}{2}$; the largest value found was $\frac{1}{2}$ itself, attained exactly at the deletions, and the random starts reached no 23-point contact configuration. This is exploration, not evidence of nonexistence.

What a proof of Theorem 5.40 has to supply is clear from Proposition 5.18: the covering estimate already uses the areas of the spherical Voronoi cells to the full, so the missing $0.083272\ldots$ has to come from their shape. The truncated volume of Section 5.11 reads the shape off pair by pair, from the lenses in which neighbouring caps overlap.

5.8. **How much of a root system a configuration can keep.** Proposition 5.48 says that a saturated configuration of 23 directions is not a bent deletion. The following makes that quantitative and turns it into a statement with no smallness in it: such a configuration can share at most 20 of its directions with any root system, so at least three of them are new. The argument is elementary and uses only the coordinates.

Take $\mathsf{R} = \{(e_i \pm e_j)/\sqrt{2} : 1 \le i < j \le 4\}$ and let $S \subseteq \mathsf{R}$. Write

$$A(S) = \{v \in S^3 : \langle v, r\rangle \le \tfrac{1}{2} \text{ for every } r \in S\} \tag{5.38}$$

for the directions that may be added to S, so that a contact configuration W with $W \cap \mathsf{R} = S$ satisfies $W \subseteq S \cup A(S)$. Call $\{i,j\}$ the *support* of the four roots $(\pm e_i \pm e_j)/\sqrt{2}$, and call a support *filled* for S when all four of its roots lie in S. The six supports pair off into three couples of complementary index pairs,

$$\{1,2\}\,|\,\{3,4\}, \qquad \{1,3\}\,|\,\{2,4\}, \qquad \{1,4\}\,|\,\{2,3\}, \tag{5.39}$$

and the argument uses only these three couples. Throughout, $t = 1/\sqrt{2}$ and $v = (x_1, x_2, x_3, x_4)$.

Lemma 5.53 (A filled support)**.** *If $\{i,j\}$ is filled for S, then every $v \in A(S)$ satisfies $|x_i| + |x_j| \le t$, and hence $x_i^2 + x_j^2 \le \frac{1}{2}$.*

Proof. The four conditions $\langle v, (\pm e_i \pm e_j)/\sqrt{2}\rangle \le \frac{1}{2}$ read $\pm x_i \pm x_j \le t$ over the four sign patterns, and the pattern matching the signs of x_i and x_j gives $|x_i| + |x_j| \le t$. Then $x_i^2 + x_j^2 \le (|x_i| + |x_j|)^2 \le t^2 = \frac{1}{2}$. ∎

Lemma 5.54 (A filled couple)**.** *If both supports of one of the couples (5.39) are filled for S, then $A(S) = \mathsf{R} \setminus S$.*

Proof. Let $\{i,j\}$ and $\{k,l\}$ be the couple and let $v \in A(S)$. By Lemma 5.53, $x_i^2 + x_j^2 \le \frac{1}{2}$ and $x_k^2 + x_l^2 \le \frac{1}{2}$, and the two sides add to $|v|^2 = 1$, so both are equalities. Equality in $x_i^2 + x_j^2 \le (|x_i| + |x_j|)^2$ forces $x_i x_j = 0$, and equality in $(|x_i| + |x_j|)^2 \le t^2$ forces $|x_i| + |x_j| = t$; so one of x_i, x_j is zero and the other is $\pm t$, and the same holds for x_k, x_l. Therefore $v = \pm t e_p \pm t e_q$ with $p \in \{i,j\}$ and $q \in \{k,l\}$, which is a root. A root in $A(S)$ cannot lie in S, since $\langle v, v\rangle = 1 > \frac{1}{2}$; and every root outside S does lie in $A(S)$, because distinct roots have inner product at most $\frac{1}{2}$. ∎

Proposition 5.55 (Twenty-two roots)**.** *If a contact configuration W satisfies $|W \cap \mathsf{R}| \ge 22$ for some root system R, then $W \subseteq \mathsf{R}$.*

Proof. Put $S = W \cap \mathsf{R}$. At most two roots are missing, and they lie in at most two of the six supports, so at least one of the three couples has both supports filled. By Lemma 5.54, $A(S) = \mathsf{R} \setminus S$, and $W \subseteq S \cup A(S) = \mathsf{R}$. ∎

At 21 the conclusion changes, and exactly one family of exceptions appears.

Proposition 5.56 (Twenty-one roots)**.** *If a contact configuration W satisfies $|W \cap \mathsf{R}| = 21$ for some root system R, then $W \subseteq \mathsf{R}$ or $|W| \le 22$.*

Proof. Let $S = W \cap \mathsf{R}$ and let ρ_1, ρ_2, ρ_3 be the missing roots. If their supports occupy at most two of the six index pairs, some couple is filled, so $A(S) = \{\rho_1, \rho_2, \rho_3\}$ by Lemma 5.54 and $W \subseteq \mathsf{R}$. Assume therefore that the three supports are distinct and meet each couple of (5.39) once. Choosing one index pair from each couple gives eight possibilities, and inspection of the list shows that four of them are *stars*, three pairs sharing an index, as with $\{1,2\},\{1,3\},\{1,4\}$, and four are *triangles*, the three pairs inside a triple of indices, as with $\{1,2\},\{1,3\},\{2,3\}$. Permuting the coordinates reduces to those two.

Let $v \in A(S)$ and suppose v is not a root. For each m the set $S \cup \{\rho_m\}$ misses only two roots, so $A(S \cup \{\rho_m\})$ consists of roots by Lemma 5.54, and therefore does not contain v. Since $v \in A(S)$, the only condition that can fail at $S \cup \{\rho_m\}$ is the one at ρ_m itself, so

$$\langle v, \rho_m \rangle > \tfrac{1}{2}, \qquad m = 1,2,3. \tag{5.40}$$

Suppose first that the supports form the star $\{1,2\},\{1,3\},\{1,4\}$. Sign changes in the coordinates $2,3,4$ are symmetries of R, so we may take $\rho_m = (\varepsilon_m e_1 + e_m)/\sqrt{2}$ with $\varepsilon_m \in \{\pm 1\}$ for $m = 2,3,4$. The supports $\{2,3\}$, $\{2,4\}$ and $\{3,4\}$ are filled, so $|x_m| \le t$ for $m = 2,3,4$ by Lemma 5.53, and (5.40) reads $\varepsilon_m x_1 + x_m > t$. If $\varepsilon_m = 1$ this gives $x_1 > t - x_m \ge 0$, and if $\varepsilon_m = -1$ it gives $x_1 < x_m - t \le 0$; the two cannot both occur, so the three signs agree, and after a sign change in the first coordinate $\rho_m = (e_1 + e_m)/\sqrt{2}$ for $m = 2,3,4$. Those three roots are pairwise at $60°$.

Suppose instead that the supports form the triangle $\{1,2\},\{1,3\},\{2,3\}$, and write $\rho_1 = (ae_1 + be_2)/\sqrt{2}$, $\rho_2 = (ce_1 + de_3)/\sqrt{2}$, $\rho_3 = (fe_2 + ge_3)/\sqrt{2}$ with signs $a, \dots, g$. The supports $\{1,4\},\{2,4\},\{3,4\}$ are filled, so $|x_m| \le t$ for $m = 1,2,3$. In coordinates (5.40) reads

$$ax_1 + bx_2 > t, \qquad cx_1 + dx_3 > t, \qquad fx_2 + gx_3 > t.$$

If $c = -a$, the sum of the first two is $bx_2 + dx_3 > 2t$, which $|x_2|, |x_3| \le t$ forbid; in the same way the first and third exclude $f = -b$, and the second and third exclude $g = -d$. Hence $a = c$, $b = f$ and $d = g$, and ρ_1, ρ_2, ρ_3 are again pairwise at $60°$. Changing the signs of the coordinates $1,2,3$ as needed makes the triple $\{(e_1 + e_2), (e_1 + e_3), (e_2 + e_3)\}/\sqrt{2}$, and the orthogonal map with matrix $\frac{1}{2}\left(\begin{smallmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{smallmatrix}\right)$, which permutes R, carries it to $\{(e_1 + e_2), (e_1 + e_3), (e_1 - e_4)\}/\sqrt{2}$; one more sign change in the fourth coordinate returns the star.

Thus in every case we may take $\rho_m = (e_1 + e_m)/\sqrt{2}$ for $m = 2,3,4$, and it remains to read off $A(S)$ for that S. The supports $\{2,3\}$, $\{2,4\}$, $\{3,4\}$ are filled, so $|x_2| + |x_3|$, $|x_2| + |x_4|$ and $|x_3| + |x_4|$ are all at most t. The maximum of $p^2 + q^2 + r^2$ over $\{p,q,r \ge 0 :\ p + q \le t,\ p + r \le t,\ q + r \le t\}$ is attained at a vertex of that polytope, and its vertices are the origin, the three points te_k and $\frac{t}{2}(1,1,1)$, with values 0, t^2 and $\frac{3}{4}t^2$. Hence

$$x_2^2 + x_3^2 + x_4^2 \le t^2 = \tfrac{1}{2}, \qquad \text{so} \qquad x_1^2 \ge \tfrac{1}{2}. \tag{5.41}$$

Each support $\{1,m\}$ keeps three of its four roots, all but $(e_1 + e_m)/\sqrt{2}$, and these give $x_1 - x_m \le t$ and $|x_m| - x_1 \le t$. If $x_1 \le -t$ the second gives $|x_m| \le t + x_1 \le 0$, so $x_m = 0$ for $m = 2,3,4$ and $x_1 = -t$, against $|v| = 1$. Therefore $x_1 \ge t$, and then $x_m \ge x_1 - t \ge 0$ for $m = 2,3,4$: on $A(S)$ the first coordinate is at least t and the other three are nonnegative. Consequently, for $v, v' \in A(S)$,

$$\langle v, v' \rangle = x_1x_1' + x_2x_2' + x_3x_3' + x_4x_4' \ \ge\ x_1x_1' \ \ge\ t^2 = \tfrac{1}{2}, \tag{5.42}$$

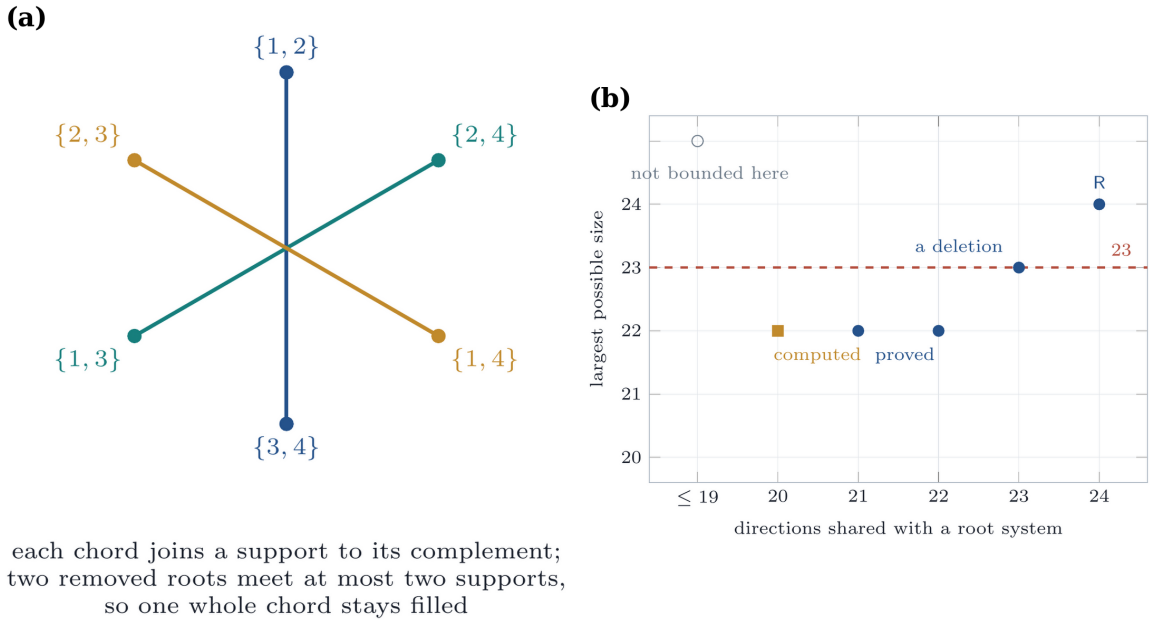

FIGURE 14. (a) The six supports of the roots, joined in the three couples of complementary index pairs. Removing two roots touches at most two supports and leaves a whole couple filled, which is Lemma 5.54; removing three can meet each couple once, and the eight ways of doing that are four stars and four triangles. (b) The largest size a contact configuration can have, against the number of directions it shares with a root system. The value 23 occurs only inside the root system, where the circumradius is exactly 2, so the saturated case is pushed to twenty directions or fewer.

with equality only if $x_1 = x_1' = t$. That makes (5.41) an equality, so (x_2, x_3, x_4) is one of the vertices te_k, and $v \in \{\rho_1, \rho_2, \rho_3\}$; the same holds for v'.

Now $W \setminus S$ lies in $A(S)$ and its elements are pairwise at 60° or more. By (5.42) any two of them are missing roots, so either every element of $W \setminus S$ is a root, in which case $W \subseteq \mathsf{R}$, or $W \setminus S$ has a single element and $|W| = 22$. ∎

Corollary 5.57 (A saturated configuration is far from every root system)**.** *A contact configuration of* 23 *directions whose cell has circumradius below* 2 *meets every root system in at most* 20 *directions.*

Proof. Let $|W| = 23$ and suppose $|W \cap \mathsf{R}| \geq 21$. By Proposition 5.55 and Proposition 5.56, either $|W| \leq 22$, which is false, or $W \subseteq \mathsf{R}$ and W is R with one root u_0 removed. In the second case the cell of W contains the point $2u_0$, by Lemma 6.1, so its circumradius is 2. ∎

Figure 14 collects the two ingredients. The bound of Corollary 5.57 separates a hypothetical saturated configuration from the root system by a fixed amount, not by an unspecified small one: three of its twenty-three directions lie outside every copy of R, so it is not reachable from a deletion by exchanging one or two directions.

Remark 5.58 (The next rung, and what it costs)*.* The same reduction can be tried at twenty roots. Of the 10626 ways to remove four roots, only four orbits under the symmetry group of R leave a cell of circumradius above 2, and in each of those the removed set is a triple pairwise at 60° together with one further root. For each of the four we minimised the

largest of the three pairwise inner products over triples of directions in $A(S)$, requiring each direction to stay an angle δ away from every root, so that the removed roots themselves are excluded. At $\delta = 0$ the value is $\frac{1}{2}$ in all four, and the minimiser is three of the removed roots; at $\delta = 2°$ the smallest of the four values is already $0.538256\ldots$, at $\delta = 5°$ it is $0.592985\ldots$, at $\delta = 10°$ it is $0.676602\ldots$, and it increases from there. Hence the doorways at twenty roots hold two directions but not three, which would push Corollary 5.57 to nineteen. This is a computation and not a proof: it searches a family of local minima and is not exhaustive. The same enumeration is carried out for one, two and three removed roots in exact integer arithmetic as a check on Proposition 5.55 and Proposition 5.56.

5.9. **The remaining case as a question about codes.** A contact configuration of m directions is the same object as a spherical code of m points on S^3 with minimal angle at least 60°, and the question is how much room a code of that size has. Write $\theta(m)$ for the largest minimal angle available to m points on S^3. Musin's theorem [Mus08] gives $\theta(25) < 60°$, and Theorem 5.25 gives $\theta(24) = 60°$, since a code of 24 points at more than 60° would be a contact configuration of that size and hence a copy of the root system, whose minimal angle is exactly 60°. The value at $m = 23$ decides the shape of the set of configurations that Theorem 5.40 is about.

Proposition 5.59 (Slack, and its absence)**.** *If $\theta(m) > 60°$, the contact configurations of m directions form a set with nonempty interior in $(S^3)^m$. If $\theta(m) = 60°$, that set has empty interior, and every contact configuration of m directions attains $\theta(m)$.*

Proof. The contact configurations of size m are exactly the m-point codes of minimal angle at least 60°, and the minimal angle is a continuous function on $(S^3)^m$. If $\theta(m) > 60°$, a code attaining $\theta(m)$ has a neighbourhood on which the minimal angle stays above 60°, and every configuration in it is a contact configuration. If instead $\theta(m) = 60°$, then a contact configuration W has minimal angle exactly 60°, so some pair w_i, w_j is at 60°; rotating w_i towards w_j through any positive angle, however small, gives a configuration at distance as small as one likes from W whose minimal angle is below 60°, and which is therefore not a contact configuration. Thus W is not an interior point. ∎

At $m \leq 22$ the first alternative is the one that holds, and the configurations at those sizes fill out an open set; that is why Corollary 5.17 has to hold uniformly over a family of positive dimension, and it does. At $m = 23$ every known computation, that of Remark 5.62 and the published tables [Slo], returns $\theta(23) = 60°$, and if that is the true value the second alternative takes over: the configurations that Theorem 5.40 is about are extremal codes, not a family with interior. That is the structural difference between 23 and every size below it, and a set with empty interior may have no saturated member. That $\theta(23) = 60°$ is not proved here; Section 5.10 shows how the codes of 23 points behave as their minimal angle is pushed towards 60° from below. Theorem 5.40 does not need it: the volume bound is proved in Section 5.12 for every contact configuration of 23 directions, saturated or not.

Theorem 5.60 (The 23-point case from a uniqueness statement)**.** *Suppose every set of* 23 *unit vectors in $\mathbb{R}^4$ with pairwise inner products at most $\frac{1}{2}$ is the image under an orthogonal map of $\mathsf{R} \setminus \{u\}$ for some root u. Then no saturated* 23*-point contact configuration exists.*

Proof. Let W be a contact configuration with $|W| = 23$, and take an orthogonal O and a root u with $W = O(\mathsf{R} \setminus \{u\})$. Every $w \in W$ is Or for a root $r \neq u$, so $\langle Ou, w\rangle = \langle u, r\rangle \leq \frac{1}{2}$, and $W \cup \{Ou\}$ is again a contact configuration. By Proposition 5.38 the circumradius of $V_c(W)$ is then at least 2, so W is not saturated. ∎

Remark 5.61 (Why a uniqueness statement and not a bound). The hypothesis of Theorem 5.60 is what Theorem 5.25 proves at 24 points, asked at 23, and one difference between the two carries the whole of the difficulty. At 24 the code has maximal size, so the semidefinite relaxation is tight there and complementary slackness forces every inner product into $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$, which is Theorem 5.30; the classification is then combinatorial. At 23 nothing is tight. No upper bound on the size of a code in terms of its minimal angle can be of any use here, whatever the method that produces it: such a bound is at least 24 at $60°$, because a code of 24 points at that angle exists, so it cannot exclude 23 points at the same angle. What is wanted is a classification at one fixed size below the maximum, which is a different kind of statement, and as far as we know it has not been attempted in this dimension.

Remark 5.62 (What the search returns). We looked for a 23-point code of minimal angle above $60°$, and for a contact configuration of 23 directions whose Gram matrix is unlike that of a deletion, by Riesz continuation: minimise $\sum_{i<j} |w_i - w_j|^{-s}$ over the sphere with s running through $4, 16, 64, 256, 1024$, then reduce the largest inner product directly. Calibration comes first. Over 120 starts the same procedure returns $0.5000071\ldots$ at $m = 24$, where the true value is $\frac{1}{2}$; $0.4980170\ldots$ at $m = 22$, a minimal angle of $60.1311\ldots°$, against the $60.1398863°$ of the published tables [Slo]; and, over 60 starts, $0.5374065\ldots$ at $m = 25$, where no contact configuration exists. Therefore the procedure locates the root system, detects the slack at 22 and the obstruction at 25, and one can read how close to $\frac{1}{2}$ it gets when $\frac{1}{2}$ is the answer. At $m = 23$, over 1500 random starts, the smallest largest inner product it reached was $0.5000078\ldots$, which is nearer to $\frac{1}{2}$ than the calibration run at $m = 24$ managed; no start went below $\frac{1}{2}$; and of the 32 outcomes that came within 5×10^{-3} of feasibility, all 32 had the inner-product multiset of a deletion, namely -1 eleven times, $-\frac{1}{2}$ eighty-eight times, 0 sixty-six times and $\frac{1}{2}$ eighty-eight times, which is an invariant of the orbit under $O(4)$. The tables [Slo] record $60.0000000°$ at 23 points, the same value as at 24, and $60.1398863°$ at 22. None of this proves the hypothesis of Theorem 5.60; it says which statement to aim at.

One finite family can be searched completely instead of sampled, and it is the natural place for a code that is not a subset of a root system. The 600-cell has 120 vertices, the unit icosians, and its inner products are $0, \pm\frac{1}{2}, \pm\cos 36°, \pm\cos 72°$ and ± 1; two vertices are closer than $60°$ exactly when they are joined by an edge, at $36°$. Its subsets with all inner products at most $\frac{1}{2}$ are therefore the independent sets of the edge graph, which is 12-regular, and the largest of them have 24 elements and are the twenty-five inscribed copies of R, five through each vertex. The question is whether a 23-element independent set can avoid all twenty-five.

Proposition 5.63 (Codes among the vertices of the 600-cell). *Let $I \subset S^3$ be the vertex set of the regular 600-cell. A subset of I whose pairwise inner products are all at most $\frac{1}{2}$ has at most 24 elements; the subsets with 24 elements are the twenty-five inscribed copies of R; and every subset with 23 elements is one of those copies with a single vertex removed. In particular no saturated contact configuration of 23 directions lies in I.*

Proof. Write $\phi = \frac{1}{2}(1 + \sqrt{5})$. Doubling the coordinates puts every vertex of the 600-cell in $\mathbb{Z}[\phi]^4$, with each coordinate one of $0, \pm 1, \pm 2, \pm\phi, \pm(\phi - 1)$, and $4\langle u, v\rangle$ is then computed exactly in $\mathbb{Z}[\phi]$; the nine values that occur are 0, ± 2, ± 4, $\pm 2\phi$ and $\pm(2\phi - 2)$, of which only 2ϕ, that is $\langle u, v\rangle = \cos 36°$, exceeds 2. Hence the sets in question are exactly the independent sets of the graph Γ on I whose edges join vertices at $36°$, and every vertex of Γ

has degree 12, its vertex figure being an icosahedron. The symmetry group of the 600-cell is transitive on I, so it suffices to enumerate the independent sets through one fixed vertex v_0. This is a finite search, carried out completely by a depth-first enumeration over bitsets, with the standard pruning that an independent set meets every clique of Γ at most once, so that a greedy partition of the remaining candidates into cliques bounds what can still be added.

The search returns no independent set of size 25 through v_0; exactly five of size 24, each of which has all its pairwise inner products in $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$ and is therefore a copy of R (by Theorem 5.25, or directly, since the 24-point contact configurations inside I are checked one by one); and exactly 115 of size 23, every one of which is contained in one of the five copies of R through v_0. Since a copy of R through v_0 has 23 other vertices and $5 \times 23 = 115$, the sets of size 23 through v_0 are precisely those five copies with one of their other vertices removed, and nothing else. Summing over the 120 vertices, $115 \times 120/23 = 600 = 25 \times 24$ independent sets of size 23 in all, one for each pair consisting of an inscribed copy of R and one of its vertices, and $5 \times 120/24 = 25$ inscribed copies.

A set of 23 vertices of one copy of R is a deletion, and is extended by the vertex it lacks, so by Proposition 5.38 its cell has circumradius 2 and it is not saturated.

The enumeration was carried out three times, independently, twice in ordinary exact arithmetic and once in Lean, where the vertex list, the nine inner-product values, the degree 12, the twenty-five cells, their inner products and the fact that five pass through each vertex are checked by the kernel from the $\mathbb{Z}[\phi]$ coordinates, and the three counts 0, 5 and 115 by the same depth-first search (Section 11). ■

Remark 5.64 (What the 600-cell decides). Proposition 5.63 settles the first family of candidates (Figure 16(a), drawn on page 64). The unit icosians form the largest finite subgroup of the unit quaternions, the binary icosahedral group of order 120; they contain the twenty-four Hurwitz units, which are R, in twenty-five different ways; and a subset of them with all inner products at most $\frac{1}{2}$ is the obvious candidate for a code built on the golden ratio rather than on $\sqrt{2}$. The proposition says that no such subset of 23 points escapes a copy of R. It does not cover codes whose coordinates lie outside $\mathbb{Q}(\sqrt{5})$, nor the continuous families that Section 5.10 finds once the minimal angle is allowed to drop below 60°, none of which has the inner-product spectrum of a subset of the 600-cell.[19] The same enumeration on the 48 unit quaternions of the binary octahedral group, the vertices of two 24-cells in dual position (Figure 15), in which two directions are closer than 60° exactly when they lie in different copies and are 45° apart, returns the two copies of R and their forty-eight deletions and nothing else; there the arithmetic is in $\mathbb{Z}[\sqrt{2}]$.

5.10. **The largest inradius as the minimal angle rises.** The remaining case is the value at one point of a function that can be computed away from that point. For $\delta \geq 0$ let

$$K(\delta) = \{W = (w_1, \dots, w_{23}) \in (S^3)^{23} : \langle w_i, w_j \rangle \leq \tfrac{1}{2} + \delta \text{ for all } i \neq j\},$$

the configurations of 23 directions whose minimal angle is at least $\arccos(\frac{1}{2} + \delta)$, and let

$$h(\delta) = \max_{W \in K(\delta)} g(W), \qquad g(W) = \min_{\theta \in S^3} \max_i \langle \theta, w_i \rangle,$$

[19] A random sampling of maximal independent sets preceded the enumeration: two million of them, drawn by greedy growth along random orders of the vertices, had sizes 10 to 22 and 24, and never 23. The enumeration replaces that observation by a statement.

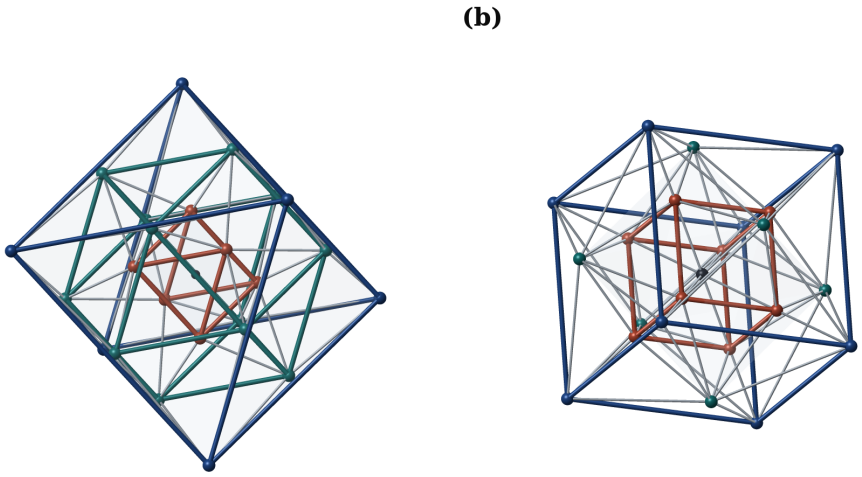

FIGURE 15. (a) The 24 contact directions of D_4 seen from a deep hole (dark), radius proportional to angle: an octahedron at 45° (clay), a cuboctahedron at 90° (teal), an octahedron at 135° (blue), and the 96 pairs at 60° as edges, grey between shells. (b) The dual 24-cell seen from the same hole and on the same scale. The hole is now itself a direction (dark), and the others form a cube at 60° (clay), an octahedron at 90° (teal) and a cube at 120° (blue); the edges are the 88 pairs at 60° that avoid the antipode of the hole, which lies at 180° and is not drawn. The two panels together carry the 48 directions of the binary octahedral group.

so that g is the function of Proposition 5.38, the cosine of the covering radius, and the inradius of $\operatorname{conv}(W)$ whenever that hull contains the origin. The contact configurations are $K(0)$, and by Proposition 5.38 a member of $K(0)$ is saturated exactly when $g > \frac{1}{2}$.

Lemma 5.65 (The limit from above)**.** *The function h is nondecreasing on $[0,\infty)$, $h(0) \geq \frac{1}{2}$, and $h(\delta) \to h(0)$ as $\delta \downarrow 0$. No saturated contact configuration of 23 directions exists if and only if $h(0) = \frac{1}{2}$; and this holds as soon as $h(\delta) \leq \frac{1}{2} + C\delta^{\kappa}$ on some interval $(0,\delta_0)$, for any constants C and $\kappa > 0$.*

Proof. The map $(\theta, W) \mapsto \max_i \langle \theta, w_i \rangle$ is continuous, and $|\max_i \langle \theta, w_i \rangle - \max_i \langle \theta, w_i' \rangle| \leq \max_i |w_i - w_i'|$, so g is continuous on $(S^3)^{23}$, indeed 1-Lipschitz in the maximum norm. Each $K(\delta)$ is a closed subset of the compact space $(S^3)^{23}$, so the maximum defining $h(\delta)$ is attained; the sets increase with δ, so h is nondecreasing; and $K(0)$ contains the deletions of a root, at which $g = \frac{1}{2}$, so $h(0) \geq \frac{1}{2}$. For the limit take $\delta_n \downarrow 0$ and $W_n \in K(\delta_n)$ with $g(W_n) = h(\delta_n)$. By compactness a subsequence converges to some W, and $W \in K(\delta_m)$ for every m, since $K(\delta_m)$ is closed and contains W_n for all $n \geq m$; hence $W \in \bigcap_m K(\delta_m) = K(0)$. By continuity $g(W) = \lim h(\delta_n)$, and $g(W) \leq h(0)$ because $W \in K(0)$, while $h(\delta_n) \geq h(0)$ by monotonicity; so the limit is $h(0)$, along every such sequence. The equivalence is the definition of saturation, and the last statement follows by letting $\delta \downarrow 0$ in the bound. ∎

Thus the hypothesis of Theorem 5.60 is the statement $h(0) = \frac{1}{2}$, and h can be looked at for $\delta > 0$, where the constraint set has interior and a local optimiser has room to move. We did this with a trust-region sequential linear programme, which maximises the smallest facet offset of $\operatorname{conv}(W)$ over tangent moves of bounded size subject to the linearised contact

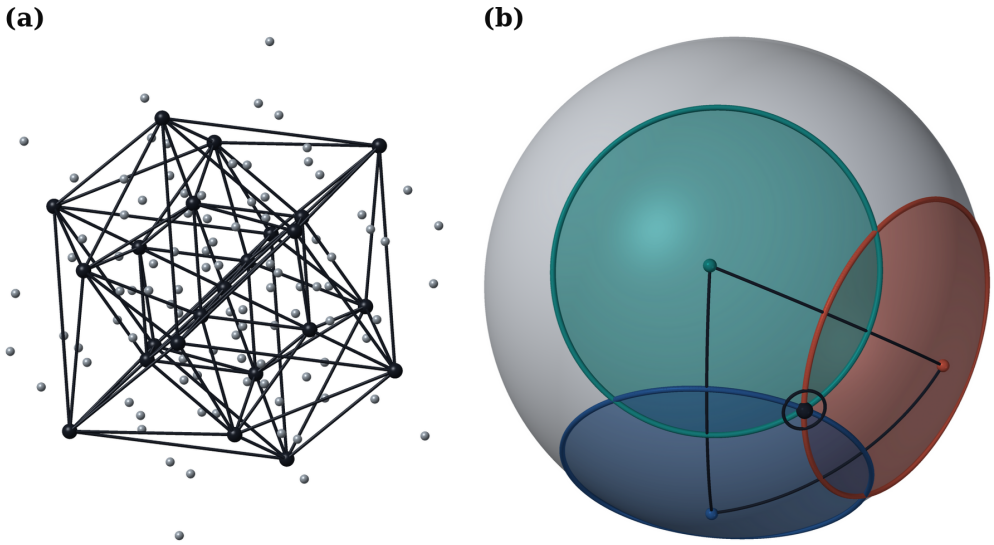

FIGURE 16. (a) The 600-cell seen from one vertex, radius proportional to angle: its vertices in shells at 36° to 144° (grey) and, darker and joined by their 88 edges, the 23 vertices of an inscribed 24-cell, the centre among them, whose twenty-fourth vertex is the antipode at 180° and is not drawn; they form a deletion. (b) The cap lemma on a great 2-sphere: three contact directions 60° apart carry caps of radius $r_* = 35.26°$ that meet in exactly one point, the circumcentre of their triangle (ringed); any smaller radius separates them.

constraints, and accepts a move only when the true inradius, read from a fresh convex hull, improves without any constraint being violated. For each δ on a decreasing schedule the starts are the twenty best configurations of the previous level, restored to the new constraint, together with a batch of fresh random starts; three passes were run, with different seeds and with 40, 60 and 80 fresh starts per level, and Table 3 records the largest inradius found at each level over the three. Every number in the table is a lower bound for $h(\delta)$ found by local search; nothing in this subsection is a proof, and nothing later uses it.

The table shows three things. First, for $\delta \geq 0.009$, that is for minimal angles up to 59.40°, the best configurations have inradius between 0.65 and 0.70: their covering radius is between 46° and 49°, so twenty-three caps of radius 49° about their directions cover the sphere, where the deletion of a root needs caps of radius 60° and the root system itself, with twenty-four directions, needs 45°. These configurations are not perturbed deletions in any sense. Their inner-product multisets have no pair at -1, where a deletion has eleven antipodal pairs, and none within 10^{-3} of $-\frac{1}{2}$ or of 0; between three and fifty-three of their pairs sit at the imposed bound $\frac{1}{2} + \delta$, and fewer than three per cent of their inner products lie within 10^{-3} of a value taken by the 600-cell, so none lies in the 600-cell either. Second, between $\delta = 0.009$ and $\delta = 0.008$ they vanish. At every level from 0.008 down the best inradius is $\frac{1}{2} + O(\delta)$, with $h(\delta) - \frac{1}{2}$ between 1.1δ and 2δ across the whole range down to $\delta = 5 \times 10^{-5}$, and the configurations attaining it are deletions of a root with their 88 tight pairs opened by amounts of order δ: every one of them keeps the eleven antipodal pairs of a deletion, and as δ decreases their inner-product multisets settle onto the deletion's, which is reached to 10^{-3} by $\delta = 3 \times 10^{-4}$. The transition is not gradual: a configuration of the first kind, restored to the constraint at $\delta = 0.008$ by the feasibility programme, does not

TABLE 3. The largest inradius g found at each level of the continuation, over three passes. The third column is the minimal angle $\arccos(\frac{1}{2}+\delta)$ imposed, the fifth the covering radius $\arccos g$ of the best configuration, and the last two the number of feasible endpoints across the three passes and how many of them are deletions of a root. At $\delta = 0$ every endpoint is a deletion and $g = \frac{1}{2}$ exactly.

δ	min. angle	$h(\delta) \geq$	cov. radius	endpoints	deletions
0.05	56.633°	0.676201	47.453°	180	0
0.03	57.995°	0.695650	45.921°	239	0
0.02	58.668°	0.686522	46.645°	159	0
0.015	59.003°	0.673135	47.691°	113	0
0.01	59.336°	0.657060	48.924°	104	0
0.009	59.403°	0.653786	49.172°	34	0
0.008	59.469°	0.508771	59.418°	2	0
0.007	59.536°	0.508968	59.405°	11	0
0.005	59.669°	0.505913	59.608°	15	0
0.003	59.801°	0.504484	59.703°	22	0
0.002	59.868°	0.502914	59.807°	26	0
0.0015	59.901°	0.502552	59.831°	34	0
0.001	59.934°	0.501392	59.908°	40	0
0.0007	59.954°	0.501123	59.926°	42	0
0.0005	59.967°	0.500878	59.942°	46	0
0.0003	59.980°	0.500480	59.968°	52	0
0.0002	59.987°	0.500378	59.975°	56	0
0.0001	59.993°	0.500202	59.987°	56	0
0.00005	59.997°	0.500081	59.995°	63	0
0	60.000°	0.500000	60.000°	51	51

reach it, because its minimal angle sits at a local maximum below 59.47° from which no local move raises it. Third, at $\delta = 0$ every feasible endpoint of every pass is a deletion of a root, with the inner-product multiset -1 eleven times, $-\frac{1}{2}$ eighty-eight times, 0 sixty-six times, $\frac{1}{2}$ eighty-eight times, and $g = \frac{1}{2}$ to fifteen digits. Figure 17 draws the table.

Configurations with a symmetry. A configuration invariant under a nontrivial element of $O(4)$ is a union of orbits of that element, and inside such a class the search has far fewer parameters and can be run from enough starts to cover it. Proposition 5.46 disposes of the transitive case. We treated every rotation of order $n \leq 12$, through angles $2\pi a/n$ and $2\pi b/n$ in two orthogonal planes with $\gcd(a, b, n) = 1$, together with the two improper involutions, a reflection in a hyperplane and $\mathrm{diag}(1, -1, -1, -1)$; for each it lists every way of writing 23 as a sum of orbit sizes, the generic orbit having n points and a point of either plane an orbit of $n/\gcd(a, n)$ or $n/\gcd(b, n)$, with at most six points on either plane since they lie on a great circle. That gives 73 orbit structures, with between 5 and 35 parameters each, and in each the continuation of Table 3 is run from sixteen random starts on the orbit representatives, through $\delta = 0.05, 0.02, 0.01, 0.005, 0.002, 0$. Eleven of the seventy-three structures reached a feasible endpoint at $\delta = 0$, sixty-nine endpoints in all, and every one of them is a deletion of a root: the structures are those with a rotation of order 2, 3 or 4 fixing a plane pointwise, a reflection with five fixed points, and $\mathrm{diag}(1, -1, -1, -1)$, which are symmetries that a deletion has, its symmetry group being the stabiliser of the deleted root in the symmetry group of R. The remaining sixty-two structures never reached the constraint at $\delta = 0$ from

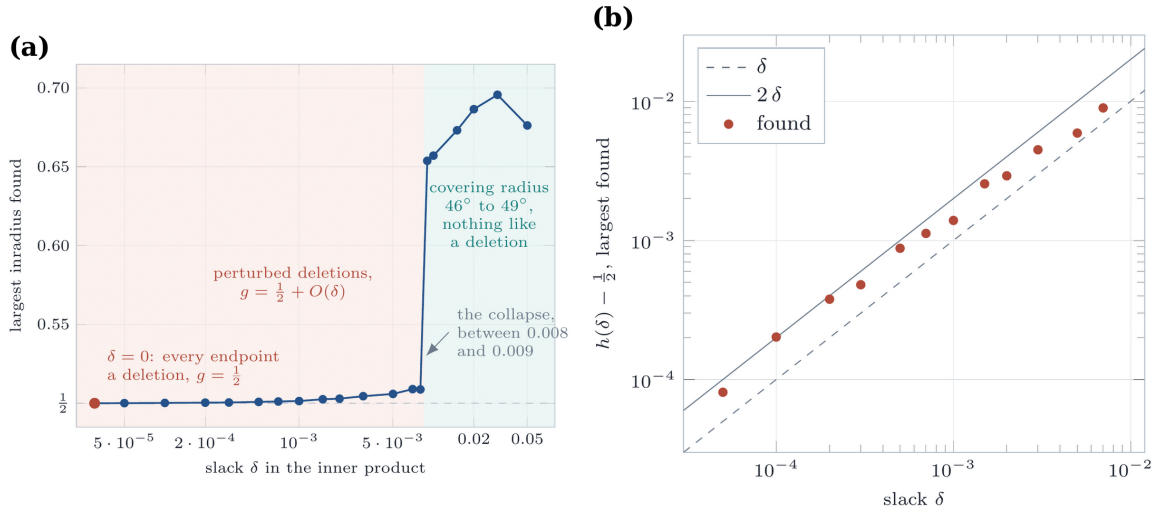

FIGURE 17. (a) The continuation of Table 3: the largest inradius found, against δ on a logarithmic axis. The plateau near $\frac{2}{3}$ runs down to $\delta = 0.009$ and is carried by configurations far from any deletion; from $\delta = 0.008$ down it collapses to $\frac{1}{2} + O(\delta)$. (b) The same table below the collapse: $h(\delta) - \frac{1}{2}$ against δ on logarithmic axes, with the lines δ and 2δ for comparison. The approach to $\frac{1}{2}$ is linear in δ, which is what a perturbed deletion gives.

any start. At $\delta = 0.01$ the best symmetric configuration has inradius 0.6599, in line with Table 3. No symmetric configuration crossed into $K(0)$ except the deletions.

Remark 5.66 (What the continuation says). The picture is the one that the hypothesis of Theorem 5.60 describes, and it locates where a proof of that hypothesis has to work. By Lemma 5.65 it is enough to show $h(\delta) \le \frac{1}{2} + C\delta$ on some interval $(0, \delta_0)$, and the data put the constant near 2 for small δ; but the same data put $h(0.009)$ above 0.65, so no such bound holds with a uniform constant beyond $\delta_0 \approx 0.008$, and an argument has to work inside a window of about half a degree in the minimal angle. That is the quantitative form of Remark 5.61: the classification at 23 points is a statement about codes of minimal angle exactly 60°, and codes at 59.3° already behave in a completely different way. A saturated configuration, if one exists, is a member of $K(0)$ with $g > \frac{1}{2}$, hence of every $K(\delta)$, and forces $h(\delta) \ge g$ for all δ; none appeared in three passes of the continuation, nor in 300 direct maximisations of g over $K(0)$ by sequential quadratic programming.[20] Of those, 94 reached a feasible endpoint, every one of them a deletion with $g = \frac{1}{2}$, and the 100 random starts, brought down through the same schedule of slacks, never reached the constraint at $\delta = 0$, which is the collapse seen from the other side. This is a search, not a proof.

5.11. **The remaining case as a strict inequality.** The preceding subsections approach Theorem 5.40 through the configuration. This subsection approaches it through the cell and arrives at an inequality between the pair angles of the configuration that proves Theorem 5.40, that the deletion of a root satisfies with room to spare, and that Section 5.12 establishes. The tools are the covering bound of Theorem 5.16 and the exact overlaps of

[20]The starts were of three kinds, taken in rotation: random points pushed apart until they satisfy the contact constraints; a deletion of a root with each direction rotated through a random angle of up to 0.05, 0.15, 0.4 or 0.8 radians; and twenty random roots completed by three random directions.

Proposition 5.23; what is added is the choice of where the integration stops, and the gain from stopping there.

Lemma 5.67 (Contact directions in a cap)**.** *If k contact directions lie in a closed cap of angular radius r on S^3, then $\sin^2 r \geq \dfrac{k-1}{2k}$. In particular two contact directions need a cap of radius at least $30°$, three need $r_* = \arcsin(1/\sqrt{3}) = 35.2644\ldots°$, and four need $r_4 = \arcsin\sqrt{3/8} = 37.7612\ldots°$.*

Proof. A point θ of the cap about c satisfies $|\theta-(\cos r)c|^2 = 1-2\cos r\,\langle\theta,c\rangle+\cos^2 r \leq \sin^2 r$, so the cap lies in the Euclidean ball of radius $\sin r$ about $(\cos r)c$. For points $x_1,\ldots,x_k$ of that ball with centroid q,

$$\begin{aligned}\sum_{i<j}|x_i-x_j|^2 &= k\sum_i |x_i-q|^2\\ &\leq k\sum_i |x_i-(\cos r)c|^2 \;\leq\; k^2\sin^2 r,\end{aligned}$$

the first step being the standard identity and the second the fact that the centroid minimises the sum of squared distances. Contact directions have $|x_i-x_j|^2 = 2-2\langle x_i,x_j\rangle \geq 1$, so $\binom{k}{2} \leq k^2\sin^2 r$. ∎

The case $k=3$ is drawn in Figure 16(b), and the lens in which the caps of two contact directions overlap, whose measure is the $\Lambda(r,\gamma)$ of (5.44) below, in Figure 18(a). The cases $k=3$ and $k=4$ are Lemma 5.22 and Proposition 5.21 again, since $\arccos\sqrt{2/3} = \arcsin\sqrt{1/3}$ and $\arccos\sqrt{5/8} = \arcsin\sqrt{3/8}$; the lemma puts them on one footing and is the reason the two radii r_* and r_4 govern what follows.

Proposition 5.68 (The cell inside a ball)**.** *Let W be a contact configuration of m directions with bounded cell, let $U(r)$ be the measure of the set of points of S^3 farther than r from every direction of W, and let $R \geq 1$. Then*

$$\operatorname{vol}(V_c(W)\cap B(R)) = \frac{1}{4}\Big[2\pi^2 + \int_0^{\arccos(1/R)} U(r)\,d(\sec^4 r)\Big] \;\leq\; \operatorname{vol}(V_c(W)), \tag{5.43}$$

and for $r < r_4$

$$U(r) = 2\pi^2 - mC(r) + \sum_{i<j}\Lambda(r,\gamma_{ij}) - \sum_{i<j<k}\Lambda_3(r;\gamma_{ij},\gamma_{ik},\gamma_{jk}), \tag{5.44}$$

where Λ_3 is the measure of the intersection of three caps of radius r about directions at the given pairwise angles, and the last sum is absent for $r < r_*$. Consequently, with

$$A_4 = \frac{1}{4}\Big[2\pi^2 + \int_0^{r_4}\big(2\pi^2 - 23\,C(r)\big)\,d(\sec^4 r)\Big] = 7.647558\ldots,$$
$$\omega_4(\gamma) = \frac{1}{4}\int_0^{r_4}\Lambda(r,\gamma)\,d(\sec^4 r),$$

and ω_3 the same integral of Λ_3, every contact configuration of 23 directions with bounded cell satisfies

$$\operatorname{vol}(V_c) \;\geq\; \operatorname{vol}(V_c\cap B(\sqrt{8/5})) \;=\; A_4 + \sum_{i<j}\omega_4(\gamma_{ij}) - \sum_{i<j<k}\omega_3(\gamma_{ij},\gamma_{ik},\gamma_{jk}). \tag{5.45}$$

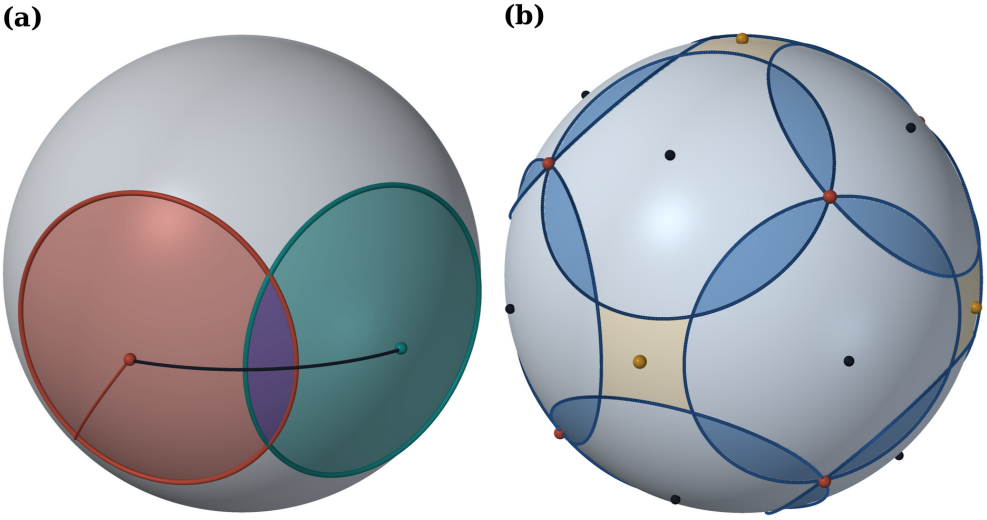

FIGURE 18. (a) Two contact directions at angle $\gamma = 60°$ and their caps of radius $r = r_*$. The lens in which the caps overlap (violet) has measure $\Lambda(r, \gamma)$, and $\omega(\gamma)$ integrates it against $d(\sec^4 r)$; the dark arc is γ and the clay arc is r. (b) The caps of radius r_* about the twelve contact directions of the same lattice, the sphere shaded by how many caps cover it: one (pale blue), two in the lenses, three at the eight triangle circumcentres (clay), and none in the six square holes (ochre).

Proof. The radial function of $V_c(W)$ in the direction θ is $\sec \delta(\theta)$ by Lemma 5.15, so that of $V_c(W) \cap B(R)$ is $\min(\sec \delta, R)$ and the volume is $\frac{1}{4} \int_{S^3} \min(\sec \delta, R)^4$. Writing $\rho = \arccos(1/R)$,

$$\min(\sec \delta, R)^4 = \sec^4 \min(\delta, \rho) = 1 + \int_0^\rho \mathbf{1}[\delta(\theta) > r] \, d(\sec^4 r),$$

and integrating over θ gives (5.43), the inequality because $V_c(W) \cap B(R) \subseteq V_c(W)$. For (5.44), $U(r)$ is $2\pi^2$ less the measure of the union of the closed caps of radius r about the directions of W, and by Lemma 5.67 no point lies in four of those caps when $r < r_4$, nor in three when $r < r_*$; the indicator of a union of sets no four of which meet is the sum of the indicators of the sets, less that of the pairwise intersections, plus that of the triple intersections, exactly. Integrating (5.44) against $d(\sec^4 r)$ over $[0, r_4]$ gives (5.45), and $\sec r_4 = \sqrt{8/5}$. ∎

The three-dimensional analogue of all this can be drawn, and Figure 18(b) and Figure 19 draw it: the Voronoi cell of the face-centred cubic lattice, whose contact directions are the twelve vertices of a cuboctahedron, cut by the ball of radius $\sqrt{3/2}$, and the caps of radius r_* about those twelve directions, which overlap in pairs and meet three at a time only at the eight triangle circumcentres; the uncovered measure there is exactly the two-term inclusion and exclusion, as it is on S^3 below r_*.

At the deletion of a root the three-point bound (5.45) is $8.140848\ldots$: the 88 tight pairs contribute $88\,\omega_4(60°) = 88 \times 0.00565947\ldots = 0.498033\ldots$, the 84 triangles of tight pairs subtract $84\,\omega_3(60°, 60°, 60°) = 84 \times 5.65 \times 10^{-5} = 0.00474\ldots$, and no other pair or triple contributes, since $\Lambda(r, \gamma) = 0$ for $\gamma \geq 2r_4 = 75.52°$. A direct quadrature of $\min(\sec \delta, \sqrt{8/5})^4$ over S^3 returns 8.14068, and the same quadrature at the full root system, with 24 directions,

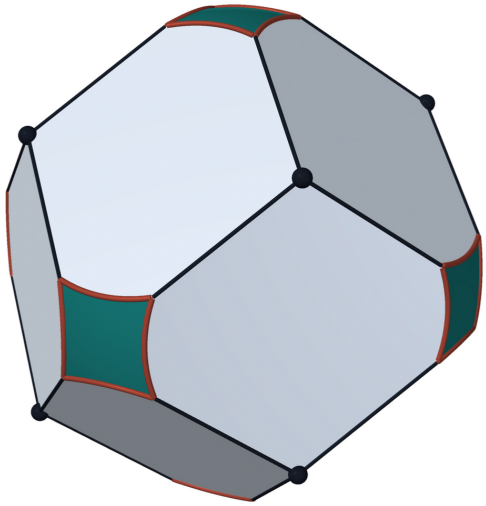

FIGURE 19. The truncated cell in dimension three: the Voronoi cell of the face-centred cubic lattice at inradius 1, a rhombic dodecahedron, cut by the ball of radius $(3/2)^{1/2}$. The ball passes through the eight three-valent vertices (dark) and replaces the six four-valent corners by spherical squares (teal). Inside the ball the cell is an exact function of the pair angles of its twelve contact directions.

returns 7.96841: there the truncation discards the deep holes at 45° and the bound falls below the true value 8, so at 24 contacts the three-point information below r_4 does not suffice; that case is settled by Theorem 5.25 instead. At 23 contacts it is enough at the one configuration known, with 0.14 to spare, and the pair-only bound (5.17) at $R = r_*$ is enough there too, with 0.034. That is the content of the following statement, which is the form in which the case of 23 contacts is proved.

Theorem 5.69 (The remaining case as an inequality between pair angles)**.** *Every contact configuration of* 23 *directions with bounded cell satisfies*

$$\operatorname{vol}(V_c) \;\geq\; \operatorname{vol}\bigl(V_c \cap B(\sqrt{3/2})\bigr) \;=\; A_* + \sum_{i<j} \omega(\gamma_{ij}), \tag{5.46}$$

where

$$A_* = \frac{1}{4}\Bigl[2\pi^2 + \int_0^{r_*} \bigl(2\pi^2 - 23\,C(r)\bigr)\, d(\sec^4 r)\Bigr] = 7.907144430\ldots$$

and ω is the weight of Proposition 5.41. Consequently $\operatorname{vol}(V_c) > 8$ *for every such configuration satisfying*

$$\sum_{i<j} \omega(\gamma_{ij}) \;>\; 8 - A_* \;=\; 0.092855570\ldots. \tag{5.47}$$

Proof. Proposition 5.68 with $R = \sec r_* = \sqrt{3/2}$: below r_* no point of S^3 lies in three of the caps, so the last sum of (5.44) is absent, and integrating against $d(\sec^4 r)$ over $[0, r_*]$ gives (5.46); this is also Proposition 5.23 at $R = r_*$. The second statement is the first with (5.47) inserted. ∎

Theorem 5.72 below proves (5.47) for every contact configuration of 23 directions, and with Theorem 5.69 that proves Theorem 5.40.

Remark 5.70 (What kind of statement this is). Inequality (5.47) needs no classification and no saturation. It is the $m = 23$ case of the 24-cell conjecture itself, in the form $\mathrm{vol}(V_c) > 8$ for every configuration of 23 contacts, saturated or not. It is also strict where the classification of Theorem 5.60 is not. That classification asks for a property of codes of minimal angle exactly $60°$, and the deletion of a root sits on the boundary of the property, with $g = \frac{1}{2}$; Remark 5.61 explains why no relaxation can detect a boundary case. Inequality (5.47) holds at the deletion with 0.034 to spare, and at every configuration reached in Section 5.10 with more: the least value of $\mathrm{vol}(V_c \cap B(\sqrt{3/2}))$ found over the configurations of Table 3, and in a direct minimisation of that quantity on $K(\delta)$, is 8.028, at $\delta = 0.05$, and the least value of $\mathrm{vol}(V_c \cap B(\sqrt{8/5}))$ is 8.129. A statement with slack is one that a relaxation can certify.

It is also linear in the pair distribution, and the two-point relaxation already delivers four fifths of it: Remark 5.45 minimises the left-hand side over every pair-angle distribution that satisfies the Bonferroni inequality at every radius and is positive definite in every degree, and finds $0.07380\ldots$ against $0.09285\ldots$. The minimising measure gives each direction nine neighbours at $62.3°$ and nearly eight at $103°$, a local picture that is consistent at any one direction and cannot be consistent at all twenty-three at once; telling the two apart requires three-point information. The semidefinite bounds of Bachoc and Vallentin [BV08] carry the distribution of triples of directions; de Laat, Leijenhorst and de Muinck Keizer [LLM24] sharpened them, at the second level, to the exact classification at 24 points of Theorem 5.30. A relaxation can fall short of a true inequality. What makes this one a good candidate is that the target is fixed, $0.092855\ldots$, that the pair angles alone reach 0.0738, and that the gap is a quantity with room on either side, not a boundary phenomenon. The next subsection carries the computation out, and the relaxation reaches the target.

5.12. **The inequality between pair angles.** The inequality (5.47) is a lower bound for a linear functional of the pair distribution of a spherical code of fixed size and fixed minimal angle, and the method built for statements of that shape is the semidefinite programming bound of Bachoc and Vallentin [BV08], which reads the distribution of triples of directions. We set it up in the form we need, prove the one lemma that turns a feasible point of the programme into an inequality for every configuration, and then describe the certificate and its verification, which is carried out in exact rational arithmetic where the objects are rational and in interval arithmetic where they are not. Throughout, $N = 23$, $u_{ij} = \langle w_i, w_j \rangle = \cos\gamma_{ij}$, and ω is the weight of Proposition 5.41, regarded as the function of the inner product $u = \cos\gamma$: increasing on $[\frac{1}{3}, \frac{1}{2}]$ from 0 to $0.00144540\ldots$ and identically 0 below $\frac{1}{3}$, since $\Lambda(r, \gamma) = 0$ for $\gamma \geq 2r_*$.

The kernels are the zonal polynomials G_k of Section 4.1 and the symmetrised three-point matrices S_k of Lemma 4.3, built at a fixed degree d from the matrices Y_k of (4.1).

From a feasible point to an inequality.

Lemma 5.71 (The certificate lemma)**.** *Let $f = \sum_{k=0}^{d} f_k G_k$ with $f_k \geq 0$ for $k \geq 1$, let $F(u, v, t) = \sum_{k=0}^{d} \langle F_k, S_k(u, v, t) \rangle$ with each F_k positive semidefinite of order $d - k + 1$, and suppose that*

$$\omega(u) + \omega(v) + \omega(t) - f(u) - f(v) - f(t) - F(u, v, t) - \frac{F(1, u, u) + F(1, v, v) + F(1, t, t)}{N - 2} \geq 0 \tag{5.48}$$

for every admissible triple, that is every $(u,v,t)\in[-1,\frac12]^3$ *with* $1+2uvt-u^2-v^2-t^2\ge 0$. *Then every contact configuration of* $N=23$ *directions satisfies*

$$\sum_{i<j}\omega(u_{ij})\ \ge\ B:=\frac{N}{2}(Nf_0-f(1))-\frac{N\,F(1,1,1)}{6(N-2)}. \tag{5.49}$$

Proof. The inner products of three distinct directions of a contact configuration form an admissible triple, since their Gram matrix is positive semidefinite with unit diagonal and its determinant is $1+2uvt-u^2-v^2-t^2$. Sum (5.48) over the $\binom{N}{3}$ triples $i<j<k$. Each pair lies in $N-2$ triples, so the ω terms give $(N-2)\sum_{i<j}\omega(u_{ij})$ and the f terms give $(N-2)\sum_{i<j}f(u_{ij})=\frac{N-2}{2}(\sum_{i,j}f(u_{ij})-Nf(1))\ge\frac{N-2}{2}(N^2f_0-Nf(1))$ by Schoenberg's theorem, since $G_0=1$. The triple sum of F and the pair sum of $F(1,u,u)$ are read off from the sum of F over all N^3 ordered triples of W, which is nonnegative by Lemma 4.3: the N triples with all three entries equal contribute $NF(1,1,1)$; the $3N(N-1)$ with exactly two equal contribute $F(1,u_{ij},u_{ij})$ each, by the symmetry of F, that is $6\sum_{i<j}F(1,u_{ij},u_{ij})$; and the $N(N-1)(N-2)$ with all entries distinct contribute $6\sum_{i<j<k}F(u_{ij},u_{ik},u_{jk})$. Hence $\sum_{i<j<k}F\ge-\sum_{i<j}F(1,u_{ij},u_{ij})-\frac{N}{6}F(1,1,1)$, while the last term of (5.48) summed over triples is exactly $\sum_{i<j}F(1,u_{ij},u_{ij})$. Putting the three parts together, $(N-2)\sum_{i<j}\omega(u_{ij})\ge\frac{N-2}{2}(N^2f_0-Nf(1))-\frac{N}{6}F(1,1,1)$, which is (5.49). ∎

The lemma uses the minimal angle only through the domain of (5.48), and the number 23 only through N. Its content is that the pair sum on the left of (5.47) is bounded below by the value of a semidefinite programme in the variables f_k and F_k, and that a feasible point of that programme with value above $8-A_*$ is a proof of Theorem 5.69's hypothesis. Such a point exists at degree 8.

Theorem 5.72 (The pair inequality)**.** *There are numbers* $f_0,\dots,f_8$ *and symmetric matrices* $F_0,\dots,F_8$, *of orders* $9,8,\dots,1$, *listed in the supplementary material, with* $f_k>0$ *for* $k\ge1$, *every* F_k *positive definite, satisfying* (5.48) *on the admissible domain, and with*

$$B=0.0929000002\ldots\ >\ 8-A_*=0.0928555702\ldots$$

Consequently every contact configuration of 23 *directions satisfies* (5.47) *with* 0.0000444 *to spare, every such configuration with bounded cell has* $\operatorname{vol}(V_c)\ge\operatorname{vol}(V_c\cap B(\sqrt{3/2}))=A_*+\sum_{i<j}\omega(\gamma_{ij})\ge A_*+B>8$, *and Theorem 5.40 holds.*

The proof is the verification of the three properties of the certificate, and we describe it in the detail that a reader who wants to repeat it would need.

Proof. The certificate. The numbers are double-precision floating-point numbers, which are dyadic rationals, and they are taken as the exact rationals they denote; nothing is rounded. All computations below in which only these rationals and the integers enter are done in exact rational arithmetic.

Positivity. The nine matrices are symmetric and positive definite: an exact $LDL^{\mathsf T}$ elimination over $\mathbb{Q}$ returns positive pivots throughout, and $f_1,\dots,f_8$ are positive. The bound B is computed exactly from (5.49); here $F(1,1,1)$ is the sum of the entries of F_0, because every entry of $Y_k(1,1,1)$ vanishes for $k\ge1$ and $T_i(1)=1$. Both steps are also checked by the Lean kernel, from the same rationals.

The constants. A_* is elementary. With $S(r)=\sec^4 r$ one has $\int r\,dS=r\sec^4 r-\tan r-\frac13\tan^3 r$ and $\int\sin 2r\,dS=\frac83\tan^3 r$, and $\sec^2 r_*=\frac32$, $\tan r_*=1/\sqrt2$, so that

$$A_*=\frac{9\pi^2}{8}-\frac{207\pi}{8}\,r_*+\frac{253\pi}{12\sqrt2},\qquad r_*=\arctan\frac{1}{\sqrt2}, \tag{5.50}$$

and $8 - A_*$ lies in $[0.092855570293, 0.092855570294]$ by interval arithmetic at forty digits. The weight ω is elementary too. The lens measure of Proposition 5.23 integrates to

$$\Lambda(r,\gamma) = 4\pi\Big[\frac{r-\gamma/2}{2} - \frac{\sin 2r - \sin\gamma}{4} - \frac{\tan(\gamma/2)}{2}\big(\sin^2 r - \sin^2(\gamma/2)\big)\Big],$$

and $\omega(\gamma) = \int_{\gamma/2}^{r_*} \Lambda(r,\gamma)\tan r\sec^4 r\,dr$ is a combination of the four primitives $\int \tan r\sec^4 r\,dr = \frac{1}{4}\sec^4 r$, $\int r\tan r\sec^4 r\,dr = \frac{1}{4}r\sec^4 r - \frac{1}{4}(\tan r + \frac{1}{3}\tan^3 r)$, $\int \sin 2r\tan r\sec^4 r\,dr = \frac{2}{3}\tan^3 r$ and $\int \sin^2 r\tan r\sec^4 r\,dr = \frac{1}{4}\tan^4 r$, with coefficients depending on γ alone. The closed form agrees with the quadrature of Proposition 5.41 to fifteen digits at every angle tested, and it is differentiated symbolically to give $d\omega/du$ and $d^2\omega/du^2$ as functions of γ, with $du = -\sin\gamma\,d\gamma$. On $(60^\circ, 2r_*)$ the function ω is analytic, and it is decreasing in γ, since $\Lambda(r,\gamma)$ is and the range of integration shrinks; so ω is increasing in u on $[\frac{1}{3},\frac{1}{2}]$.

Tables. At 12001 angles γ_i equally spaced from $2r_*$ down to 60°, interval arithmetic gives an upper bound u_i of $\cos\gamma_i$, a lower bound of $\omega(\gamma_i)$ and an enclosure of $\omega'(u)$ at $u = \cos\gamma_i$; consecutive u_i differ by at most 1.44×10^{-5}, and $|\omega''(u)| \le 0.289712$ on the whole range, by interval evaluation of the second derivative on 3000 subintervals. For $u \ge u_i$ the monotonicity gives $\omega(u) \ge \omega(\gamma_i)$, and the mean value theorem gives $|\omega'(u) - \omega'(\cos\gamma_i)| \le 0.289712\,(u - \cos\gamma_i)$.

A simpler form. The bracket of the integrand vanishes at the lower limit $r = \gamma/2$, so differentiating under the integral sign leaves only the coefficients of $\sec^4 r$ and $\tan^2 r\sec^2 r$, whose derivatives in γ are rational in u; the result, and the closed form it integrates to, are

$$\omega'(u) = \frac{\pi}{16}\,\frac{(3u-1)^2}{(1+u)^2\sqrt{1-u^2}}, \qquad \omega(u) = 4\pi\Big[\tfrac{9}{32}\arctan\frac{t_*-\tau}{1+t_*\tau} - \tfrac{1}{96}\big(4\tau^3 - 24\tau + 11\sqrt{2}\big)\Big],$$

with $\tau = \tan\frac{\gamma}{2} = ((1-u)/(1+u))^{1/2}$ and $t_* = \tan r_* = 1/\sqrt{2}$, for $\frac{1}{3} \le u < 1$. The first shows at once that ω increases and vanishes to second order at $u = \frac{1}{3}$; a symbolic computation checks that the second has the first as its derivative and vanishes at $\frac{1}{3}$, and that both agree with the form above to 10^{-15}. The Lean check below evaluates them.

The inequality (5.48). The polynomial $P(u,v,t) = f(u) + f(v) + f(t) + F(u,v,t) + (F(1,u,u) + F(1,v,v) + F(1,t,t))/21$ is expanded exactly, from the recurrences for T_i, P_k and G_k, into 449 monomials of total degree at most 16 with rational coefficients; it is symmetric in u, v, t, so (5.48) need only be verified on the ordered part $-1 \le u \le v \le t \le \frac{1}{2}$ of the domain, and it reads $Q := 1000(\omega(u) + \omega(v) + \omega(t)) - P \ge 0$ after the scaling by 1000 under which the certificate was computed. The verification is a branch and bound over boxes, in interval arithmetic with every operation rounded outward. On a box with centre c and half-widths ρ_1, ρ_2, ρ_3 whose u, v and t ranges lie inside $(\frac{1}{3},\frac{1}{2}]$, Taylor's theorem gives

$$Q \;\ge\; Q(c) - \sum_v |\partial_v Q(c)|\rho_v - \frac{1}{2}\sum_{v,w}\Big(\sup_{\text{box}}|\partial_v\partial_w Q|\Big)\rho_v\rho_w,$$

where $Q(c)$ and $\partial_v Q(c)$ are bounded through the tables and the exact polynomial evaluated at the point c, and the second derivatives through the bound on ω'' and the interval evaluation of the second derivatives of P over the box. A variable whose range is not inside $(\frac{1}{3},\frac{1}{2}]$ has its ω term replaced by 0, which is a lower bound, with derivatives 0. A box on which the right-hand side is nonnegative is verified; a box on which $1 + 2uvt - u^2 - v^2 - t^2$ is negative throughout is outside the domain and is discarded; any other box is bisected along the direction that contributes most to the Taylor remainder. Starting from $[-1,\frac{1}{2}]^3$, the

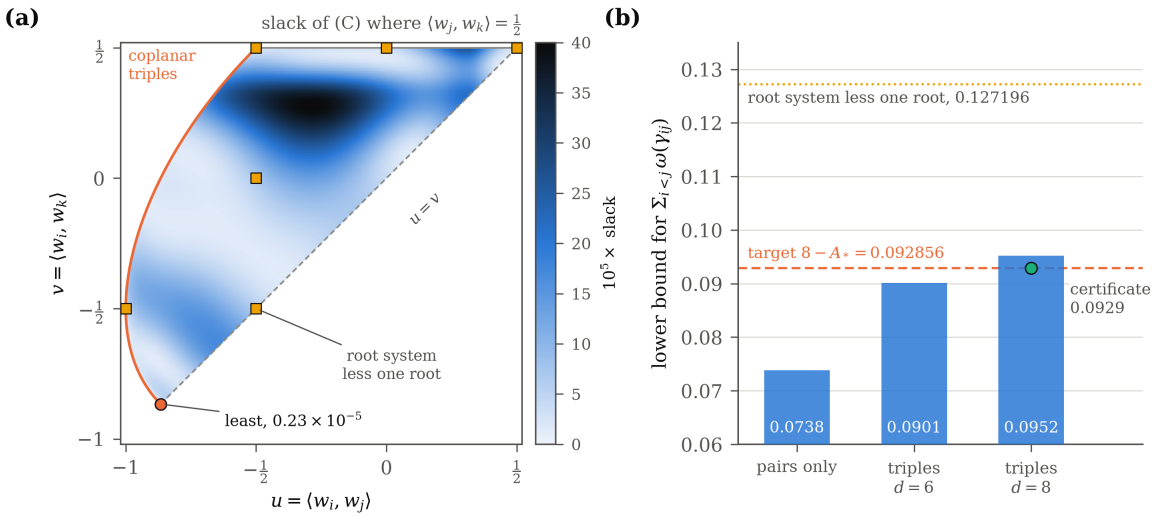

FIGURE 20. (a) The certificate of Theorem 5.72: the slack of the condition (5.48), the amount by which its left-hand side exceeds 0, on the slice of the admissible domain in which one pair of the triple is at 60°. It is positive throughout and largest, about 4×10^{-4}, in the interior. It is least, 2.3×10^{-6}, at the corner $(-\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2})$ where the two arcs of coplanar triples meet the diagonal, the triple in which the third direction lies opposite the bisector of the pair at 60°; a search of the whole domain finds no smaller value. The six squares are the triples with a pair at 60° that occur in the root system less one root. (b) The values of the relaxations against the target $8 - A_*$: the pair angles alone (Remark 5.45), the three-point relaxation at degrees 6 and 8, and the certificate, at which the bound is fixed at 0.0929 and the least slack is maximised; the dotted line is the pair sum 0.127196 of the root system less one root.

process ends after 44 levels with 313 780 boxes verified and 13 517 discarded (Figure 21(b)), and no box reaching the minimum width 10^{-6} undecided.

The same branch and bound was carried out twice more in Lean. The first run expands P from the certificate independently, multiplies it by 315 so that every coefficient becomes an integer over a power of two, and runs the subdivision with the same Taylor form in dyadic arithmetic, integers and exponents only, with no rounding at any point; it takes from outside nothing but the tables of bounds for ω, ω' and ω'', processes 421 881 boxes, fewer than the first run because its nested evaluation of the polynomials encloses more tightly than a sum of monomials, and subdivides down to the width 2^{-20} where it has to. The second run removes the tables as well: in outward-rounded 256-bit interval arithmetic, with arctan from its series and π from Machin's formula, it evaluates the simpler forms above and A_*, tabulates ω and ω' on the grid of step 2^{-16} with $|\omega''| \le 0.2856$, checks that A_* exceeds $8 - 0.0928555703$, and repeats the branch and bound with that table, passing on 419 913 boxes. In both runs Lean states as a theorem that the subdivision returns true and settles it by evaluating compiled code, so these two steps trust Lean's compiler as well as its kernel. Hence (5.48) holds on the admissible domain, Lemma 5.71 gives (5.47) for every contact configuration of 23 directions, and Theorem 5.69 gives the rest. ∎

Remark 5.73 (How the certificate was found, and how tight it is). The programme of Lemma 5.71, with (5.48) imposed on a sample of admissible triples and B maximised (Figure 20), is solved by an interior-point method; the sample is refined four times by adding the triples at which the current solution violates (5.48) most, on a check over about 1.15 million further triples, and the value reported is reduced by $\binom{23}{3}/21$ times the largest violation that remains, so that it is a bound over the whole check. At degree 6 the value is 0.09011, that is 97.0 per cent of $8 - A_*$; at degree 8 it is 0.09523, or 102.6 per cent. The pair angles alone, in Remark 5.45, gave 79.5 per cent; the triples supply the rest. The certificate of Theorem 5.72 is then obtained by fixing B at 0.0929 and maximising instead the least slack of (5.48) over the sample, with $F_k - 10^{-7} I$ and $f_k - 10^{-7}$ constrained to be positive semidefinite so that the solution survives being read as exact rationals: the least slack is 1.35×10^{-5} on the final sample of 21 648 triples and 1.30×10^{-5} on the check, against $\omega(60^\circ) = 0.00145$. On the whole domain it is smaller, 2.30×10^{-6}, at the coplanar triple $(-\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, \frac{1}{2})$, in which two directions are at 60° and the third lies opposite their bisector. Neither the sample nor the check came near that point; the branch and bound of the proof covers every admissible triple and decides it as it decides the rest. The solution has two further features. The two-point part is almost inert, $f_0 = 0.000927$ and $f_1, \dots, f_8$ below 3×10^{-8}, so it is the matrices F_k that carry the argument. And the inequality is far from tight at the deletion of a root, where the pair sum is 0.1272 against $B = 0.0929$; the relaxation had room, which is the situation Remark 5.70 asked for. Finally, the checker itself was tested on a certificate it should reject: adding 10^{-5} to f_0, which raises P by 3×10^{-5} and so exceeds the slack, produces a certificate for which the same programme exhibits the admissible triple $(-\frac{1}{16}, -\frac{1}{16}, -\frac{1}{16})$ at which (5.48) fails.

Remark 5.74 (What is verified where). The proof of Theorem 5.72 rests on three kinds of computation. The exact rational steps, positivity of the matrices and the value of B, are verified by the Lean kernel. The expansion of P is an exact rational computation, carried out independently in Lean and outside it, and the two expansions agree with a floating-point evaluation of the same expressions at random triples to 10^{-13}. The branch and bound is run three times, once in interval arithmetic on double-precision numbers with outward rounding after every operation, and twice in Lean as described, the second time with every constant computed inside Lean. In Lean the statement that the subdivision succeeds is settled by running the compiled program, so it relies on the compiler as well as on the kernel. The mathematics that makes the subdivision a proof, the Taylor form and the monotonicity of ω, is written out above, and the Taylor form is applied only on boxes inside the open interval where ω is analytic.

5.13. **Twenty-four points at positive slack.** The equality argument of Lemma 5.27 needs the inner products to be at most $\frac{1}{2}$, and it does not apply to a code whose inner products exceed $\frac{1}{2}$ by a little. Theorem 7.9 needs that case, with a quantitative conclusion, and the certificate gives it, because the only terms of the identities of Proposition 5.29(iii) that can change sign outside $\mathcal{I}_4$ are those whose weight carries a factor $(u+1)(\frac{1}{2} - u)$ with $u > \frac{1}{2}$. Statements of this kind, in which a code that is nearly optimal is forced near the optimal one, are proved by Böröczky and Glazyrin [BG24] for codes whose optimality follows from the linear programming bound, among them the minimal vectors of E_8 and of the Leech lattice. For D_4 that bound is not sharp, and the argument has to start from the second level of the hierarchy.

Theorem 5.75 (Twenty-four points, approximately)**.** *Let $0 \le \kappa \le 2 \cdot 10^{-26}$, and let $C \subset S^3$ be a set whose pairwise inner products are at most $\frac{1}{2} + \kappa$. Then $|C| \le 24$. If $|C| = 24$, every*

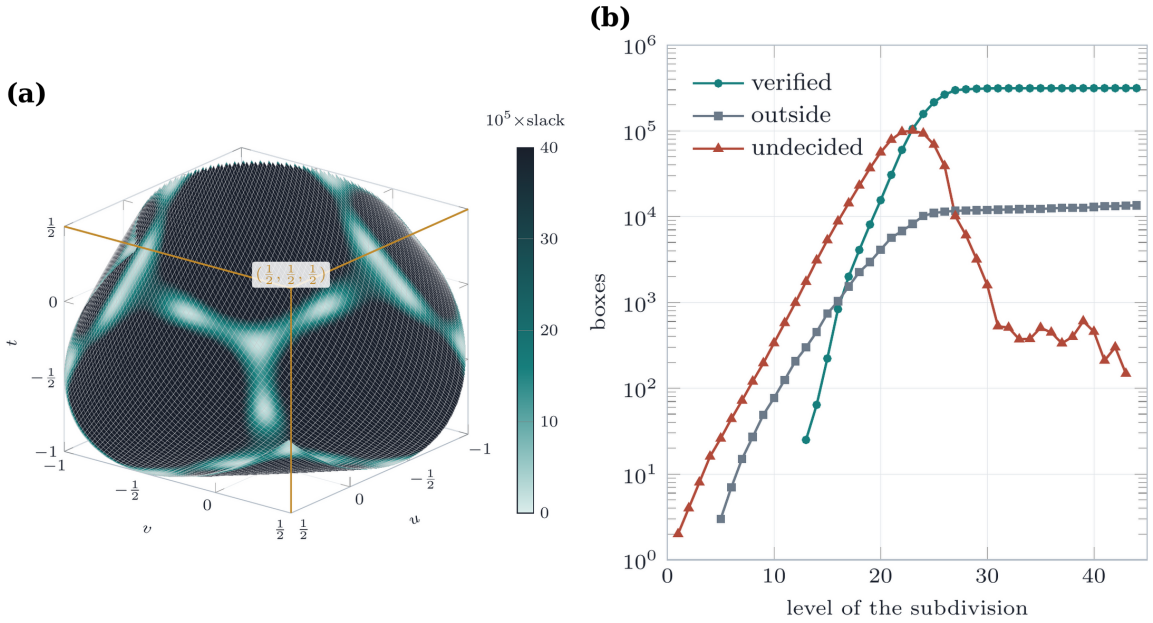

FIGURE 21. (a) The domain of the certificate condition: the admissible region of Lemma 5.71 in three dimensions, where the triples (u, v, t) of pairwise inner products of three contact directions fill the part of the cube $[-1, \frac{1}{2}]^3$ enclosed by the surface of coplanar triples. The surface is coloured by the slack of (5.48). The slack is small, from about 1.2×10^{-5}, along the curves where two of the three directions are nearly antipodal, and least, 2.3×10^{-6}, at the triple $(-\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, \frac{1}{2})$ and its permutations. (b) The branch and bound of the proof of Theorem 5.72, level by level: the boxes live, the boxes discarded on the enclosure, and the width reached.

pairwise inner product of C lies within $3 \cdot 10^{-4}$ of $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$, and there are an orthogonal map R and a labelling $C = \{w_1, \dots, w_{24}\}$ with $\sum_i |w_i - Ra_i/\sqrt{2}|^2 \le 0.0113^2$.

Proof. The set C is finite, since its points are at angular distance at least $\arccos(\frac{1}{2} + \kappa)$ from one another. The maps $\psi_{\lambda,a}$ of (5.20) are defined on every subset of S^3 of at most two points, whatever its inner product, so Lemma 5.28 makes the certificate kernel positive on the subsets of C of size at most 2, and the identities of Proposition 5.29(iii) hold at every subset Q of C, being identities between polynomials. The first chain of the proof of Lemma 5.27, with Proposition 5.29(ii), gives

$$\sum_{\substack{Q \subseteq C \\ 2 \le |Q| \le 4}} \sigma_{|Q|}(Q) \;\le\; 24 - |C|. \tag{5.51}$$

Each σ_k is a sum of terms $c\,w\,\sigma$ with $c \ge 0$, $\sigma = m^{\mathsf{T}} M m \ge 0$ and w one of the weights of Proposition 5.29(iii). The constants, the Gram determinants and the principal minors are nonnegative at every set of real vectors. Put $p(u) = (u+1)(\frac{1}{2} - u)$; it lies in $[0, \frac{9}{16}]$ for $u \in [-1, \frac{1}{2}]$ and in $[-a, 0)$ for $u \in (\frac{1}{2}, \frac{1}{2} + \kappa]$, with $a = \kappa(\frac{3}{2} + \kappa)$, so an elementary symmetric function e_r of the n values of p at the inner products of Q is at least $-\binom{n}{r} a (\frac{9}{16})^{r-1}$, and is nonnegative unless Q has an inner product above $\frac{1}{2}$. Hence $\sigma_k(Q) \ge -aB_k$ for $k = 3, 4$, where $B_k = \sum_r \binom{n_k}{r} (\frac{9}{16})^{r-1} S_{k,r}$, $n_3 = 3$, $n_4 = 6$, and $S_{k,r}$ bounds the sum of $c\,\sigma$ over the

terms of σ_k with weight e_r. We bound these sums from the deposited data in ball arithmetic. The data give each M as TXT^{T} with X positive definite, so $m^{\mathsf{T}}Mm \le \lambda_{\max}(X)\,|T^{\mathsf{T}}m|^2$; the program bounds $\lambda_{\max}(X)$ by the trace or the largest absolute row sum of X, and each entry of $T^{\mathsf{T}}m$ by the ℓ^1 norm of its coefficients in the tensor Chebyshev basis, which is at least its supremum on $[-1,1]^6$; it finds $B_3 \le 8.75\cdot 10^7$ and $B_4 \le 7.34\cdot 10^9$. For a pair, $\sigma_2(\frac12)=0$ and $|\sigma_2'|\le 0.226$ on $[\frac12,\frac12+10^{-3}]$, so $\sigma_2(u)\ge -0.226\,\kappa$ at an inner product above $\frac12$. Therefore the left side of (5.51) is at least $-E(|C|,\kappa)$, where

$$E(m,\kappa)=\tbinom{m}{2}\,0.226\,\kappa+\tbinom{m}{3}\,aB_3+\tbinom{m}{4}\,aB_4.$$

If $|C|\ge 25$, the same holds for 25 of its points, and (5.51) gives $E(25,\kappa)\ge 1$; but $E(25,2\cdot 10^{-26})<10^{-11}$. Hence $|C|\le 24$.

Let $|C|=24$ and let u be an inner product of C at most $\frac12$. Every other term of (5.51) is at least the lower bound just found for it, these lower bounds add up to at least $-E(24,\kappa)$, and the right side is 0; so $\sigma_2(u)\le E(24,\kappa)\le 2.35\cdot 10^{-12}$. By Proposition 5.29(iv), σ_2 is divisible by $f(u)=(u+1)(u+\frac12)^2u^2(\frac12-u)$, and the quotient is at least $2.62\cdot 10^{-4}$ on $[-1,\frac12]$ (an exact division, and a mean value bound in ball arithmetic on 30 000 intervals; the division and the bound are repeated in Lean in exact rationals, with a second-order Taylor form on 135 intervals). Between consecutive zeros f has a single critical point, by Rolle's theorem applied to its derivative of degree 5, which vanishes at $-\frac12$ and 0; so at distance at least $\delta=3\cdot 10^{-4}$ from its zeros f is at least its least value at the six points $-1+\delta$, $-\frac12\pm\delta$, $\pm\delta$, $\frac12-\delta$, and there $\sigma_2\ge 2.94\cdot 10^{-12}$ (in Lean the bound $f\ge 1.12\cdot 10^{-8}$ is checked on the three ranges directly, by the same branch and bound, giving $\sigma_2\ge 2.93\cdot 10^{-12}$, which suffices). Thus u lies within δ of $\{-1,-\frac12,0,\frac12\}$, and an inner product above $\frac12$ lies within $\kappa<\delta$ of $\frac12$. Lemma 5.76 gives the rest. ∎

Lemma 5.76 (From approximate inner products to the root system)**.** *Let $w_1,\dots,w_{24}$ be unit vectors of $\mathbb{R}^4$ with Gram matrix G, and let $\delta\le 3\cdot 10^{-4}$. If every off-diagonal entry G_{ij} lies within δ of a number $g_{ij}\in\{-1,-\frac12,0,\frac12\}$, then the matrix g with these entries and 1 on the diagonal is the Gram matrix of the normalised roots in some order, and, with the roots numbered in that order, for an orthogonal map R,*

$$\Big(\sum_i |w_i-Ra_i/\sqrt2|^2\Big)^{1/2}\;\le\;\frac{2D}{\sqrt{6-D}}+\frac{\sqrt{24}\,D}{6},\qquad D=23\delta,$$

which is below 0.0113 *at $\delta=3\cdot 10^{-4}$.*

Proof. A row of G has at most one entry near -1 off the diagonal: two would belong to vectors within $\sqrt{2\delta}$ of $-w_i$, hence within $2\sqrt{2\delta}$ of each other, with inner product at least $1-4\delta>\frac12+\delta$. Therefore a row of $2G$ restricted to five columns has at most two entries of absolute value above $1+2\delta$, both at most 2, and its norm is at most $\rho=\sqrt{8+3(1+2\delta)^2}$. For a $k\times k$ submatrix, $k\le 5$, expanding the determinant of $2g=2G-2\Delta$ multilinearly in its rows and bounding every term by Hadamard's inequality,

$$|\det(2g)_{IJ}-\det(2G)_{IJ}|\;\le\;(\rho+2\delta\sqrt{k})^k-\rho^k\;<\;1,$$

the last at $k=5$ being 0.81. The matrix $2g$ is integral, so each of its minors is an integer. Every 5×5 minor of $2G$ vanishes, since G has rank at most 4; so every 5×5 minor of $2g$ vanishes, and $2g$ has rank at most 4. Every principal minor of $2G$ is nonnegative; so every principal minor of $2g$ of order at most 4 is an integer above -1, hence nonnegative, those of higher order vanish, and $2g$ is positive semidefinite. It is therefore the Gram matrix of 24 vectors of $\mathbb{R}^4$ of norm $\sqrt2$ with pairwise inner products in $\{-2,-1,0,1\}$, distinct because

no inner product is 2, and Lemma 5.26 makes g the Gram matrix $V^{\mathsf T}V$ of the normalised roots $V = (v_1, \dots, v_{24})$ in some order.

For the distance, let $W = (w_1, \dots, w_{24})$ and $\Delta = G - g$, so $\|\Delta\| \le D$ by the row sums. Since $VV^{\mathsf T} = 6I$, the matrix $P = g/6$ is the orthogonal projection onto the row space of V. If $Gx = \lambda x$ with $\lambda > 0$, then $\lambda(I-P)x = (I-P)\Delta x$, since $(I-P)g = 0$, and the four positive eigenvalues of G are at least $6 - D$ by Weyl's inequality; so $\|W(I-P)\|_F^2 = \operatorname{tr}(I-P)G(I-P) \le 4D^2/(6-D)$. Put $M = WV^{\mathsf T}/6$, so that $WP = MV$. From $PGP = g + P\Delta P$ and $VV^{\mathsf T} = 6I$ follows $M^{\mathsf T}M = I + V\Delta V^{\mathsf T}/36$, so $\|M^{\mathsf T}M - I\| \le D/6$, and the orthogonal factor R of the polar decomposition $M = R(M^{\mathsf T}M)^{1/2}$ has $\|M - R\| \le D/6$. Hence $\|W - RV\|_F \le \|W(I-P)\|_F + \|M-R\|\,\|V\|_F \le 2D/\sqrt{6-D} + \sqrt{24}D/6$. ∎

Neither statement is sharp. The bound E counts every triple and every quadruple of the 24 points as if each had an inner product above $\frac12$, and bounds each sum of squares on the whole cube $[-1,1]^6$; and the lemma needs the inner products within $3\cdot10^{-4}$ of the four values, far closer than the conclusion should require, because it deduces integrality from a bound on each determinant separately. For Theorem 7.9 it suffices that every constant is explicit.[21]

Remark 5.77 (The ceiling of Theorem 5.75). No sharpening of B_3 and B_4 carries the proof of Theorem 5.75 to a slack of order 10^{-2}. The pairs above $\frac12$ contribute $\binom{24}{2}\,0.226\,\kappa = 62.4\,\kappa$ to $E(24,\kappa)$ whatever B_3 and B_4 are, and an inner product near the double zero 0 is pinned to within δ only when $2.62\cdot10^{-4} f(\delta) > E(24,\kappa)$, with $f(\delta) \approx \delta^2/8$; so the proof needs $\kappa < 5.2\cdot10^{-7}\delta^2$ even with $B_3 = B_4 = 0$. At the tolerance $\delta = 3\cdot10^{-4}$ of Lemma 5.76 that is $4.7\cdot10^{-14}$, and since the four target values are $\frac12$ apart, no rounding lemma can allow $\delta \ge \frac14$, which caps κ at $9.2\cdot10^{-9}$ (Figure 23(b)). The square root comes from the double zeros of σ_2 at $-\frac12$ and 0, and every kernel whose bound is exactly 24 has them, since $\sigma_2 \ge 0$ vanishes at those inner products of the root system. A form of Theorem 5.25 at slack of order 10^{-2} therefore cannot come from this kernel, and Proposition 7.10 shows that a kernel computed afresh on the enlarged domain $[-1, \frac12 + s]$ with a margin on pairs cannot supply it either.

What the three-point bound of Section 5.12 can do on the enlarged domain is the cardinality half of that statement, with a slack twenty-three orders of magnitude larger than in Theorem 5.75.

Theorem 5.78 (The kissing number is stable)**.** *Let $s_0 = 0.008$. Every set of points of S^3 whose pairwise inner products are at most $\frac12 + s_0$ has at most 24 elements. Consequently, in a packing of unit balls in $\mathbb{R}^4$, at most 24 other centres lie within distance $2/\sqrt{1-2s_0} = 2.0161\dots$ of any centre.*[22]

Proof. Let $t = \frac12 + s_0$, let $f(u) = \sum_{k=1}^{d} f_k G_k(u)$ with $f_k \ge 0$, and let $F(u,v,w) = \sum_{k=0}^{d} \langle F_k, S_k(u,v,w)\rangle$ with every F_k positive semidefinite, in the notation of Section 4.1. For a finite $C \subset S^3$ with inner products $u_{xy} \le t$, Lemma 4.1 gives $\sum_{x,y\in C} f(u_{xy}) \ge 0$ and

[21] The constant that dominates is the sum of squares of the four-point constraint with weight e_1, the sum of the six values $(u+1)(\frac12 - u)$, for which the bound is $1.22\cdot10^{9}$; with the $\binom{24}{4} = 10\,626$ quadruples and the factor 6 of e_1, it makes $E(24,\kappa)$ about $1.2\cdot10^{14}\kappa$. Against it stands the value $2.9\cdot10^{-12}$ of σ_2 at distance $3\cdot10^{-4}$ from a double zero, which is quadratic in the distance. The ratio of the two is the $2\cdot10^{-26}$ of the theorem.

[22] Two centres at distances $d_1, d_2 \in [2,\rho]$ from c and at least 2 apart see each other from c at inner product at most $(d_1^2 + d_2^2 - 4)/(2d_1d_2)$, which increases in each d_i on that range, and so at most $1 - 2/\rho^2$. This equals $\frac12 + s_0$ exactly when $\rho^2 = 4/(1-2s_0)$. The directions of the centres within ρ therefore form a set of the kind in the first sentence.

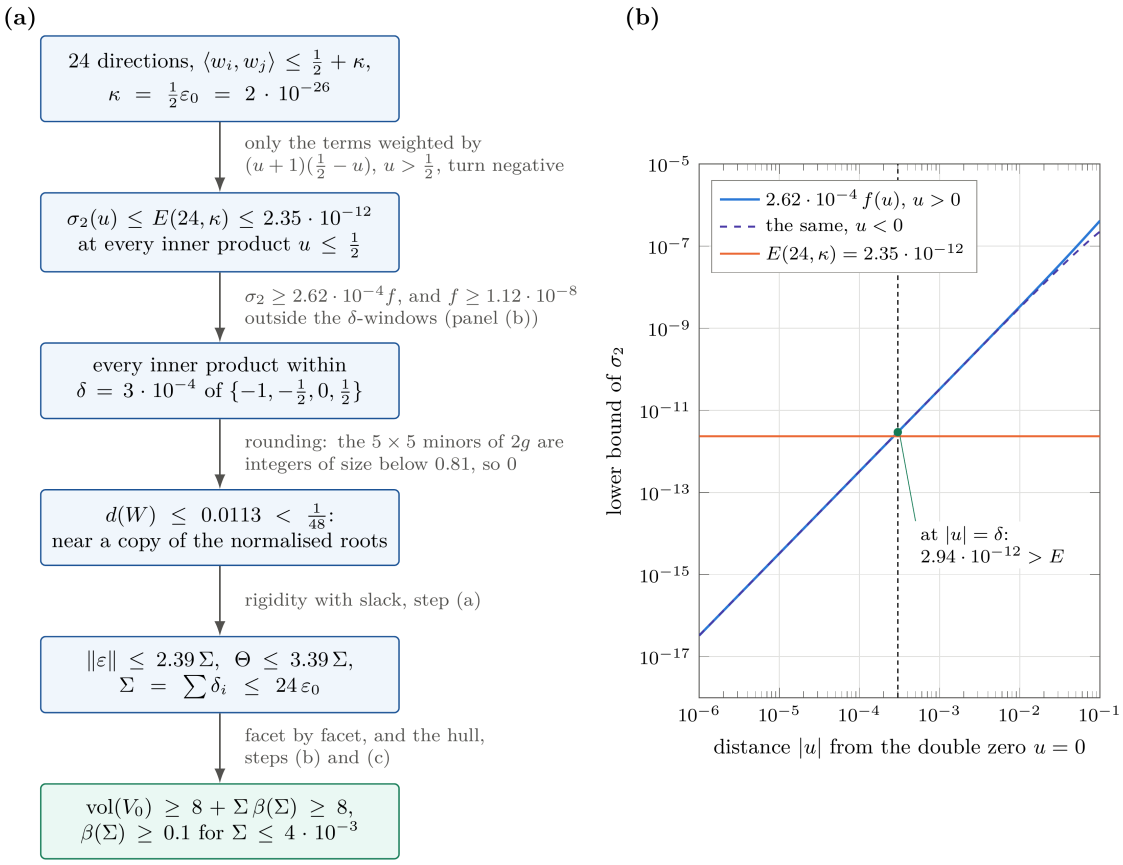

FIGURE 22. Where $\varepsilon_0 = 4 \cdot 10^{-26}$ comes from. (a) The chain of explicit bounds that replaces the compactness argument of Theorem 7.9, with the number each step produces. Twenty-four directions with inner products at most $\frac{1}{2} + \kappa$ make every sign-changing sum-of-squares term of the certificate at least $-E$ in total (Theorem 5.75); so $\sigma_2 \le E$ at each inner product up to $\frac{1}{2}$, which pins it near a zero of σ_2; Lemma 5.76 turns that into a distance from the root system; the rigidity of Theorem 5.51, with slack, bounds the displacement by the sum S of the distances beyond contact; and the facet, moment and hull estimates close the argument. (b) To scale: the lower bound $2.62 \cdot 10^{-4} f(u)$ of σ_2 near its double zero $u = 0$, for $u > 0$ and $u < 0$, against the level $E(24,\kappa) = 2.35 \cdot 10^{-12}$. The two cross at $|u| \approx 2.7 \cdot 10^{-4}$, inside the window $|u| \le \delta = 3 \cdot 10^{-4}$, where the bound is already $2.94 \cdot 10^{-12}$. The other double zero, $-\frac{1}{2}$, gives the same picture; at the simple zeros -1 and $\frac{1}{2}$ the bound is linear and the window is far wider than needed.

Lemma 4.3 gives $\sum_{x,y,z \in C} F(u_{xy}, u_{xz}, u_{yz}) \ge 0$. Separating the terms with coincident points as in the proof of Lemma 5.71 and adding,

$$|C|(f(1) + F(1,1,1)) + \sum_{x \ne y} (f(u_{xy}) + 3F(1, u_{xy}, u_{xy})) + \sum_{x,y,z \text{ distinct}} F(u_{xy}, u_{xz}, u_{yz}) \;\ge\; 0,$$

the sums running over ordered pairs and triples. Suppose that

$$\text{(i)} \quad f(u) + 3F(1,u,u) + 1 \;\le\; e_1 \qquad \text{for } u \in [-1, t],$$

$$\text{(ii)}\quad F(u,v,w) \;\le\; e_2 \qquad \text{for admissible } (u,v,w)\in[-1,t]^3.$$

Then $(|C|-1)(1-e_1)-(|C|-1)(|C|-2)e_2 \le f(1)+F(1,1,1)$. A set of 25 or more points contains one of exactly 25, so $|C|\le 24$ as soon as $24(1-e_1)-552e_2 > f(1)+F(1,1,1)$.

The certificate has degree $d=10$. It was found by solving the corresponding programme in floating point, with (i) and (ii) imposed at sampled points and refined in rounds, $f(1)+F(1,1,1)$ held at 23.96 and the least slack maximised, and its entries are read as the exact rationals they denote. We establish, with no floating point beyond outward-rounded interval arithmetic and with the polynomial and branch-and-bound routines of Section 5.12: that every f_k is nonnegative and every F_k positive definite, by exact $LDL^{\mathsf T}$ factorisation; the value $f(1)+F(1,1,1)=23.959999949\ldots$, exactly; (i) with $e_1=10^{-6}$, by branch and bound on the exact univariate polynomial; and (ii) with $e_2=7\cdot10^{-5}$ on the ordered admissible domain $-1\le u\le v\le w\le t$, $1+2uvw-u^2-v^2-w^2\ge 0$, which suffices because F is symmetric, every box being decided by a second-order Taylor bound about its centre, 2.7 million boxes in all. Both domains are run up to the least double at or above t. Then $24(1-e_1)-552e_2 = 23.961336 > 23.96$. The check is repeated in Lean, in exact dyadic arithmetic, from the certificate entries alone: Lean expands F and $f+3F(1,\cdot,\cdot)+1$ from the recurrences (756 monomials, as in the first expansion), and checks the positivity, the value of the bound, the inequality at $|C|=25$ and both branch and bounds, (i) on 2 957 intervals (Section 11).

For the second statement, let y_i be centres at distances $d_i \le D = 2/\sqrt{1-2s_0}$ from $c=0$. Two of them satisfy $|y_i-y_j|\ge 2$, so their directions have inner product at most $(d_i^2+d_j^2-4)/(2d_id_j)$, which increases in each d_i on $[2,D]$ and is therefore at most $1-2/D^2 = \frac12+s_0$. ∎

Musin's theorem [Mus08] is the case $s_0=0$; the content of Theorem 5.78 is that the count does not jump there, the first slack at which a twenty-fifth point could enter being bounded away from 0 by an explicit amount. Two certificates of degree 8, at $s_0=0.005$ and $s_0=0.0065$, pass the same check, and Figure 23(a) shows why the method stops soon after $s_0=0.008$: the three-point bound at degree 10 on the enlarged domain rises by about 97 per unit of s from about 24.13 at $s=0$, so no certificate of this kind reaches much beyond $s=0.009$, while a numerical search finds 25 points of S^3 with pairwise inner products at most 0.53743, a minimal angle of 57.49°, so the true threshold lies below $s=0.0375$. The theorem does not describe the shape of a set of 24 points with slack s_0, and Section 7.1 needs that shape.

The same method excludes twenty-six points at a larger slack, where the three-point bound at degree 10 is about 25.68.

Theorem 5.79 (Twenty-six points)**.** *Let $s_1=0.01468$. Every set of points of S^3 whose pairwise inner products are at most $\frac12+s_1$ has at most 25 elements. Consequently, in a packing of unit balls in $\mathbb{R}^4$, at most 25 other centres lie within distance $2/\sqrt{1-2s_1} = 2.0300\ldots$ of any centre.*

Proof. Let $t=\frac12+s_1$. Under (i) and (ii) of the proof of Theorem 5.78, the inequality obtained there, applied to a set of exactly 26 points, reads $25(1-e_1)-600e_2 \le f(1)+F(1,1,1)$, and a set of 26 or more points contains one of exactly 26. The certificate has degree 10. It was found as the one of Theorem 5.78, with $f(1)+F(1,1,1)$ held at 24.8, and it is checked in the same way: positivity by exact $LDL^{\mathsf T}$ factorisation; the value $f(1)+F(1,1,1)=24.799999928\ldots$, exactly; (i) with $e_1=10^{-6}$ on 790 intervals; and (ii) with $e_2=2.9\cdot10^{-4}$ on the ordered admissible domain, 1.33 million boxes in all. Then

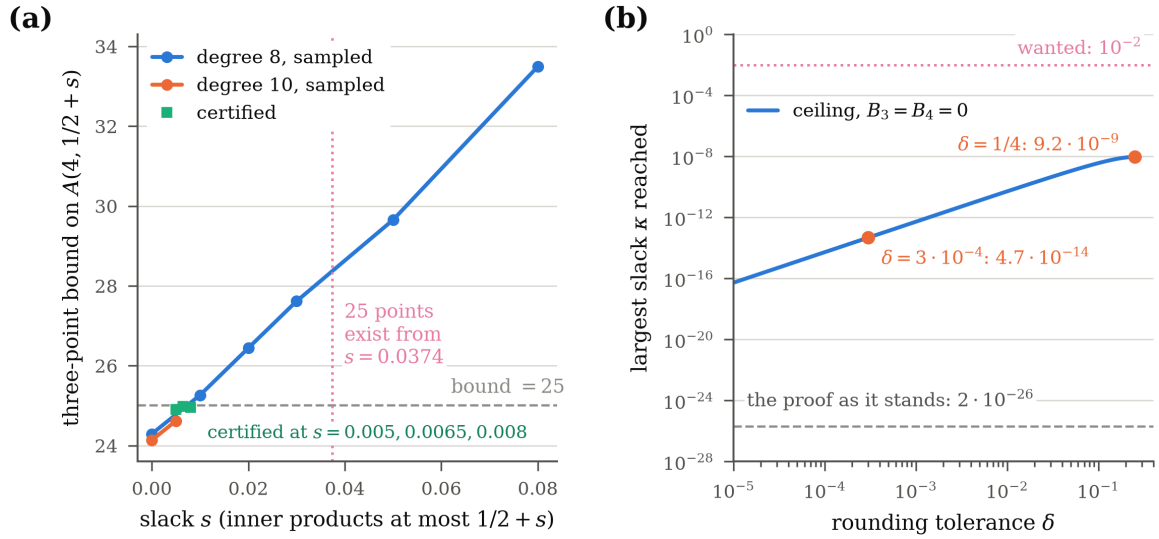

FIGURE 23. The two halves of Theorem 5.25 at positive slack. (a) The three-point bound on the number of points of S^3 with inner products at most $\frac{1}{2}+s$, from sampled programmes of degree 8 and 10 (each sampled value is slightly optimistic), with the three certified values: degree 8 at $s = 0.005$ and 0.0065, degree 10 at $s = 0.008$, the one Theorem 5.78 uses. It crosses 25 near $s = 0.009$; a 25-point code exists from $s = 0.03743$ on. (b) How far the proof of Theorem 5.75 could be pushed if the triple and quadruple terms cost nothing ($B_3 = B_4 = 0$): the largest slack κ it reaches, against the tolerance δ that a rounding lemma would allow (Remark 5.77). The curve grows like δ^2 and stops below 10^{-8}; the proof as it stands gives $2 \cdot 10^{-26}$.

$25(1-e_1) - 600e_2 = 24.825975 > 24.8$. The check is repeated in Lean from the certificate entries alone, as for Theorem 5.78, with the inequality taken at $|C| = 26$. The second statement follows from the first exactly as in Theorem 5.78. ∎

Remark 5.80 (Where the certificates of Theorems 5.78 and 5.79 are tight). In both certificates the two-point part is almost absent, with $f(1)$ below 10^{-4}, and the bound is carried by the three-point matrices: $F(1,1,1) = 23.95993$ at $s_0 = 0.008$ and 24.79993 at $s_1 = 0.01468$. In floating point both polynomials of the proof are negative on their whole domains. The largest value of $f(u) + 3F(1,u,u) + 1$ is $-1.5 \cdot 10^{-5}$ at s_0 and $-1.8 \cdot 10^{-4}$ at s_1, attained at the touching inner product $u = t$. The largest value of F is $-1.3 \cdot 10^{-5}$ at $(-0.477, t, t)$ and $-8.8 \cdot 10^{-6}$ at $(-0.470, t, t)$: two pairs at the touching angle and the third pair nearly opposite, the three directions lying in one plane at s_1 and nearly so at s_0. This is the triple $(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ of three consecutive points of a hexagon of the root system, compressed to the inner product t. At the inner products of the root system itself both polynomials are negative, the first between $-9.3 \cdot 10^{-3}$ and $-3.9 \cdot 10^{-5}$ at s_0, the second at most $-5.3 \cdot 10^{-5}$, so neither is tight there and the certificates do not single out the root system: they bound the size of a code by 24.96 and 25.8, and integrality does the rest. The room in the final inequalities, 0.04 and 0.2, goes almost entirely into the thresholds that the branch and bound proves: $552e_2 = 0.0386$ and $600e_2 = 0.174$, against $(|C|-1)e_1$ of about $2.5 \cdot 10^{-5}$. With the floating-point maxima in place of e_1 and e_2 the left sides would be 24.0072 and 25.0053.

6. Positivity for every deviation direction

This section proves Theorem 1.6: the defect $\Delta(\varphi,\eta)$ of a single deviation is nonnegative at every point of the full two-sphere of deviation directions and at every tilt. The defect is first rewritten as the deficiency of a cap cut from one fixed polytope; that polytope is then enclosed in a cross-polytope, for which the extremal cap can be determined in closed form. The combinatorial type of the perturbed cell, which changes many times as the deviation varies, plays no part. The defect itself, as an explicit function region by region, is computed in Appendix B, and nothing here depends on it.

6.1. **The defect as a cap deficiency.** Throughout, $u_0 = \alpha_0/\sqrt{2}$ is the undeviated contact direction, $u_1 = \cos\varphi\, u_0 + \sin\varphi\,\eta$ the deviated one, and

$$Q \;=\; \bigcap_{i\neq 0}\{x\in\mathbb{R}^4 : \langle x,u_i\rangle \le 1\}$$

the intersection of the half-spaces of the twenty-three contact directions that stay at roots. The local cell at the all-contact corner is $Q\cap\{\langle x,u_1\rangle\le 1\}$, so

$$\Delta(\varphi,\eta) \;=\; \mathrm{vol}(Q\cap\{\langle x,u_1\rangle\le 1\}) - 8. \tag{6.1}$$

Write $p=\langle x,u_0\rangle$, and let N denote the eight roots making an angle of 60° with α_0, so that $\langle u_0,u_i\rangle = \frac12$ for $i\in N$.

Lemma 6.1 (Structure of Q)**.** *Let $F\subset u_0^\perp$ be the octahedral facet of $\mathcal{V}$ at u_0, translated to the origin, so that the facet is u_0+F. Then F is a regular octahedron whose six vertices are $\pm v_1,\pm v_2,\pm v_3$, where v_1,v_2,v_3 are three pairwise orthogonal contact directions orthogonal to u_0; in particular (u_0,v_1,v_2,v_3) is an orthonormal basis of $\mathbb{R}^4$, $\mathrm{vol}_3(F)=\frac43$, and*

$$F = \Big\{y\in u_0^\perp : \textstyle\sum_i |\langle y,v_i\rangle| \le 1\Big\}.$$

Moreover $Q\cap\{p\ge 1\}$ is the pyramid Pyr *with apex $2u_0$ and base u_0+F, of volume $\frac13$, while $Q\cap\{p\le 1\}=\mathcal{V}$. Hence $\mathrm{vol}(Q)=8+\frac13=\frac{25}{3}$.*

Proof. Six roots are orthogonal to α_0, and they fall into three orthogonal pairs $\pm\alpha_{i_1},\pm\alpha_{i_2},\pm\alpha_{i_3}$; put $v_j=\alpha_{i_j}/\sqrt{2}$. Each $u_0\pm v_j$ satisfies $\langle u_0\pm v_j,u_i\rangle\le 1$ for every root and meets six of those constraints with equality, so the six points $u_0\pm v_j$ are vertices of $\mathcal{V}$ lying on the facet at u_0; since a facet of the 24-cell is a regular octahedron with six vertices, these are all of them. A regular octahedron with vertices $\pm v_j$ in an orthonormal frame is $\{y:\sum_j|\langle y,v_j\rangle|\le 1\}$ and has volume $\frac{2^3}{3!}=\frac43$.

For $i\in N$ set $m_i = u_i-\frac12 u_0$, so $m_i\perp u_0$ and $|m_i|=\frac{\sqrt3}{2}$. Writing $x=pu_0+y$ with $y\perp u_0$, the constraint $\langle x,u_i\rangle\le 1$ reads $\langle y,m_i\rangle\le 1-\frac{p}{2}$. At $p=1$ these eight constraints cut out exactly F, so at height p they cut out $(2-p)F$; that is,

$$\bigcap_{i\in N}\{\langle x,u_i\rangle\le 1\} \;=\; \{pu_0+y \;:\; y\in(2-p)F\}. \tag{6.2}$$

Let $1\le p\le 2$ and $y\in(2-p)F\subseteq F$. The other fifteen constraints are then slack: for the root $-\alpha_0$ one has $\langle x,-u_0\rangle=-p\le -1$; for a root α_j orthogonal to α_0, $\langle x,u_j\rangle=\langle y,\pm v_k\rangle\le 1$ because $\max_{y\in F}\langle y,v_k\rangle=1$; and for a root α_j with $\langle u_0,u_j\rangle=-\frac12$, necessarily $u_j=-u_i$ for some $i\in N$, whence $\langle x,u_j\rangle=-\frac{p}{2}+\langle y,-m_i\rangle\le -\frac12+\frac12=0$. Thus on $\{p\ge 1\}$ the set Q agrees with Equation (6.2), which there is the cone over u_0+F with apex $2u_0$, of volume $\int_1^2\mathrm{vol}_3((2-p)F)\,dp=\frac43\int_0^1\lambda^3 d\lambda=\frac13$. Finally $\mathcal{V}\subseteq\{p\le 1\}$, and below that hyperplane Q and $\mathcal{V}$ are cut out by the same constraints apart from the one at α_0, which is inactive there. ∎

Proposition 6.2 (Cap form of the defect)**.** *For a unit vector u put* $\mathrm{Cap}(u) = \{x \in Q : \langle x, u\rangle \geq 1\}$*. Then*

$$\Delta(\varphi, \eta) \;=\; \tfrac{1}{3} - \mathrm{vol}(\mathrm{Cap}(u_1)).$$

Proof. By Lemma 6.1, $\mathrm{vol}(Q) = \frac{25}{3}$, and Equation (6.1) gives $\Delta = \mathrm{vol}(Q) - \mathrm{vol}(\mathrm{Cap}(u_1)) - 8$. ∎

The deviation parameters enter only through the cutting half-space: Q is one fixed polytope with 25 vertices, and the question is how much of it a half-space at unit distance from the origin can cut off. At $u_1 = u_0$ the answer is $\frac{1}{3}$, the pyramid.

6.2. **A cross-polytope around Q.** Let (u_0, v_1, v_2, v_3) be the orthonormal frame of Lemma 6.1 and write $x_0 = \langle x, u_0\rangle$, $x_j = \langle x, v_j\rangle$ for the coordinates in that frame.

Lemma 6.3 (Enclosure)**.** *$Q \subseteq B$, where*

$$B = \{x \in \mathbb{R}^4 : |x_0| + |x_1| + |x_2| + |x_3| \leq 2\}$$

is the cross-polytope of radius 2 in the frame (u_0, v_1, v_2, v_3).

Proof. Sixteen roots make an angle of $60°$ or $120°$ with α_0, and they come in eight antipodal pairs. If α_i is one of the eight at $60°$ then $u_i = \frac{1}{2}u_0 + \frac{\sqrt{3}}{2}n_i$ with n_i a unit vector orthogonal to u_0; since the eight vectors n_i are the outer normals of the eight faces of the regular octahedron F, they must be $(\pm v_1 \pm v_2 \pm v_3)/\sqrt{3}$, one for each choice of signs. Hence

$$\langle x, u_i\rangle = \tfrac{1}{2}(x_0 \pm x_1 \pm x_2 \pm x_3),$$

and the eight constraints $\langle x, u_i\rangle \leq 1$ say exactly that $x_0 \pm x_1 \pm x_2 \pm x_3 \leq 2$ for all eight sign patterns, that is, $x_0 + |x_1| + |x_2| + |x_3| \leq 2$. The eight roots at $120°$ are the negatives of these, and give $-x_0 + |x_1| + |x_2| + |x_3| \leq 2$. All sixteen roots are distinct from α_0, so all sixteen constraints are among those defining Q. ∎

The enclosure enlarges the volume, $\mathrm{vol}(B) = \frac{32}{3}$ against $\mathrm{vol}(Q) = \frac{25}{3}$, but not the cap in the critical direction: the half-space $\{x_0 \geq 1\}$ cuts from B the cone over the octahedron $\{\sum_{j\geq 1} |x_j| \leq 1\}$ with apex $2u_0$, of volume exactly $\frac{1}{3}$, which is the pyramid it cuts from Q. Figure 24(a) shows the three bodies in section.

6.3. **The extremal cap of the cross-polytope.** We now determine the largest volume that a half-space at unit distance from the centre can cut from B. Write $B_1^4 = \{\|x\|_1 \leq 1\}$, so $B = 2B_1^4$, and let e_0, e_1, e_2, e_3 be the frame vectors.

Lemma 6.4 (Cap formula)**.** *Let u be a unit vector with coordinates u_k in the frame, and set $a_k = u_k^2$, so that $a_0 + a_1 + a_2 + a_3 = 1$. Define*

$$g(a) = \begin{cases} a\,(2\sqrt{a} - 1)^4, & a \geq \frac{1}{4}, \\ 0, & a \leq \frac{1}{4}. \end{cases}$$

Then

$$\mathrm{vol}(B \cap \{\langle x, u\rangle \geq 1\}) \;=\; \tfrac{1}{3}\, g[a_0, a_1, a_2, a_3],$$

where $g[\,\cdot\,]$ denotes the third-order divided difference of g at the four nodes.[23]

[23]Nothing about divided differences is used beyond the following, which is therefore recorded here. For a function h and pairwise distinct nodes $y_0, \ldots, y_n$, the divided difference is defined by $h[y_0] = h(y_0)$ and

$$h[y_0, \ldots, y_n] = \frac{h[y_1, \ldots, y_n] - h[y_0, \ldots, y_{n-1}]}{y_n - y_0},$$

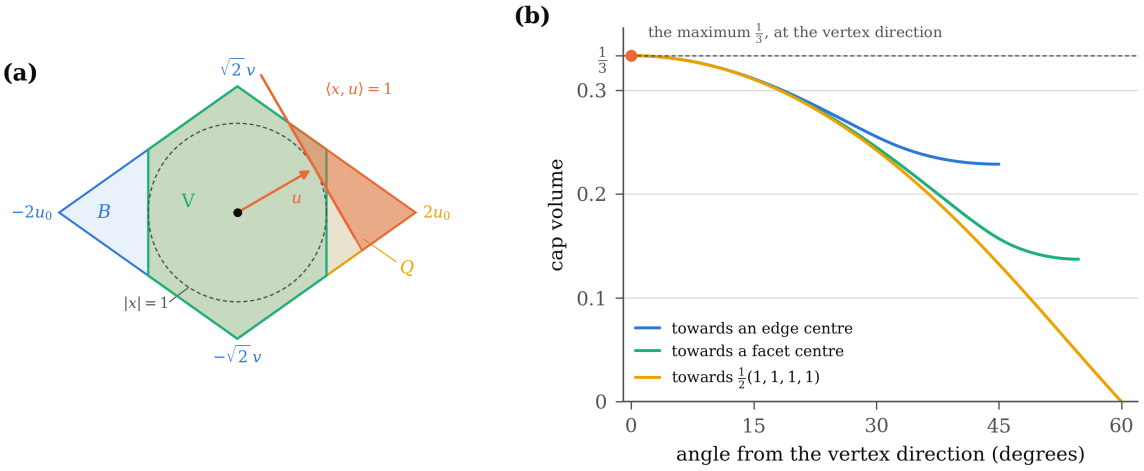

FIGURE 24. (a) The section of the reference cell $\mathcal{V}$, of Q, and of the cross-polytope B by the plane spanned by the base direction u_0 and one transverse direction. Coordinates are exact: the section of $\mathcal{V}$ is a hexagon, that of Q the same hexagon with its right edge replaced by the two edges meeting at the apex $2u_0$, and that of B a rhombus through the same apex and the same two vertices $\pm\sqrt{2}\,v$. The dashed circle is the unit sphere, on which every contact hyperplane is tangent, and the shaded triangle is the section of the cap cut by a half-space at unit distance whose normal is tilted 30° from u_0. (b) The volume of the cap cut from the cross-polytope by a half-space at unit distance, along three great circles leaving a vertex direction: towards an edge centre, towards a facet centre, and towards $\frac{1}{2}(1,1,1,1)$, where the half-space meets B only in its four positive vertices and the cap volume falls to zero. The maximum $\frac{1}{3}$ of Theorem 6.6 is attained at the vertex direction, marked.

Proof. Both sides are unchanged by a signed permutation of the coordinates, so we may assume $u_k \geq 0$ for all k. Split B into the sixteen simplices $S_\sigma = \operatorname{conv}\{0, 2\sigma_0 e_0, \dots, 2\sigma_3 e_3\}$, $\sigma \in \{\pm 1\}^4$, each of volume $\frac{2^4}{4!} = \frac{2}{3}$. For a 4-simplex Δ with vertices $p_0, \dots, p_4$ at which a linear form f takes distinct values $f_0, \dots, f_4$, one has

$$\operatorname{vol}(\Delta \cap \{f \geq s\}) = \operatorname{vol}(\Delta) \sum_{i=0}^{4} \frac{(f_i - s)_+^4}{\prod_{j \neq i}(f_i - f_j)}. \tag{6.3}$$

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and unwinding the recursion gives the closed form $h[y_0, \dots, y_n] = \sum_i h(y_i) \big/ \prod_{j \neq i}(y_i - y_j)$, from which it is visibly a symmetric function of the nodes. Three consequences are used below. First, if h is C^n then $h[y_0, \dots, y_n] = h^{(n)}(\xi)/n!$ for some ξ in the smallest interval containing the nodes; in particular the divided difference extends continuously to coincident nodes, with $h[y, \dots, y] = h^{(n)}(y)/n!$, so all the formulas below may be read at coincident nodes by continuity. Second, if $h^{(n-1)}$ is nondecreasing then $h[y_0, \dots, y_n] \geq 0$ for all nodes, immediate by the mean-value form where h is C^n and in general by uniform approximation, which is the case we need, our g being C^3 with g''' nondecreasing but not C^4 at $a = \frac{1}{4}$. Third, adding a node differentiates: $\partial h[y_0, \dots, y_n]/\partial y_i = h[y_0, \dots, y_n, y_i]$. These three facts turn a statement about the fourth derivative of one variable into a monotonicity statement in four.

[24] On S_σ the form $\langle \cdot, u\rangle$ takes the value 0 at the origin and $2\sigma_k u_k$ at $2\sigma_k e_k$. With $s = 1$ the vertex 0 contributes nothing, and a vertex $2\sigma_k e_k$ contributes only when $\sigma_k = +1$ and $2u_k > 1$. Summing the contribution of the k-th vertex over the eight sign choices of the remaining coordinates gives

$$\sum_{\sigma_l = \pm 1,\, l \neq k} \prod_{l \neq k} \frac{1}{2u_k - 2\sigma_l u_l} = \prod_{l \neq k} \Big(\frac{1}{2u_k - 2u_l} + \frac{1}{2u_k + 2u_l} \Big) = \prod_{l \neq k} \frac{u_k}{u_k^2 - u_l^2},$$

so that

$$\operatorname{vol}(B \cap \{\langle x, u\rangle \geq 1\}) = \frac{2}{3} \sum_k \frac{(2u_k - 1)_+^4}{2u_k} \prod_{l \neq k} \frac{u_k}{u_k^2 - u_l^2} = \frac{1}{3} \sum_k \frac{(2u_k - 1)_+^4 \, u_k^2}{\prod_{l \neq k} (u_k^2 - u_l^2)}.$$

The k-th summand is $g(a_k) / \prod_{l \neq k} (a_k - a_l)$, and the sum is the divided difference. Coincident nodes are handled by continuity, both sides being continuous in u. ∎

The form of this answer has a classical explanation. A divided difference of a truncated power represents a B-spline, and the volume of a slice of a simplex is such a quantity; the identification goes back to Curry and Schoenberg [CSc66], and the next lemma rests on it.

Two properties of g drive everything that follows. Writing $t = \sqrt{a}$ for $a \geq \frac{1}{4}$, repeated differentiation gives

$$\begin{aligned} g' &= (2t-1)^3 (6t-1), \qquad g'' = \frac{6(2t-1)^2 (4t-1)}{t}, \\ g''' &= 96 - \frac{60}{t} + \frac{3}{t^3}, \end{aligned} \tag{6.4}$$

and $g'''' = \dfrac{3(20t^2 - 3)}{2t^5}$.

Lemma 6.5 (Monotonicity)**.** *g is of class C^3 on $[0, \infty)$, its third derivative is nondecreasing, and consequently $g[a_0, a_1, a_2, a_3]$ is nondecreasing in each node.*

Proof. All three expressions in Equation (6.4) vanish at $t = \frac{1}{2}$, so g, g', g'', g''' all vanish at $a = \frac{1}{4}$ and match the identically zero branch; hence $g \in C^3$. For $t \geq \frac{1}{2}$ one has $20t^2 - 3 \geq 2 > 0$, so $g'''' > 0$ there, while $g''' \equiv 0$ on $[0, \frac{1}{4}]$; therefore g''' is nondecreasing on $[0, \infty)$. A function whose $(n-1)$st derivative is nondecreasing has all nth-order divided

[24] Equation (6.3) is the only fact about simplices the proof uses, and it has a short derivation. Put $\Delta = \{\lambda \in \mathbb{R}^5 : \lambda_i \geq 0,\ \sum \lambda_i = 1\}$ in barycentric coordinates, so that $f = \sum_i f_i \lambda_i$ and the uniform probability measure on Δ pushes forward under f to the distribution of $\sum_i f_i \lambda_i$ for λ uniform on the simplex. The coordinates λ_i are the normalised spacings of four independent uniform points on an interval, and the density of $\sum f_i \lambda_i$ is 4! times the fourth-order divided difference of $t \mapsto (t - x)_+^3 / 3!$ in the nodes f_i, which is the Curry–Schoenberg B-spline with those knots [CSc66]. Integrating that density from s upwards raises the truncated power by one degree and produces exactly the displayed sum. Equivalently, and with no probability: both sides of Equation (6.3) are, as functions of s, piecewise polynomials of degree 4 that vanish for $s \geq \max_i f_i$ and agree with $\operatorname{vol}(\Delta)$ for $s \leq \min_i f_i$, with matching derivatives at every breakpoint, so they agree identically. For d-simplices the exponent 4 is replaced by d throughout.

differences nonnegative, so $g[y_0,\dots,y_4]\ge 0$ for any five nodes.[25] Since

$$\frac{\partial}{\partial a_i}\,g[a_0,a_1,a_2,a_3]=g[a_0,a_1,a_2,a_3,a_i],$$

the divided difference is nondecreasing in each node. ∎

Theorem 6.6 (Extremal cap of the cross-polytope)**.** *For every unit vector $u\in\mathbb{R}^4$,*

$$\operatorname{vol}(B\cap\{\langle x,u\rangle\ge 1\})\ \le\ \tfrac{1}{3},$$

with equality if and only if $u=\pm e_k$ for some k.

Figure 24(b) shows the cap volume along three great circles leaving a vertex direction, computed from the enumerated cell and independent of the argument below.

Proof. By Lemma 6.4 the claim is that $\Phi(a):=g[a_0,a_1,a_2,a_3]\le 1$ on the simplex $a_k\ge 0$, $\sum_k a_k=1$, with equality only at the vertices. Let $S=\{k:a_k>\frac14\}$; since the nodes sum to 1, $|S|\le 3$. We may relabel so that $a_0\ge a_1\ge a_2\ge a_3$.

Case $|S|=0$. All four nodes lie in $[0,\frac14]$, where g vanishes identically together with all its derivatives, so $\Phi(a)=0$.

Case $|S|=1$. Only a_0 exceeds $\frac14$, and again g vanishes to infinite order at the other three nodes, so the Newton form of the divided difference collapses to a single term:

$$\Phi(a)=\frac{g(a_0)}{(a_0-a_1)(a_0-a_2)(a_0-a_3)}.$$

Write $c=\sqrt{a_0}\in(\frac12,1]$. The nodes a_1,a_2,a_3 range over the polytope $P_c=\{\beta\in[0,\frac14]^3:\beta_1+\beta_2+\beta_3=1-c^2\}$, and $\beta\mapsto\sum_l\log(a_0-\beta_l)$ is concave, so $\prod_l(a_0-\beta_l)$ attains its minimum over P_c at a vertex of P_c, that is, at a point where at most one coordinate lies strictly between 0 and $\frac14$. Three ranges of c occur.

If $c\ge\frac{\sqrt3}{2}$ the extreme point is $(1-c^2,0,0)$ and the required inequality $g(a_0)\le\prod_l(a_0-\beta_l)$ reads $(2c-1)^4\le c^2(2c^2-1)$, that is,

$$(1-c)\,(14c^3-18c^2+7c-1)\ \ge\ 0.$$

The cubic has a single real root, at $c=0.7410\ldots<\frac{\sqrt3}{2}$, and is positive beyond it, so the product is nonnegative on $[\frac{\sqrt3}{2},1]$ and vanishes only at $c=1$.

If $\frac{1}{\sqrt2}\le c\le\frac{\sqrt3}{2}$ the extreme point is $(\frac14,\frac34-c^2,0)$ and the inequality reads $(2c-1)^4\le(c^2-\frac14)(2c^2-\frac34)$, that is, $\frac{1}{16}(2c-1)\,q(c)\ge 0$ with

$$q(c)=-112c^3+200c^2-102c+13.$$

Here $q'(c)=-2(168c^2-200c+51)$ has its roots at $\frac{50\pm\sqrt{358}}{84}$, of which only $\frac{50+\sqrt{358}}{84}=0.8205\ldots$ lies in $[\frac{1}{\sqrt2},\frac{\sqrt3}{2}]$. Therefore q rises and then falls on that interval and attains its minimum at an endpoint, where $q(\frac{1}{\sqrt2})=113-79\sqrt2$ and $q(\frac{\sqrt3}{2})=163-93\sqrt3$. Both are positive, since $113^2=12769>12482=2\cdot 79^2$ and $163^2=26569>25947=3\cdot 93^2$.

[25] If $g[a_0,a_1,a_2,a_3]$ were Schur-convex in $(a_0,\dots,a_3)$, Theorem 6.6 would reduce in one step to evaluating $g[\,\cdot\,]$ at the majorisation maximum $(1,0,0,0)$. The criterion for Schur-convexity of a symmetric divided difference of this form [MOA11, Ch. 3] is $g^{(5)}\ge 0$, and $g^{(5)}(t^2)=\frac{45}{4}t^{-7}(1-4t^2)$ is negative at every $t>\frac12$, which is the whole interior of the support. The one place the fifth derivative is positive is the point mass at $a=\frac14$, where g'''' jumps from 0 to 96. Monotonicity in each node separately, which needs only $g''''\ge 0$, is therefore the strongest tool of this kind that the function supports, and it is why the proof below splits into cases instead of reducing to a single extremal configuration.

If $\frac{1}{2} \le c \le \frac{1}{\sqrt{2}}$ the extreme point is $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2} - c^2)$, the required inequality becomes $16(c-\frac{1}{2})^4 c^2 \le 2(c-\frac{1}{2})^3(c+\frac{1}{2})^3$, that is, $8(c-\frac{1}{2})c^2 \le (c+\frac{1}{2})^3$, that is, $r(c) \ge 0$, where

$$r(c) = 8\Big((c+\tfrac{1}{2})^3 - 8(c-\tfrac{1}{2})c^2\Big) = -56c^3 + 44c^2 + 6c + 1.$$

Again $r'(c) = -168c^2 + 88c + 6$ has a single root, $\frac{22+\sqrt{736}}{84} = 0.5849\ldots$, in $[\frac{1}{2}, \frac{1}{\sqrt{2}}]$, so r attains its minimum on that interval at an endpoint; $r(\frac{1}{2}) = 8$ and $r(\frac{1}{\sqrt{2}}) = 23 - 11\sqrt{2} > 0$, the latter because $23^2 = 529 > 242 = 2 \cdot 11^2$.

In all three ranges $\Phi(a) \le 1$, with equality only when $c = 1$, that is $a = (1,0,0,0)$ and $u = \pm e_0$.

Case $|S| \ge 2$. Here $a_0 \ge a_1 \ge \frac{1}{4}$, so a lies in

$$R = \Big\{a : \ a_0 \ge a_1 \ge a_2 \ge a_3 \ge 0, \ \sum_k a_k = 1, \ a_1 \ge \tfrac{1}{4}\Big\}.$$

This is a polytope of dimension three, and at each of its vertices three of the five inequalities $a_0 \ge a_1$, $a_1 \ge a_2$, $a_2 \ge a_3$, $a_3 \ge 0$, $a_1 \ge \frac{1}{4}$ hold with equality. Solving the ten such systems and keeping the solutions that lie in R gives the five vertices

$$(\tfrac{3}{4}, \tfrac{1}{4}, 0, 0), \quad (\tfrac{1}{2}, \tfrac{1}{2}, 0, 0), \quad (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}, 0), \quad (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, 0), \quad (\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}).$$

Three facts bound Φ on R.

First, Φ is an average of g'''. Let λ be uniformly distributed on the simplex $\{\lambda \in \mathbb{R}^4 : \lambda_k \ge 0, \ \sum_k \lambda_k = 1\}$ and put $T = \sum_k a_k \lambda_k$. Then

$$\Phi(a) = \tfrac{1}{6}\,\mathbb{E}\, g'''(T). \tag{6.5}$$

[26]

Second, $g'''(a) \le 96\,(a - \frac{1}{4})_+$ for every $a \ge 0$. Both sides vanish for $a \le \frac{1}{4}$. On $[\frac{1}{4}, \infty)$ the function g''' is concave, since its second derivative $g^{(5)} = \frac{45}{4}\, a^{-7/2}(1 - 4a)$ is negative there, so it lies below its tangent at $\frac{1}{4}$. By Equation (6.4), $g'''(\frac{1}{4}) = 0$ and $g''''(\frac{1}{4}) = 96$, so that tangent is $96(a - \frac{1}{4})$.

Third, $\mathbb{E}\lambda_k = \frac{1}{4}$ for each k, so $\mathbb{E}T = \frac{1}{4}$ because the nodes sum to 1, and $\mathbb{E}(T - \frac{1}{4})_+ = \frac{1}{2}\,\mathbb{E}\,|T - \frac{1}{4}| = \frac{1}{2}\,\mathbb{E}\,|\sum_k (a_k - \frac{1}{4})\lambda_k|$. The last expression is an average of absolute values of linear functions of a, hence a convex function of a.

Together,

$$\Phi(a) \ \le \ 16\,\mathbb{E}(T - \tfrac{1}{4})_+,$$

and the right side, being convex, is largest on R at a vertex. To evaluate it there, apply Equation (6.5) to $h(x) = (x - \frac{1}{4})_+^4$, whose third derivative is $24(x - \frac{1}{4})_+$; this gives $16\,\mathbb{E}(T - \frac{1}{4})_+ = 4\,h[a_0, a_1, a_2, a_3]$. The function h vanishes on $[0, \frac{1}{4}]$ together with its first three

[26] This is the Hermite–Genocchi formula, and it follows from the footnote to Equation (6.3). For a C^3 function h and a point c below every node, Taylor's formula with integral remainder writes $h(x)$ as a polynomial of degree two plus $\int_c^\infty h'''(s)\,\frac{1}{2}(x-s)_+^2\,ds$. The polynomial has third divided difference 0, and for distinct nodes the third divided difference of $x \mapsto \frac{1}{2}(x-s)_+^2$ is $\frac{1}{6}$ times the density $f_T(s)$ of $T = \sum_k a_k\lambda_k$ at s, the Curry–Schoenberg B-spline of that footnote in dimension three [CSc66]. So $h[a_0, a_1, a_2, a_3] = \frac{1}{6}\int h'''(s)\, f_T(s)\, ds = \frac{1}{6}\,\mathbb{E}\, h'''(T)$, and coincident nodes follow by continuity. The function g is C^3 by Lemma 6.5, so the formula applies to it.

derivatives, so, as in the case $|S| = 1$, only the nodes above $\frac{1}{4}$ contribute:

$$h[\tfrac{3}{4},\tfrac{1}{4},0,0] = \frac{h(\frac{3}{4})}{\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{3}{4}} = \frac{2}{9}, \quad h[\tfrac{1}{2},\tfrac{1}{2},0,0] = \frac{d}{dx}\frac{h(x)}{x^2}\Big|_{x=\frac{1}{2}} = \frac{3}{16},$$
$$h[\tfrac{1}{2},\tfrac{1}{4},\tfrac{1}{4},0] = \frac{h(\frac{1}{2})}{\frac{1}{4}\cdot\frac{1}{4}\cdot\frac{1}{2}} = \frac{1}{8}, \quad h[\tfrac{1}{3},\tfrac{1}{3},\tfrac{1}{3},0] = \frac{1}{2}\frac{d^2}{dx^2}\frac{h(x)}{x}\Big|_{x=\frac{1}{3}} = \frac{27}{256},$$

and $h[\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4}] = 0$. The largest is $\frac{2}{9}$, so $\Phi(a) \le \frac{8}{9} < 1$ on all of R, and the case $|S| \ge 2$ contains no point of equality. ∎

Remark 6.7 (An independent check of the case $|S| \ge 2$). The maximum of Φ on R is $0.76583\ldots$, attained at $(\frac{3}{4},\frac{1}{4},0,0)$, against the bound $\frac{8}{9} = 0.888\ldots$ just proved. The case can also be settled by a finite check that uses only Lemma 6.5: Φ is nondecreasing in each node, so on any box $\prod_k [l_k, h_k]$ it is at most its value at the upper corner. Bisecting a uniform grid of 216 cubes of side $\frac{1}{8}$, and refining only where needed, covers R by 303 boxes whose upper corner values are rational numbers at most 1, the largest being $0.99755\ldots$. The nodes at a corner are taken at rational points t_k with t_k^2 at least the corner coordinate, and ties are broken upward, both of which only raise the bound. The check is exact rational arithmetic throughout and is repeated in Lean, which finds the same 303 boxes and the same largest value and also checks the three polynomial inequalities of the case $|S| = 1$.

6.4. **The single-deviation theorem.**

Theorem 6.8 (Direction-of-deviation positivity)**.** *For every unit deviation direction η orthogonal to u_0 and every tilt $\varphi \in [0,\pi]$,*

$$\Delta(\varphi,\eta) \;\ge\; 0,$$

with equality if and only if $\varphi = 0$.

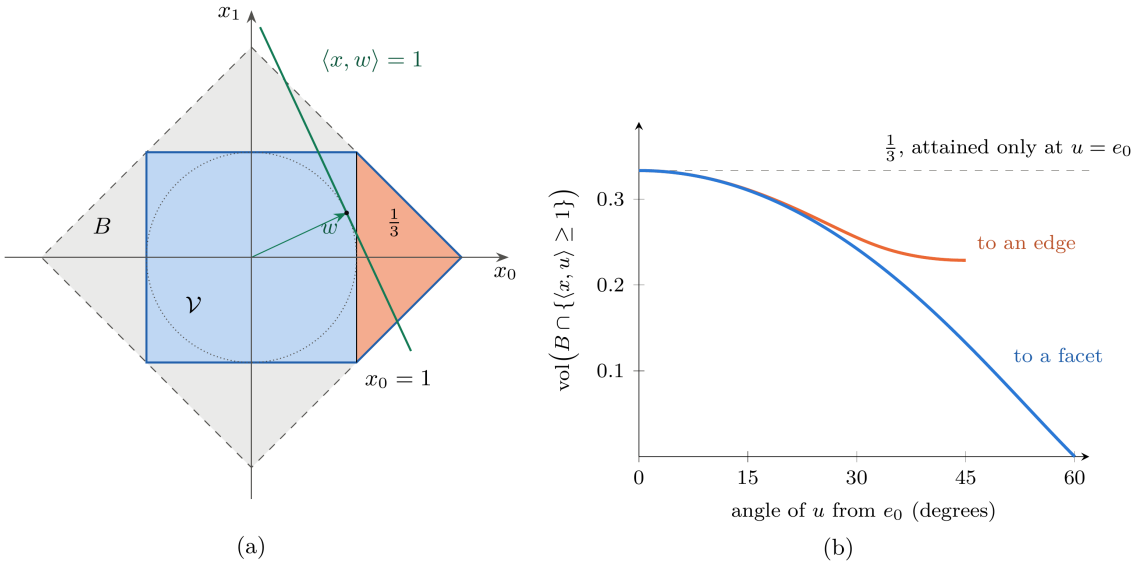
Proof. By Lemma 6.3, $\mathrm{Cap}(u_1) \subseteq B \cap \{\langle x, u_1\rangle \ge 1\}$, so Theorem 6.6 gives $\mathrm{vol}(\mathrm{Cap}(u_1)) \le \frac{1}{3}$, and Proposition 6.2 turns this into $\Delta \ge 0$.

For the equality case, suppose $\Delta(\varphi,\eta) = 0$, so $\mathrm{vol}(\mathrm{Cap}(u_1)) = \frac{1}{3}$. Then the cap of B also has volume $\frac{1}{3}$, so $u_1 = \pm e_k$ by Theorem 6.6. If $u_1 = -u_0$ then $\mathrm{Cap}(u_1) = \{x \in Q : \langle x, u_0\rangle \le -1\}$, which lies in a hyperplane because $-\alpha_0$ is a root and Q therefore lies in $\{\langle x, u_0\rangle \ge -1\}$; if $u_1 = \pm v_j$ then $\pm v_j$ is itself one of the contact directions defining Q, so again the cap lies in a hyperplane and has volume 0. In each of these cases $\Delta = \frac{1}{3}$, not 0. The remaining possibility is $u_1 = u_0$, that is $\varphi = 0$, where the cap is the pyramid of Lemma 6.1 and $\Delta = 0$. ∎

Theorem 1.6 can now be proved in the form stated in the introduction.

Proof of Theorem 1.6. Let $u_1,\ldots,u_n$ be the directions from c of the active neighbours and $d_1,\ldots,d_n$ their distances. Two active neighbours cannot lie on one ray from c, since they would be less than $2\sqrt{2}-2 < 2$ apart; so the directions are distinct. By hypothesis all but at most one of them are normalised roots of R. If every direction is a root direction, let u_0 be any normalised root and $u_1 = u_0$; otherwise let u_0 be a normalised root that no direction takes, which exists by hypothesis, and u_1 the deviating direction, and write $u_1 = \cos\varphi\, u_0 + \sin\varphi\, \eta$ with $\varphi \in [0,\pi]$ and η a unit vector orthogonal to u_0. By Lemma 2.9, $\mathrm{vol}(V_c^{\mathrm{act}})$ is at least the volume of $P = \bigcap_i \{z : \langle z - c, u_i\rangle \le 1\}$, the same half-spaces moved to distance 1. Every half-space of P is one of those defining $Q \cap \{\langle x, u_1\rangle \le 1\}$ about c, so P contains that body, and by Equation (6.1) and Theorem 6.8

$$\mathrm{vol}(V_c^{\mathrm{act}}) \;\ge\; \mathrm{vol}(P) \;\ge\; 8 + \Delta(\varphi,\eta) \;\ge\; 8.$$



(a) (b)

FIGURE 25. The cap argument of Section 6. (a) The plane $x_2 = x_3 = 0$ in the frame of four orthogonal roots, the deviating root being e_0, to scale. The cross-polytope B (dashed) meets it in a square, the polytope Q of the twenty-three undeviated roots in the pentagon outlined in blue, and the 24-cell $\mathcal{V}$ in the inner square; the dotted circle is the unit sphere. The half-space $x_0 \geq 1$ cuts the same piece off Q and off B (orange), of volume $\frac{1}{3}$ in $\mathbb{R}^4$, so that $\operatorname{vol} Q = 8 + \frac{1}{3}$. A deviated contact replaces it by a half-space $\langle x, w\rangle \geq 1$ tangent to the unit sphere (green). (b) The volume that $\langle x, u\rangle \geq 1$ cuts off B, from Lemma 6.4, along two arcs of unit normals u leaving the vertex direction e_0: towards the midpoint of an edge, $u = (\cos t, \sin t, 0, 0)$, and towards the centre of a facet, $u = (\cos t, s, s, s)$ with $s = \sin t/\sqrt{3}$. Both stay below $\frac{1}{3}$ for $t > 0$, as Theorem 6.6 asserts for every u. In the plane of panel (a) alone the comparison would go the other way, which is why the theorem needs the four-dimensional cap formula.

Suppose equality holds throughout. Then $\Delta(\varphi, \eta) = 0$, so $\varphi = 0$ and $u_1 = u_0$ by Theorem 6.8; then $Q \cap \{\langle x, u_0\rangle \leq 1\}$ is the 24-cell $c + \mathcal{V}$ by Lemma 6.1. If one of the other 23 normalised roots were missing from the active directions, P would contain the 24-cell together with the pyramid over the facet of that root, of volume $\frac{1}{3}$ by Lemma 6.1 applied at that root, and could not have volume 8. So P is the 24-cell. Finally, every facet of the 24-cell is a regular octahedron of positive 3-volume, so moving any of its facet hyperplanes outward strictly increases the volume; hence $V_c^{\mathrm{act}} = P$ forces every $d_i = 2$, and the active neighbours form $c + \sqrt{2}\,\mathsf{R}$. Conversely that configuration has cell $c + \mathcal{V}$, of volume 8. ∎

The argument covers every combinatorial type of the perturbed cell at once, because the cap inequality is a statement about a single polytope and makes no reference to which facets the deviating half-space cuts. The enclosure $Q \subseteq B$ adds $\frac{7}{3}$ to the volume of Q, more than a quarter of it, and yet loses nothing at the critical direction, so the estimate is sharp; the same argument applied to the 24-cell in place of Q, or to a smaller enclosing body, would not have this property.

6.5. **A quantitative form near the base direction.** Theorem 6.8 gives a sign and no more. For small and moderate tilts one can say how large the defect is, by enclosing Q in a cone instead of a cross-polytope. The eight roots at 60° alone cut out

$$K \;=\; \{x : x_0 + |x_1| + |x_2| + |x_3| \le 2\},$$

a cone with apex $2u_0$, and $Q \subseteq B \subseteq K$. The cap of a cone is a union of simplices, so it has a closed form.

Proposition 6.9 (Cone bound)**.** *Let η be a unit vector orthogonal to u_0, with coordinates $\eta_j = \langle \eta, v_j\rangle$ in the frame of Lemma 6.1, and write $c = \cos\varphi$, $\sigma = \sin\varphi$. If $\sigma \max_j |\eta_j| < c$, then*

$$\Delta(\varphi,\eta) \;\ge\; \frac{1}{3} - \frac{c^2\,(2c-1)_+^4}{3\prod_{j=1}^3 (c^2 - \sigma^2\eta_j^2)}. \tag{6.6}$$

The right-hand side equals $\varphi^2/3 + O(\varphi^4)$ as $\varphi \to 0$, and it is strictly positive for every such η if and only if $c > c_0$, where $c_0 = 0.7410861\ldots$ is the unique real root of $14c^3 - 18c^2 + 7c - 1$; that is, for $\varphi < 42.17598\ldots^\circ$.

Proof. Every root at 60° to α_0 is one of the twenty-three, so $Q \subseteq K$ by the first half of the proof of Lemma 6.3, and $\mathrm{vol}(\mathrm{Cap}(u_1)) \le \mathrm{vol}(K \cap \{\langle x, u_1\rangle \ge 1\})$.

Put $w = 2u_0 - x$. Then K becomes the cone $C = \{w : w_0 \ge |w_1| + |w_2| + |w_3|\}$ with apex at the origin, and the half-space $\{\langle x, u_1\rangle \ge 1\}$ becomes $\{\langle w, u_1\rangle \le 2c - 1\}$, since $\langle 2u_0, u_1\rangle = 2c$. If $2c - 1 \le 0$ the intersection is at most the apex, because $\langle w, u_1\rangle \ge w_0\,(c - \sigma \max_j |\eta_j|)$ on C, which is positive at every $w \ne 0$ of C under the stated hypothesis; then the cap has volume 0 and $\Delta = \frac{1}{3}$, which is (6.6). Assume $2c - 1 > 0$.

Intersecting C with an orthant $\{\sigma_j w_j \ge 0\}$, $\sigma \in \{\pm 1\}^3$, gives the simplicial cone generated by $g_0 = e_0$ and $g_j = e_0 + \sigma_j e_j$, and $\det(g_0, g_1, g_2, g_3) = \sigma_1\sigma_2\sigma_3$. Truncating it by $\{\langle w, u_1\rangle \le h\}$, $h = 2c - 1$, gives the simplex with vertices 0 and $h\, g_i / \langle g_i, u_1\rangle$, of volume

$$\frac{h^4}{24} \cdot \frac{1}{\langle g_0, u_1\rangle \prod_{j=1}^3 \langle g_j, u_1\rangle}, \qquad \langle g_0, u_1\rangle = c, \quad \langle g_j, u_1\rangle = c + \sigma_j\sigma\eta_j.$$

Summing over the eight sign patterns factorises:

$$\begin{aligned} \mathrm{vol}\big(C \cap \{\langle w, u_1\rangle \le h\}\big) &= \frac{h^4}{24c} \prod_{j=1}^3 \Big(\frac{1}{c + \sigma\eta_j} + \frac{1}{c - \sigma\eta_j}\Big) \\ &= \frac{h^4}{24c} \prod_{j=1}^3 \frac{2c}{c^2 - \sigma^2\eta_j^2} = \frac{c^2 h^4}{3\prod_j (c^2 - \sigma^2\eta_j^2)}. \end{aligned}$$

With $h = 2c - 1$ this is the subtracted term in (6.6), and Proposition 6.2 converts the cap bound into the stated bound on Δ.

For the expansion, $c = 1 - \varphi^2/2 + O(\varphi^4)$ gives $2c - 1 = 1 - \varphi^2 + O(\varphi^4)$ and $c^2 = 1 - \varphi^2 + O(\varphi^4)$, so the numerator $c^2(2c-1)^4$ is $1 - 5\varphi^2 + O(\varphi^4)$. Since $\sigma^2 = \varphi^2 + O(\varphi^4)$ and $\sum_j \eta_j^2 = 1$,

$$\prod_{j=1}^3 (c^2 - \sigma^2\eta_j^2) = \prod_{j=1}^3 (1 - \varphi^2 - \varphi^2\eta_j^2) + O(\varphi^4) = 1 - 4\varphi^2 + O(\varphi^4),$$

so the quotient is $\frac{1}{3}(1 - 5\varphi^2)(1 + 4\varphi^2) + O(\varphi^4) = \frac{1}{3} - \frac{\varphi^2}{3} + O(\varphi^4)$, and the right-hand side of (6.6) is $\varphi^2/3 + O(\varphi^4)$.

Finally, the bound is positive exactly when $c^2(2c-1)^4 < \prod_j (c^2 - \sigma^2\eta_j^2)$. Writing $a_j = \eta_j^2$, the constraint set is the simplex $\{a \ge 0,\ \sum a_j = 1\}$, and $a \mapsto \sum_j \log(c^2 - \sigma^2 a_j)$ is a sum

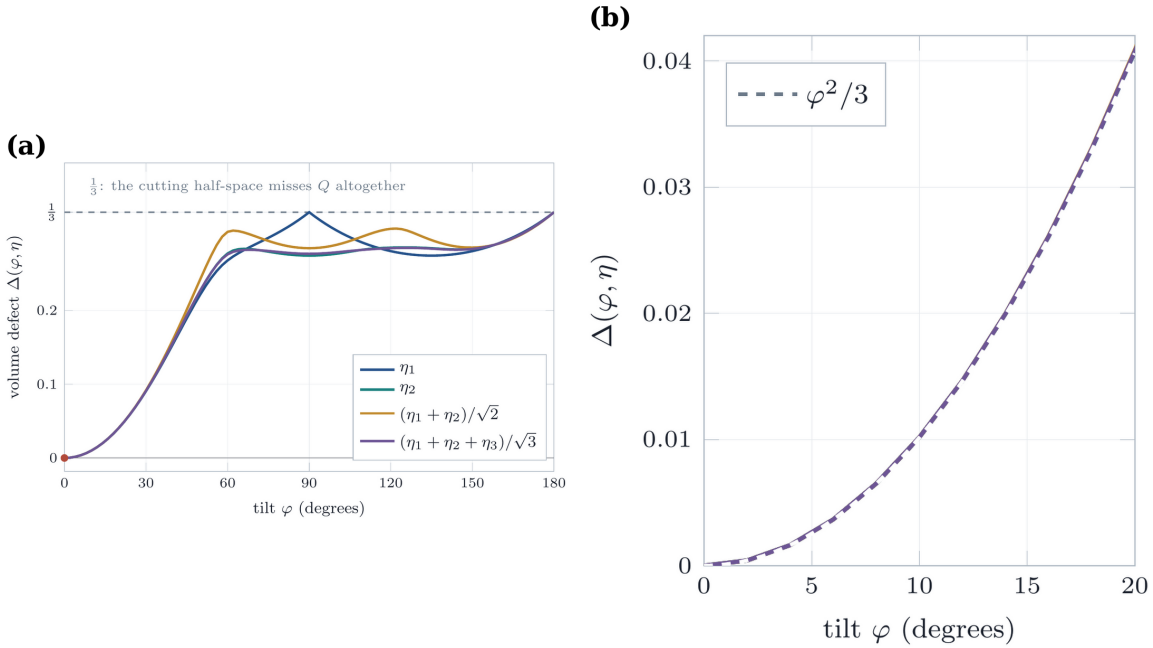


FIGURE 26. (a) The volume defect $\Delta(\varphi, \eta)$ over the whole range of tilts, for four deviation directions spanning the transverse sphere: two of the orthonormal frame directions and two diagonal combinations of them. The defect is nonnegative throughout and vanishes only at $\varphi = 0$, as Theorem 6.8 asserts, and it returns to the value $\frac{1}{3}$ at $\varphi = \pi$, where the cutting half-space misses Q entirely and the cell is exactly Q itself. The intermediate maxima and minima are where the deviating half-space starts or stops cutting a vertex of Q. The curves are computed from directly enumerated cell volumes, by a route independent of the proof. (b) The same four curves at small tilt, against the second-order rate $\varphi^2/3$ of Proposition 6.9. At plotting accuracy the four are indistinguishable from that rate and from each other.

of concave functions, hence concave, so its minimum over the simplex is at a vertex. The worst case is therefore $\eta = \pm v_j$, where the product is $(c^2 - \sigma^2)c^4 = (2c^2 - 1)c^4$, and the condition becomes $(2c - 1)^4 < c^2(2c^2 - 1)$. Expanding,

$$c^2(2c^2 - 1) - (2c - 1)^4 = (1 - c)(14c^3 - 18c^2 + 7c - 1),$$

which for $c \in (0, 1)$ is positive exactly when $14c^3 - 18c^2 + 7c - 1 > 0$. The cubic has one real root $c_0 = 0.7410861\ldots$ and is negative below it and positive above, so the bound is positive for all η precisely when $c > c_0$. ∎

The threshold c_0 is the same cubic that governs the first case of Theorem 6.6, because both come from comparing $(2c - 1)^4$ with $c^2(2c^2 - 1)$, once for the cone and once for the divided difference at a single node. Beyond $42.17598\ldots^\circ$ the cone is too generous an enclosure and (6.6) goes negative, which is exactly where the cross-polytope argument is needed. Figure 26(a) plots the defect over the whole range and Figure 26(b) against that second-order rate.

7. Neighbours that do not touch

Sections 5 and 6 concern contact configurations, in which every centre that can cut the cell touches c. This section treats the others. It proves the bound when their distances pass a covering criterion (Proposition 7.1), when at most twenty-three centres lie within $\sqrt{6}$ and near the root system (Section 7.1), computes the second order of the cell volume at the root system (Section 7.3), and reduces what is left of Conjecture 1.3 to two explicit statements (Section 7.4).

Call c *shell-free* if every centre within $2\sqrt{2}$ of it lies at distance exactly 2. The volume of a cell inside a ball is read off the sphere by the radial form of Lemma 5.15; the version needed here allows the centres to be at any distance.

Proposition 7.1 (The distance criterion)**.** *Let c be a centre of a unit-ball packing of $\mathbb{R}^4$ and $d_1 \le d_2 \le \dots$ the distances from c of the other centres within $2\sqrt{2}$. Then*

$$\operatorname{vol}(V_c) \;\ge\; \operatorname{vol}(V_c \cap B(c,\sqrt{2})) \;\ge\; \Phi(d_1, d_2, \dots),$$

$$\Phi(d_1, d_2, \dots) = \frac{1}{4}\Big[2\pi^2 + \int_0^{\pi/4} \Big(2\pi^2 - \sum_i C\big(\arccos(\tfrac{d_i}{2}\cos r)\big)\Big)_+ \, d(\sec^4 r)\Big],$$

where $C(\rho) = \pi(2\rho - \sin 2\rho)$ is the measure of a cap of angular radius ρ on S^3 and the i-th term is omitted when $\frac{d_i}{2}\cos r \ge 1$. Φ depends on the distances alone and is non-decreasing in each of them. For m contacts and no other centre within $2\sqrt{2}$, $\Phi > 8.044$ for every $m \le 22$.

Proof. Put $c = 0$. The cell is star-shaped about 0, with radial function $\rho(\theta) = \min_y(|y|/2)/\langle \theta, u_y\rangle$ over the centres y with $\langle \theta, u_y\rangle > 0$, and $\rho \ge 1$. A centre beyond $2\sqrt{2}$ contributes a value at least $\sqrt{2}$, so $\min(\sqrt{2}, \rho)$ is determined by the centres within $2\sqrt{2}$, and $\operatorname{vol}(V_c \cap B(\sqrt{2})) = \frac{1}{4}\int_{S^3} \min(\sqrt{2},\rho)^4 = \frac{1}{4}\big[2\pi^2 + \int_0^{\pi/4} U(r)\, d(\sec^4 r)\big]$, $U(r)$ being the measure of $\{\theta : \rho(\theta) > \sec r\}$. The complement of that set is the union over the centres of $\{\theta : \langle \theta, u_i\rangle \ge \frac{d_i}{2}\cos r\}$, a cap of angular radius $\arccos(\frac{d_i}{2}\cos r) \le r$, and so $U(r) \ge \big(2\pi^2 - \sum_i C(\cdot)\big)_+$. A cap shrinks as its centre moves out, which is the monotonicity. For m contacts the bound is the covering estimate of Theorem 5.16 stopped at 45°, which changes it only for $m \le 11$, where it stays above 11.5; its value at $m = 22$ is evaluated in ball arithmetic. ∎

Lemma 7.2 (Twenty-four contacts leave no room)**.** *A centre with 24 contacts has no other centre within $2\sqrt{2}$, and its Voronoi cell is the 24-cell.*

Proof. By Theorem 5.25 the contact directions are the normalised roots of a copy of R. For a unit vector θ whose absolute coordinates in the frame of Section 2 are $a \ge b \ge c' \ge d'$, the largest inner product with a normalised root is $(a+b)/\sqrt{2}$, and $(a+b)^2 \ge a^2+b^2+c'^2+d'^2 = 1$ because $2ab \ge 2b^2 \ge c'^2 + d'^2$; so every direction is within 45° of a contact direction, which is to say that the covering radius of the root directions is 45°. A centre y at distance d from c and at an angle of at most 45° from the contact $2w$ has $|y - 2w|^2 \le d^2 + 4 - 2\sqrt{2}\,d$, which is at least 4 only when $d \ge 2\sqrt{2}$. The contacts cut out the 24-cell, of circumradius $\sqrt{2}$, and by Lemma 2.7 no centre beyond $2\sqrt{2}$ cuts it. ∎

Proof of Theorem 1.1. If $\Phi(d_1, d_2, \dots) > 8$ this is Proposition 7.1, with strict inequality. Let c be shell-free with m contacts, whose directions form a contact configuration W; by Lemma 2.7, $V_c \cap B(c,\sqrt{2}) = V_c(W) \cap B(c,\sqrt{2})$. If $m \le 22$, Proposition 7.1 gives $\operatorname{vol}(V_c) > 8.044$. If $m = 24$, Lemma 7.2 makes V_c the 24-cell, of volume 8. Let $m = 23$. The cell $V_c(W)$

TABLE 4. The distance criterion for m contacts and k further centres at a common distance d: $\Phi > 8$, and so $\mathrm{vol}(V_c) > 8$, as soon as d is at least the value shown (in ball arithmetic).

m	k	$d \geq$	m	k	$d \geq$
22	1	2.1676	22	2	2.2340
21	2	2.0705	21	3	2.1267
20	3	2.0451	18	6	2.0560
19	4	2.0332	12	12	2.0267
23	1	never			

is bounded, since a direction θ with $\langle \theta, w \rangle \leq 0$ for every $w \in W$ would make $W \cup \{\theta\}$ a contact configuration of 24 directions, hence a copy of the root directions by Theorem 5.25, whose deletions have bounded cells. The proof of Theorem 5.40 bounds the part of that cell inside the smaller ball, $\mathrm{vol}(V_c(W) \cap B(c, \sqrt{3/2})) > 8$ (Theorem 5.72), whatever the circumradius, and $V_c \cap B(c, \sqrt{3/2}) = V_c(W) \cap B(c, \sqrt{3/2})$; so $\mathrm{vol}(V_c) > 8$. Equality occurs only for $m = 24$, where the contacts form a copy of $\sqrt{2}\,\mathsf{R}$. ■

Theorem 1.1 leaves Conjecture 1.3, a centre that is not shell-free and whose distances fail the criterion, and Proposition 7.1 places that case precisely. At most twenty-two centres within $2\sqrt{2}$ never fail it, and neither does any profile in which the centres that do not touch are far enough out; Table 4 gives the thresholds for a few profiles, certified in ball arithmetic. Twenty-three contacts and any further centre within $2\sqrt{2}$ always fail it, since the criterion counts caps and the case of 23 contacts needs the pair terms of Section 5.11. The remaining case is therefore a centre crowded by at least twenty-three neighbours within $2\sqrt{2}$, among them near-contacts, centres a little farther than 2. The packing of 49 balls of Section 2.6 is settled by the criterion, with $\Phi > 18.33$.

The remaining case is tight. Take the deletion of a root, the 23 other normalised roots at distance 2, and put one further centre at distance $2 + \delta$ on the axis of the deleted root, $0 < \delta < 2\sqrt{2} - 2$. This is a packing, and by Lemma 6.1 its cell is the 24-cell together with the part of the pyramid over the vacant facet below the hyperplane at distance $1 + \delta/2$, of volume $\frac{25}{3} - \frac{1}{3}(1 - \frac{\delta}{2})^4$. As $\delta \to 0$ this tends to 8 and the configuration to the root system, so a proof of Conjecture 1.3 has to be sharp near the 24-cell, as the proof for contact configurations is.

The case was also searched. We minimised the exact volume of the cell over configurations of 22 to 26 centres around a fixed one, with the packing constraints imposed exactly and the exact gradient of the volume, from starts that crowd the centre with near-contacts: the root system with every root pushed out and turned a little and centres added in its holes, deletions of a root with up to three centres near the deleted one, and random configurations in the shell. Without further constraint, over 240 starts, 224 of the 239 feasible endpoints brought their non-contacts into contact or moved them beyond $2\sqrt{2}$, the least volume found being 8, at the root system; the other 15 kept a centre in the shell, and the least volume among them was 8.3362. With one neighbour held at a prescribed distance $2 + \delta$, for eight values of δ from 0.001 to 0.8 and 60 starts each, the least volume found was, for every $\delta \leq 0.4$, that of the family above, $\frac{25}{3} - \frac{1}{3}(1 - \frac{\delta}{2})^4$, to nine digits; at $\delta = 0.8$ it was 8.2837, with 22 contacts, a little below that value; the least over all eight was 8.00067, at $\delta = 0.001$, and no feasible endpoint fell below 8 (Figure 27(c)). The search enters no proof. What a proof needs, for caps of unequal radii, is what the proof of Theorem 5.40 supplies for equal

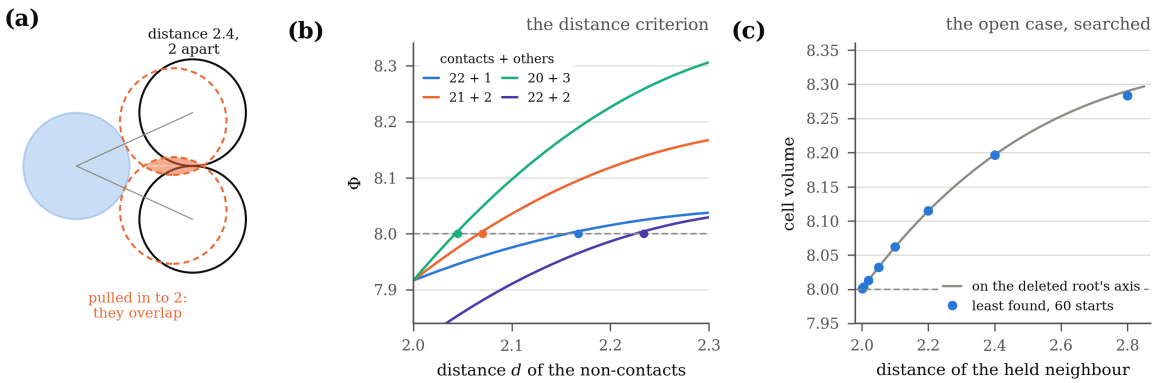


FIGURE 27. Neighbours that do not touch. (a) A two-dimensional section, to scale: two centres at distance 2.4 from the central one and 2 from each other, a packing; pulled in to distance 2 they overlap (shaded), so the configuration at the all-contact corner is not a packing, and Lemma 2.9 gives a lower bound and not a contact configuration. (b) The distance criterion of Proposition 7.1 for m contacts and k further centres at a common distance d; it passes 8 at the thresholds of Table 4 (dots). (c) The remaining case is tight: the volume $\frac{25}{3} - \frac{1}{3}(1-\frac{\delta}{2})^4$ of the cell of the deletion of a root with one further centre at distance $2+\delta$ on the deleted root's axis, which tends to 8 as $\delta \to 0$, and the least volume found by local minimisation from 60 starts with one neighbour held at distance $2+\delta$. None is below 8.

ones, an exact account of the pairs, with the three-point method of [BV08] carried from the sphere to a shell; Section 7.1 does that for at most twenty-three centres within $\sqrt{6}$.

7.1. **Further reductions near the contact regime.** This subsection narrows Conjecture 1.3 in three directions. It removes every centre with at most twenty-three others within $\sqrt{6}$, at any distances; every centre whose contacts contain most of a root system; and every centre around which no other centre lies at a distance between $2+\varepsilon_0$ and $\sqrt{6}$, for $\varepsilon_0 = 4\cdot 10^{-26}$, which is a local form of the twenty-four-cell conjecture at the root system, for all packings, with an explicit neighbourhood. The tools are three elementary facts about packings that hold whatever the distances, and, for the first, the certificate of Theorem 5.72 carried from contacts to centres at any distance. Throughout, $c = 0$, and the half-space of a centre z is $H_z = \{x : \langle x, z\rangle \le |z|^2/2\}$. The identities in the proofs of Lemmas 7.3 and 7.6 and Proposition 7.4 and the combinatorics of Corollary 7.5 are checked twice, in exact arithmetic and in Lean's kernel (Section 11).

Lemma 7.3 (Holes)**.** *Let W be the set of contact directions of c, so that $2w$ is a centre for every $w \in W$. Every other centre z satisfies $|z| \ge 4\langle z/|z|, w\rangle$ for every $w \in W$, and so $|z| \ge 4g(W)$, where $g(W) = \min_{\theta\in S^3} \max_{w\in W}\langle \theta, w\rangle$ is the cosine of the covering radius of W on S^3.*

Proof. $4 \le |z-2w|^2 = |z|^2 - 4\langle z, w\rangle + 4$. ∎

For the root system $g = 1/\sqrt{2}$, and the lemma returns the bound $2\sqrt{2}$ of Lemma 7.2. A centre that does not touch c can come close to it only in a deep hole of its contacts: at distance $2+\delta$ only where the contact directions leave a cap of angular radius $\arccos(\frac{1}{2}+\frac{\delta}{4})$ empty (Figure 28(b)).

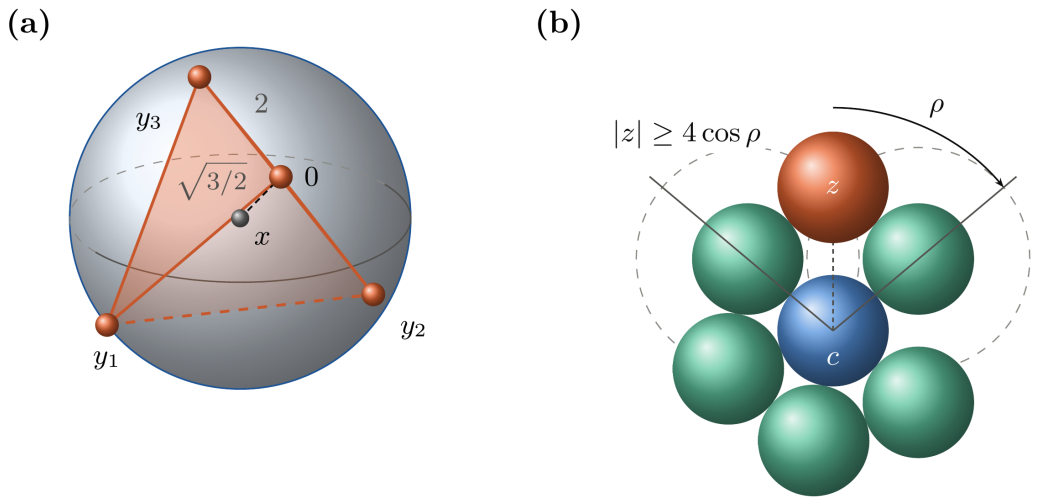


FIGURE 28. Two elementary facts behind Section 7.1, to scale. (a) Lemma 7.6: four points pairwise at least 2 apart do not fit in a ball of radius less than $\sqrt{3/2}$, and the regular tetrahedron of edge 2 in the ball of radius $\sqrt{3/2}$ is the extreme case; so no point x of the open ball $B(\sqrt{3/2})$ about the centre 0 is closer to three other centres than to 0. (b) Lemma 7.3 in a plane section: the balls of the contacts (green) leave a hole of angular radius ρ, here 50°, and a further centre z can enter it only at a distance $|z| \geq 4\cos\rho$ from c, where it meets the circles of radius 2 (dashed) about the two contacts at the rim.

Proposition 7.4 (The inversion hull)**.** *Let Y be any set of centres other than c, and let $K(Y)$ be the convex hull of 0 and the points $4y/|y|^2$, $y \in Y$, the images of the centres under inversion in the sphere of radius 2. Every centre $z \notin Y \cup \{0\}$ has $K(Y) \subseteq H_z$. Consequently*

$$V_c \;\supseteq\; V(Y) \cap K(Y), \qquad V(Y) = \bigcap_{y \in Y} H_y.$$

Proof. H_z is convex and contains 0, so it suffices to test the points $q = 4y/|y|^2$, and for these

$$\frac{|z|^2}{2} - \langle q, z \rangle = \frac{(|z|^2 - 4)(|y|^2 - 4) + 4\,(|y - z|^2 - 4)}{2|y|^2} \;\geq\; 0,$$

since centres are at least 2 apart and at least 2 from 0. ∎

The bound depends on Y alone: whatever the rest of the packing does, it does not cut into $V(Y) \cap K(Y)$ (Figure 29(b) shows the three-dimensional analogue). With Y the twenty-four neighbours of a centre near the root system it is sharp to first order.

Corollary 7.5 (Contacts that contain most of a root system)**.** *Let $U = \{\alpha/\sqrt{2} : \alpha \in \mathsf{R}'\}$ be the normalised roots of a copy R' of R, and $Q = \{x : \langle x, u \rangle \leq 1,\ u \in U\}$ its 24-cell. Every vertex of Q is the sum $u + u'$ of exactly three pairs of orthogonal elements of U. If the contact directions W of c contain, for every vertex of Q, one of its three pairs, then $Q \subseteq V_c$, so $\mathrm{vol}(V_c) \geq 8$, with equality only if $W = U$. The hypothesis holds whenever W contains all but at most two elements of U, and whenever it contains all but three, unless the three missing ones are pairwise at 60° with a common vertex (96 of the 2024 triples).*

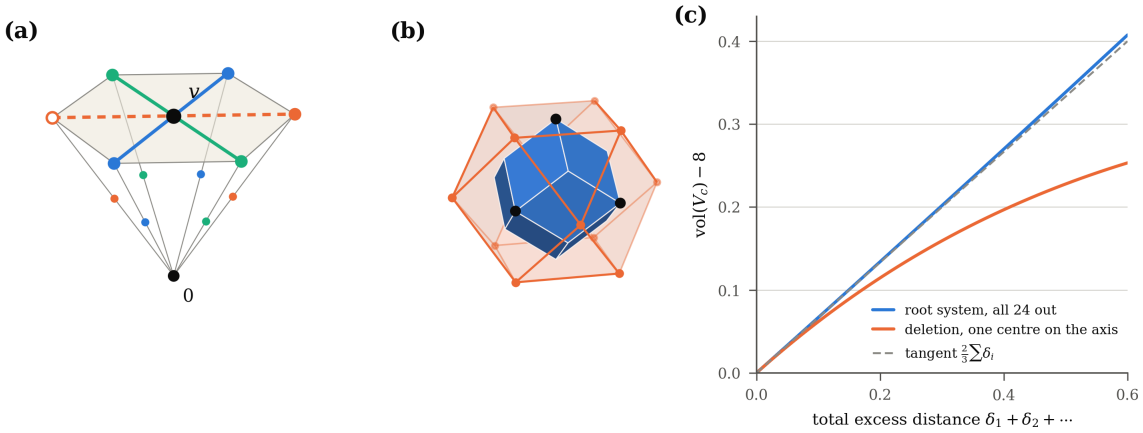


FIGURE 29. (a) Corollary 7.5: a vertex v of the 24-cell and its three orthogonal pairs of normalised roots, in the linear projection of $\mathbb{R}^4$ that keeps the vertex direction vertical and sends the other three coordinate directions to horizontal unit vectors at 120°. The six roots at v lie on a hexagon, the points $2u$ on a larger one, and v is the midpoint of the chord from $2u$ to $2u'$ of each pair, so it lies in the inversion hull of any contact set that keeps one pair. One root is missing (hollow), its chord is lost (dashed), and v keeps the other two. (b) Proposition 7.4 in three dimensions: the Voronoi cell of the twelve contacts of the face-centred cubic packing, a rhombic dodecahedron (blue), lies in the convex hull of the points $2u$ (orange) and touches its boundary only at its six four-valent vertices (black), as the 24-cell touches the hull of the points $2u$ at its 24 vertices; every other centre leaves that hull whole. (c) Theorem 7.9 to first order: the exact volume excess of the root system pushed out uniformly, $8(1+\delta/2)^4 - 8$ with $\sum \delta_i = 24\delta$, and of the deletion with one centre at $2+\delta$ on the deleted axis, $\frac{1}{3} - \frac{1}{3}(1-\frac{\delta}{2})^4$, against the total excess distance; both are tangent at the root system to $\frac{2}{3}\sum \delta_i$.

Proof. A vertex $v = u + u'$ with $u, u' \in W$ is the midpoint of the points $2u$ and $2u'$, which $K(2W)$ contains, so Q, the convex hull of its vertices, lies in $K(2W)$, and by Proposition 7.4 in the half-space of every centre that is not a contact. It lies in the half-space $\{\langle x, w\rangle \le 1\}$ of every contact too: for $w \in U$ because $v \in Q$, and for $w \notin U$ because $\langle v, w\rangle = \langle u, w\rangle + \langle u', w\rangle \le \frac{1}{2} + \frac{1}{2}$, the contact directions being pairwise at least 60° apart. So $Q \subseteq V_c$. If some $u \in U$ is not in W, then $\langle u, w\rangle < 1$ for every contact direction w, and a centre z with $|z| > 2$ has $\langle (1+t)u, z\rangle \le (1+t)|z| < |z|^2/2$ as soon as $2(1+t) < |z|$; only finitely many centres lie within 3 of c, so for small $t > 0$ a neighbourhood of $(1+t)u$ lies in V_c and outside Q, and $\mathrm{vol}(V_c) > 8$. If $W \supseteq U$, then $W = U$ by the kissing number and $V_c = Q$ by Lemma 7.2. The combinatorial statements are finite and are checked in integer arithmetic: the 72 orthogonal pairs, three at each of the 24 vertices; no set of one or two roots meets all three pairs at a vertex; exactly 96 sets of three do, the triples of pairwise 60° roots at a common vertex, which are also the triples whose deletion raises the circumradius of the cell to $\sqrt{6}$. The same statements are proved again in Lean's kernel. Figure 29(a) shows one vertex and its three pairs. ∎

Lemma 7.6 (No three centres cut the same point of $B(\sqrt{3/2})$)**.** *If $|x| < \sqrt{3/2}$, at most two centres y satisfy $|x-y| < |x|$. Hence, for $R \le \sqrt{3/2}$,*

$$\operatorname{vol}(V_c \cap B(R)) = \operatorname{vol} B(R) - \sum_y S_R(|y|) + \sum_{\{y,y'\}} \operatorname{vol}(B(R) \setminus (H_y \cup H_{y'})),$$

where $S_R(d)$ is the volume of the part of $B(R)$ beyond a hyperplane at distance $d/2$ from 0, and the sums run over the centres other than c, only those closer than $2R$ contributing.

Proof. If three centres y_1, y_2, y_3 did, the four points $0, y_1, y_2, y_3$, pairwise at least 2 apart, would lie in the closed ball of radius $|x|$ about x (Figure 28(a)). For four points p_i with centroid g, $\sum_{i<j} |p_i - p_j|^2 = 4\sum_i |p_i - g|^2 \le 4\sum_i |p_i - x|^2$, so $24 \le 16|x|^2$, against $|x| < \sqrt{3/2}$.[27] The formula is inclusion and exclusion, whose terms of order three vanish. ∎

For contact configurations this is the absence of triple overlaps on which Theorem 5.72 rests; the lemma says that it holds for every packing.

Corollary 7.7 (At most twenty-two centres within $\sqrt{6}$)**.** *Put $S = S_{\sqrt{3/2}}$ and $\Psi(d_1, d_2, \dots) = \frac{9}{8}\pi^2 - \sum_i S(d_i)$. For every centre, $\operatorname{vol}(V_c) \ge \Psi(d_1, d_2, \dots)$. In particular, if at most 22 centres other than c lie within $\sqrt{6}$ of c, then $\operatorname{vol}(V_c) > 8.046$.*

Proof. Take $R = \sqrt{3/2}$ in Lemma 7.6 and drop the pair terms, which are nonnegative. S decreases and vanishes for $d \ge \sqrt{6}$, so at most twenty-two centres within $\sqrt{6}$ give $\operatorname{vol}(V_c) \ge \frac{9}{8}\pi^2 - 22S(2) = 8.04610\dots$, in ball arithmetic, and again in interval arithmetic in Lean. ∎

Ψ is a second distance criterion beside Φ, with thresholds within 0.003 of those of Table 4. The count is taken within $\sqrt{6}$ rather than within $2\sqrt{2}$, so that any number of centres between $\sqrt{6}$ and $2\sqrt{2}$ is allowed.

Theorem 7.8 (At most twenty-three centres within $\sqrt{6}$)**.** *If at most 23 centres other than c lie within $\sqrt{6}$ of c, then $\operatorname{vol}(V_c) > 8$, whatever their distances from c and from one another.*

With all twenty-three at distance 2 this is Theorem 5.40; here they need not touch c. A centre that moves out from distance 2 to d returns to $B(\sqrt{3/2})$ the part $S(2) - S(d)$ of its cap, and it takes from each pair term it shares only the part of the moving section that the other cap covers. The proof lets the certificate of Theorem 5.72 use one eleventh of what is returned in each triple of centres, and shows that this covers everything the certificate loses when two directions come closer than 60°, which the packing allows only to centres that have moved out (Figures 30 and 31).

Proof. With at most 22 such centres this is Corollary 7.7. Let there be exactly 23, $y_1, \dots, y_{23}$, and write $d_i = |y_i| \in [2, \sqrt{6})$, $w_i = y_i/d_i$, $h_i = d_i/2$ and $u_{ij} = \langle w_i, w_j \rangle$. Put $R = \sqrt{3/2}$, let $S = S_R$ be as in Corollary 7.7, and let $s(d) = S(2) - S(d)$, which vanishes at $d = 2$ and increases. For $h, h' \in [1, R)$ and $-1 \le x < 1$ let $\Pi(h, h', x)$ be the volume of the part of $B(R)$ beyond two hyperplanes at distances h and h' from 0 whose unit normals have inner product x. The pair term of y_i and y_j in Lemma 7.6 is $\Pi(h_i, h_j, u_{ij})$, and $\Pi(1, 1, u) = \omega(u)$ is the weight of Section 5.12. With $A_* = \frac{9}{8}\pi^2 - 23S(2)$, Lemma 7.6 reads

$$\operatorname{vol}(V_c) \;\ge\; \operatorname{vol}(V_c \cap B(R)) \;=\; A_* + \sum_i s(d_i) + \sum_{i<j} \Pi(h_i, h_j, u_{ij}). \tag{7.1}$$

[27] The radius cannot be enlarged. Four points pairwise at distance 2, such as c and three contacts of D_4 at 60° from one another, are the vertices of a regular tetrahedron of edge 2, whose circumradius is $2\sqrt{6}/4 = \sqrt{3/2}$, and at its circumcentre x, at distance $\sqrt{3/2}$ from c, the bisecting hyperplanes of the three contacts meet. The two inequalities of the proof are then equalities: $g = x$, and every $|p_i - x|$ is $\sqrt{3/2}$.

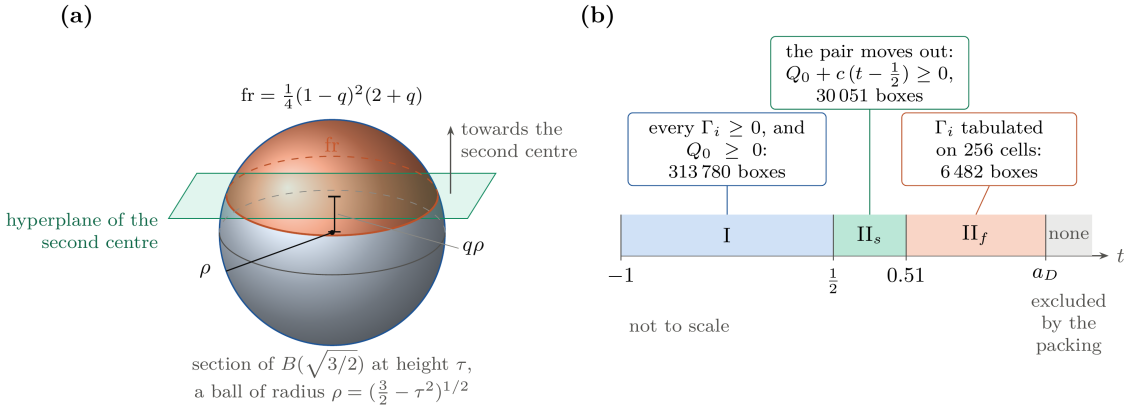


FIGURE 30. The proof of Theorem 7.8. (a) The share fr: the section of $B(\sqrt{3/2})$ by the hyperplane of one centre is a ball of dimension three, and the hyperplane of a second centre cuts from it a cap; moving the first hyperplane out removes from the pair term exactly this cap, and from the cap term of the centre the whole section. (b) The three ranges of the largest inner product t of three directions, with what settles each; beyond a_D the packing leaves no room.

Step 1: a centre far out. Let $D = 2.1648$. In ball arithmetic $s(D) = 0.0928820\ldots$, which exceeds $8 - A_* = 0.0928555\ldots$; so if some $d_i \geq D$, the right side of (7.1) exceeds 8. From now on every $d_i < D$. The packing condition $|y_i - y_j| \geq 2$ reads

$$u_{ij} \;\leq\; a(d_i, d_j) := \frac{d_i^2 + d_j^2 - 4}{2 d_i d_j} \;\leq\; a(D, D) = 0.573229\ldots =: a_D, \tag{7.2}$$

and a increases in each argument on $[2, D]^2$.

Step 2: a certificate for centres at mixed distances. Let P and B be the polynomial and the bound of Theorem 5.72, so that (5.48) reads $\omega(u) + \omega(v) + \omega(t) \geq P(u, v, t)$, and for three of the centres put

$$Q_{ijk} = \Pi(h_i, h_j, u_{ij}) + \Pi(h_i, h_k, u_{ik}) + \Pi(h_j, h_k, u_{jk}) \\ + \tfrac{1}{11}\big(s(d_i) + s(d_j) + s(d_k)\big) - P(u_{jk}, u_{ik}, u_{ij}).$$

Each pair of the 23 centres lies in 21 triples and each centre in $231 = 21 \cdot 11$,[28] so the sum of the Q_{ijk} over all triples is $21\big(\sum_i s(d_i) + \sum_{i<j} \Pi(h_i, h_j, u_{ij})\big) - \sum_{i<j<k} P(u_{jk}, u_{ik}, u_{ij})$. The proof of Lemma 5.71 uses of the directions only that they are $N = 23$ unit vectors: Schoenberg's theorem for the terms in f and Lemma 4.3 for the terms in F give $\sum_{i<j<k} P(u_{jk}, u_{ik}, u_{ij}) \geq (N-2)B = 21B$ for any such vectors. Hence, if every $Q_{ijk} \geq 0$, then $\sum_i s(d_i) + \sum_{i<j} \Pi(h_i, h_j, u_{ij}) \geq B$, and (7.1) gives $\mathrm{vol}(V_c) \geq A_* + B > 8$.

Step 3: three variables. Q_{ijk} is symmetric in the three centres, because P is; number them $1, 2, 3$ so that $t = u_{12} \geq v = u_{13} \geq u = u_{23}$. For $1 \leq \tau < R$ the section $\{x \in B(R) : \langle x, w\rangle = \tau\}$ is a ball of dimension three and radius $\rho = (R^2 - \tau^2)^{1/2}$, of volume $A(\tau) = \frac{4\pi}{3}\rho^3$. A second hyperplane $\{x : \langle x, w'\rangle = h'\}$ with $\langle w, w'\rangle = x$ meets it in a plane at signed

[28] This fixes the weight $\frac{1}{11}$: it is the one with which each centre's term $s(d_i)$ enters the sum over triples with the same factor, 21, as each pair term.

distance $q\rho$ from its centre, $q = (h' - \tau x)/((1-x^2)^{1/2}\rho)$, so the share of the section beyond the second hyperplane is

$$\mathrm{fr}(\tau, h', x) = \tfrac{1}{4}(1-q)^2(2+q) \quad \text{if } q < 1, \qquad \mathrm{fr}(\tau, h', x) = 0 \quad \text{if } q \geq 1,$$

the volume $\frac{\pi}{3}\eta^2(3\rho - \eta)$ of a cap of height $\eta = (1-q)\rho$ divided by that of the ball (Figure 30(a)). By Fubini's theorem $\partial_h \Pi(h, h', x) = -A(h)\,\mathrm{fr}(h, h', x)$, and $\frac{d}{dh}s(2h) = A(h)$. On the range that occurs, $h, h' \in [1, D/2]$ and $x \leq a_D$, the share decreases in h' and in τ and increases in x: q increases in h'; its derivative in x has the sign of $h'x - \tau < 0$; and $\frac{d}{d\tau}\log q = -\frac{x}{h'-\tau x} + \frac{\tau}{R^2-\tau^2} > 0$, since $\frac{x}{1-\tau x} \leq \frac{a_D}{1-(D/2)a_D} = 1.5103\ldots < 2 \leq \frac{\tau}{R^2-\tau^2}$. Raising first h_i and then h_j from 1, and bounding each share by its value with the other height at 1, gives $\Pi(h_i, h_j, x) \geq \omega(x) - \varphi(h_i, x) - \varphi(h_j, x)$ with $\varphi(h, x) = \int_1^h A(\tau)\,\mathrm{fr}(\tau, 1, x)\,d\tau$. As $s(d_i) = \int_1^{h_i} A$, it follows that

$$Q_{123} \;\geq\; Q_0(u, v, t) + \Gamma_1(h_1) + \Gamma_2(h_2) + \Gamma_3(h_3), \tag{7.3}$$

where $Q_0 = \omega(u) + \omega(v) + \omega(t) - P(u, v, t)$ is the quantity that (5.48) asserts to be nonnegative,

$$\Gamma_1(h) = \int_1^h A(\tau)\Big(\frac{1}{11} - \mathrm{fr}(\tau, 1, t) - \mathrm{fr}(\tau, 1, v)\Big)d\tau,$$

and Γ_2 and Γ_3 are the same with (t, u) and (v, u), the inner products at the centre concerned, in place of (t, v). It remains to show that the right side of (7.3) is nonnegative for all heights that (7.2) allows. By (7.2), $t \leq a_D$, and there are three ranges of t (Figure 30(b)).

(I) $t \leq \frac{1}{2}$. Every share is at most $\mathrm{fr}(1, 1, \frac{1}{2}) = 0.023710\ldots < \frac{1}{22}$, so every $\Gamma_i \geq 0$, and $Q_0 \geq 0$ is (5.48), which Theorem 5.72 proves.

(II_s) $\frac{1}{2} < t \leq 0.51$. The integrands are now at least $r = \frac{1}{11} - 2\,\mathrm{fr}(1, 1, 0.51) \geq 0.037900$, so $\Gamma_1 + \Gamma_2 + \Gamma_3 \geq r\,(s(d_1) + s(d_2))$. The identity

$$\tfrac{1}{2} + \tfrac{1}{4}(d_1 + d_2 - 4) - a(d_1, d_2) = \frac{(d_1 - 2)(d_2 - 2)(d_1 + d_2 + 2)}{4 d_1 d_2}$$

turns (7.2) for the pair y_1, y_2 into $(d_1 - 2) + (d_2 - 2) \geq 4(t - \frac{1}{2})$. The function $x \mapsto s(2 + x)$ vanishes at 0 and is concave, its derivative $\frac{1}{2}A(1 + \frac{x}{2})$ decreasing; so it is subadditive and lies above its chords, and $s(d_1) + s(d_2) \geq s(2 + 4(t - \frac{1}{2})) \geq 4\kappa(t - \frac{1}{2})$ with $\kappa = s(2.04)/0.04 \geq 0.696362$. Hence $Q_{123} \geq Q_0(u, v, t) + c\,(t - \frac{1}{2})$ with $c = 4r\kappa \geq 0.105569$. The branch and bound of the proof of Theorem 5.72, with this linear term added to the Taylor form and ω tabulated up to $u = a_D$, shows the right side nonnegative on the slab: it verifies 30 051 boxes and discards 3 408 outside the domain.[29]

(II_f) $0.51 \leq t \leq a_D$. The branch and bound bounds Q_0 on each box as before, and $\Gamma_1 + \Gamma_2 + \Gamma_3$ from below by a table: each Γ_i is integrated in ball arithmetic over 256 cells of $h \in [1, D/2]$, with the share taken at the lower end of the cell and at the upper ends of the box's inner products; it is replaced by its right-hand minimum $\min_{h' \geq h} \Gamma_i(h')$, which is nondecreasing in h, and then minimised over the triples of cells whose upper ends satisfy (7.2) at the lower ends of the box's inner products. It verifies 6 482 boxes and discards 1 242 outside the domain.

Thus every $Q_{ijk} \geq 0$, and Step 2 concludes. The constants of Steps 1 and 3, the two branch and bounds and a rerun of the verification of (5.48) are computed in ball and interval arithmetic; the identity for a is also proved in Lean, and the two branch and bounds are

[29] The certificate alone does not reach past the face $t = \frac{1}{2}$. At $u = v = -\frac{\sqrt{3}}{2}$, where (5.48) comes nearest to equality on that face, Q_0 falls by about 0.080 per unit of t beyond it (Figure 31(b)). Two directions closer than 60° belong to centres that have moved out, and that gains at least $c = 0.1056$ per unit of t.

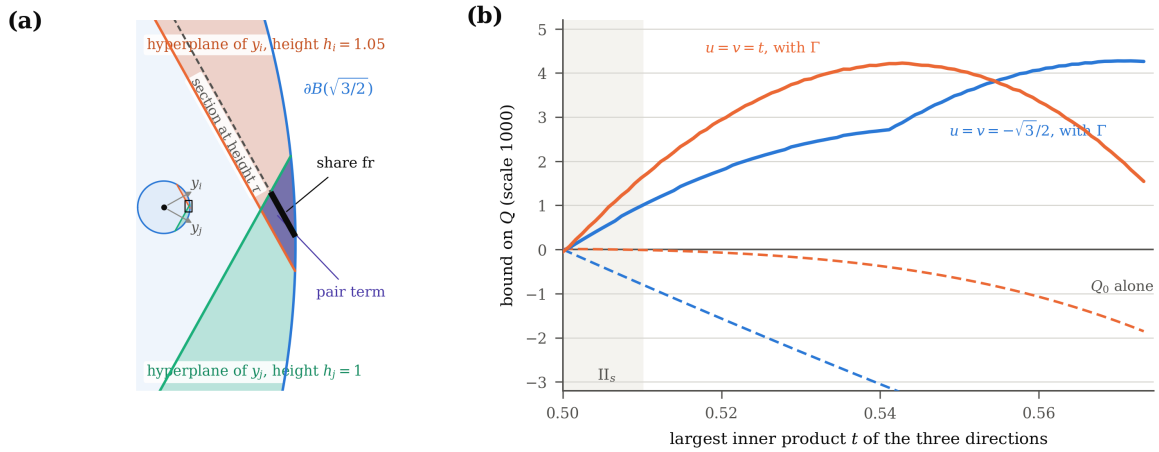


FIGURE 31. (a) A plane section through c and two centres at the packing boundary $|y_i - y_j| = 2$, with $d_i = 2.10$ and $d_j = 2$, so that the cosine $a(d_i, d_j) = 0.525$ of their angle exceeds $\frac{1}{2}$, which two contacts cannot do; the inset shows the whole section and the window. Beyond the two hyperplanes lie the caps (orange, green) and their overlap, the pair term (violet). The section of $B(\sqrt{3/2})$ at height τ (dashed) meets the second cap in the share fr (thick). To scale. (b) The bound along two lines past the face $t = \frac{1}{2}$ where contacts end: Q_0 alone (dashed) turns negative, while Q_0 plus the lower bound for $\Gamma_1 + \Gamma_2 + \Gamma_3$ that the proof uses (solid) stays positive. Blue: $u = v = -\frac{\sqrt{3}}{2}$, where (5.48) comes nearest to equality on the face; orange: $u = v = t$. The shaded strip is the range II_s. The scale is that of the computation, 1000 times the volume.

repeated in Lean in exact dyadic arithmetic, over 39 399 and 9 545 boxes, with the shares, the packing bounds and the envelopes of the Γ_i computed there and only the tables of ω, the constant c and the enclosures of the cell integrals taken from ball arithmetic, and then a second time, over 39 551 and 9 545 boxes, with those computed in Lean as well, together with $s(D) > 8 - A_*$, $a_D/(1 - \frac{D}{2}a_D) < 2$ and $\mathrm{fr}(1, 1, \frac{1}{2}) < \frac{1}{22}$ (Section 11). ■

A centre with 23 contacts may have further centres near it. If the contact directions have covering radius at most $\arccos(\sqrt{6}/4) = 52.2388°$, every other centre is at distance at least $4\cos 52.2388° = \sqrt{6}$ by Lemma 7.3, and Theorem 7.8 applies; a deletion of a root is Corollary 7.5. A contact configuration of 23 directions whose cell has circumradius at least 2 is a deletion (Proposition 5.38 and Theorem 5.25). Therefore a centre with 23 contacts and a cell of volume at most 8 would need a saturated contact configuration, with a hole of angular radius between 52.24° and 60° holding a twenty-fourth centre closer than $\sqrt{6}$.

Why the count stops at twenty-three. At the root system twenty-four centres lie within $\sqrt{6}$, and the right side of Lemma 7.6 is $\frac{9}{8}\pi^2 - 24S(2) + 96\,\omega(\frac{1}{2}) = 7.906940\ldots$, the volume of the 24-cell inside $B(\sqrt{3/2})$: the cell reaches out to radius $\sqrt{2}$, and the ball misses 0.093 of it (Figure 32).[30] Hence with twenty-four or more centres within $\sqrt{6}$ no lower bound for

[30] The 96 is the number of pairs of roots at 60°, eight at each of the 24 roots; pairs at 90° or more have disjoint caps in $B(\sqrt{3/2})$, since the caps of two contacts overlap only when their angle is below $2\arctan(1/\sqrt{2}) = 70.53°$. The missing volume sits at the 24 vertices of the 24-cell, the points $r + r'$ at radius

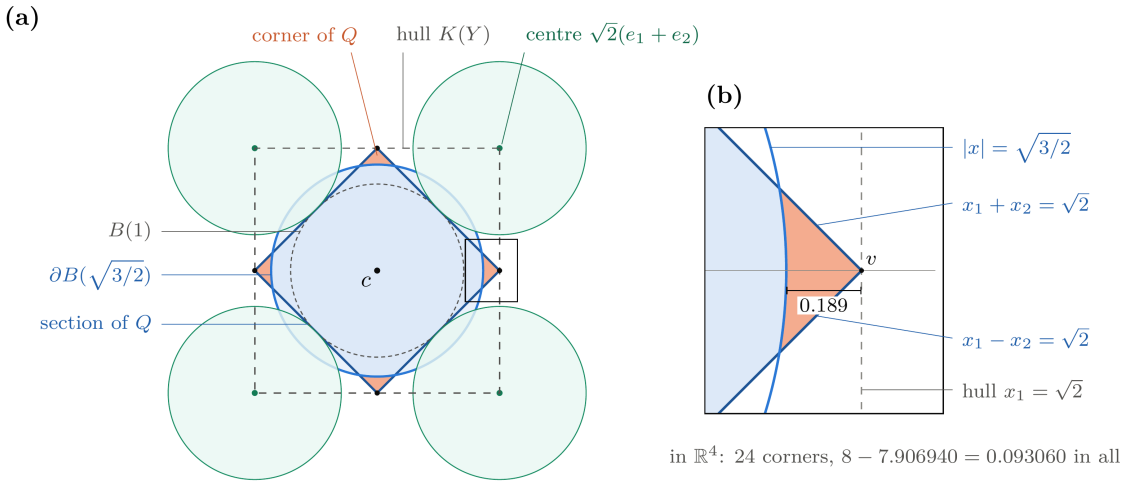


FIGURE 32. Why the pair terms stop at twenty-three centres, to scale. (a) The section of the 24-cell Q by the plane $x_3 = x_4 = 0$: the square $|x_1|+|x_2| \le \sqrt{2}$, whose sides touch the unit ball and whose corners are vertices of Q at radius $\sqrt{2}$. The ball $B(\sqrt{3/2})$, inside which Lemma 7.6 makes the cell an exact function of the pairs, cuts the corners off (orange). The four centres of the plane, $\sqrt{2}(\pm e_1 \pm e_2)$ at distance 2, with their unit balls, and their inversion hull (dashed), which meets Q only at its vertices. (b) One corner, magnified: the vertex $v = \sqrt{2}\,e_1$, the two sides through it and the side $x_1 = \sqrt{2}$ of the hull; the corner is $\sqrt{2} - \sqrt{3/2} = 0.189$ deep. In $\mathbb{R}^4$ the 24 corners hold $8 - 7.906940 = 0.093060$ of the volume of Q.

$\mathrm{vol}(V_c \cap B(\sqrt{3/2}))$ reaches 8, and a proof has to look beyond that ball. Near the root system the next theorem does, through the inversion hull.

Theorem 7.9 (The conjecture near the contact regime)**.** *Let $\varepsilon_0 = 4 \cdot 10^{-26}$. If no centre of a unit-ball packing of $\mathbb{R}^4$ lies at a distance from c strictly between $2 + \varepsilon_0$ and $\sqrt{6}$, then $\mathrm{vol}(V_c) \ge 8$, with equality only if the contacts of c form a copy of $\sqrt{2}\,\mathsf{R}$ and V_c is the regular 24-cell they cut out. The same conclusion holds whenever exactly 24 centres lie within $\sqrt{6}$ of c, at distances $2 + \delta_i$ with $\sum_i \delta_i \le 4 \cdot 10^{-3}$, and their directions W satisfy $d(W) \le \frac{1}{48}$, with d as in Theorem 5.51.*

In particular the D_4 configuration is a strict local minimum of the cell volume among all packings, in an explicit neighbourhood: every packing whose twenty-four centres within $\sqrt{6}$ of c are that close to $\sqrt{2}\,\mathsf{R}'$, for some copy R' of R, and whose other centres are unrestricted, has $\mathrm{vol}(V_c) \ge 8$, with equality only at $\sqrt{2}\,\mathsf{R}'$ itself, and to first order the excess is $\frac{2}{3}\sum \delta_i$ (Figure 29(c)). The two constants differ greatly in size. The neighbourhood of the second statement comes from the rigidity of Theorem 5.51 and from elementary estimates for the facets of a polytope near the 24-cell, and uses no certificate. The constant ε_0 is chosen so that Theorem 5.75 places twenty-four directions with inner products at most $\frac{1}{2} + \frac{1}{2}\varepsilon_0$ inside that neighbourhood, and it inherits the size of the sums of squares of the certificate (Figure 22).

$\sqrt{2}$, in 24 congruent corners of volume 0.003878 each; in the plane of the two orthogonal vertices $\sqrt{2}\,e_1$ and $\sqrt{2}\,e_2$ each corner is a right-angled triangle of depth $\sqrt{2} - \sqrt{3/2} = 0.189$ with its tip cut off by an arc.

Proof. Step 1: the first statement follows from the second. Let Y be the set of centres within $\sqrt{6}$ of c; by hypothesis every one of them is at a distance $d_i = 2+\delta_i$ with $0 \le \delta_i \le \varepsilon_0$. If $|Y| \le 23$, Theorem 7.8 gives $\operatorname{vol}(V_c) > 8$. Otherwise the packing condition gives $\langle w_i, w_j\rangle \le (d_i^2+d_j^2-4)/(2d_id_j)$ for the directions, and the right side, which increases in d_i and in d_j, is at most $1-2/(2+\varepsilon_0)^2 \le \frac12+\frac12\varepsilon_0$. By Theorem 5.75, with $\kappa = \frac12\varepsilon_0$, $|Y| = 24$ and $d(W) \le 0.0113 < \frac{1}{48}$; and $\sum_i \delta_i \le 24\varepsilon_0$. Thus the second statement applies.

Step 2: the second statement. Rotate and relabel so that $d(W)$ is attained at $R = I$ and the identity labelling, write u_i for the normalised roots and $y_i = d_iw_i$ for the partner of $2u_i$, put $\varepsilon_i = w_i - u_i$, $e_i = |\varepsilon_i|$, $\|\varepsilon\|^2 = \sum_i e_i^2 = d(W)^2$, $S = \sum_i \delta_i$, $e = \max_i e_i$, $\delta = \max_i \delta_i$ and $\Theta = e+\delta$.

(a) *Rigidity with slack:* $\|\varepsilon\| \le 2.38\,(1+\delta)S$ *for* $S \le 4\cdot 10^{-3}$. The proof of Theorem 5.51 goes through with the contact condition $\langle w_i, w_j\rangle \le \frac12$ on the 96 pairs of T replaced by the packing condition $|y_i - y_j| \ge 2$, which reads $\langle w_i, w_j\rangle \le \frac12 + \kappa_{ij}$ with $\kappa_{ij} = (d_i^2+d_j^2-d_id_j-4)/(2d_id_j) \le \frac14(\delta_i+\delta_j)(1+\delta)$. Therefore $g = \Lambda\tau$ satisfies $g_p \le a_p := \kappa_{ij} + \frac14(e_i^2+e_j^2) + e_ie_j$ on every pair $p = \{i,j\}$ of T, and the constant stress still makes the entries of g sum to 0. Hence the positive parts of g are at most a entrywise, the negative parts sum to at most $A = \sum_T a_p$, and by (5.36)

$$\|g\|_2^2 \;\le\; \|a\|_2^2 + \|g\|_\infty A \;\le\; \|a\|_2^2 + \tfrac{\sqrt{11}}{4}\|g\|_2 A, \qquad \|g\|_2 \;\le\; \tfrac{\sqrt{11}}{8}A + \bigl(\tfrac{11}{64}A^2 + \|a\|_2^2\bigr)^{1/2}.$$

Every index lies in eight pairs of T, so $A \le 2(1+\delta)S + 6\|\varepsilon\|^2$ and $\|a\|_2 \le (1+\delta)\|\delta\|_2 + 6\|\varepsilon\|^2 \le (1+\delta)S + 6\|\varepsilon\|^2$. With $\|\tau\|_2 \le \|g\|_2$ by (5.35), and $\|\varepsilon\| \le \|\tau\|_2 + \frac12\|\varepsilon\|^2$, this gives $\|\varepsilon\| \le 2.129(1+\delta)S + 9.49\|\varepsilon\|^2$. For $\|\varepsilon\| \le 1/48$ the last term is at most $0.198\|\varepsilon\|$, so $\|\varepsilon\| \le 2.66(1+\delta)S$; fed back once, $\|\varepsilon\| \le 2.129(1+\delta)S/(1-9.49\cdot 2.66(1+\delta)S)$, which is at most $2.38(1+\delta)S$ for $S \le 4\cdot 10^{-3}$. Consequently $\|\varepsilon\| \le 2.39\,S$ and $\Theta \le \|\varepsilon\| + S \le 3.39\,S \le 1.4\cdot 10^{-2}$. The argument of Theorem 5.51 passed from $\|g\|_1$ to $\|g\|_2$ directly; bounding the positive and the negative parts separately halves the constant.

(b) *The volume to second order, facet by facet.* Move the data linearly, $w_i(t) = u_i + t\varepsilon_i$ and $h_i(t) = 1 + t\delta_i/2$ for $0 \le t \le 1$, and let $P(t) = \{x : \langle x, w_i(t)\rangle \le h_i(t)\}$, so that $P(0) = Q$ and $P(1) = V(Y)$. Each facet $F_i(t)$ of $P(t)$ moves with normal velocity $(\dot h_i - \langle x, \dot w_i\rangle)/|w_i|$, so the derivative of $\operatorname{vol} P(t)$ is $\sum_i \int_{F_i(t)} (\frac12\delta_i - \langle x, \varepsilon_i\rangle)\, dA/|w_i(t)|$. Parametrise $F_i(t)$ by its projection along u_i onto the hyperplane of F_i^0: $x = u_i + z + s(z)u_i$ with $z \perp u_i$ and $s(z) = q_i(\frac12\delta_i + \frac12 e_i^2 - \langle z, \varepsilon_i\rangle)$, $q_i = t/(1-\frac12 te_i^2)$, as $\langle x, w_i(t)\rangle = h_i(t)$ requires. Let $G_i(t) \subset u_i^\perp$ be the image and $m_i(t) = \int_{G_i(t)} z\, dz$. The area element of the hyperplane is $|w_i(t)|/(1-\frac12 te_i^2)$ times dz, and, as $\langle u_i, \varepsilon_i\rangle = -\frac12 e_i^2$, $\frac12\delta_i - \langle x, \varepsilon_i\rangle = (\frac12\delta_i + \frac12 e_i^2 - \langle z, \varepsilon_i\rangle)/(1-\frac12 te_i^2)$; so, exactly,

$$\frac{d}{dt}\operatorname{vol} P(t) = \sum_i \frac{(\frac12\delta_i + \frac12 e_i^2)\operatorname{vol} G_i(t) - \langle \varepsilon_i, m_i(t)\rangle}{(1-\frac12 te_i^2)^2}. \tag{7.4}$$

At $t = 0$, G_i is the regular octahedron O_i with vertices $\pm u_k$ at the six roots orthogonal to u_i: its volume is $\frac43$, its circumradius 1, $m_i(0) = 0$, and its faces lie on the planes $\langle z, u_j\rangle = \frac12$ of the eight neighbours $j \sim i$, those with $\langle u_i, u_j\rangle = \frac12$, at distance $1/\sqrt{3}$ from 0. Its sections parallel to a face at distance $c \le 1/\sqrt{3}$ from 0 have area $\frac{3\sqrt3}{4}(1-c^2)$, and so those of λO_i have area $\frac{3\sqrt3}{4}(\lambda^2 - c^2)$.

For $t > 0$, $G_i(t)$ is cut out by the affine functions $\phi_j(z) = \langle x, w_j(t)\rangle - h_j(t)$, $j \ne i$. A first homothety locates it. $P(t)$ lies in the ball of radius $r_1 = \sqrt2(1+\frac12\delta)/(1-\sqrt2 e)$, because $\langle x, u_j\rangle \le h_j + |x|e$ and $\max_j\langle x, u_j\rangle \ge |x|/\sqrt2$; so a point x of $F_i(t)$ has $|\langle x, u_i\rangle - 1| \le \eta_1 = \frac12\delta + r_1 e$, and then $\langle z, u_j\rangle \le \frac12 + \eta_2$ with $\eta_2 = \eta_1 + \frac12\delta + r_1 e$, which puts $G_i(t)$ in RO_i,

$R = 1 + 2\eta_2$. For a neighbour j, with $x_0 = u_i + z$ and $\langle u_j, \varepsilon_j\rangle = -\frac{1}{2}e_j^2$,

$$\phi_j(z) = \langle z, u_j\rangle - \tfrac{1}{2} + \Delta_j(z), \qquad \Delta_j(z) = t\langle x_0 - u_j, \varepsilon_j\rangle - \tfrac{1}{2}t(e_j^2 + \delta_j) + (\tfrac{1}{2} + t\langle u_i, \varepsilon_j\rangle)\, s(z),$$

and $|x_0 - u_j|^2 = \frac{1}{4} + |z - u_j + \frac{1}{2}u_i|^2$ is at most R^2 on the part of RO_i beyond the plane of face j and at most $1 + \sqrt{3}D$ at depth D beneath it in O_i. So, with

$$D_{ij} = Re_j + \tfrac{1}{2}e_j^2 + \tfrac{1}{2}\delta_j + \frac{(\frac{1}{2} + e_j)(\frac{1}{2}\delta_i + \frac{1}{2}e_i^2 + Re_i)}{1 - \frac{1}{2}e_i^2}, \qquad \theta_{ij} = \frac{2D_{ij}}{\sqrt{3}\,(1 - e_j)},$$

a point of $G_i(t)$ outside O_i lies within $t\theta_{ij}$ beyond the plane of some face j, and a point of O_i outside $G_i(t)$ lies within $t\theta_{ij}$ beneath the plane of some face j or in the cap $\{\langle z, u_k\rangle \geq 1 - tD_\perp\}$ of O_i at a vertex u_k, of volume $\frac{2}{3}(tD_\perp)^3$, where $D_\perp = \sqrt{5}e + \frac{1}{2}e^2 + \frac{1}{2}\delta + e(\frac{1}{2}\delta + \frac{1}{2}e^2 + e)/(1 - \frac{1}{2}e^2)$ bounds $|\Delta_k|/t$ on O_i for the orthogonal roots k in the same way; the roots at inner product $-\frac{1}{2}$ and -1 with u_i never cut RO_i. In particular $G_i(t) \subseteq \lambda O_i$ with $\lambda = 1 + \sqrt{3}\max\theta_{ij}$. Put $\theta_i = \sum_{j\sim i}\theta_{ij}$, $a_{\text{in}} = \frac{3\sqrt{3}}{4}(1 - (1/\sqrt{3} - \max\theta_{ij})^2)$ and $a_{\text{out}} = \frac{3\sqrt{3}}{4}(\lambda^2 - \frac{1}{3})$, the largest section areas of the slabs just found. Since $\int_{O_i} z\,dz = 0$, the moment is at most $\int_{G_i\setminus O_i}|z| + \int_{O_i\setminus G_i}|z|$, and

$$\operatorname{vol} G_i(t) \;\geq\; \tfrac{4}{3} - a_{\text{in}}\, t\theta_i - 4(tD_\perp)^3, \qquad |m_i(t)| \;\leq\; (a_{\text{in}} + R\, a_{\text{out}})\, t\theta_i + 4(tD_\perp)^3.$$

The denominators in (7.4) lie between 1 and $\bar\gamma = (1 - \frac{1}{2}e^2)^{-2}$, so integrating over t,

$$\begin{aligned}\operatorname{vol} V(Y) - 8 \;\geq\; \tfrac{2}{3}(S + \|\varepsilon\|^2) - \tfrac{1}{2}a_{\text{in}}\sum_i(\tfrac{1}{2}\delta_i + \tfrac{1}{2}e_i^2)\theta_i - \tfrac{1}{2}\bar\gamma\,(a_{\text{in}} + Ra_{\text{out}})\sum_i e_i\theta_i \\ - (\tfrac{1}{2}S + \tfrac{1}{2}\|\varepsilon\|^2 + \bar\gamma\sqrt{24}\,\|\varepsilon\|)D_\perp^3.\end{aligned}$$

Each facet is charged only with the perturbations of its own eight neighbours, and the neighbour relation is the edge graph of the 24-cell, regular of degree 8; so $\sum_i\sum_{j\sim i} e_ie_j \leq 8\|\varepsilon\|^2$ and $\sum_i\sum_{j\sim i} e_i\delta_j \leq 8\|\varepsilon\|\,\|\delta\|_2$, and with $D_{ij} \leq R'e_j + \frac{1}{2}\delta_j + c_3(\frac{1}{2}\delta_i + R'e_i)$, $R' = R + \frac{1}{2}e$, $c_3 = (\frac{1}{2} + e)/(1 - \frac{1}{2}e^2)$,

$$\sum_i e_i\theta_i \leq k_1(8(1+c_3)R'\|\varepsilon\|^2 + 4(1+c_3)\|\varepsilon\|S), \quad \sum_i \delta_i\theta_i \leq k_1(8(1+c_3)R'S\|\varepsilon\| + 4(1+c_3)S^2),$$

$k_1 = 2/(\sqrt{3}(1 - e))$, using $\|\delta\|_2 \leq S$. Every loss is thus of second order in S, through $\|\varepsilon\| \leq 2.39\,S$, with no factor of the size of the largest perturbation.

(c) *The hull costs only a fourth-order amount.* For $x \in V(Y)$, $\langle x, u_i\rangle \leq h_i + |x|e_i$, so $V(Y) \subseteq (1 + \theta)Q$ with $\theta = (\frac{1}{2}\delta + \sqrt{2}e)/(1 - \sqrt{2}e) \leq 1.5\,\Theta$. The points $4w_i/d_i$ spanning $K(Y)$ lie within $\delta_i + 2e_i \leq 2\Theta$ of $2u_i$, and a convex hull of points within r of the vertices of a polytope of inradius $\sqrt{2}$ about 0 contains its homothet by $1 - r/\sqrt{2}$, so $K(Y) \supseteq (1 - \theta')\operatorname{conv}\{2u_i\}$ with $\theta' \leq \sqrt{2}\,\Theta$. The polytope $\operatorname{conv}\{2u_i\}$ is the 24-cell $\{x : \langle x, v\rangle \leq 2$ for the 24 vertices v of $Q\}$, and Q lies in it and meets its boundary only at those vertices. Hence $(1 + \theta)Q \setminus (1 - \theta')\operatorname{conv}\{2u_i\}$ lies in the union of 24 caps, one at each vertex, cut off by a hyperplane at depth $s = \sqrt{2}(\theta + \theta')$ below the vertex $(1 + \theta)v$. The tangent cone of Q at $v = \sqrt{2}e_1$ is $\{|x_k| \leq \sqrt{2} - x_1,\ k = 2, 3, 4\}$, whose section at depth s is a cube of side $2s$, so each cap lies in a pyramid of volume $2s^4$, and $\operatorname{vol}(V(Y) \setminus K(Y)) \leq 48s^4 \leq 13\,958\,\Theta^4$.

By Proposition 7.4, (b), (c) and (a), $\operatorname{vol}(V_c) \geq \operatorname{vol}(V(Y) \cap K(Y)) \geq 8 + S\cdot\beta(S)$, where $\beta(S)$ is $\frac{2}{3}$ less the losses of (b) and the hull term $13\,958\,\Theta^4$ of (c), all divided by S and evaluated at the bound of (a) for $\|\varepsilon\|$, $e = \|\varepsilon\|$, $\delta = S$ and $\Theta = e + \delta$. With these constants frozen, the losses are $\alpha\|\varepsilon\|^2 + \beta'\|\varepsilon\| + \gamma'$ with $\alpha > \frac{2}{3}$, so the largest $\|\varepsilon\|$ is the worst case; and every part of $\beta(S)$ other than $\frac{2}{3}$ is increasing in S. We evaluate β in exact rational

arithmetic, and again in Lean, and find $\beta(4 \cdot 10^{-3}) \geq 0.1$: at that point the moment term costs $0.39\,S$, the volume term $0.04\,S$, the caps $2 \cdot 10^{-4} S$ and the hull $0.12\,S$ (and β stays positive up to $S = 4.5 \cdot 10^{-3}$). Thus $\mathrm{vol}(V_c) > 8$ unless $S = 0$. If $S = 0$, then $\kappa = 0$, so $\varepsilon = 0$ by (a), $Y = \sqrt{2}\,\mathsf{R}$, and $V_c \supseteq V(Y) \cap K(Y) = Q$ with $\mathrm{vol}(V_c) = 8$ only if $V_c = Q$. ■

7.2. **What remains near the contact regime.** Together with Proposition 7.1, the results above confine Conjecture 1.3 to a centre c with all of the following: at least 24 other centres within $\sqrt{6}$; at least one of them at a distance between $2 + \varepsilon_0$ and $\sqrt{6}$; $\Phi \leq 8$ and $\Psi \leq 8$; and contacts that do not contain one orthogonal pair of roots at each vertex of a 24-cell. By Proposition 7.23 the bound at such a centre, and with it Conjecture 1.3, follows from two statements: the volume bound (G) for exactly twenty-four centres within $\sqrt{6}$, and the bound (C) on the pair terms for twenty-five or more. Four statements would give both.

(i) *The shape of twenty-four directions at positive slack.* Any 24 unit vectors of $\mathbb{R}^4$ with pairwise inner products at most $\frac{1}{2} + s$, for s of order 10^{-2}, satisfy $d(W) \leq \frac{1}{48}$. This is proved for $s \leq 2 \cdot 10^{-26}$ (Theorem 5.75), and its cardinality half for $s \leq 0.008$ (Theorem 5.78). It is false from $s = 0.01698$ on (Proposition 7.21), well before $s = 0.0374$, where 25-point codes exist.

(ii) *The volume between the explicit neighbourhood and the pair terms.* For exactly 24 centres within $\sqrt{6}$ with $d(W) \leq \frac{1}{48}$, $\mathrm{vol}(V_c) \geq 8$ for $\sum \delta_i$ from $4 \cdot 10^{-3}$, where Theorem 7.9 ends, up to about 0.155, where the right side of Lemma 7.6 reaches 8. It consists of two inequalities: $\mathrm{vol}\, V(Y) - 8 \geq \frac{2}{3} S - \frac{1}{2} S^2$, which Section 7.3 establishes at second order and proves for pure push-outs, and a bound for the part of $V(Y)$ outside the inversion hull, which part (c) of the proof of Theorem 7.9 gives as $13\,958\,\Theta^4$. With Proposition 7.17 the first can be replaced by $\mathrm{vol}(Q_w) \geq 8$ near the root system, whose second order is Proposition 7.18; by Proposition 7.19 that bound is attained along closed curves of direction sets that violate the packing constraint (Remark 7.20).

(iii) *Twenty-three contacts and one near-contact.* Along the ray of one centre pushed out the pair terms take over only at $\delta = 0.197$, where the eight tight pairs of that centre have slack $\delta/4 = 0.049$, beyond the reach of (i). What that ray needs is a volume bound for a saturated 23-point configuration with a near-contact, which Theorem 5.72 does not give: its margin is $4.4 \cdot 10^{-5}$, and a near-contact at $2 + \delta$ removes $S(2 + \delta) \approx 0.14$.

(iv) *Certificates for the other configurations.* The right side of Lemma 7.6 is at least 8 for 24 centres within $\sqrt{6}$ away from the root system, and for each count from 25 to 28. It is proved for 23 by Theorem 7.8, at 29 and 30 by Theorems 7.32 and 7.35, from 31 on by Theorem 7.25, and at 28 by Propositions 7.37 and 7.38; it is open from 25 to 27.

By Proposition 7.21, statement (i) fails at slack 0.017, so for twenty-four centres (i) and (ii) cannot be proved one after the other over the whole range, and the case has to be treated through the single volume statement (G).

Three computations locate the difficulty; none of them is used in a proof. First, minimising the right side of Lemma 7.6 over configurations of exactly M centres within $\sqrt{6}$, with the exact gradient and with starts that carry the best configuration of one count on to the next, gives the root system at $M = 24$, at $7.906940\ldots$, and least values 8.264, 8.393, 8.517, 8.592, 8.776 and 8.917 for $M = 25, \ldots, 30$, rising, with occasional small drops, to 10.895 at $M = 43$ (Figure 34). For $M = 24$ every one of twelve starts that ends at a local minimum ends at the root system. All 196 local minima found for $M \geq 25$ lie at or above 8.264; every one of them has some of its centres on the sphere of radius $\sqrt{6}$, where their caps are empty,

and the least one for each count has at most 22 strictly inside.[31] For $M \geq 44$ no packing was found: from 300 random starts for each count up to 49, the least squared violation of the constraints grows from 0.14 at 44 to 6.5 at 49, while 42 and 43 centres fit with most of them on that sphere. At most 49 centres fit within $\sqrt{6}$ in any case.[32] The configurations with 25 to 27 centres therefore have room to spare; what (C) lacks for them is a relaxation that detects it. Second, at twenty-four centres the size of the gap can be read off: the pair terms reach 8 only at $\sum \delta_i = 0.155$ along the root system pushed out evenly, and at $\delta = 0.197$ with one centre pushed out and the others in contact, while the explicit neighbourhood of Theorem 7.9 ends at $\sum \delta_i = 4 \cdot 10^{-3}$, and its rigidity estimate (a) stops near $8 \cdot 10^{-3}$, so no sharpening of its constants bridges that factor. Letting the truncation radius grow with the distances does not change this. Four points pairwise at least 2 apart within $|x|$ of a point x force $16|x|^2 \geq 12 + d_1^2 + d_2^2 + d_3^2$ for the distances of the three centres other than 0, so the pair terms stay exact in a slightly larger ball once the centres move out; along the root system pushed out evenly that moves the crossing only from 0.1551 to 0.1473. Third, the shape half of (i) does not follow from the kernel of Theorem 5.30 at moderate slack (Remark 5.77), nor from a second-level kernel computed afresh on the enlarged domain $[-1, \frac{1}{2} + s]$: such a kernel with bound exactly 24 would vanish on every small perturbation of the root system, and with a margin in its two-point constraint the root system itself uses up the margin (Proposition 7.10).

Proposition 7.10 (The root system uses up the margin)**.** *Let $s > 0$ and $w(u) = (u+1)(u+\frac{1}{2})^2 u^2 (\frac{1}{2} + s - u)$, which is nonnegative on $[-1, \frac{1}{2} + s]$, and let N and $\mu > 0$ be such that every set C of 24 points of S^3 with inner products at most $\frac{1}{2} + s$ satisfies*

$$\mu \sum_{\{x,y\} \subset C} w(\langle x, y \rangle) \;\leq\; N - 24,$$

and the chain of Lemma 5.27 gives this for a kernel of the second level with the margin μw in its two-point constraint. Then $N - 24 \geq 36\mu s$. Consequently the bound $w(u) \leq (N-24)/\mu$ that the inequality yields for a single inner product holds at every $u \in [-1, \frac{1}{2} + s]$ as soon as $36s \geq \max w$; at $s = 0.008$ one has $36s = 0.288$ and $\max w < 0.0198$.

Proof. The normalised roots of R form such a set C, their inner products being -1, $-\frac{1}{2}$, 0 and $\frac{1}{2}$. The weight vanishes at the first three and equals $\frac{3}{2} \cdot 1 \cdot \frac{1}{4} \cdot s = \frac{3}{8} s$ at $\frac{1}{2}$, and each root has eight neighbours at $60°$, so 96 of the 276 pairs have inner product $\frac{1}{2}$ and the sum of the weights is $96 \cdot \frac{3}{8} s = 36s$. The comparison $36s \geq \max w$ at $s = 0.008$ needs only crude estimates. Put $b = \frac{1}{2} + s$. On $[0, b]$, $(u+1)(u+\frac{1}{2})^2 \leq (1+b)(\frac{1}{2}+b)^2$, and $u^2(b-u) \leq \frac{4}{27} b^3$, the largest value of $u^2(b-u)$; at $s = 0.008$ the product is below 0.0298. On $[-1, 0]$, $(u+1)(b-u) \leq \frac{1}{4}(1+b)^2$ by the inequality of arithmetic and geometric means, and $|u(u+\frac{1}{2})| \leq \frac{1}{2}$, so $w \leq \frac{1}{16}(1+b)^2 < 0.143$. Both are below $36s = 0.288$. A finer

[31] Adding a centre never raises the right side of Lemma 7.6: its cap loses to each other cap the lens they share, and these lenses are disjoint, since no point of $B(\sqrt{3/2})$ lies in three caps, so together they are at most the cap. The least value can therefore rise from M to $M+1$ only because no further centre fits in the least configurations with M centres, and making room for one pushes the centres near c outward.

[32] Two centres at distances $d, d' \in [2, \sqrt{6})$ from c and at least 2 apart have directions with inner product at most $(d^2 + d'^2 - 4)/(2dd')$, which increases in d and d' and so stays below $\frac{2}{3}$. The directions therefore form a spherical code in S^3 of angle $\arccos \frac{2}{3} = 48.19°$, and the linear programming bound of [DGS77] applies: a polynomial of degree 13 with nonnegative Gegenbauer coefficients and constant coefficient 1, which has no root in $[-1, \frac{2}{3}]$ by Sturm's theorem and is negative at 0, takes the value $49.577\ldots$ at 1. The three-point bound does no better, 49.5 at degree 8. The largest packing the survey finds has 43 centres, 29 of them within 10^{-4} of the sphere of radius $\sqrt{6}$.

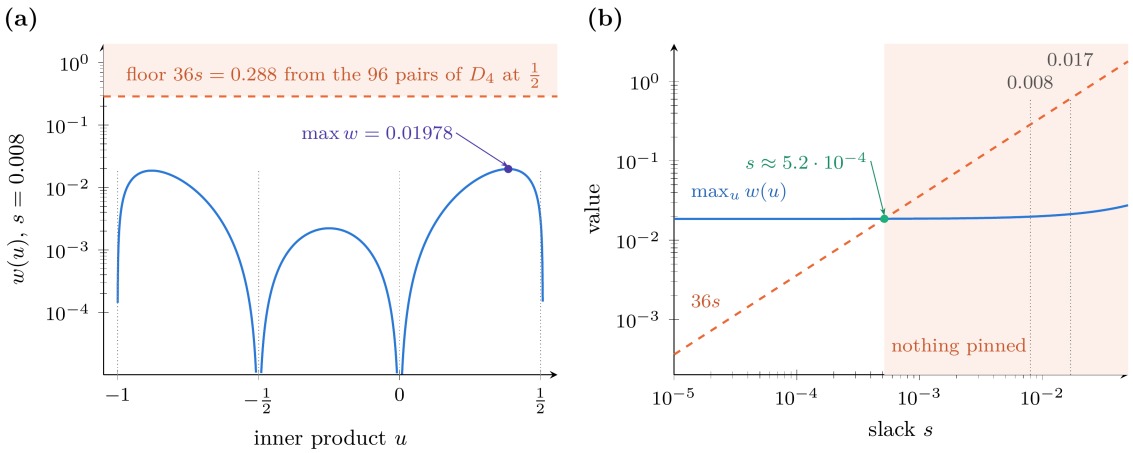


FIGURE 33. The margin of the second level against the root system (Proposition 7.10). (a) The weight w at $s = 0.008$ on a logarithmic scale, with its zeros at -1, $-\frac{1}{2}$ and 0 and at $\frac{1}{2}+s$; the largest value is 0.01978, near $u = 0.3855$, and the 96 pairs of the root system at $\frac{1}{2}$ force $(N - 24)/\mu \geq 36s = 0.288$ (shaded). No inner product is excluded. (b) The largest value of w and the floor $36s$ as functions of the slack. They cross near $s = 5.2 \cdot 10^{-4}$; to the right of the crossing, which contains the slacks 0.008 of Theorem 5.78 and 0.017 of Proposition 7.21, the margin excludes nothing.

evaluation, in ball arithmetic on $2 \cdot 10^5$ subintervals of $[-1, 0.508]$, gives $\max w \leq 0.019779$; Figure 33 shows both quantities. ∎

Near the double zero at 0 one has $w(u) = \frac{1}{8}(1 + 2s)u^2 + O(u^3)$, so even for small s the window that the margin leaves about 0 has half-width of order $(288s)^{1/2}$, the square root of Remark 5.77 once more. A weight that also vanishes at $\frac{1}{2}$ is not used up by the root system, but no pointwise reading of it can be sharper than the codes that occur allow, and those are the subject of the next remark.

Remark 7.11 (Twenty-four points at slack 0.008). Local maximisation from the normalised roots, under the constraint that every inner product be at most 0.508, finds 24-point codes in which one inner product is 0.1231, at distance at least 0.123 from each of -1, $-\frac{1}{2}$, 0 and $\frac{1}{2}$, and others at -0.0876 and -0.5394; the root-sum-square distance of these codes from the nearest rotated copy of the normalised roots is between 0.134 and 0.151, and the largest distance of a single direction from its root is 0.064. Hence at slack 0.008 no statement that pins every inner product near the four values of the root system is true. What Corollary 7.28 asks for is weaker, and these codes meet each of its three forms: every direction within 0.093 of a root, a root-sum-square distance below 0.228, and a sum of squared deviations of the inner products from those of the labelled roots of at most 0.3059. The code at distance 0.151 has that sum equal to 0.1377, and for it the bound in the proof of Corollary 7.28(iii) gives 0.1516 in place of 0.1512. These values are computed in floating point. An earlier numerical exploration of kissing configurations in four dimensions, through the Hopf map, is in [AP13].

7.3. **The second order at the root system.** Statement (ii) of the list above asks for the volume of the cell of twenty-four centres near the root system beyond the explicit

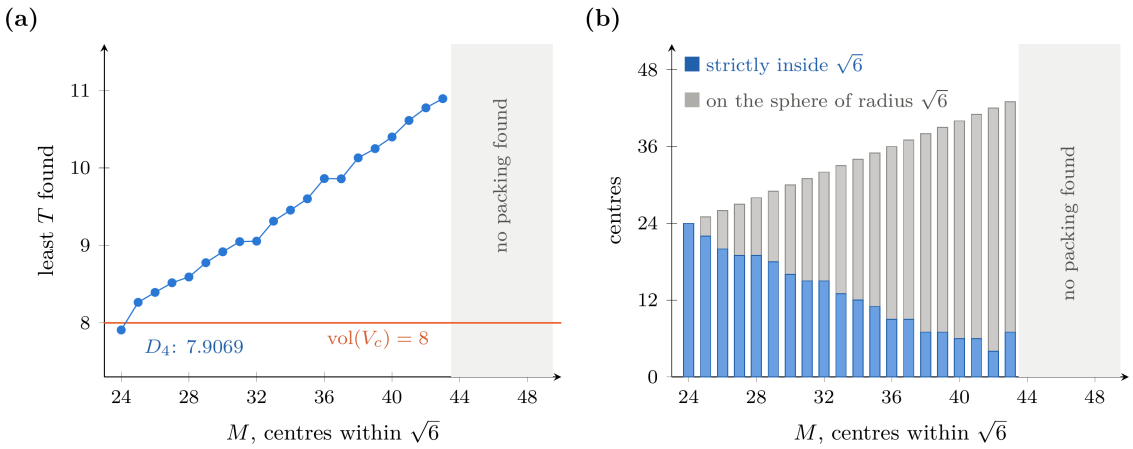


FIGURE 34. The right side of Lemma 7.6 for every count of centres within $\sqrt{6}$, computed by local minimisation in floating point. (a) The least value found for M centres, $M = 24, \dots, 43$, against the level 8; at $M = 24$ it is the root system, $7.906940\dots$. For $M \geq 44$ no packing was found, in 300 random starts for each count. (b) How the configurations of (a) meet the count: the centres strictly inside the sphere of radius $\sqrt{6}$, and those on it, whose caps are empty.

neighbourhood of Theorem 7.9. Its step (b) charges the tilts of the directions a loss of order $10^2 S^2$, from $73S^2$ at $S = 10^{-3}$ to $98S^2$ at $S = 4 \cdot 10^{-3}$, against the gain $\frac{2}{3}S$, and that stops it. This subsection computes the second order exactly, certifies its minimum on the packing cone, integrates the push-outs exactly, and says at its end what separates the result from statement (ii).

Lemma 7.12 (Second variation of the volume of a 4-polytope)**.** *Let $P(t) = \{x : \langle x, n_i(t)\rangle \leq h_i(t)\}$ be a bounded polytope of $\mathbb{R}^4$ with unit normals $n_i(t)$ moving along geodesics of S^3, $\dot{n}_i(0) = \tau_i \perp n_i(0)$, and $h_i(t) = h_i + t\eta_i$. Write $v_i(x) = \eta_i - \langle x, \tau_i\rangle$ for the normal velocity of facet i at $t = 0$, F_i for the facets, $F_{ij} = F_i \cap F_j$ for the 2-faces, and θ_{ij} for the angle between n_i and n_j on a 2-face. Then $t \mapsto \mathrm{vol}\, P(t)$ is twice continuously differentiable, with*

$$\frac{d}{dt}\,\mathrm{vol}\, P\Big|_0 = \sum_i \int_{F_i} v_i \, dA,$$

$$\frac{d^2}{dt^2}\,\mathrm{vol}\, P\Big|_0 = \sum_i h_i |\tau_i|^2 \,\mathrm{vol}_3 F_i + \sum_{i<j} \frac{1}{\sin\theta_{ij}} \int_{F_{ij}} \left(2v_i v_j - \cos\theta_{ij}\,(v_i^2 + v_j^2)\right) dA_2.$$

Proof. The first formula is the transport theorem. For the second, follow each facet by the purely normal motion $\dot{x} = v_i n_i$, which keeps x on the moving hyperplane and, that hyperplane being flat, preserves the area element to first order; along it $Dv_i/Dt = \ddot{h}_i - \langle x, \ddot{n}_i\rangle = |\tau_i|^2\langle x, n_i\rangle = |\tau_i|^2 h_i$ on F_i, since $\ddot{n}_i = -|\tau_i|^2 n_i$ on a geodesic. The 2-face F_{ij} moves, relative to the material of facet i and along the conormal ν_i of F_i at that face, at speed $(v_j - v_i\cos\theta_{ij})/\sin\theta_{ij}$, because $n_j = \cos\theta_{ij}\, n_i + \sin\theta_{ij}\,\nu_i$ there; the transport theorem on the moving facet then gives $\frac{d}{dt}\int_{F_i} v_i = \int_{F_i} |\tau_i|^2 h_i + \sum_j \int_{F_{ij}} v_i(v_j - v_i\cos\theta_{ij})/\sin\theta_{ij}$, and adding the two contributions of each 2-face gives the formula. A 2-face that appears or disappears

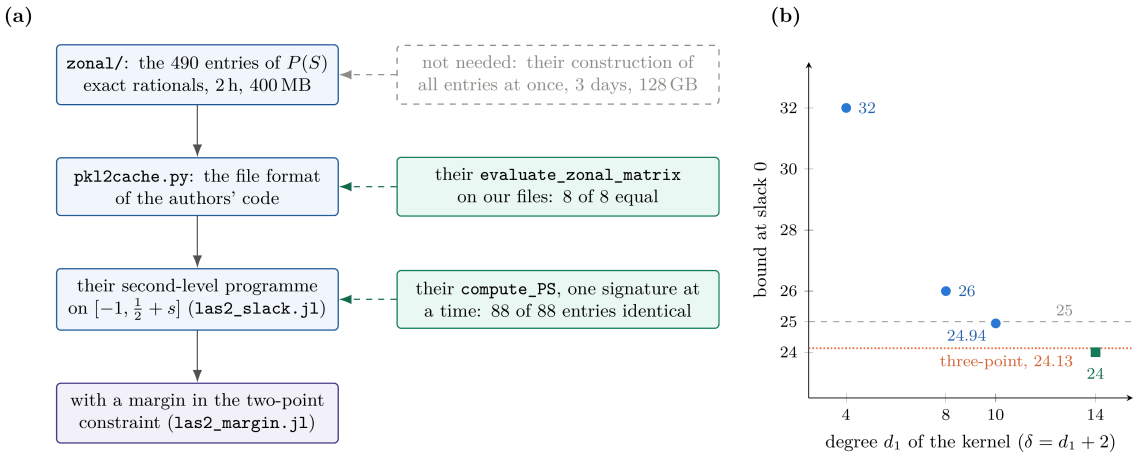


FIGURE 35. The second level of the hierarchy at slack 0. (a) The zonal matrices of Section 5.5.3 replace the construction of [LLM24] in their second-level programme, and the two checks against their entries. (b) The bound of the programme at slack 0 against the degree d_1 of the kernel, with $\delta = d_1 + 2$: 32 at $(4, 6)$, 26 at $(8, 10)$, 24.942 at $(10, 12)$, and the certificate's 24 at $(14, 16)$; beside the three-point bound at degree 10, 24.13, and the level 25 that a bound on the number of points has to beat.

under the motion does so with area tending to 0, so both formulas are continuous across combinatorial changes; the second derivative is therefore continuous, and $\operatorname{vol} P$ is C^2.[33] ■

Proposition 7.13 (The Hessian at the root system)**.** *At* $Q = \{x : \langle x, u_i\rangle \le 1\}$*, in the coordinates* $\xi = (\tau, \eta) \in \mathbb{R}^{72} \times \mathbb{R}^{24}$*,* $\tau_i \perp u_i$*, with* $\delta_i = 2\eta_i$ *the push-out of the* i*-th centre, the volume of* $V(Y)$ *has the expansion*

$$\operatorname{vol} V(Y) \;=\; 8 + \tfrac{2}{3}\sum_i \delta_i + \tfrac{1}{2}\,H(\xi,\xi) + O(|\xi|^3),$$

where H *is the form of Lemma 7.12 with* $\operatorname{vol}_3 F_i = \frac{4}{3}$*,* $\theta_{ij} = 60^\circ$ *and* F_{ij} *the* 96 *equilateral triangles of side* $\sqrt{2}$*; by the midpoint rule, exact for quadratics,*

$$H(\xi,\xi) = \tfrac{4}{3}\sum_i |\tau_i|^2 + \sum_{\{i,j\}\in T} \frac{1}{3}\sum_m \big(2v_i(m)v_j(m) - \tfrac{1}{2}(v_i(m)^2 + v_j(m)^2)\big),$$

m running over the midpoints of the edges of F_{ij}*. The form is indefinite: in orthonormal coordinates its least eigenvalue is* -8*, of multiplicity two, followed by* $-6.194\ldots$ *of multiplicity eight; its largest is* 4*, simple, and its kernel has dimension* 15*, the six rotations and nine further flat directions. On the push-outs alone* ($\tau = 0$) *it is* $H(\eta,\eta) = 2\sum_T \eta_i\eta_j - 4\sum_i \eta_i^2$*, with matrix* $\mathrm{Adj} - 4I$ *for the edge graph of the* 24*-cell, and*

$$H(\eta,\eta) + \Big(\sum_i \delta_i\Big)^2 = 2\sum_T \eta_i\eta_j + 8\sum_{i<j}\eta_i\eta_j \;\ge\; 0 \qquad (\eta \ge 0), \tag{7.5}$$

with equality exactly when at most one η_i *is nonzero.*

[33] In dimension n the k-th variation is an integral over the $(n-k)$-faces and the first jumps sit at the vertices, in the n-th derivative: the area of a polygon is only C^1 in its edge data, and the volume of a polytope of $\mathbb{R}^4$ is C^3.

Proof. The expansion is Taylor's formula for the C^2 function of Lemma 7.12; the first derivative is $\frac{4}{3}\sum_i \eta_i$ because the centroid of each facet lies on its normal. The data are those of the regular 24-cell. With $\tau = 0$ each triangle contributes $2\eta_i\eta_j - \frac{1}{2}(\eta_i^2 + \eta_j^2)$, and each vertex of the edge graph lies on eight edges, which gives $H(\eta, \eta)$; adding $(\sum_i \delta_i)^2 = 4\sum_i \eta_i^2 + 8\sum_{i<j} \eta_i\eta_j$ gives (7.5). The spectrum is computed in floating point, and the form is checked against second differences of the exact volume of the polytope, computed afresh by intersecting halfspaces, to $6 \cdot 10^{-8}$. ∎

Proposition 7.14 (The worst second-order loss on the packing cone)**.** *Let $\mathcal{C}$ be the first-order feasible cone of the packing constraints, $(\Lambda\tau)_{ij} \le \frac{1}{2}(\eta_i + \eta_j)$ on the 96 tight pairs and $\eta_i \ge 0$. Then*

$$m \;:=\; \min\{H(\xi,\xi) :\; \xi \in \mathcal{C},\; \textstyle\sum_i \delta_i = 1\} \;=\; -1,$$

attained on the ray of a single centre pushed out, where $\mathrm{vol} = \frac{25}{3} - \frac{1}{3}(1 - \frac{\delta}{2})^4 = 8 + \frac{2}{3}\delta - \frac{1}{2}\delta^2 + \frac{1}{6}\delta^3 - \frac{1}{48}\delta^4$ *exactly. Consequently, for packings whose tilts are of order S, as estimate (a) of the proof of Theorem 7.9 makes them when $d(W) \le \frac{1}{48}$, the second-order model of the volume is*

$$\mathrm{vol}\, V(Y) - 8 \;\ge\; \tfrac{2}{3}S - \tfrac{1}{2}S^2 + O(S^3), \qquad S = \sum_i \delta_i,$$

positive up to $S = \frac{4}{3}$, against the 0.155 *at which the pair terms of Lemma 7.6 take over.*

Proof. $m \le -1$ is the ray. For $m \ge -1$ it suffices that $H + cc^{\mathsf{T}}$, $c = 2\mathbf{1}_\eta$, be copositive on $\mathcal{C} = \{\xi : B\xi \ge 0\}$, B the 120 rows of the cone, since $c \cdot \xi = \sum_i \delta_i$; and a certificate is

$$H + cc^{\mathsf{T}} \;=\; P + B^{\mathsf{T}} N B, \qquad P \succeq 0, \quad N \ge 0 \text{ entrywise},$$

since then $\xi^{\mathsf{T}}(H + cc^{\mathsf{T}})\xi = \xi^{\mathsf{T}}P\xi + (B\xi)^{\mathsf{T}}N(B\xi) \ge 0$ on $\mathcal{C}$. Such a pair exists, and it is verified with no floating point: in the scaled integral model of Q (roots $(\pm1, \pm1, 0, 0)$, vertices $(\pm2, 0, 0, 0)$ and $(\pm1, \pm1, \pm1, \pm1)$, rational bases of the tangent spaces, $t_i = \sqrt{2}\,\tau_i$) the form H, the rows B and c are rational; N is symmetric, constant on the 43 orbits of the signed-permutation group on unordered pairs of constraints, with nonnegative rational values, 35 of them positive; and P is assembled exactly and is positive semidefinite by an exact LDL^{T}, whose 30 zero pivots are the push-outs and the rotations. The certificate was found as the dual of the Shor relaxation with the products of the cone constraints, whose value is -1, averaged over the group and made exact. That the rows B are the first-order packing constraints, the derivative of $|y_i - y_j|^2$ being $8(B\xi)_{ij}$ on the tight pairs, is checked against the exact distances to $2 \cdot 10^{-9}$. The verification is repeated in Lean, where H, B and c are built from the root system in the proof assistant itself (Section 11). ∎

Remark 7.15 (What the second order gives for (ii), and what it does not). The expansion says that the second-order loss on the cone is exactly the loss on the deleted root's axis, $\frac{1}{2}S^2$: the tilts, which step (b) of Theorem 7.9 charges at order $10^2 S^2$, cost nothing more at second order than the pushes do. Three things separate this from statement (ii). Tilting is not monotone: there are feasible packings near the root system whose cell is smaller than that of the same distances at the roots, in 103 of 237 trials and by up to $3.6 \cdot 10^{-5}$, and by $0.117\,S^2$ at second order in the worst push pattern, so the bound does not reduce to monotonicity in the support numbers. No bound on the remainder is proved: on the 300 packings of Figure 36(b), with S up to 0.48, the exact volume satisfies

$$\mathrm{vol}\, V(Y) - 8 - \tfrac{2}{3}S + \tfrac{1}{2}S^2 \;\ge\; 0$$

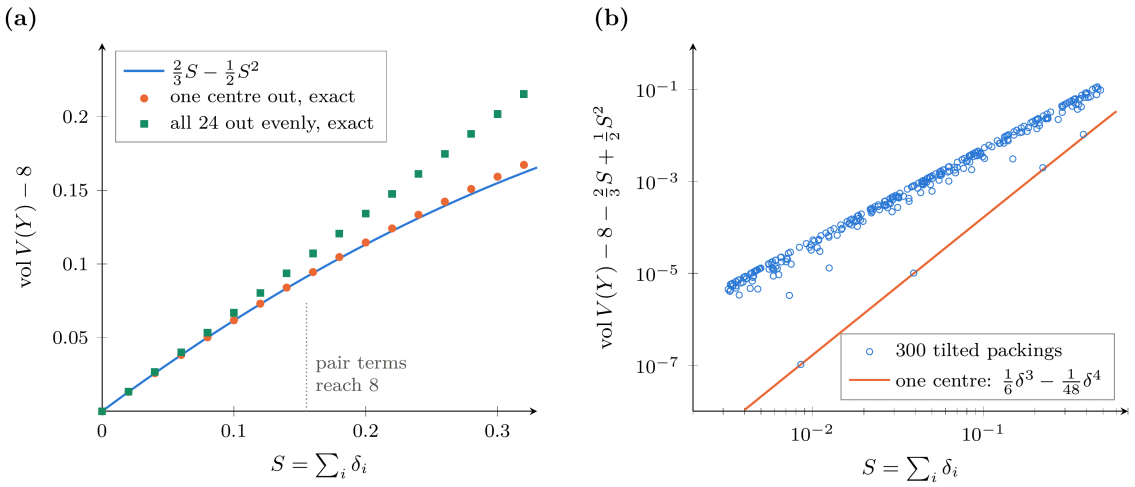


FIGURE 36. The second order at the root system. (a) $\operatorname{vol} V(Y) - 8$ against $S = \sum_i \delta_i$: the second-order model $\frac{2}{3}S - \frac{1}{2}S^2$, which is the worst case on the packing cone (Proposition 7.14), the exact value with one centre pushed out, on which it is sharp to second order, and the exact value with all twenty-four pushed out evenly; the dotted line is where the pair terms of Lemma 7.6 reach 8 along the even push. Step (b) of the proof of Theorem 7.9 ends at $S = 4.5 \cdot 10^{-3}$. (b) What is left over, $\operatorname{vol} V(Y) - 8 - \frac{2}{3}S + \frac{1}{2}S^2$, on 300 packings near the root system with tilts along the edges of the cone of up to $5.8\,S$ (computed in floating point), and exactly on the one-centre ray, $\frac{1}{6}\delta^3 - \frac{1}{48}\delta^4$: it is positive on every one, and least on that ray.

in every case, with the minimum on the one-centre family, and the third-order remainder $\operatorname{vol} -8 - \frac{2}{3}S - \frac{1}{2}H(\xi,\xi)$ lies between $-0.004\,S^3$ and $+0.166\,S^3$; the statement that carries (ii) to the pair terms, and well past them, is that the displayed inequality holds for every packing with $d(W) \le \frac{1}{48}$, and a proof through a bound on the remainder has to control the cubic term to the size it has on the tested packings, $0.004\,S^3$, where the estimates of the proof of Theorem 7.9 have constants of order 10^2 already at second order. In such a proof the extremal family is the one-centre ray, on which everything is explicit. And the inequality concerns $V(Y)$, while far centres may cut off the part of $V(Y)$ outside the inversion hull $K(Y)$ (Proposition 7.4), which part (c) of the proof of Theorem 7.9 bounds by $13\,958\,\Theta^4$. That bound is far from the truth: along the ray of one centre pushed out and along the even push, $\operatorname{vol}(V(Y) \setminus K(Y))$ stays below 10^{-6} up to $S = 0.197$, where the bound is 21, and it grows as the seventh power of S on the first ray and as the fourth on the second. Statement (ii) needs a hull estimate of that quality beside the inequality above.

Lemma 7.16 (Pure push-outs, exactly)**.** *For* $0 \le \eta_i \le 1$,

$$\operatorname{vol}\{x : \langle x, u_i\rangle \le 1 + \eta_i \ \forall i\} \ \ge \ 8 + \frac{1}{3}\sum_i (1 - (1-\eta_i)^4),$$

with equality whenever at most one η_i *is nonzero. In particular, with* $S = 2\sum_i \eta_i$*, the right side is at least* $8 + \frac{2}{3}S - \frac{1}{2}S^2$*.*

Proof. Along $h(t) = \mathbf{1} + t\eta$ the volume has derivative $\sum_i \eta_i A_i(t)$, $A_i(t)$ the 3-volume of the i-th facet. That facet is $\{x : \langle x, u_i\rangle = 1 + t\eta_i,\ \langle x, u_j\rangle \le 1 + t\eta_j \ (j \ne i)\}$, which

contains the facet with the other support numbers lowered to 1, so $A_i(t) \geq \frac{4}{3}(1-t\eta_i)^3$ by the one-centre formula $\mathrm{vol} = \frac{25}{3} - \frac{1}{3}(1-s)^4$ at $h = \mathbf{1} + se_i$. Integrating, $\mathrm{vol} - 8 \geq \sum_i \eta_i \int_0^1 \frac{4}{3}(1-t\eta_i)^3 dt = \frac{1}{3}\sum_i (1-(1-\eta_i)^4)$. For the last claim, $1-(1-\eta)^4 = 4\eta - 6\eta^2 + \eta^3(4-\eta)$ and $\sum \eta_i^2 \leq (\sum \eta_i)^2$. ∎

Proposition 7.17 (The push-outs integrated out)**.** *Let $w_1, \dots, w_{24}$ be unit vectors, $Q_w = \{x : \langle x, w_j\rangle \leq 1 \ \forall j\}$ their contact cell, A_i^w the 3-volume of its i-th facet, and $s_i = \max\{\langle x, w_i\rangle : \langle x, w_j\rangle \leq 1 \ \forall j \neq i\} - 1$ the height of the apex of the i-th deletion above that facet. Then for $0 \leq \eta_i \leq s_i$,*

$$\mathrm{vol}\{x : \langle x, w_i\rangle \leq 1 + \eta_i \ \forall i\} \ \geq \ \mathrm{vol}(Q_w) + \sum_i A_i^w \, \frac{s_i}{4}\Big(1 - \big(1 - \tfrac{\eta_i}{s_i}\big)^4\Big).$$

Proof. As in Lemma 7.16, the derivative along $\mathbf{1}+t\eta$ is $\sum_i \eta_i A_i(t)$ and $A_i(t) \geq a_i(t\eta_i)$, where $a_i(s)$ is the 3-volume of the section of the deletion $\{\langle x, w_j\rangle \leq 1, \ j \neq i\}$ by the hyperplane $\langle x, w_i\rangle = 1 + s$. The deletion is convex, so by the Brunn–Minkowski inequality $a_i(s)^{1/3}$ is concave on $[0, s_i]$, and it is nonnegative at s_i; hence $a_i(s) \geq (1 - s/s_i)^3 a_i(0)$ with $a_i(0) = A_i^w$. Integrate. ∎

At the root system $s_i = 1$ and $A_i^w = \frac{4}{3}$, and Proposition 7.17 is Lemma 7.16. In general it removes the push-outs from the problem: the inequality of Remark 7.15 reduces to one involving only the three functionals $\mathrm{vol}(Q_w)$, A_i^w and s_i of the tilted directions, and the packing constraints $\langle w_i, w_j\rangle \leq \frac{1}{2} + \frac{1}{2}(\eta_i + \eta_j) + O(\eta^2)$ that tie the tilts to the pushes. At second order that is Proposition 7.14; beyond it, what statement (ii) asks for is bounds on the three functionals to second order in the tilts with explicit remainders. The difficulty lies in the tilts, not the pushes.[34]

Setting the push-outs to zero isolates the tilts. The block of H on the 72 tangent coordinates is the Hessian of $\mathrm{vol}(Q_w)$, the volume of the contact cell, as a function of the directions alone, with no packing constraint.

Proposition 7.18 (The tilt block)**.** *The restriction of the form H of Proposition 7.13 to pure tilts, $\eta = 0$, is positive semidefinite. Its kernel has dimension 15: the six infinitesimal rotations and nine further directions, none of them an infinitesimal strain $\tau_i = Su_i - \langle u_i, Su_i\rangle u_i$ with S symmetric.*

Proof. An exact LDL^{T} factorisation of the 72×72 block over $\mathbb{Q}$ has 57 positive pivots, the least $\frac{5}{66}$, and 15 zero pivots, each with the rest of its row zero; the kernel is computed exactly, contains the six rotation fields, and meets the nine strain fields only in 0. The factorisation is repeated in Lean. ∎

Thus the root system is a critical point of $\mathrm{vol}(Q_w)$ at which no tilt lowers the volume at second order. By Proposition 7.17, a lower bound $\mathrm{vol}(Q_w) \geq 8$ on an explicit neighbourhood of the root system would give the first inequality of statement (ii) for every Y whose directions lie in it, with no packing constraint and no rigidity estimate, leaving the hull bound and statement (i) to place the directions there. The neighbourhood cannot be the whole space of direction sets. Without the packing constraint the contact cell can be smaller than the 24-cell: a set of 24 unit directions with inner products up to 0.579 has a contact cell with 24 facets, 99 vertices and volume 7.96553, the same to eight digits whether computed directly or as a quarter of its surface area.

[34] On 63 tilted packings the least excess of the volume over the bound of Proposition 7.17 is $3.9 \cdot 10^{-6}$; Lemma 7.16 holds on 400 random push patterns, and the one-centre formula agrees with the computed volume to 10^{-14}.

The lower bound, if true, is not strict. The kernel in Proposition 7.18 contains the tangent of a closed curve of direction sets, none of them a rotation of the root system except its endpoints, along which the contact cell keeps volume exactly 8.

Proposition 7.19 (A closed curve of contact cells of volume 8)**.** *Write $\mathbb{R}^4 = P \oplus P'$, with P the plane of an A_2 hexagon of roots, and $e(a) = (\cos a, \sin a)$ in orthonormal coordinates of either plane. Up to a rotation, the 24 unit roots are $(e(60^\circ m), 0)$, $m = 0, \dots, 5$, and the eighteen vectors $(\sigma e(a), \sqrt{1-\sigma^2}\, e(b))$ with $\sigma = 1/\sqrt{3}$ and (a, b) in $\{30^\circ, 150^\circ, 270^\circ\} \times \{0^\circ, 120^\circ, 240^\circ\}$ or in $\{90^\circ, 210^\circ, 330^\circ\} \times \{60^\circ, 180^\circ, 300^\circ\}$. For $0 \le \theta \le \pi/3$ let $w(\theta)$ be the same list with the first six turned to $(e(\theta + 60^\circ m), 0)$ and σ replaced by*

$$\sigma(\theta) = \frac{4C}{4C^2 + 3}, \qquad C = \cos(\tfrac{\pi}{6} - \theta).$$

Then $\operatorname{vol} Q_{w(\theta)} = 8$ for every θ. The sets $w(0)$ and $w(\pi/3)$ are the root system; for $0 < \theta < \pi/3$ the largest inner product in $w(\theta)$ is at least $4C^2/(4C^2+3) > \frac{1}{2}$. Therefore $w(\theta)$ is not a rotation of the root system, and it violates the packing constraint. For fixed θ and any other $\sigma \in (0, \frac{\sqrt{3}}{2})$ the contact cell has volume above 8.

Proof. Write $x = (p, q)$. The first six constraints say that p lies in the regular hexagon Hex_θ of inradius 1 with edge normals $e(\theta + 60^\circ m)$. For such p the other eighteen say $\langle q, e(b)\rangle \le h_+$ for $b \in \{0^\circ, 120^\circ, 240^\circ\}$ and $\langle q, e(b)\rangle \le h_-$ for $b \in \{60^\circ, 180^\circ, 300^\circ\}$, with $h_\pm = (1 - \sigma m_\pm(p))/\sqrt{1-\sigma^2}$, where m_+ and m_- are the support functions of the triangles $\{e(30^\circ), e(150^\circ), e(270^\circ)\}$ and $\{e(90^\circ), e(210^\circ), e(330^\circ)\}$. Since $m_\pm \le |p| \le 2/\sqrt{3}$ and $\sigma < \sqrt{3}/2$, both are positive. The slice over p is the intersection of two opposite equilateral triangles of inradii h_+ and h_-. If $h_+ \le 2h_-$ and $h_- \le 2h_+$ it is a hexagon of area $\sqrt{3}\,(4h_+h_- - h_+^2 - h_-^2)$; otherwise it is the smaller triangle, whose area exceeds that expression by $\sqrt{3}\,(h_+ - 2h_-)^2$ or $\sqrt{3}\,(h_- - 2h_+)^2$. Integrating over Hex_θ, of area $2\sqrt{3}$,

$$(1-\sigma^2)\operatorname{vol} Q \;\ge\; 12 - 2\sqrt{3}\,\sigma J_1 + \sqrt{3}\,\sigma^2 J_2,$$

where $J_1 = \int (m_+ + m_-)$ and $J_2 = \int (4m_+m_- - m_+^2 - m_-^2)$ over Hex_θ, with equality when no slice is a triangle. Rotation by 60° maps Hex_θ to itself and exchanges m_+ and m_-, so J_1 and J_2 are six times their integrals over the sector between the rays at 30° and 90°. There $m_+ = \langle p, e(30^\circ)\rangle$ and $m_- = \langle p, e(90^\circ)\rangle$ are linear, and the sector of Hex_θ is the union of the triangles 0, $e(30^\circ)/C$, v and 0, v, $e(90^\circ)/C$, where $v = \frac{2}{\sqrt{3}} e(\theta + 30^\circ)$ is the vertex of Hex_θ in it. Integrating over the two triangles gives

$$J_1 = \frac{\sqrt{3}\,(4C^2+3)}{3C}, \qquad J_2 = \frac{\sqrt{3}\,(16C^4 - 8C^2 + 9)}{12\,C^2},$$

and then $4C^2\big(12 - 2\sqrt{3}\,\sigma J_1 + \sqrt{3}\,\sigma^2 J_2 - 8(1-\sigma^2)\big) = \big(4C - \sigma(4C^2+3)\big)^2$. Hence $\operatorname{vol} Q \ge 8$, with equality exactly when $\sigma = \sigma(\theta)$ and no slice is a triangle. At $\sigma(\theta)$ the conditions $h_\pm \le 2h_\mp$ are linear in p on each triangle. At its vertices they reduce to $4C^2 \ge 3$ and to $3 \mp 2\sqrt{3}\sin 2\varphi \ge 0$ with $\varphi = \frac{\pi}{6} - \theta$, or hold trivially, and both hold for $|\varphi| \le \frac{\pi}{6}$. The inner product of $(e(\theta), 0)$ with $(\sigma e(30^\circ), \cdot)$ is $\sigma C = 4C^2/(4C^2+3)$, which exceeds $\frac{1}{2}$ exactly when $4C^2 > 3$. At $\theta = 0$ the Gram matrix is that of the unit roots under an explicit labelling, and $w(\pi/3) = w(0)$ as a set. The identities, and the Gram matrix at $\theta = 0$, are checked symbolically and again in Lean; as an independent check, $|\operatorname{vol} Q_{w(\theta)} - 8|$ is below 10^{-14} at thirteen points in double precision and below 10^{-48} at three with 50-digit arithmetic. ∎

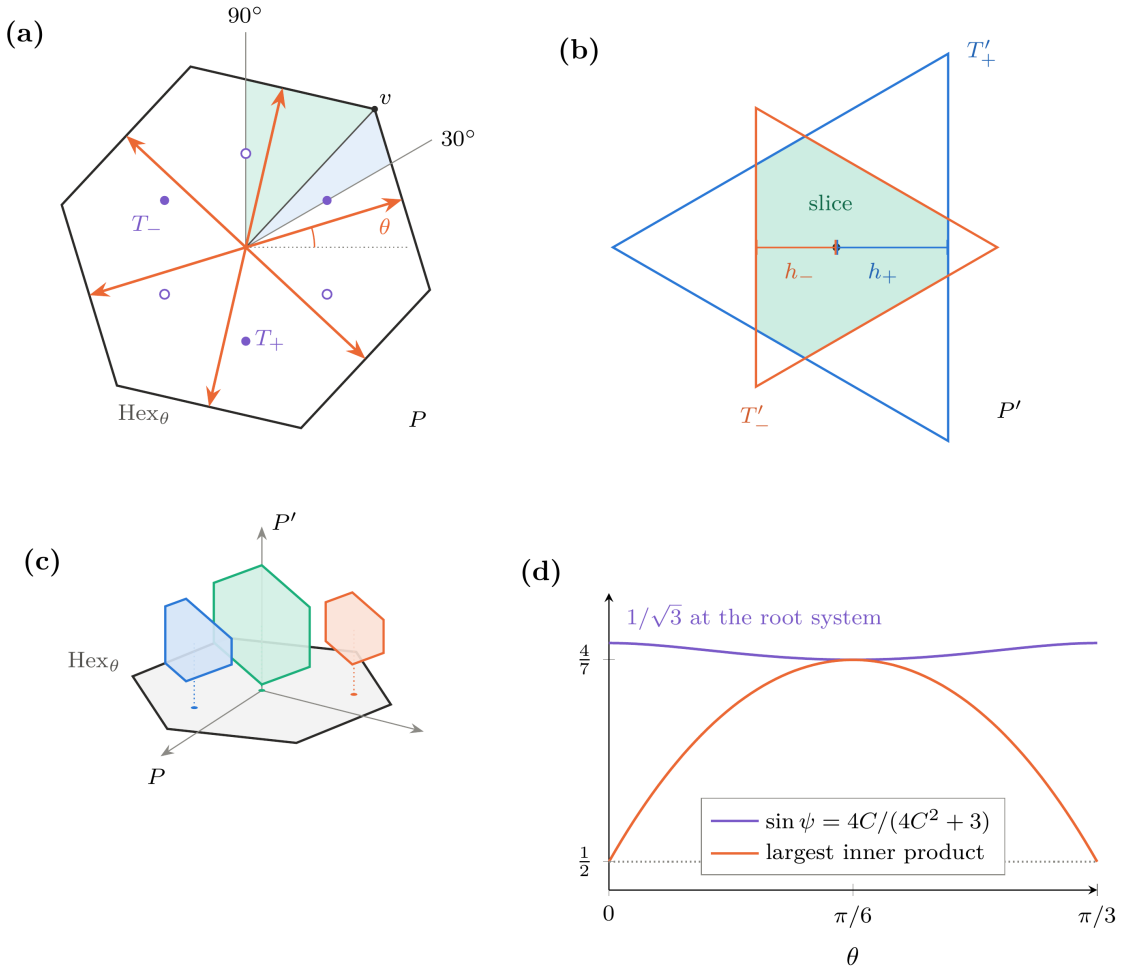


FIGURE 37. The curve of Proposition 7.19, drawn at $\theta = 17°$. (a) The plane P: the six roots of the hexagon turned by θ from the reference direction (dotted), the hexagon Hex_θ of inradius 1 that they cut out, the directions in P of the other eighteen roots, at radius σ (filled: the triangle T_+ at $30°, 150°, 270°$; open: T_-), and the sector between the rays at $30°$ and $90°$, split at the vertex v into the two triangles (blue and green) over which J_1 and J_2 are computed. (b) The slice over a point of Hex_θ: the intersection of two opposite triangles of inradii h_+ and h_- in P', a hexagon while $h_+ \le 2h_-$ and $h_- \le 2h_+$. (c) The cell as the union of its slices over Hex_θ, with the exact slices over three points; the two-dimensional slices are drawn in one plane transverse to P. (d) Along the curve: the tilt $\sin\psi$ of the eighteen roots and the largest inner product, which exceeds $\frac{1}{2}$ for $0 < \theta < \pi/3$ and reaches $\frac{4}{7}$ at $\theta = \pi/6$.

Remark 7.20 (What the curve means). Through each of the 16 hexagons of A_2 roots there is such a curve. So the root system is not a strict local minimum of $\mathrm{vol}(Q_w)$ up to rotations, and no positivity of higher-order terms can prove $\mathrm{vol}(Q_w) \ge 8$ near it. Along straight lines tk in the flat directions the volume grows at fourth order, $(\mathrm{vol}\, Q_w - 8)/t^4$ between 0.05 and

0.23 for random unit k. A minimisation that lets the other 57 tilt coordinates move finds this curve. At the points $\theta = 0.1$, 0.3 and $\pi/6$ of it the Hessian of $\mathrm{vol}(Q_w)$ on the 72 tangent directions is positive semidefinite, with seven zero eigenvalues (the rotations and the curve) and the next 0.0055, 0.031 and 0.045. Thus numerically the curve is the floor of a valley. Its contact cells have 78 vertices, six of them on six facets, where the 24-cell has 24. Along the curve the directions are not a contact configuration, since the largest inner product reaches $\frac{4}{7}$ at $\theta = \pi/6$. In a packing such directions need push-outs, which Proposition 7.17 turns into volume above 8. A proof of $\mathrm{vol}(Q_w) \geq 8$ near the root system therefore has to control the volume near these curves, or use the packing constraint there.

Away from the root system the bound $\mathrm{vol}(Q_w) \geq 8$ can hold only for a small slack. We minimised $\mathrm{vol}(Q_w)$ over 24 unit directions with inner products at most $\frac{1}{2}+s$, from 48 random starts spread into codes of that slack and eight perturbations of the root system, with the exact gradient $-A_i^w c_i$ (c_i the centroid of the i-th facet). For $s = 0.005$ and 0.01 every start that reaches a code ends at the root system or on a curve of Proposition 7.19, with volume 8. For $s = 0.02$ the codes it finds away from the root system have larger cells, the least 8.0679. For $s = 0.03$ and 0.035 they have cells of volume 7.99802 and 7.99288. Codes of that slack are the directions of packings only with push-outs, and these compensate for the difference many times over, so the bound needed is on the whole cell, not on Q_w (Remark 7.22).

Statement (i) of Section 7.2 fails well before the slack at which 25-point codes appear.

Proposition 7.21 (A second twenty-four-point code)**.** *There are* 24 *points of* S^3 *with rational coordinates whose pairwise inner products are at most* $\frac{1}{2} + \frac{849}{50000} = 0.51698$*, and whose set* W *has* $d(W) \geq \frac{57}{500}$*, with* d *as in Theorem 5.51.*

Proof. The points are listed in the supplementary data; each is the inverse stereographic image of a rational point of $\mathbb{R}^3$, so its norm is exactly 1. In exact arithmetic, and again in Lean, every two of them have inner product at most $\frac{1}{2} + \frac{849}{50000}$, and the inner product $-0.72817\ldots$ of the ninth and eleventh is at distance at least $\frac{57}{250}$ from each of $-1, -\frac{1}{2}, 0, \frac{1}{2}, 1$. If an orthogonal map and a relabelling moved the points by e_i onto the normalised roots, with $\sum e_i^2 = d(W)^2$, every inner product would lie within $e_i + e_j \leq 2d(W)$ of one of those five values; so $d(W) \geq \frac{57}{500}$. ∎

The same code appears from most starting points. Of 64 random configurations, spread into codes of slack 0.03 and then pushed to the least possible largest inner product, three return to the root system, and the other 59 all stop at the same value 0.516978, a code with no antipodal pair, 84 pairs at that inner product and five kinds of point. Its contact cell has volume 8.0948; realised as the directions of a packing with all centres at one distance, it needs that distance to be at least 2.0349, which leaves a cell of volume at least 8.674. Therefore it does not threaten Conjecture 1.3, but it limits how the conjecture can be proved: twenty-four directions of slack 0.017 or more need not lie near the root system, and any argument that first places them there stops at that slack.

Remark 7.22 (What the twenty-four-centre case needs). Let Y be the centres within $\sqrt{6}$ of c and $T(Y)$ the right side of Lemma 7.6. Whatever the rest of the packing does, $\mathrm{vol}(V_c) \geq T(Y)$ by that lemma and $\mathrm{vol}(V_c) \geq \mathrm{vol}(V(Y) \cap K(Y))$ by Proposition 7.4. Hence the case of Conjecture 1.3 with exactly twenty-four centres within $\sqrt{6}$ follows from

(G) $\mathrm{vol}(V(Y) \cap K(Y)) \geq 8$ for every packing set Y of 24 centres with $2 \leq |y| < \sqrt{6}$ and $T(Y) \leq 8$, with equality only at the root system,

a statement about 24 points in which statements (i) and (ii) of Section 7.2 enter together, and which Proposition 7.21 does not touch. The condition $T(Y) \leq 8$ cannot be dropped.

When centres move out towards $\sqrt{6}$ the inversion hull shrinks, and there is a packing with $\mathrm{vol}(V(Y) \cap K(Y)) = 4.750$; there $T(Y) = 10.82$. With the condition, the constrained minimisation (under the packing constraints and $T(Y) \le 8$) finds nothing below 8. None of 40 random configurations can be brought into the region $T(Y) \le 8$, in line with the survey of Section 7.2, where every local minimum of T with 24 centres is the root system. Of 64 starts made inside it, with tilts up to 0.3 and up to four centres pushed out, one as far as 2.19, the 20 that stay in the region all end at the root system with volume 8. No proof of (G) is known: it is the part of Conjecture 1.3 that the pair terms and the inversion hull leave between them, and Remark 7.43 sets out what a proof needs.

For twenty-five or more centres within $\sqrt{6}$ the pair terms alone suffice from thirty-one centres on (Theorem 7.25), and with triples of typed directions added they suffice at thirty and at twenty-nine (Theorems 7.32 and 7.35), and at twenty-eight (Propositions 7.37 and 7.38). From twenty-five to twenty-seven they are not proved to suffice, although the least value of T found numerically for such configurations is 8.264 (Figure 34). Together with (G), the counts from twenty-five to twenty-seven are all that stands between the theorems of this section and Conjecture 1.3.

Proposition 7.23 (The two statements that remain)**.** *Let $T(Y)$ denote the right side of Lemma 7.6 for a set Y of centres within $\sqrt{6}$ of c. Suppose that*

(G) *for every packing set Y of exactly 24 centres with $2 \le |y| < \sqrt{6}$ and $T(Y) \le 8$ one has $\mathrm{vol}(V(Y) \cap K(Y)) \ge 8$, with equality only if Y is a copy of $\sqrt{2}\,\mathsf{R}$ about c; and*

(C) *for every packing set Y of at least 25 centres with $2 \le |y| < \sqrt{6}$ one has $T(Y) > 8$.*

Then Conjecture 1.3 holds: every Voronoi cell of a unit-ball packing of $\mathbb{R}^4$ has volume at least 8, with equality only for the 24-cell of a copy of $\sqrt{2}\,\mathsf{R}$.

Proof. Let Y be the set of centres other than c within $\sqrt{6}$ of it. Only these have nonempty caps in $B(\sqrt{3/2})$, so Lemma 7.6 gives $\mathrm{vol}(V_c) \ge \mathrm{vol}(V_c \cap B(\sqrt{3/2})) = T(Y)$, and Proposition 7.4 gives $\mathrm{vol}(V_c) \ge \mathrm{vol}(V(Y) \cap K(Y))$. If $|Y| \le 23$, then $\mathrm{vol}(V_c) > 8$ by Theorem 7.8. If $|Y| \ge 25$, then $\mathrm{vol}(V_c) \ge T(Y) > 8$ by (C). If $|Y| = 24$ and $T(Y) > 8$, then $\mathrm{vol}(V_c) > 8$. If $|Y| = 24$ and $T(Y) \le 8$, then $\mathrm{vol}(V_c) \ge 8$ by (G), and equality forces Y to be a copy of $\sqrt{2}\,\mathsf{R}$; then c has 24 contacts, and V_c is the 24-cell by Lemma 7.2. ∎

7.4. **The statements (G) and (C).** This subsection proves (C) for every count from twenty-eight on, shows that twenty-four centres close to a root system leave no room for another within $\sqrt{6}$, proves (G) along the push-outs of the root system, and then sets out what (G), and (C) for the counts from twenty-five to twenty-seven, still require. Throughout, Y is the set of centres y with $2 \le |y| < \sqrt{6}$, $\hat{y} = y/|y|$, $R = \sqrt{3/2}$, and

$$U(Y) = \sum_{y \in Y} S(|y|) - \sum_{\{y,y'\} \subset Y} \Pi\left(\tfrac{|y|}{2}, \tfrac{|y'|}{2}, \langle \hat{y}, \hat{y}' \rangle\right)$$

is the volume of the union of the caps of Lemma 7.6, Π being the pair term of the proof of Theorem 7.8. So $T(Y) = \frac{9}{8}\pi^2 - U(Y)$, and (C) asks that $U(Y) < \frac{9}{8}\pi^2 - 8 = 3.10330\ldots$ whenever $|Y| \ge 25$.

Lemma 7.24 (The pair term)**.** *Let $1 \le h_1, h_2 < R$ and $-1 < u < 1$. Put $\gamma = \arccos u$, $\alpha_k = \arccos(h_k/R)$, $L = \max(-\alpha_1, \gamma - \alpha_2)$, $U = \min(\alpha_1, \gamma + \alpha_2)$, let ϕ_c be the angle of the vector $(h_1 \sin\gamma,\ h_2 - h_1\cos\gamma)$, clipped to $[L, U]$, and let $G(h, x) = R^4 x - 2R^2h^2 \tan x + h^4(\tan x + \frac{1}{3}\tan^3 x)$. Then*

$$\Pi(h_1, h_2, u) = \tfrac{\pi}{4}\big(G(h_2, \phi_c - \gamma) - G(h_2, L - \gamma) + G(h_1, U) - G(h_1, \phi_c)\big)$$

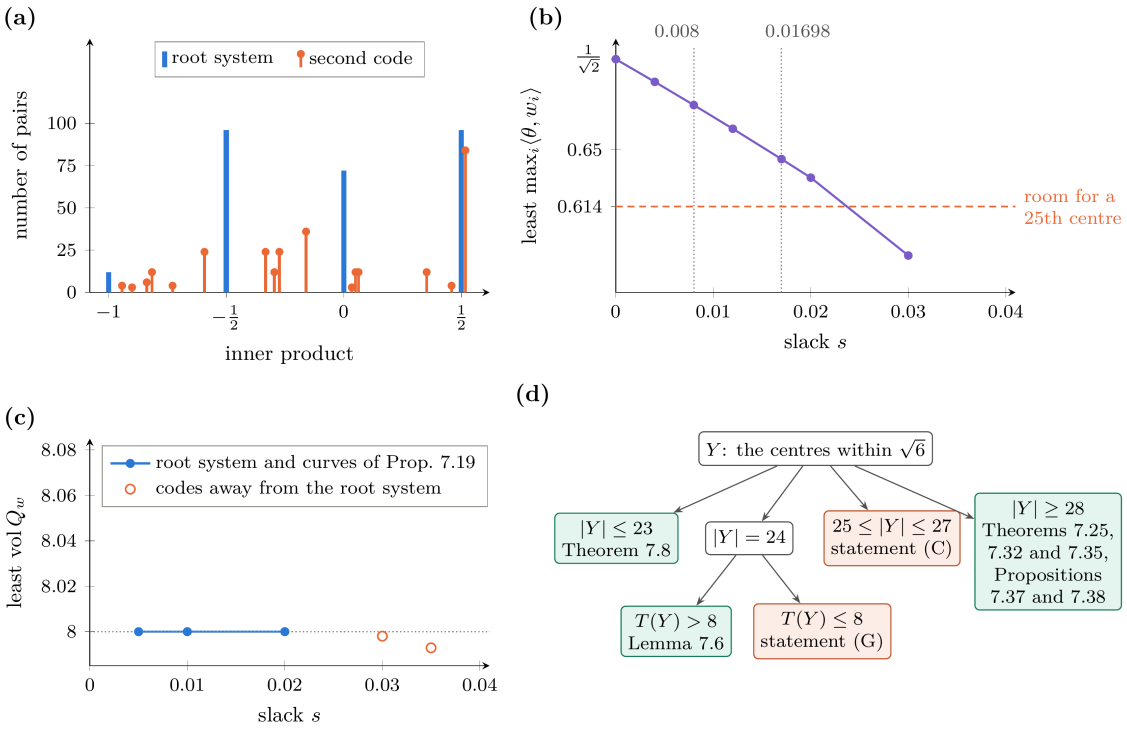


FIGURE 38. Twenty-four-point codes at positive slack, and the two statements that remain. (a) The inner products of the 276 pairs of the root system and of the second code of Proposition 7.21. (b) The deepest hole of a 24-point code of slack s, the least value of $\max_i\langle\theta, w_i\rangle$ found; a further centre within $\sqrt{6}$ beside twenty-four within 2.0161 needs 0.6141 or less, which the codes found reach only near $s = 0.025$, beyond the slack 0.008 of Theorem 5.78 and the slack 0.01698 of the second code. (c) The least contact-cell volume found over codes of slack s. (d) The cases of Proposition 7.23; the green boxes are proved, the orange ones are statement (G) and statement (C) for twenty-five to twenty-seven centres, (C) being proved from twenty-nine on by Theorems 7.25, 7.32 and 7.35 and at twenty-eight by Propositions 7.37 and 7.38. Panels (b) and (c) are floating-point minimisations, evidence and not bounds.

if $L < U$, *and* $\Pi(h_1, h_2, u) = 0$ *otherwise. Moreover* Π *is nonincreasing in* h_1 *and in* h_2, *and nondecreasing in* u.

Proof. Write $x = (p, q)$ with p in the plane P of the two normals w_1, w_2 and $q \perp P$. The two conditions $\langle x, w_k\rangle > h_k$ involve p alone, and for fixed p the condition $|x| < R$ leaves a disc of area $\pi(R^2 - |p|^2)$ for q; so $\Pi = \int \pi(R^2 - |p|^2)\,dp$ over the part of P beyond both lines. In polar coordinates $p = \rho(\cos\phi, \sin\phi)$, with w_1 at $\phi = 0$ and w_2 at $\phi = \gamma$, the admissible ρ run from $\rho_m(\phi) = \max\bigl(h_1/\cos\phi,\, h_2/\cos(\phi - \gamma)\bigr)$ to R, and $\int_{\rho_m}^{R} \pi(R^2 - \rho^2)\rho\,d\rho = \frac{\pi}{4}(R^2 - \rho_m^2)^2$. This is positive exactly for $L < \phi < U$. The two terms of the maximum cross once, at $\tan\phi = (h_2 - h_1\cos\gamma)/(h_1\sin\gamma)$; the second line binds before the crossing and the first after it, and $\int \frac{\pi}{4}(R^2 - h^2\sec^2 x)^2\,dx = \frac{\pi}{4}G(h, x)$. For the monotonicity, read the same integral circle by circle. On the circle of radius ρ the admissible angles form the intersection

of the arcs $(-\beta_1, \beta_1)$ and $(\gamma - \beta_2, \gamma + \beta_2)$, where $\beta_k = \arccos(h_k/\rho) < \frac{\pi}{2}$; the second arc lies in $(-\frac{\pi}{2}, \frac{3\pi}{2})$ and meets the first only inside $(-\frac{\pi}{2}, \frac{\pi}{2})$, so the length of the intersection is $\max(0, \min(\beta_1, \gamma+\beta_2) - \max(-\beta_1, \gamma-\beta_2))$, which does not increase when h_1, h_2 or $\gamma \in [0, \pi]$ increases. The weight $\pi(R^2 - \rho^2)\rho$ depends on none of them. ∎

Theorem 7.25 (Thirty-one or more centres)**.** *If at least* 31 *centres other than* c *lie within* $\sqrt{6}$ *of* c*, then* $U(Y) < 3.0854$*, and so* $\mathrm{vol}(V_c) \geq T(Y) > 8.0179$*. In particular, statement (C) of Proposition* 7.23 *holds for every packing set of at least* 31 *centres.*

Proof. For $2 \leq d \leq \sqrt{6}$ let $p(d) = (T_0(x), \dots, T_r(x))$, where the T_a are the Chebyshev polynomials and $x = (2d - c_1)/c_2$ for rationals c_1, c_2 close to $2 + \sqrt{6}$ and $\sqrt{6} - 2$; let U_k be the Chebyshev polynomials of the second kind, which are the zonal polynomials of S^3 [DGS77].[35] For symmetric matrices $A_0, \dots, A_D$ put

$$K(y, y') = \sum_{k=0}^{D} \frac{U_k(\langle \hat{y}, \hat{y}' \rangle)}{k+1}\, p(|y|)^{\mathsf{T}} A_k\, p(|y'|).$$

Suppose that $A_1, \dots, A_D$ are positive semidefinite and that

(a) $K(y, y') \leq \Pi(\frac{|y|}{2}, \frac{|y'|}{2}, \langle \hat{y}, \hat{y}' \rangle)$ whenever $2 \leq |y|, |y'| \leq \sqrt{6}$ and $\langle \hat{y}, \hat{y}' \rangle \leq a(|y|, |y'|)$, the condition (7.2) for $|y - y'| \geq 2$;

(b) $\begin{pmatrix} A_0 & z \\ z^{\mathsf{T}} & t \end{pmatrix}$ is positive semidefinite for some vector z and number t;

(c) $S(d) + \frac{1}{2}\sum_k p(d)^{\mathsf{T}} A_k p(d) - z \cdot p(d) \leq m$ for $2 \leq d \leq \sqrt{6}$, with $m < 0$.

Then Theorem 4.5, with $p(|y|)$ in place of $p(y)$, $s_y = S(|y|)$ and $\pi_{yy'} = \Pi(\frac{|y|}{2}, \frac{|y'|}{2}, \langle \hat{y}, \hat{y}' \rangle)$, gives by (a), (b) and (c)

$$U(Y) \leq \sum_{y \in Y} \Big(S(|y|) + \tfrac{1}{2} K(y, y) - z \cdot p(|y|) \Big) + \tfrac{t}{2} \leq |Y|\, m + \tfrac{t}{2},$$

and the right side decreases as $|Y|$ grows. The certificate has $D = 12$ and $r = 4$, $m = -0.072797\dots$, and $31m + \frac{t}{2} = 3.08537\dots$, below $\frac{9}{8}\pi^2 - 8$. The semidefinite programme that finds it imposes (a) at about 41 000 sampled triples with a margin of 10^{-5}; the proof does not use the samples. The matrices are rounded to dyadic rationals and are positive semidefinite by an exact LDL^{T}, and t is set to $z^{\mathsf{T}} A_0^{-1} z$ rounded up, which makes (b) exact. For (a), K is a polynomial in $(|y|, |y'|, u)$; its tensor Bernstein coefficients[36] on $[2, d_{\max}]^2 \times [-1, a(d_{\max}, d_{\max})]$, with $d_{\max} = 2.4494898 > \sqrt{6}$, are computed exactly and subdivided at midpoints, and on each box the largest coefficient bounds K, while by Lemma 7.24 Π is at least its value at the corner with the larger distances and the smaller inner product, computed from the closed form in ball arithmetic [Joh17]. Boxes with $|y| > |y'|$ (K and Π are symmetric) or lying wholly beyond $u = a(|y|, |y'|)$ are set aside by exact rational tests, and the branch and bound closes (a) with 191 740 boxes. Condition (c) is checked in the same way on 356 intervals, S being decreasing, and the final inequality in ball arithmetic.

[35] The harmonic polynomials of degree k on S^3 form a space of dimension $(k+1)^2$, and for an orthonormal basis (Y_m) of it the addition theorem reads $\sum_m Y_m(x) Y_m(x') = \frac{k+1}{2\pi^2} U_k(\langle x, x' \rangle)$. Hence $\sum_{y,y'} a_y a_{y'} U_k(\langle \hat{y}, \hat{y}' \rangle) \geq 0$ for all real a_y, which is Lemma 4.1, and the factor $1/(k+1)$ makes each zonal term equal to 1 at $u = 1$, since $U_k(1) = k + 1$.

[36] On an interval $[a, b]$ a polynomial of degree n is a combination of the polynomials $\binom{n}{k}(x-a)^k (b-x)^{n-k}/(b-a)^n$, $0 \leq k \leq n$, which are nonnegative and sum to 1. It therefore lies between the least and the largest of its coefficients in that basis, and on halving the interval the gap between these bounds and its true range shrinks quadratically. On a box the basis is the product of the bases of the coordinates.

The whole check is repeated in Lean from the entries of the certificate alone, in exact dyadic arithmetic: the coefficients of K are rounded up once to multiples of 2^{-320}, which can only raise K because the Bernstein basis is nonnegative, every subdivision is then carried out without rounding, and Π, S and π are bounded in outward-rounded interval arithmetic with 256 fractional bits. It closes (a) with the same 191 740 boxes and (c) with the same 356 intervals (Section 11.2). ∎

Remark 7.26 (Where the certificate of Theorem 7.25 is tight). The bound $|Y|m+\frac{t}{2}$ is affine in the number of centres, with slope $m = -0.072797\ldots$ and value $\frac{t}{2} = 5.342081\ldots$ at $|Y| = 0$, so it crosses $\frac{9}{8}\pi^2 - 8$ at $|Y| = 30.75$; that is why the certificate starts at 31, and why at 30 it gives only 3.15817. By complementary slackness the measures at which the programme attains its value are carried by the points where the certificate is tight, and those points can be read off. In floating point, the function in (c) comes within $3 \cdot 10^{-5}$ of m only at the two ends of the distance range, $d = 2$ and $d = \sqrt{6}$, and lies at least $4.6 \cdot 10^{-4}$ below m in between, $1.7 \cdot 10^{-3}$ at $d = 2.2$. For the three pairs of end distances the slack of (a) falls to the margin 10^{-5} of the programme at the touching inner product $u = a(|y|, |y'|)$ and at three further inner products: $u = -0.821$, -0.426 and -0.121 for two centres at distance 2; $u = -0.770$, -0.430 and -0.067 for one at 2 and one at $\sqrt{6}$; and $u = -0.581$, -0.499 and 0.041 for two at $\sqrt{6}$, where $u = -1$ is tight as well. Off the ends the slack of (a) is at least $2.6 \cdot 10^{-4}$. So the relaxation is extremal on a distribution of centres at distance 2 and at distance $\sqrt{6}$ whose pairs sit at these four or five inner products, the nodes of a Delsarte-type bound, and not at the inner products -1, $-\frac{1}{2}$, 0 and $\frac{1}{2}$ of the root system. Below 31 the value of the programme lies above the largest union that the search of Figure 34 finds (Table 5), so the measures on which it is extremal are not those of the packings found.

With at most 23 centres within $\sqrt{6}$ the bound holds by Theorem 7.8, and with 31 or more by Theorem 7.25 (Figure 39); Theorems 7.32 and 7.35 below settle thirty and twenty-nine, Propositions 7.37 and 7.38 settle twenty-eight, and what remains of (C) is the range from 25 to 27.

Remark 7.27 (Where the kernel stops). The argument is Delsarte's linear programming method [DGS77], with the distance from c carried along as a continuous label, and it is blind in the same way: a kernel on pairs alone bounds the kissing number of $\mathbb{R}^4$ by 25 and not 24 [OS79]. Table 5 gives, for each count, the least value of $|Y|m + \frac{t}{2}$ that such a certificate reaches, against the largest union the search of Figure 34 finds. Below 31 the kernel cannot separate the configurations that exist from those it allows; at 25 it does not even reach the value $3.19636\ldots$ of the root system, which has only twenty-four centres. Adding to the programme the bound of Theorem 5.78, at most 24 centres within 2.0161, leaves these values unchanged. A kernel on triples in the manner of [BV08] does better. With the distances as continuous labels, each anchor weighted by a nonnegative polynomial in its own distance, its programme at thirty centres and angular degree 6 stays at 3.10344 after three rounds of cutting planes, above 3.10330, and each of its inequalities on triples would have to be verified to within about 10^{-8} on a six-dimensional domain, since it enters the bound once for each of the $30 \cdot 29 \cdot 28 = 24\,360$ ordered triples. Giving each direction only the range of its distance as a type (Lemma 4.4) keeps the distances in the inequalities on pairs, which stay three-dimensional, and leaves the inequalities on triples in the three inner products alone; at angular degree 8 this closes the last case at thirty (Theorem 7.32) and the whole of twenty-nine (Theorem 7.35).

TABLE 5. Every count of centres within $\sqrt{6}$ from 25 to 49: the best two-point bound on the union of the caps (floating point, sampled constraints, degree 14 in the angle and 5 in the distance), the largest union found by the search of Figure 34, and the least T found; *none* means that the search found no packing with that count. (C) requires the union to stay below $\frac{9}{8}\pi^2 - 8 = 3.10330$, which the bound gives from 31 on; at 31 the certificate of Theorem 7.25, solved with a margin, proves 3.0854.

$\lvert Y\rvert$	bound	union	T	$\lvert Y\rvert$	bound	union	T
25	3.330	2.839	8.264	38	2.386	0.973	10.131
26	3.360	2.711	8.393	39	2.261	0.853	10.250
27	3.307	2.586	8.517	40	2.125	0.703	10.400
28	3.253	2.512	8.592	41	1.979	0.490	10.614
29	3.199	2.327	8.776	42	1.821	0.326	10.777
30	3.144	2.186	8.917	43	1.647	0.208	10.895
31	3.080	2.055	9.048	44	1.456	none	none
32	3.003	2.049	9.054	45	1.244	none	none
33	2.920	1.790	9.313	46	1.001	none	none
34	2.828	1.648	9.455	47	0.735	none	none
35	2.727	1.502	9.602	48	0.459	none	none
36	2.619	1.240	9.863	49	0.174	none	none
37	2.506	1.243	9.860				

Twenty-four centres close to a root system leave no room for a twenty-fifth. The proof of Lemma 7.2 shows that every unit vector makes an angle of at most 45° with some normalised root. The following versions keep a margin, measured direction by direction, in root-sum-square, and through the inner products alone.

Corollary 7.28 (No room beside a near root system)**.** *Let $2 \le \rho < \sqrt{6}$, put $\tau = a(\sqrt{6}, \rho)$ and $h = 1/\sqrt{2} - \tau$, and let 24 centres $y_1, \dots, y_{24}$ lie within ρ of c. Suppose that, for a labelling of the roots a_i and, in* (i) *and* (ii)*, an orthogonal map σ, one of the following holds:*

(i) $|\hat{y}_i - \sigma(a_i)/\sqrt{2}| < h$ *for every i;*
(ii) $\sum_i |\hat{y}_i - \sigma(a_i)/\sqrt{2}|^2 < 6h^2$*;*
(iii) $\rho = 2/\sqrt{1 - 0.016}$ *and* $\sum_{i<j} \big(\langle \hat{y}_i, \hat{y}_j\rangle - \frac{1}{2}\langle a_i, a_j\rangle\big)^2 \le 0.3059$.

Then no other centre lies within $\sqrt{6}$ of c. For $\rho = 2/\sqrt{1 - 0.016} = 2.0161\dots$ one has $\tau = 0.614038\dots$ and $h = 0.093068\dots$, so (i) *asks for less than* 0.0930 *in each direction and* (ii) *for a root-sum-square distance below $\sqrt{6}\,h = 0.22796\dots$. If 24 centres lie within $2/\sqrt{1 - 4\cdot 10^{-26}}$ of c, no other centre lies within $\sqrt{6}$ of c.*

Proof. Let z be a further centre with $|z| < \sqrt{6}$. By (7.2), and since a increases in each argument on $[2, \sqrt{6}]^2$, $\langle \hat{z}, \hat{y}_i\rangle \le a(|z|, |y_i|) \le \tau$ for every i.

(i) The covering radius of the normalised roots is 45°, so some i has $\langle \hat{z}, \sigma(a_i)/\sqrt{2}\rangle \ge 1/\sqrt{2}$, and then $\langle \hat{z}, \hat{y}_i\rangle > 1/\sqrt{2} - h = \tau$, a contradiction.

(ii) Put $t_i = \langle \hat{z}, \sigma(a_i)/\sqrt{2}\rangle$. Since $\hat{z}$ is a unit vector, $|\hat{y}_i - \sigma(a_i)/\sqrt{2}| \ge t_i - \langle \hat{z}, \hat{y}_i\rangle \ge t_i - \tau$, so

$$\sum_i |\hat{y}_i - \sigma(a_i)/\sqrt{2}|^2 \;\ge\; \sum_i (t_i - \tau)_+^2. \tag{7.6}$$

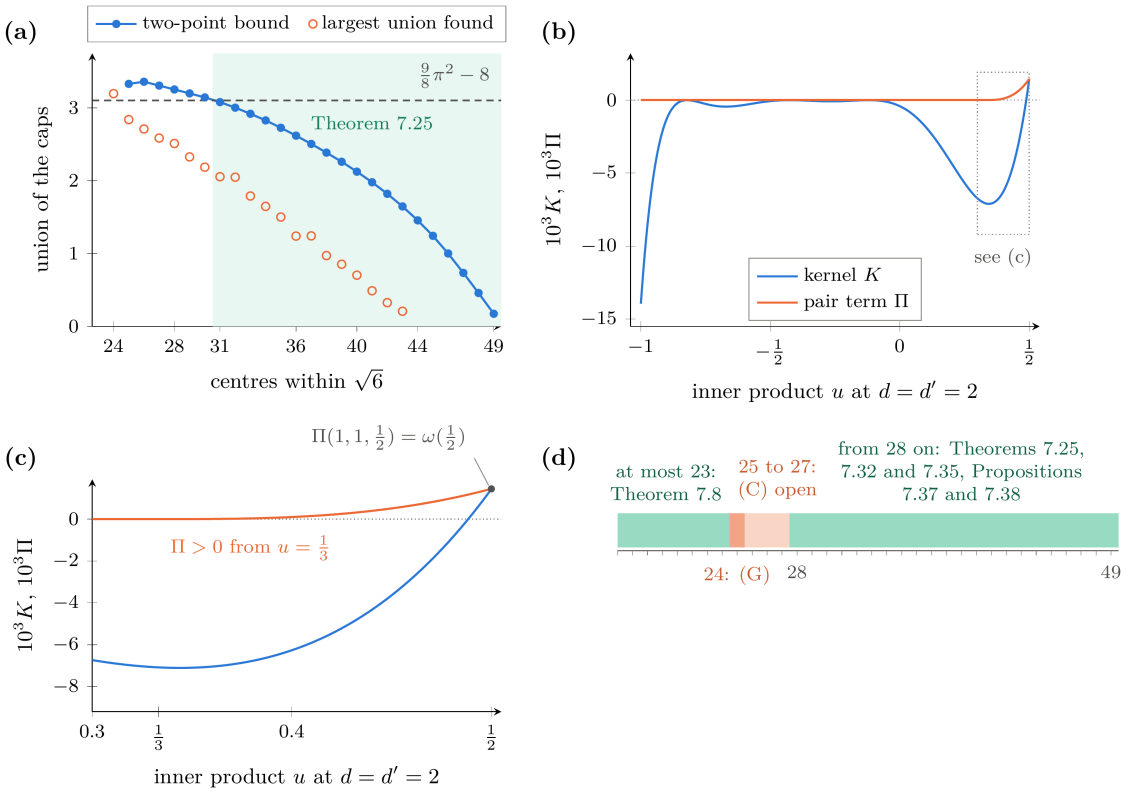


FIGURE 39. Statement (C) from thirty centres on. (a) The best two-point bound on the union of the caps of Lemma 7.6 for M centres within $\sqrt{6}$, and the largest union found by the search of Figure 34 (at 24, the root system); (C) asks for values below $\frac{9}{8}\pi^2 - 8$, and Theorem 7.25 proves it from 31 on (shaded). (b) The kernel K of the certificate of Theorem 7.25 and the pair term Π at $d = d' = 2$, times 10^3. Condition (a) of the proof asks that $K \le \Pi$: the kernel comes within 10^{-5} of 0 at $u = -0.82$, -0.43 and -0.12, and nearly meets Π at $u = \frac{1}{2}$, the inner product of neighbouring roots ($1.4354 \cdot 10^{-3}$ against $1.4454 \cdot 10^{-3}$). (c) The same near $u = \frac{1}{2}$; Π is positive from $u = \frac{1}{3}$, where the two caps first meet. (d) The counts of centres within $\sqrt{6}$ and what settles each; thirty and twenty-nine are settled by the kernel on typed triples of Theorems 7.32 and 7.35, and twenty-eight by Propositions 7.37 and 7.38. Panel (a) is floating point, except the value 3.0854 at 31, which is proved.

The normalised roots are the vectors $(\pm e_j \pm e_k)/\sqrt{2}$, and expanding over them gives $\sum_i t_i^2 = 6$ and $\sum_i t_i^4 = 3$ for every unit vector: the root system is a spherical 5-design. The roots come in pairs $\pm a$, and at most one member of a pair has $t_i > 0$; so the right side of (7.6) is $\sum_k (s_k - \tau)_+^2$ over twelve numbers $s_k \in [0, 1]$, one for each pair, with $\sum_k s_k^2 = 3$ and $\sum_k s_k^4 = \frac{3}{2}$. Put $\gamma = 2\sqrt{2}\,h - 4h^2$, $\beta = \sqrt{2}\,h - \gamma$ and $q(s) = (s - \tau)_+^2 - \beta s^2 - \gamma s^4$. On $[0, \tau]$, $q(s) \ge -s^2(\beta + \gamma\tau^2)$, and $\beta + \gamma\tau^2 = -2h^2(1 - 3\sqrt{2}\,h + 2h^2) < 0$. On $[\tau, 1]$, q is a quartic with $q(1/\sqrt{2}) = q'(1/\sqrt{2}) = 0$ by the choice of β and γ, so $q(s) = (s - 1/\sqrt{2})^2 r(s)$ with r quadratic. Its leading coefficient is $-\gamma < 0$, and it is positive at both ends, since $r(\tau) = -\tau^2(\beta + \gamma\tau^2)/h^2$ and $(1 - 1/\sqrt{2})^2\, r(1) = (1 - \tau)^2 - \beta - \gamma = (h - \frac{1}{\sqrt{2}})(h - \frac{3\sqrt{2}-4}{2})$.

These signs hold for $0 < h < \frac{3\sqrt{2}-4}{2} = 0.1213\ldots$, and here $h \le 1/\sqrt{2} - a(\sqrt{6}, 2) = 0.0947\ldots$; the identities and the signs are checked again in Lean. Thus $q \ge 0$ on $[0,1]$, and

$$\sum_k (s_k - \tau)_+^2 \;\ge\; \beta \sum_k s_k^2 + \gamma \sum_k s_k^4 \;=\; 3\beta + \tfrac{3}{2}\gamma \;=\; 6h^2,$$

against (ii). The bound is attained: at a deep hole of the root system six of the t_i equal $1/\sqrt{2}$ and the others are at most 0.

(iii) Let X and V be the 24×4 matrices with rows $\hat{y}_i$ and $a_i/\sqrt{2}$, so that $V^{\mathsf{T}}V = 6I$ and $P = VV^{\mathsf{T}}/6$ is the orthogonal projection onto the column space of V. Put $A = X^{\mathsf{T}}X$, $E = X^{\mathsf{T}}(I - P)X$, $e = \operatorname{tr} E$ and $\alpha = \|A - 6I\|$, all norms being Frobenius norms. Since $\operatorname{tr} A = 24$ and $\|V^{\mathsf{T}}X\|^2 = 6\operatorname{tr}(X^{\mathsf{T}}PX) = 144 - 6e$,

$$\|XX^{\mathsf{T}} - VV^{\mathsf{T}}\|^2 = \|A\|^2 + 144 - 2\|V^{\mathsf{T}}X\|^2 = \alpha^2 + 12e,$$

and the left side is twice the sum in (iii), the diagonals being equal. For an orthogonal Q, $\|X - VQ\|^2 = e + \|PX - VQ\|^2 = e + 6\|M/6 - Q\|^2$ with $M = V^{\mathsf{T}}X$, and the least value over Q is $e + \frac{1}{6}\sum_k (\mu_k - 6)^2$, where the μ_k are the singular values of M. As $M^{\mathsf{T}}M = 6(A - E)$, $\mu_k = \sqrt{6\lambda_k}$ for the eigenvalues λ_k of $A - E$, and these lie within $\|A - E - 6I\| \le \alpha + e$ of 6, because $\|E\| \le \operatorname{tr} E$ for $E \succeq 0$. Hence

$$\min_Q \|X - VQ\|^2 = e + \sum_k \bigl(\sqrt{\lambda_k} - \sqrt{6}\bigr)^2 \;\le\; e + \frac{(\alpha + e)^2}{\bigl(\sqrt{6 - \alpha - e} + \sqrt{6}\bigr)^2}.$$

Put $\kappa = 2 \cdot 0.3059 = 0.6118$, so that $\alpha^2 + 12e \le \kappa$ under (iii). Then $\alpha \le \sqrt{\kappa}$ and $\alpha + e \le \alpha + \frac{1}{12}(\kappa - \alpha^2) \le \sqrt{\kappa} < 0.7822$, the middle expression being increasing in α for $\alpha < 6$. So the denominator is at least $D_0 = (\sqrt{5.2178} + \sqrt{6})^2 > (2.2842 + 2.4494)^2 > 22.4069$, and, using $e \le \frac{1}{12}(\kappa - \alpha^2)$ in the first term and $e \le \frac{\kappa}{12}$ in the square,

$$e + \frac{(\alpha + e)^2}{D_0} \;\le\; \frac{\kappa - \alpha^2}{12} + \frac{(\alpha + \kappa/12)^2}{D_0} \;=\; \frac{\kappa}{12} + \frac{\kappa^2}{144 D_0} + \frac{\kappa}{6 D_0}\,\alpha - \Bigl(\frac{1}{12} - \frac{1}{D_0}\Bigr)\alpha^2.$$

This is a concave quadratic in α, and its largest value, $\frac{\kappa}{12} + \frac{\kappa^2}{144 D_0} + \frac{\kappa^2}{144 D_0^2}\bigl(\frac{1}{12} - \frac{1}{D_0}\bigr)^{-1}$, is below 0.05124 for every $D_0 \ge 22.4069$. Hence $\min_Q \|X - VQ\|^2 < 0.0515$, while $6h^2 > 0.05197$ at $\rho = 2/\sqrt{1 - 0.016}$. Therefore (ii) holds with $\sigma(x) = Q^{\mathsf{T}}x$ for the minimising Q. (Splitting $[0, \kappa/12]$ into 200 intervals of e and bounding the square roots by rationals gives the same conclusion in exact arithmetic, and this is checked again in Lean.)

For the last statement, the directions of the 24 centres have pairwise inner products at most $a(\rho, \rho) = \frac{1}{2} + 2 \cdot 10^{-26}$, so Theorem 5.75 places them within 0.0113 of a copy of the normalised roots, and (i) applies, as $0.0113 < 1/\sqrt{2} - a(\sqrt{6}, 2.0001)$. ∎

Parts (ii) and (iii) compare the directions with the roots through a labelling. The next statement needs none: it asks only that the twenty-four directions be close to a spherical 5-design, measured by the Gegenbauer sums

$$S_k = \sum_{i,j=1}^{24} G_k\bigl(\langle \hat{y}_i, \hat{y}_j \rangle\bigr), \qquad k = 1, \ldots, 5,$$

where $G_k = U_k/(k+1)$, with U_k the Chebyshev polynomial of the second kind, is the Gegenbauer polynomial of S^3 normalised by $G_k(1) = 1$. Each S_k is nonnegative, and all five vanish for the root system in every position, since it is a spherical 5-design.

Proposition 7.29 (No room from the design defects)**.** *Let 24 centres $y_1, \dots, y_{24}$ lie within $\rho = 2/\sqrt{1-0.016}$ of c. If*

$$\tfrac{1}{963}\,(103\,S_1 + 191\,S_2 + 280\,S_3 + 230\,S_4 + 159\,S_5) < 0.2678,$$

then no other centre lies within $\sqrt{6}$ of c.

Proof. Write $c_1, \dots, c_5$ for the weights $103/963, \dots, 159/963$, and let $p = \sum_{k=0}^{5} p_k G_k$ with

$$(p_0, \dots, p_5) = \Big(\tfrac{39954677}{453230000000},\ \tfrac{41}{89653},\ \tfrac{29}{33806},\ \tfrac{17}{14206},\ \tfrac{95}{98183},\ \tfrac{52}{86257}\Big).$$

Then $p(t) < 0$ for $-1 \le t \le t_0$, where $t_0 = 0.614039$. Put

$$A(t) = 10^{-8}(26327 + 59724\,t + 112736\,t^2), \qquad B(t) = 10^{-8}(260109 + 4473253\,t + 5671392\,t^2),$$

and $R = -p - (1+t)A^2 - (t_0 - t)B^2$. The polynomial R has degree at most 5, constant term $4.99996\ldots \cdot 10^{-8}$, and its other five coefficients below $4.5 \cdot 10^{-10}$ in absolute value, so $R(t) > 4.8 \cdot 10^{-8}$ for $|t| \le 1$. For $-1 \le t \le t_0$ the first two terms of

$$-p(t) = (1+t)A(t)^2 + (t_0 - t)B(t)^2 + R(t)$$

are nonnegative, hence $p(t) < 0$.[37] Moreover $t_0 > \tau = a(\sqrt{6}, \rho)$, since $\tau^2 = (2+\rho^2)^2/(24\rho^2) = 139129/369000 < t_0^2$.

Let z be a further centre with $|z| < \sqrt{6}$. As in the proof of Corollary 7.28, $\langle \hat{z}, \hat{y}_i \rangle \le \tau$ for every i, so $\sum_i p(\langle \hat{z}, \hat{y}_i \rangle) < 0$. On the other hand, the addition formula $G_k(\langle x, y \rangle) = h_k^{-1} \sum_m Y_{km}(x) Y_{km}(y)$, with $h_k = (k+1)^2$ and an orthonormal basis Y_{km} of the harmonics of degree k, and the Cauchy–Schwarz inequality give

$$\Big|\sum_i G_k(\langle \hat{z}, \hat{y}_i \rangle)\Big| = h_k^{-1} \Big|\sum_m Y_{km}(\hat{z}) \sum_i Y_{km}(\hat{y}_i)\Big| \le \sqrt{S_k},$$

because $\sum_m Y_{km}(\hat{z})^2 = h_k$ and $\sum_m \big(\sum_i Y_{km}(\hat{y}_i)\big)^2 = h_k S_k$. Hence

$$\sum_i p(\langle \hat{z}, \hat{y}_i \rangle) \ \ge\ 24 p_0 - \sum_{k=1}^{5} |p_k| \sqrt{S_k} \ \ge\ 24 p_0 - \Big(\sum_k \frac{p_k^2}{c_k}\Big)^{1/2} \Big(\sum_k c_k S_k\Big)^{1/2}.$$

In exact arithmetic $(24 p_0)^2 > 0.26785 \sum_k p_k^2 / c_k$, so under the hypothesis the right side is positive, a contradiction. The sign of p, the inequality $0.614039 > \tau$, the bound 0.26785 and the vanishing of $S_1, \dots, S_5$ for the root system are checked again in Lean. ∎

Remark 7.30 (What (C) needs from twenty-five to twenty-seven centres)*.* Let Y be a packing set of M centres, $25 \le M \le 28$, with $U(Y) \ge 3.10330$. At $M = 28$ there is no such set (Proposition 7.38), but what follows holds there as well and shows what the kernels had to get past. Since $U(Y) \le \sum_y S(|y|)$ and S falls steeply, from $S(2) = 0.13896$ to $S(2.0161) = 0.12733$, $S(2.1) = 0.07581$ and $S(2.2) = 0.03340$, most of the union must come from centres close to c: if at most 22 centres lay within ρ, then $U(Y) \le 22 S(2) + (M - 22) S(\rho)$, which is below 3.10330 once $(M - 22) S(\rho) < 0.0461$, that is, once $\rho \ge 2.313$ for $M = 28$ and $\rho \ge 2.268$ for $M = 25$. Hence at least 23 centres lie within 2.313, and by Theorem 5.78 at most 24 within 2.0161. The two cases differ in kind.

[37] The quadratics A and B solve $-p - 5 \cdot 10^{-8} = (1+t)A^2 + (t_0 - t)B^2$, six equations for their six coefficients, rounded to eight decimals, and R absorbs the rounding. Both nearly vanish at -0.7256 and -0.0633, where p has two pairs of complex roots, 0.007 and 0.010 from the real axis, and comes within 10^{-7} of zero; the only real root of p is 0.61474. That is why the representation needs eight decimals. A Taylor bound about the midpoint of each of 29 subintervals of $[-1, t_0]$, in rational arithmetic, gives the sign of p independently.

With 24 centres close to c, the others must find room beside them, and Corollary 7.28 says there is none as soon as the 24 directions are within 0.093 each of a root system, or within 0.228 of one in root-sum-square. A centre within $\sqrt{6}$ beside twenty-four centres within 2.0161 needs a hole of their directions at inner product 0.6141 or less; the local searches behind Figure 38(b) find no 24-point code of slack up to 0.02 with a hole deeper than 0.632, and at slack 0.008 none deeper than 0.678. What this case needs is therefore the shape half of Theorem 5.78 in a weak form, with a tolerance of 0.228 in root-sum-square, 0.0465 in root mean square, in place of the $\frac{1}{48}$ in root mean square of Theorem 7.9. At $M = 28$ the kernels settle this case without it (Proposition 7.38). Proposition 7.21 does not contradict it, since the deepest hole of its code lies at inner product 0.659 (over 3000 local minimisations). The relaxations at hand do not give it. A three-point certificate on the directions does not detect holes. The value of a labelled certificate is a polynomial in the number n of directions of slack 0.008, and it excludes n where it is negative. Without a further direction, no certificate of degree 10 excludes $n = 24.9$, and none of degree 12 excludes $n = 24.7$; with a further direction at inner product at most 0.6141 with each, none of degree 7, 10 or 12 excludes $n = 24$, and none of degree 10 or 12 excludes even $n = 24.5$. The sampling of the constraints can only lower these values, so a positive value is no artefact of it.[38] Measured in points, the further direction would have to count for at least 0.7 of a point, and these certificates give it at least half a point less; the kernel of Theorem 7.25, which uses pairs only, is weaker still. The second level of the hierarchy determines twenty-four points exactly [LLM24], but not with a margin on pairs, which the root system itself uses up (Proposition 7.10). At this slack the statement cannot pin the inner products one by one, since they need not lie near those of a root system (Remark 7.11), but by Corollary 7.28(iii) a bound on the sum of their squared deviations suffices, and Proposition 7.29 asks for less still: a bound on five weighted design defects, with no labelling. The codes of slack 0.008 found by local search reach 0.0905 against the 0.2678 it allows (twelve starts). The second level does not give that bound from its value alone: at slack 0.008 and the degrees $(14, 16)$ its value is 24.555 (Section 7.1), which leaves more than half a point of room. Nor does the programme with the number of points fixed at 24, at most $\binom{24}{2} + \frac{1}{100}$ pairs, and the weighted defect as its objective. At the degrees $(14, 16)$ its moment side reads 1.44 at a relative infeasibility of $4 \cdot 10^{-4}$, unchanged over its last two iterations, with $S_1, \dots, S_5$ near 0.24, 0.24, 0.46, 0.73 and 6.4, and for these moments no polynomial of degree at most 11 in place of p in the proof of Proposition 7.29 excludes a further centre. At this slack the relaxation does not confine twenty-four points closely enough to a root system for the defects to settle the case.

With 23 centres close to c, the case is Theorem 7.8 with two or more further centres, and Theorem 5.72 has a margin of only $4.4 \cdot 10^{-5}$ there, against $S(2.2) = 0.0334$ for each further centre within 2.2; so the further centres must be compensated by their own pair terms with the 23. Packings leave room here: minimised with at least 23 centres held within ρ, T does not come below 8.353, 8.472, 8.681 and 8.781 for $M = 25, \dots, 28$. The relaxations do not see that room. The root system with one centre pushed out along its ray to a distance between 2.0161 and 2.197, together with a fictitious centre near $\sqrt{6}$ in one of its holes, has a union of caps at least $\frac{9}{8}\pi^2 - 8$; the fictitious centre needs inner products only up to 0.612 with the others, and neither the kernel of Theorem 7.25 nor a three-point kernel on the directions alone tells this configuration from a packing. Nor do the directions alone settle the case.

[38] With the normalisation $c_{AA} = -1$ and total trace at most 1000, the least values at $n = 24$ fall from 0.127 at degree 10 to 0.0096 at degree 12. They fall as fast when the further direction may lie at inner product 0.71, where the root system with a direction in one of its deep holes meets every constraint, from 0.144 to 0.0116, so the fall does not point towards exclusion.

The argument of Corollary 7.28(ii), run with twenty-three directions and two further centres near $\sqrt{6}$, needs the directions within root-sum-square 0.132 of a rotated root system with one root removed, the least budget being spent at two directions 24.1° on either side of the removed root. Twenty-three-point codes of slack 0.008 lie up to 0.225 from the nearest such system, and a second family of twenty-three-point codes, far from every subset of the root system, reaches slack 0.00811, just above 0.008. A proof of this case has to use the distances, through T, as (G) does. What the pair terms give is Lemma 7.31: when $T \le 8$, twenty-four of the centres lie within 2.444, so that the case of twenty-three close centres is again a question about the twenty-four closest.

The values in this remark other than those of Corollary 7.28, Proposition 7.29, and Theorem 5.78 are optima of semidefinite programmes or results of local searches, computed in floating point. They show where the methods stop and are used in no proof.

Lemma 7.31 (Twenty-four centres within 2.444)**.** *Let Y be a packing set of M centres with $2 \le |y| < \sqrt{6}$, where $25 \le M \le 33$, and suppose that $T(Y) \le 8$. Then at least 24 centres of Y lie within 2.444 of c. If Z consists of the 24 centres of Y closest to c and d_{25} is the distance of the next, then $T(Z) \le 8 + (M-24)\,S(d_{25})$.*

Proof. For $X \subseteq Y$ the union of the caps of Y is at most that of X together with the caps of the centres of $Y \setminus X$, so

$$T(X) \;\le\; T(Y) + \sum_{y \in Y \setminus X} S(|y|). \tag{7.7}$$

If at most 22 centres lie within 2.444, then, S being decreasing, $U(Y) \le \sum_y S(|y|) \le 22S(2) + 11S(2.444) = 3.05722\ldots < \frac{9}{8}\pi^2 - 8$, and $T(Y) > 8$. Suppose that exactly 23 lie within 2.444, and let X be these. The proof of Theorem 7.8 bounds $T(X)$ from below for any 23 centres at distances in $[2, \sqrt{6})$: if one of them is at distance $D = 2.1648$ or more, Step 1 gives $T(X) \ge A_* + s(D)$, and otherwise Step 2 gives $T(X) \ge A_* + B$. As $s(D) = 0.0928820\ldots < B$, in either case $T(X) \ge A_* + s(D) = 8 + 2.6491\ldots \cdot 10^{-5}$. The other $M - 23 \le 10$ centres lie beyond 2.444, where $S(2.444) = 2.5324\ldots \cdot 10^{-6}$, and (7.7) gives $T(Y) \ge T(X) - 10\,S(2.444) \ge 8 + 1.1 \cdot 10^{-6} > 8$. Thus at least 24 centres lie within 2.444. The last statement is (7.7) with $X = Z$, the $M - 24$ other centres being at distance at least d_{25}. The constants are computed in ball arithmetic at 200 bits; Figure 40 shows them. ∎

With 34 centres the same argument would need $11\,S(2.444)$, which exceeds the room $2.6491 \cdot 10^{-5}$ of Theorem 7.8; that count is settled by Theorem 7.25 in any case.

The kernel of Theorem 7.25 can be sharpened by splitting into cases according to how many centres lie within given distances of c. At thirty centres this leaves only configurations of the kind that the first part of Remark 7.30 describes, near-contacts with further centres in their holes, and those a kernel on pairs does not separate from configurations that do not exist. A kernel on triples of directions does, once each direction carries the range of its distance from c as a type. Such kernels are those of Lemma 4.4, and they enter the proofs through (4.3).

Theorem 7.32 (Thirty centres)**.** *If exactly 30 centres other than c lie within $\sqrt{6}$ of c, then $T(Y) > 8$. With Theorem 7.25, statement (C) of Proposition 7.23 holds for every packing set of at least 30 centres.*

Proof. Let $N(\rho)$ be the number of centres within ρ of c; by Theorem 5.78, $N(2.0161) \le 24$. Call a *case* a set of bounds $\ell_j \le N(\rho_j) \le h_j$ at finitely many rational radii $2 < \rho_1 < \cdots <$

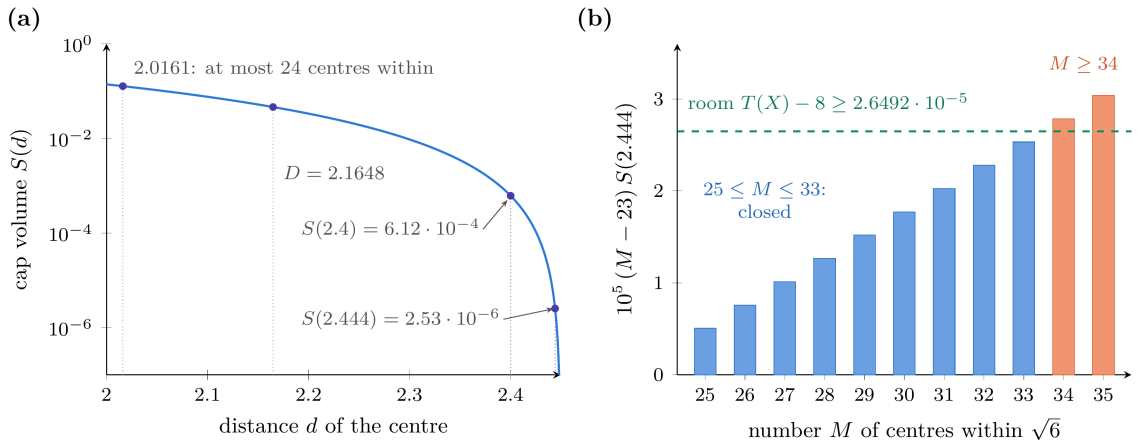


FIGURE 40. The constants of Lemma 7.31. (a) The volume $S(d)$ of the cap of a centre at distance d, which vanishes at $\sqrt{6}$, with the radii of the argument: 2.0161 of Theorem 5.78, D of Step 1 of Theorem 7.8, 2.4 of Theorem 7.32 and 2.444. (b) The caps of the $M - 23$ centres beyond 2.444 against the room that Theorem 7.8 leaves above 8 for the twenty-three closest: they fit under it for $25 \leq M \leq 33$ and not from $M = 34$.

$\rho_J < \sqrt{6}$, always including $N(2.0161) \leq 24$. The radii cut $[2, d_{\max}]$ into bins; let n_b be the number of centres in bin b and e_b its lower end. Take a certificate with properties (a) and (b) of the proof of Theorem 7.25 and, in place of (c), bounds

$$f(d) = S(d) + \tfrac{1}{2}\textstyle\sum_k p(d)^{\mathsf{T}} A_k p(d) - z \cdot p(d) \leq m_b \qquad \text{on each bin } b,$$

with no condition on the signs of the m_b. The chain of inequalities in that proof gives $U(Y) \leq \sum_b n_b m_b + \frac{t}{2}$. If $U(Y) \geq \frac{9}{8}\pi^2 - 8$, then also $\sum_b n_b S(e_b) \geq \sum_y S(|y|) \geq U(Y) \geq \frac{9}{8}\pi^2 - 8$, because S decreases and the pair terms are nonnegative; so for every $s \geq 0$

$$U(Y) \leq \sum_b n_b(m_b + sS(e_b)) - s(\tfrac{9}{8}\pi^2 - 8) + \tfrac{t}{2},$$

and $T(Y) > 8$ follows once the largest value of the right side over the integer vectors (n_b) that the case allows, with $\sum_b n_b = 30$, is below $\frac{9}{8}\pi^2 - 8$. That largest value is found exactly by dynamic programming over the bins.

Split first at $N(2.1) \leq 21$ against $N(2.1) \geq 22$, then the second case at $N(2.4) \leq 24$ against $N(2.4) \geq 25$, then $N(2.1) \geq 22$, $N(2.4) \geq 25$ at $N(2.1) \leq 23$ against $N(2.1) \geq 24$, and $22 \leq N(2.1) \leq 23$, $N(2.4) \geq 25$ at $N(2.2) \leq 23$ against $N(2.2) \geq 24$. The five cases so obtained cover every packing, and four of them carry certificates, with $D = 12$ and $r = 4$: the case $N(2.1) \leq 21$ with bound 3.0617, the two cases with $22 \leq N(2.1) \leq 23$ and $N(2.4) \geq 25$ with 3.0992 and 3.0968, and the case $N(2.1) \geq 24$, $N(2.4) \geq 25$ with 3.0845. Each of them is checked as in Theorem 7.25: exact positivity, the branch and bound of (a) over the whole admissible domain with Π in ball arithmetic, the bin bounds by Bernstein coefficients (an interval that meets two bins held to the smaller bound), and the dynamic programme and the final inequality in ball arithmetic. The four branch and bounds close 185 million boxes, 134 million of them in the tightest case, and evaluate Π in ball arithmetic at 29.6 million of them.

In the fifth case, $N(2.1) \geq 22$ and $N(2.4) \leq 24$, the pairs do not suffice (Remark 7.33), and the triples of Lemma 4.4 are added. Suppose that $T(Y) \leq 8$ in this case. By Proposition 7.34 below, whose certificates are of the kind just described and do not use this theorem, at least 22 centres lie within 2.05, 23 within 2.15 and 24 within 2.25. So exactly 24 lie within 2.25, none in $(2.25, 2.4]$, and six in $(2.4, \sqrt{6})$. Give a centre the type a, b or f as its distance from c lies in $[2, 2.05]$, $(2.05, 2.25]$ or $(2.4, \sqrt{6})$, and give its direction $\hat{y}$ the same type; distinct centres have distinct directions, since two centres on one ray from c are less than $\sqrt{6} - 2$ apart. With the bins $[2, 2.05]$, $(2.05, 2.15]$, $(2.15, 2.25]$ and $(2.4, \sqrt{6})$, the vector (n_b) of bin counts is one of

$$(22, 1, 1, 6), \quad (22, 2, 0, 6), \quad (23, 0, 1, 6), \quad (23, 1, 0, 6), \quad (24, 0, 0, 6), \tag{7.8}$$

and it fixes the number N_{st} of pairs and N_{str} of triples of centres of each combination of types. Theorem 4.5, used without (a), gives

$$U(Y) \leq \sum_y f(|y|) + \tfrac{t}{2} + \sum_{\{y,y'\}} \bigl(K(y, y') - \Pi_{yy'}\bigr), \qquad \Pi_{yy'} = \Pi\bigl(\tfrac{|y|}{2}, \tfrac{|y'|}{2}, \langle \hat{y}, \hat{y}' \rangle\bigr),$$

and adding (4.3) for the typed directions gives

$$\begin{aligned} U(Y) \leq \sum_y \bigl(f(|y|) + q_{\tau(y)}\bigr) + \tfrac{t}{2} + \sum_{\{y,y',y''\}} R_{\tau(y)\tau(y')\tau(y'')} \\ + \sum_{\{y,y'\}} \bigl(K(y, y') - \Pi_{yy'} + Q_{\tau(y)\tau(y')}(\langle \hat{y}, \hat{y}' \rangle)\bigr), \end{aligned}$$

with R evaluated at the three inner products of the directions. Suppose that $f \leq m_b$ on bin b; that $K(y, y') - \Pi_{yy'} + Q_{st}(\langle \hat{y}, \hat{y}' \rangle) \leq \kappa_{st}$ whenever $|y|$ lies in the range of type s, $|y'|$ in that of type t, and $\langle \hat{y}, \hat{y}' \rangle \leq a(|y|, |y'|)$; and that $R_{str} \leq \kappa_{str}$ at every (u, v, w) for which the matrix with rows $(1, u, v)$, $(u, 1, w)$, $(v, w, 1)$ is positive semidefinite and each of u, v, w is at most the value of a at the upper ends of the two ranges concerned. Every pair and every triple of centres meets these conditions, because $a(d, d')$ increases in each argument on $[2, \sqrt{6}]$: its derivative in d has the sign of $d^2 - d'^2 + 4 > 0$. Then

$$U(Y) \leq \sum_b n_b (m_b + q_{\tau(b)}) + \tfrac{t}{2} + \sum_{s \leq t} N_{st}\, \kappa_{st} + \sum_{s \leq t \leq r} N_{str}\, \kappa_{str}, \tag{7.9}$$

whatever the signs of the constants, and it remains to see that the right side is smaller than $\frac{9}{8}\pi^2 - 8$ at each vector (7.8).

The certificate has $D = 12$ and $r = 4$ in the pair kernel and $d = 8$ in (4.2). Its matrices are rounded to dyadic rationals; the A_k and the matrix of (b) are positive semidefinite by exact LDL^{T} factorisations, as before, and the f_k and F_k^s, shifted by 2^{-40} times the identity, are positive definite by the same test. The constants κ and m_b are the largest values found on dense samples and by local maximisation, raised by $2 \cdot 10^{-5}$ for the pairs, $5 \cdot 10^{-6}$ for the triples and $2 \cdot 10^{-6}$ for the bins, and each is then proved. The six inequalities on pairs are checked as (a) in the proof of Theorem 7.25, the polynomial $K + Q_{st}$ being bounded above by tensor Bernstein coefficients and Π below at a corner of each box in ball arithmetic; they close 47 646 boxes. The bins take 4131 intervals. Each R_{str} is a polynomial in (u, v, w); on a box it is bounded by its value at the centre and the first- and second-order Taylor terms, in interval arithmetic, boxes on which the determinant of the Gram matrix is negative throughout are set aside, and when two or three types coincide the inner products are ordered, as the symmetry of R allows. The nine branch and bounds close 8 275 056 boxes. The right side of (7.9), evaluated in ball arithmetic, is largest at $(22, 1, 1, 6)$, where it is 3.08721, below $\frac{9}{8}\pi^2 - 8 = 3.10330\ldots$. Hence $T(Y) > 8$ in the fifth case as well. ∎

Remark 7.33 (Why the pairs do not suffice at thirty). In the fifth case of the proof of Theorem 7.32 a configuration with $T(Y) \le 8$ would consist of two shells: twenty-four centres within 2.25, which by (7.7) carry by themselves a union of caps of at least 3.0996, and six beyond 2.4, with caps of volume at most $S(2.4) = 0.00061$ each. Each far centre would sit in a hole of the 22 to 24 directions within 2.1, at inner product at most $a(2.1, \sqrt{6}) = 0.6231$ with every one of them, and two far centres would be at least 48.2° apart. Such holes are what a kernel on pairs does not see. The optimum of the two-point programme on the subcase $22 \le N(2.1) \le 23$, $N(2.2) = N(2.4) = 24$ is 3.1112, no single further split at the radii 2.0161, 2.05, 2.15, 2.2, 2.25, 2.3 or 2.35 brings both halves below 3.1097, and on the subcase $N(2.2) \le 23$ degrees up to 16 in the angle and 6 in the distance give 3.1012, too close to 3.10330 for a certificate with a usable margin. With the triples of Lemma 4.4 at degree 8 the optimum of the programme on the whole case falls to 2.941, and the certificate of the proof, rounded and with its margins, gives 3.08721.[39]

Below thirty centres the tree of Theorem 7.32 no longer closes its cases, but a single split still does, at several radii, and at thirty it supplies the counts on which the fifth case of that proof rests. The counts it gives say where the centres of a counterexample to (C) must lie.

Proposition 7.34 (Where the centres lie from twenty-five to thirty)**.** *Let Y be a packing set of exactly M centres with $2 \le |y| < \sqrt{6}$, where $25 \le M \le 30$, and suppose that $T(Y) \le 8$. Then for each radius ρ at which Table 6 has an entry in the row of M, at least that many centres of Y lie within ρ of c. At least 24 centres of Y lie within 2.25 of c if $M = 30$, within 2.3 if $M = 29$, and within 2.35 if $M = 27$ or 28; and at least 8 lie within 2.0161 if $M = 29$.*

At $M = 30$ and $M = 29$ the proposition enters the proofs of Theorems 7.32 and 7.35, which then show that no such Y exists.

Proof. For an entry n at the radius ρ, take in the proof of Theorem 7.32 the tree with M centres in place of thirty and the single split $N(\rho) \le n - 1$ against $N(\rho) \ge n$. Its first case carries a certificate of degree 12 in the angle and 4 in the distance, and the largest value the certificate gives over the count vectors of that case lies below $\frac{9}{8}\pi^2 - 8$; the largest of these values is 3.10276, at $M = 27$ and $\rho = 2.35$. Thus $T(Y) > 8$ in the first case, and $N(\rho) \ge n$ whenever $T(Y) \le 8$. An entry that repeats the one to its left needs no certificate of its own. The last statement is the split $N(\rho) \le 23$ against $N(\rho) \ge 24$ at $\rho = 2.25$, 2.3 and 2.35, and $N(2.0161) \le 7$ against $N(2.0161) \ge 8$ at $M = 29$. Each certificate is found and checked as those of Theorem 7.32, with the samples of the programme refined along the antipodal pairs, where the caps are disjoint and Π vanishes.[40] ∎

The caps alone give the counts in parentheses. Since $\sum_y S(|y|) \ge U(Y) \ge \frac{9}{8}\pi^2 - 8$ when $T(Y) \le 8$, and S decreases, at most k centres can lie within ρ only if $kS(2) + (M-k)S(\rho) \ge \frac{9}{8}\pi^2 - 8$. The pair terms add most where the centres are most crowded. With twenty-nine centres the caps force only two within 2.05 and fifteen within 2.1, and the certificates nineteen and twenty-two (Figure 41).

[39] The optima quoted here are those of the semidefinite programmes in floating point, with sampled constraints; they are not bounds on U and only show where each method ends. The value of the certificate is a bound, proved as described.

[40] The margin by which the kernel is held below Π costs more here than with twenty-four centres. It is given up at each of the $\binom{M}{2}$ pairs, so that at $M = 29$ a margin of $5 \cdot 10^{-5}$ raises the bound by about 0.02. The margins are therefore chosen case by case, between $2.5 \cdot 10^{-6}$ and $5 \cdot 10^{-5}$, according to the room that the floating-point optimum of the case leaves below the level.

TABLE 6. Proposition 7.34. For M centres within $\sqrt{6}$ of c with $T(Y) \leq 8$, the least number of centres within ρ of c that the case certificates prove, and in parentheses the least number that the caps alone force. A count within one radius holds within every larger one. The row 24 restates Proposition 7.42, and at 2.1 the row 30 is the first split in the proof of Theorem 7.32; an entry with only a number in parentheses means that no count above that number is proved.

	radius ρ			
M	2.05	2.1	2.15	2.2
24	20 (18)	22 (21)	23 (22)	23 (22)
25	17 (15)	21 (20)	22 (21)	23 (22)
26	15 (12)	20 (18)	22 (21)	23 (22)
27	14 (9)	20 (17)	21 (20)	22 (21)
28	(6)	21 (16)	22 (19)	23 (21)
29	19 (2)	22 (15)	23 (19)	23 (21)
30	22 (0)	22 (14)	23 (18)	23 (20)

Three limits remain. At 2.0161, where the case of twenty-three close centres of Remark 7.30 is decided, the single splits give little: they prove eight centres within 2.0161 at $M = 29$, while the optimum of the programme on the first case already exceeds the level when that case asks for twelve there, for three at $M = 25$ and 28, and for one at $M = 26$ and 27. Twenty-four centres lie within 2.444 by Lemma 7.31; a single split brings that radius down to 2.25 at $M = 30$, 2.3 at $M = 29$ and 2.35 at $M = 27$ and 28, but not below 2.4 for $M = 25$ or 26, where the optimum on the first case is 3.1124 and 3.1068. And, as for (G), the counts say nothing about the directions: the case of twenty-four close centres needs each of them within 0.093 of a root system, and the case of twenty-three a certificate that tells a packing from the fictitious configurations of Remark 7.30, except at $M = 28$, where Proposition 7.38 settles both.

The counts of Proposition 7.34 are enough at twenty-nine centres: with them in place of the tree of Theorem 7.32, the kernel on typed triples closes the case at once.

Theorem 7.35 (Twenty-nine centres)**.** *If exactly* 29 *centres other than c lie within $\sqrt{6}$ of c, then $T(Y) > 8$.*

Proof. Suppose that $T(Y) \leq 8$. By Proposition 7.34 at least 8 centres lie within 2.0161, 19 within 2.05, 22 within 2.1, 23 within 2.15 and 24 within 2.3, and by Theorem 5.78 at most 24 lie within 2.0161. Give a centre the type a, b or f as its distance from c lies in $[2, 2.05]$, $(2.05, 2.3]$ or $(2.3, \sqrt{6})$, and give its direction the same type. With the seven bins $[2, 2.0161]$, $(2.0161, 2.05]$, $(2.05, 2.1]$, $(2.1, 2.15]$, $(2.15, 2.2]$, $(2.2, 2.3]$ and $(2.3, \sqrt{6})$, the integer vectors (n_b) with $\sum_b n_b = 29$ that meet these bounds number 23 396. They fall into 51 classes by the numbers of centres of each type, and the class fixes the numbers N_{st} of pairs and N_{str} of triples of each combination of types. The derivation of (7.9) applies word for word, with these bins and types, and it remains to see that its right side is below $\frac{9}{8}\pi^2 - 8$ at each of the 23 396 vectors.

The certificate again has $D = 12$ and $r = 4$ in the pair kernel and $d = 8$ in (4.2), and it is made exact and checked as in the proof of Theorem 7.32. The room is smaller than at thirty, and the margins are smaller with it: $5 \cdot 10^{-6}$ for the pairs and 10^{-6} for

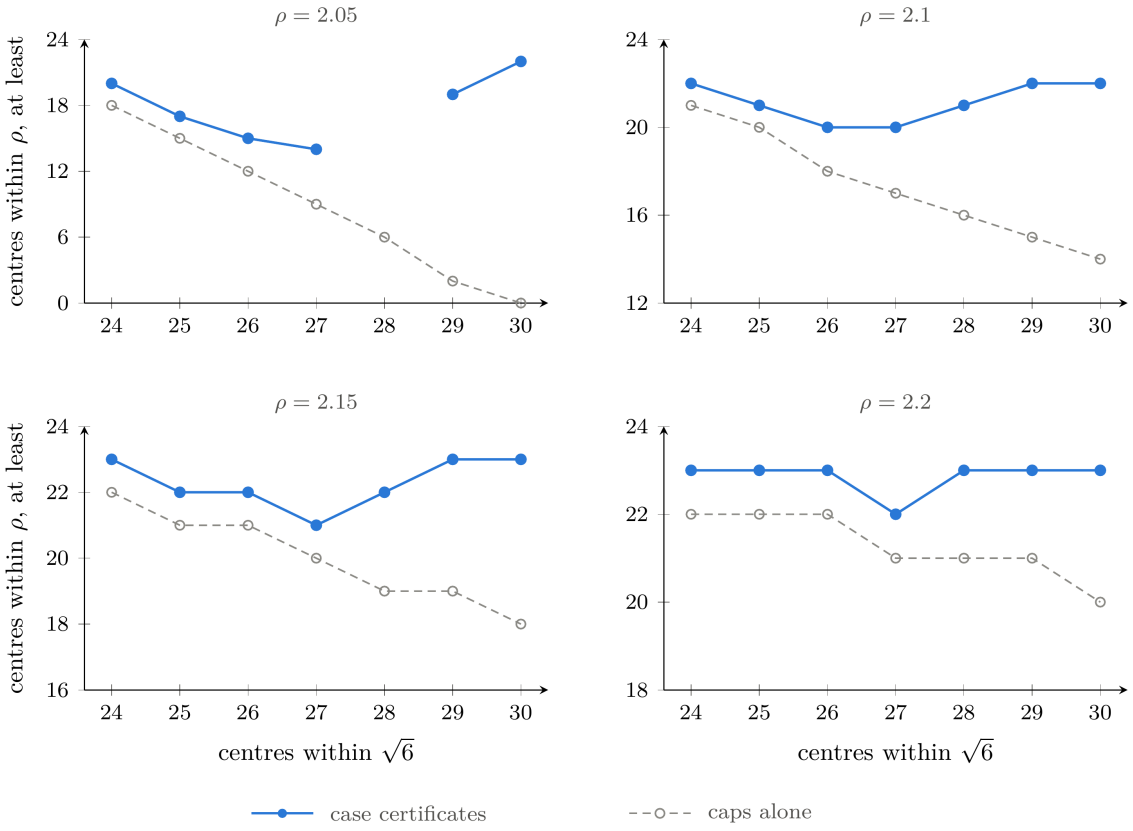


FIGURE 41. Proposition 7.34: for M centres within $\sqrt{6}$ of c with $T(Y) \le 8$, the least number within 2.05, 2.1, 2.15 and 2.2 of c, as the case certificates prove it (filled marks) and as the caps alone force it (open marks). At $M = 24$ the counts are those of Proposition 7.42, and at $M = 30$ and $\rho = 2.1$ that of the first split in the proof of Theorem 7.32.

the triples and for the bins, each given up at every one of the 406 pairs or 3654 triples. The six inequalities on pairs close 21 895 boxes, the bins take 17 762 intervals, and the ten inequalities on triples, one for each combination of types, close 16 301 757 boxes. The right side of (7.9), evaluated in exact rational arithmetic, is largest at $(19, 0, 4, 0, 0, 5, 1)$, where it is 3.10147, below $\frac{9}{8}\pi^2 - 8 = 3.10330\ldots$. ∎

Remark 7.36 (Where the certificates of Theorems 7.32 and 7.35 are tight). At twenty-nine centres the seven bin bounds m_b agree to within 10^{-4}: in floating point the function f of the proof of Theorem 7.32 stays within $2.2 \cdot 10^{-4}$ of -1.0883 on the whole of $[2, \sqrt{6}]$. Up to $3 \cdot 10^{-3}$, the right side of (7.9) therefore depends only on the numbers n_{a}, n_{b} and n_{f} of centres of the three types, and it is largest at $(n_{\mathsf{a}}, n_{\mathsf{b}}, n_{\mathsf{f}}) = (19, 9, 1)$, where six of the 23 396 vectors share the largest value, 3.10147, to within 10^{-6}. With the number n_{a} of centres within 2.05 held fixed and the other bounds of Proposition 7.34 kept, the largest value falls by 0.005 to 0.007 for each centre added between 10 and 22: it is 3.1442 at $n_{\mathsf{a}} = 12$, 3.1082 at 18, 3.1015 at 19 and 3.0872 at 21. The certificate thus needs the count $N(2.05) \ge 19$ of Proposition 7.34 exactly; with 18 in its place the bound would exceed $\frac{9}{8}\pi^2 - 8$. At $(19, 9, 1)$ the bound is the sum of -24.215 from the centres, 18.390 from $\frac{t}{2}$, 5.257 from the pairs and

3.670 from the triples. Each inequality on pairs stays within 10^{-6} of its largest value along whole intervals of inner products, for two centres at distance 2 on $[-0.631, 0.029]$ and near -1, -0.82 and $\frac{1}{2}$, while the inequalities on triples are largest at isolated triples, three of them (of types aaa, abb and bbf) with the three directions in one plane through c.

In the fifth case of Theorem 7.32 the numbers of centres of each type are nearly fixed, 22 to 24 of type a, at most 2 of type b and 6 of type f, and the certificate is fitted to them. At $(22, 1, 1, 0, 6)$, where the bound 3.08720 is largest, the triples of types aaa, aaf and aff contribute 136.7, -319.7 and 145.8, all the pairs together 31.29 and all the triples -35.63, against 9.185 from $\frac{t}{2}$ and -1.762 from the centres. A certificate with such cancellations serves only the counts it was made for. The bound is largest at the fewest centres within 2.05 and within 2.15 that the case allows, 22 and 23, and it falls as centres move inward, to 3.0369 when all twenty-four lie within 2.05.

The certificate behind the level L_{26} of Proposition 8.1, the largest of the levels, rests on two counts. Its largest value, 3.32906, is at $(n_{\mathsf{a}}, n_{\mathsf{b}}, n_{\mathsf{f}}) = (15, 9, 2)$, with exactly the fifteen centres within 2.05 of Proposition 7.34 and the twenty-four within 2.444 of Lemma 7.31. With fourteen centres within 2.05 it would be 3.33156, and with twenty-three within 2.444 it would be 3.3590. In the other direction it falls by 0.003 to 0.008 for each further centre within 2.05, to 3.30716 at twenty and 3.27836 at twenty-four. A larger count within 2.05 at $M = 26$ would therefore lower L_{26}; below 3.3274 the level at $M = 25$ would be the largest.

At twenty-eight centres the kernel of Theorem 7.35 does not reach the level on the whole case (Remark 7.39), but it does once the number of centres within 2.0161 is bounded.

Proposition 7.37 (Twenty-eight centres, few of them close)**.** *Suppose that exactly* 28 *centres other than c lie within $\sqrt{6}$ of c. If at most* 14 *of them lie within* 2.0161 *of c, or at most* 15 *lie within* 2.0161 *and at most* 3 *beyond* 2.35*, or at most* 16 *lie within* 2.0161 *and none beyond* 2.35*, then $T(Y) > 8$.*

Proof. Suppose that $T(Y) \le 8$. By Proposition 7.34 at least 21 centres lie within 2.1, 22 within 2.15, 23 within 2.2 and 24 within 2.35, and the caps alone force 6 within 2.05 (Table 6). Give a centre the type a, b or f as its distance from c lies in $[2, 2.05]$, $(2.05, 2.35]$ or $(2.35, \sqrt{6})$, and give its direction the same type. With the seven bins $[2, 2.0161]$, $(2.0161, 2.05]$, $(2.05, 2.1]$, $(2.1, 2.15]$, $(2.15, 2.2]$, $(2.2, 2.35]$ and $(2.35, \sqrt{6})$, the integer vectors (n_b) with $\sum_b n_b = 28$ that meet these bounds number 62 865 with $n_1 \le 14$, 62 063 with $n_1 \le 15$ and $n_7 \le 3$, and 27 423 with $n_1 \le 16$ and $n_7 = 0$. The derivation of (7.9) applies word for word, with these bins and types, and it remains to see that its right side is below $\frac{9}{8}\pi^2 - 8$ at each of these vectors.

The certificate has $D = 12$ and $r = 4$ in the pair kernel and $d = 8$ in (4.2), and it is made exact and checked as in the proof of Theorem 7.32, except that the matrix of (b) is rounded to dyadic rationals as a whole, as for the level L_{24} of Proposition 8.1: its negative eigenvalues are set to zero, 2^{-30} is added to its diagonal, and an exact LDL^{T} factorisation shows it positive semidefinite. The margins are 10^{-6} for the pairs, $5 \cdot 10^{-8}$ for the triples and 10^{-7} for the bins, each given up at every one of the 378 pairs or 3276 triples, with two exceptions. For two centres of type a the margin is $5 \cdot 10^{-6}$: that inequality comes within 10^{-5} of its largest value all over the face of its domain where the two centres are at distance 2, so a smaller margin would need small boxes on the whole face. For three centres of type f the margin is $1.005 \cdot 10^{-5}$: the inequality of that kind has a narrow peak which the samples miss by about $3 \cdot 10^{-6}$, and a smaller margin leaves no room to close the boxes near it. The six inequalities on pairs close 3 532 617 boxes, 3 167 158 of them for two centres of type a,

and evaluate Π in ball arithmetic at 3 172 187 of them; the bins take 8511 intervals, and the ten inequalities on triples close 17 006 005 boxes.

Narrower bins give smaller constants. Three centres in the bins b, b' and b'' have inner products at most $a(\rho_b, \rho_{b'})$, $a(\rho_b, \rho_{b''})$ and $a(\rho_{b'}, \rho_{b''})$, where ρ_b is the upper end of bin b, because a increases in each argument; the inequality on triples of their kind therefore holds on that smaller set with a smaller constant, and the same goes for pairs. With the same margins, the triples of centres in the bins $(1, 3, 7)$ and $(1, 6, 7)$ get a constant of at most $3.1835 \cdot 10^{-4}$ in place of $3.2107 \cdot 10^{-4}$, those in $(3, 3, 3)$ at most $5.3504 \cdot 10^{-4}$ in place of $5.4310 \cdot 10^{-4}$, those in $(1, 1, 3)$ at most $5.6123 \cdot 10^{-4}$ in place of $5.6188 \cdot 10^{-4}$, and the pairs in the bins $(3, 6)$ and $(6, 6)$ at most $6.59465 \cdot 10^{-2}$ and $6.59657 \cdot 10^{-2}$ in place of $6.59661 \cdot 10^{-2}$; these inequalities close 9 432 949 further boxes. In (7.9) each pair or triple then counts with the constant of its bins where one is given, and with that of its types otherwise. The right side, evaluated in exact rational arithmetic, is largest at $(14, 0, 9, 0, 0, 1, 4)$ among the vectors with $n_1 \le 14$, where it is 3.099739, at $(15, 0, 8, 0, 0, 2, 3)$ among those with $n_1 \le 15$ and $n_7 \le 3$, where it is 3.102197, and at $(16, 0, 7, 0, 0, 5, 0)$ among those with $n_1 \le 16$ and $n_7 = 0$, where it is 3.102468, all below $\frac{9}{8}\pi^2 - 8 = 3.10330\ldots$. None of the inequalities checked above involves the counts, so the one certificate and these checks serve all three sets of vectors. ∎

The certificate of Proposition 7.37 has degree 8 in the triples, and the programmes of degree 10 for the larger sets of vectors leave violations between their samples that are too large to check (Remark 7.39). A convex combination of certificates is again a candidate certificate, since the matrices that have to be positive semidefinite form a convex set and every other condition is checked afresh for the combination; certificates of degree 10 found for single values of n_1, each from the samples of the one before, combine to reach further.

Proposition 7.38 (Twenty-eight centres)**.** *If exactly* 28 *centres other than* c *lie within* $\sqrt{6}$ *of* c*, then* $T(Y) > 8$*.*

Proof. The derivation is that of Proposition 7.37, with the same types and bins, over the 73 722 integer vectors (n_b) with $\sum_b n_b = 28$ that meet the same bounds and $n_1 \le 21$. The certificate has $D = 12$ and $r = 4$ in the pair kernel and $d = 10$ in (4.2). It is a convex combination of fourteen certificates, found for the case at twenty-eight centres or parts of it and three for twenty-six centres (Appendix D), and it is made exact and checked as that of Proposition 7.37, the matrix of (b) again rounded as a whole. The margins are $2 \cdot 10^{-6}$ for the pairs, $5 \cdot 10^{-7}$ for the triples and $5 \cdot 10^{-7}$ for the bins, except that for two centres of type a the margin is $5 \cdot 10^{-6}$, for the reason given in the proof of Proposition 7.37. The six inequalities on pairs close 120 628 boxes, 70 020 of them for two centres of type a, and evaluate Π in ball arithmetic at 72 761 of them; the bins take 1270 intervals, and the ten inequalities on triples close 168 808 634 boxes. The right side of (7.9), evaluated in exact rational arithmetic, is largest at $(21, 4, 2, 0, 0, 0, 1)$, where it is 3.103041, below $\frac{9}{8}\pi^2 - 8 = 3.10330\ldots$.

For $n_1 = 22$ a second combination is used, over the 451 vectors with $n_1 = 22$ that meet the same bounds: with the vectors of smaller n_1 left to the first, its weights can favour the certificates found for twenty-two close centres. It is a convex combination of nine certificates, eight found for the case at twenty-eight centres or parts of it and one for twenty-six centres, made exact and checked in the same way. The margins are 10^{-6} for the pairs, two centres of type a included, $2 \cdot 10^{-7}$ for the triples and 10^{-7} for the bins, except that for two centres of type a and one of type f the margin is $6 \cdot 10^{-7}$: the inequality of that kind has a narrow peak which the samples miss by about $4 \cdot 10^{-7}$. The six inequalities on pairs close 64 474 boxes, 21 433 of them for two centres of type a, and evaluate Π in ball arithmetic at 24 195 of them;

the bins take 2339 intervals, and the ten inequalities on triples close 207 350 728 boxes. The right side of (7.9), evaluated in exact rational arithmetic, is largest at $(22, 0, 2, 0, 0, 0, 4)$, where it is 3.102969, below $\frac{9}{8}\pi^2 - 8$.

For $n_1 = 24$ a third combination is used in the same way, over the 126 vectors with $n_1 = 24$ that meet the same bounds. It is a convex combination of six certificates found for the case at twenty-eight centres or parts of it, with nine tenths of the weight on two found for twenty-three close centres (Appendix D), made exact and checked in the same way. The margins are 10^{-7} for the pairs, two centres of type a included, $3 \cdot 10^{-8}$ for the triples and 10^{-7} for the bins, except that for the triples with at least two centres of type b or at least two of type f the margin is $1.03 \cdot 10^{-6}$, and for one centre of each type $5.3 \cdot 10^{-7}$: at the vectors where the bound is largest these kinds have no triples, or 26, and the larger margins shorten their branch and bound. The six inequalities on pairs close 29 367 boxes, 9835 of them for two centres of type a, and evaluate Π in ball arithmetic at 11 253 of them; the bins take 7156 intervals, and the ten inequalities on triples close 117 675 808 boxes. The right side of (7.9), evaluated in exact rational arithmetic, is largest at $(24, 2, 0, 0, 0, 1, 1)$, where it is 3.103206, below $\frac{9}{8}\pi^2 - 8$.

For $n_1 = 23$ and $n_7 \le 3$ a fourth combination is used in the same way, over the 246 vectors with $n_1 = 23$ and $n_7 \le 3$ that meet the same bounds. It is a convex combination of eight certificates found for the case at twenty-eight centres or parts of it, with 0.62 of the weight on two found for twenty-two close centres and 0.35 on four found for twenty-three (Appendix D), made exact and checked in the same way. The margins are $5 \cdot 10^{-7}$ for the pairs, two centres of type a included, 10^{-7} for the triples and 10^{-7} for the bins, except that for two centres of type a and one of type f the margin is $6 \cdot 10^{-7}$, since that inequality peaks on the face where an a and the f direction are at their largest inner product 0.6176, and the samples miss that peak by $2 \cdot 10^{-7}$, and for three centres of type f it is $1.1 \cdot 10^{-6}$, since these vectors have at most one such triple. The six inequalities on pairs close 37 565 boxes, 15 489 of them for two centres of type a, and evaluate Π in ball arithmetic at 17 493 of them; the bins take 2719 intervals, and the ten inequalities on triples close 162 022 640 boxes. The right side of (7.9), evaluated in exact rational arithmetic, is largest at $(23, 0, 0, 0, 0, 4, 1)$, where it is 3.102775, below $\frac{9}{8}\pi^2 - 8$.

With $n_1 = 23$ and $n_7 = 4$ the one remaining centre lies in one of the bins 2 to 6. The fourth combination settles bin 2 as well: none of its inequalities involves the counts, and the right side of (7.9) at $(23, 1, 0, 0, 0, 0, 4)$, evaluated in exact rational arithmetic, is 3.102361. For each of the vectors $(23, 0, 1, 0, 0, 0, 4)$, $(23, 0, 0, 1, 0, 0, 4)$ and $(23, 0, 0, 0, 1, 0, 4)$ a combination of its own is used, with the constants taken on the bins that the vector occupies, as in the proof of Proposition 7.37: at such a vector every pair and every triple of centres lies in bins 1, 7 and the one further bin b, each inequality is checked on the pairs or triples of those bins with a constant of its own, and an inequality whose pairs or triples all lie in the listed combinations of bins is not needed on the whole ranges of its types. Each of the three combinations has one certificate found for twenty-six centres and eleven, eleven and ten found for the case at twenty-eight centres or parts of it, with between 0.57 and 0.68 of the weight on rounds for twenty-two close centres and between 0.30 and 0.41 on rounds for twenty-three (Appendix D), and they are made exact and checked in the same way. For $b = 3$ the margins are 10^{-7} for the pairs, 10^{-8} for the triples and 10^{-7} for the bins; for $b = 4$ and $b = 5$ they are $5 \cdot 10^{-7}$, 10^{-7} and 10^{-7}. Each is raised, by between 10^{-8} and 10^{-6}, for the triples of five or six combinations of bins: those at which a dense search puts the largest value of the inequality above its largest sampled value, and those of which the vector has few triples. The inequalities on pairs close 23 258, 13 678 and 12 261 boxes for $b = 3$, 4

and 5, and evaluate Π in ball arithmetic at 13 352, 10 559 and 9499 of them; the bins take 2748, 2676 and 2816 intervals, and the inequalities on triples close 127 221 931, 125 799 088 and 124 117 379 boxes. The right side of (7.9), evaluated in exact rational arithmetic, is 3.103281, 3.101860 and 3.102128 at the three vectors, below $\frac{9}{8}\pi^2 - 8$.

The last vector, $(23, 0, 0, 0, 0, 1, 4)$, needs one step more. Its combination is found in the same way, from one certificate found for twenty-six centres and twelve found for the case at twenty-eight centres or parts of it, with 0.60 of the weight on rounds for twenty-two close centres and 0.38 on rounds for twenty-three (Appendix D), and it is made exact and checked in the same way, on the triples of the bins $(1,1,1)$, $(1,1,6)$, $(1,1,7)$, $(1,6,7)$ and $(1,7,7)$ and on the pairs of the bins $(1,1)$, $(1,6)$, $(1,7)$ and $(6,7)$, with margins 10^{-7} for the pairs, 10^{-8} for the triples and 10^{-7} for the bins. With these constants alone the right side of (7.9) at that vector is 3.10336, just above $\frac{9}{8}\pi^2 - 8$. What settles it is a count of the 1012 triples of two centres in bin 1 and one in bin 7, which uses two facts about directions. Let $y \in S^3$, and let $x_1, \dots, x_k \in S^3$ have pairwise inner products at most $c_0 \in (0,1)$. First, if $\langle x_i, y\rangle \le s_1$ for every i, where $s_1 < 0$ and $s_1^2 > c_0$, then $k \le (1-c_0)/(s_1^2 - c_0)$: the sum $S = \sum_i x_i$ has $|S|^2 \le k + k(k-1)c_0$ and $\langle S, -y\rangle \ge k|s_1|$, so $k^2 s_1^2 \le |S|^2$ gives $k s_1^2 \le 1 + (k-1)c_0$. Second, if $s_2 \le \langle x_i, y\rangle \le \tau$ for every i, where $s_2 \ge 0$, $s_2^2 < c_0$ and $\tau < 1$, then $k(1 - \sqrt{(1+g)/2}) \le 2$ with $g = (c_0 - s_2^2)/(1-\tau^2)$: write $x_i = \gamma_i y + (1-\gamma_i^2)^{1/2} p_i$ with p_i a unit vector orthogonal to y; for $i \ne j$,

$$\langle p_i, p_j\rangle = \frac{\langle x_i, x_j\rangle - \gamma_i\gamma_j}{\sqrt{(1-\gamma_i^2)(1-\gamma_j^2)}} \le g,$$

since, where the numerator is positive, it is at most $c_0 - s_2^2$ and the denominator is at least $1 - \tau^2$, so on the two-sphere of unit vectors orthogonal to y the caps of angular radius $\frac{1}{2}\arccos g$ about the p_i have disjoint interiors, and each has area $2\pi(1 - \sqrt{(1+g)/2})$. Take $c_0 = a(2.0161, 2.0161) = 0.50795$, the largest inner product of two close directions, $\tau = a(2.0161, \sqrt{6}) = 0.61403$, that of a close and a far direction, $s_1 = -0.7944$ and $s_2 = 0.58$: for each far centre, at most 3 close directions have inner product at most -0.7944 with its direction, and at most 9 have inner product at least 0.58. The inequality for two close centres and one far centre has the constant $2.747585 \cdot 10^{-4}$ on its bins, but it comes close to that value only where the inner products of the two close directions with the far one are both near the largest value 0.614 or both near -0.813. Its polynomial is symmetric in the two close centres, coefficient by coefficient, and it is checked with a second constant, $2.746103 \cdot 10^{-4}$, on the triples in which the first of these inner products lies in $[-0.7944, 0.58]$, and with a third, $2.742084 \cdot 10^{-4}$, on those in which the first is at most -0.7944 and the second at least 0.58; by the symmetry the second constant bounds every triple with one of the two inner products in $[-0.7944, 0.58]$, and the third every triple with one of them at most -0.7944 and the other at least 0.58. For a far centre with $k \le 3$ close directions at inner product at most -0.7944 and $l \le 9$ at least 0.58, at most $\binom{k}{2} + \binom{l}{2}$ of the 253 pairs of close centres take the first constant, kl the third and the rest the second, and in (7.9) the 253 triples with that far centre count with the largest such sum, which is reached at $k = 0$ and $l = 9$. The inequalities on pairs close 21 697 boxes, and evaluate Π in ball arithmetic at 12 191 of them; the bins take 2713 intervals, and the inequalities on triples close 200 740 860 boxes. The right side of (7.9), with the triples of two close centres and one far centre so counted and evaluated in exact rational arithmetic, is 3.103231, below $\frac{9}{8}\pi^2 - 8$.

By Theorem 5.78 at most 24 centres lie within 2.0161, so $n_1 \le 24$, and by Proposition 7.34 $n_7 \le 4$; the combinations above therefore cover every vector that meets the bounds, and $T(Y) > 8$. ∎

Remark 7.39 (The last vector at twenty-eight centres). Without its last step, the proof of Proposition 7.38, together with Proposition 7.34 and Theorem 5.78, leaves a set Y of 28 centres with $T(Y) \leq 8$ only one vector of bin counts, $(23, 0, 0, 0, 0, 1, 4)$: exactly 23 centres within 2.2, all of them within 2.0161, exactly 24 within 2.35 and four beyond. Since $a(d, d')$ increases in each argument, the directions of such a Y, ordered by the distance of their centres from c, would satisfy $\langle \hat{y}_i, \hat{y}_j \rangle \leq a(\rho_i, \rho_j)$, where ρ_i is 2.0161 for the first twenty-three, 2.35 for the next one, and $\sqrt{6}$ for the last four. That last step could therefore also come from a statement about 28 points of S^3: no such points exist. Local searches find none. With the twenty-one distances within 2.1 as further variables and a reward for each centre at 2.0161, local minimisation from 2000 random starts reaches twelve centres at 2.0161, with room 10^{-5} in every inequality, once, and thirteen never; from the configuration with twelve, a thirteenth centre moved in to 2.0161 misses by $4.2 \cdot 10^{-3}$, and 600 random perturbations of it, each followed by the same search, end again at twelve, far from the twenty-three that the statement needs. A search of this kind is evidence and not a proof. The statement is of the kind that Remark 7.30 describes for twenty-four close centres: the centres beyond 2.35, up to four, need holes of the twenty-three close directions at inner product at most $a(2.0161, \sqrt{6}) = 0.6141$, and the typed three-point bound of Lemma 4.4, applied to the directions alone, does not see holes. At degrees 8 and 10 it does not exclude even the directions of twenty-three centres at 2.0161, one at 2.35 and four at $\sqrt{6}$, which the statement excludes in particular: the optimum of its programme is 0.

The kernel of Proposition 7.37 does not reach the whole case either. On its samples the solver returns 3.10668 for the programme at $d = 8$, above the level, with its solution flagged as inaccurate, and at $d = 10$ values between 3.0917 and 3.0951 over five rounds of refinement, each ended by a numerical error, while the solutions violated the inequalities on triples between the samples by more than 10^{-4}; a proof with 3276 triples needs about $2 \cdot 10^{-6}$. The largest value of the degree-10 kernel over the count vectors lies at twenty-two centres within 2.0161, one in $(2.05, 2.1]$, one in $(2.2, 2.35]$ and four beyond 2.35, where the searches above find no configuration, and asking for at most 20 centres within 2.0161 lowers it by less than $2 \cdot 10^{-4}$. Kernels of degree 8 set up for the cases that remain do no better than that of Proposition 7.37. With the constants that its programme finds on its own samples, the certificate of Proposition 7.37 gives 3.09042 over the vectors with $n_1 \leq 15$ and $n_7 = 4$ and 3.09607 over those with $n_1 \leq 16$, while programmes of degree 8 for these two sets of vectors reach 3.08978 and 3.09593 on their samples; at $n_1 \leq 13$ the constants of the check put the bound 0.0154 above its value with the sampled ones. The convex combinations of Proposition 7.38 go further: they settle $n_1 = 24$, $n_1 = 23$ with at most three centres beyond 2.35, and the five vectors with $n_1 = 23$ and $n_7 = 4$, four of them with a combination for each vector and constants on its bins. A single combination for the five vectors does not reach the level: in floating point, with the largest value of each inequality found by local maximisation and by a dense search, and margins 10^{-7} for the pairs and 10^{-8} for the triples, the best one found gives 3.10408 there, and about 3.1040 with no margins at all. At the fifth vector, $(23, 0, 0, 0, 0, 1, 4)$, where four centres need holes of the twenty-three close directions, the combination of its own gives 3.10336 in the same way, and 3.10329 with no margins, too close to the level for a check. Two further rounds of the programme for that vector, with the type b narrowed to its bin, reach 3.0796 and 3.0834 on their samples but 3.664 and 3.273 at the largest values of their inequalities between them. Splitting the bin of the close centres, at 2.002 for instance, does not help either: the inequality for two close centres and one far centre has its largest value, $2.74748 \cdot 10^{-4}$, at an interior point of its domain, with inner products 0.4288 between the close directions and -0.8130 between each of them

and the far one, and every such split keeps that point. What settles the vector is the count in the last step of the proof of Proposition 7.38: that inequality comes within $1.4 \cdot 10^{-7}$ of its largest value only where the far direction is nearly opposite both close directions or at nearly the largest inner product with both, and the two facts about directions used there limit how many close directions can be in either position. With that count the combination gives 3.103231, in exact arithmetic. The bound at that vector therefore comes from the directions after all, through how often a far direction can have close directions at either end of its range, and not from a certificate that detects holes.

Statement (G) can be proved in full along one family: the root system with its centres pushed out along their own rays, by any amounts the region $T \le 8$ allows. Lemma 7.16 gives the gain, and it remains to control the part of $V(Y)$ that the inversion hull cuts away, which lies in small pyramids at the vertices of the 24-cell.

Theorem 7.40 ((G) along the push-outs)**.** *Let $u_1, \dots, u_{24}$ be the normalised roots of a copy of* R*, let Y consist of the 24 centres $(2+\delta_i)u_i$ with $\delta_i \ge 0$, and suppose that $T(Y) \le 8$. Then every δ_i is below 0.1971, and*

$$\operatorname{vol}(V(Y) \cap K(Y)) \;\ge\; 8 + \tfrac{1}{5} \sum_i \delta_i .$$

In particular (G) holds for Y, with equality only when every $\delta_i = 0$.

Proof. Put $\eta_i = \delta_i/2$ and $Q = \{x : \langle x, u_i\rangle \le 1 \ \forall i\}$, the 24-cell, so that $V(Y) = \{x : \langle x, u_i\rangle \le 1 + \eta_i \ \forall i\}$ and $K(Y) = \operatorname{conv}\{0,\, 2u_i/(1+\eta_i)\}$.

The size of the push-outs. Moving a centre out along its ray shrinks its cap in $B(\sqrt{3/2})$, and with it the union of the caps, so T does not decrease in any δ_i. Hence $T(Y)$ is at least the value of T on the root system with only the largest δ_i kept. On that ray

$$T(\delta) = \tfrac{9}{8}\pi^2 - 23S(2) - S(2+\delta) + 88\,\Pi(1,1,\tfrac{1}{2}) + 8\,\Pi(1,1+\tfrac{\delta}{2},\tfrac{1}{2}),$$

since caps whose normals meet at 90° or more do not meet in $B(\sqrt{3/2})$, and in ball arithmetic, with Lemma 7.24, $T(0.1971) = 8.0000152\ldots > 8$. Therefore every $\delta_i < 0.1971$ and $\eta_i < 0.09855$.

The gain. By Lemma 7.16, $\operatorname{vol} V(Y) \ge 8 + \frac{1}{3}\sum_i (1 - (1-\eta_i)^4)$.

The loss. Let $\eta = \max_i \eta_i$ and $\lambda = (1+\eta)^2$, so $\lambda - 1 < 0.207$. The hull $K_0 = \operatorname{conv}\{2u_i\}$ is the 24-cell $\{x : \langle x, \hat{v}\rangle \le \sqrt{2}\}$, v running over the vertices of Q, and its facet in the direction of v is the octahedron with vertices $2u$ for the six roots u through v, centred at v. Since $V(Y) \subseteq (1+\eta)Q$ and $K(Y) \supseteq (1+\eta)^{-1}K_0$, a point x of $V(Y) \setminus K(Y)$ has $(1+\eta)x \in \lambda Q$ and $\langle (1+\eta)x, \hat{v}\rangle > \sqrt{2}$ for some vertex v. In coordinates with $v = \sqrt{2}e_1$, the six facets of Q through v are $x_1 \pm x_j \le \sqrt{2}$, $j = 2,3,4$, so $|x_j| < (\lambda - 1)x_1 < \frac{1}{3}x_1$. Let η_v be the largest η_u over the six roots u through v. As $x \in V(Y)$, $x_1 \pm x_j \le \sqrt{2}(1+\eta_v)$. As $x \notin K(Y)$, x is not in the pyramid from 0 over the octahedron with vertices $2u/(1+\eta_v)$, which $K(Y)$ contains because $2u/(1+\eta_u)$ lies farther out on the same ray; that octahedron lies in the hyperplane $x_1 = \sqrt{2}/(1+\eta_v)$ and contains every point of it with $|x_j|$ below a third of $\sqrt{2}/(1+\eta_v)$, so the ray through x meets it and $x_1 > \sqrt{2}/(1+\eta_v)$. Hence x lies in the truncated cone $\{\sqrt{2}/(1+\eta_v) < x_1,\ |x_j| \le \sqrt{2}(1+\eta_v) - x_1\}$, whose section at $x_1 = \sqrt{2}(1+\eta_v) - s$ is a cube of side $2s$; its volume is $\int_0^{h_v} 8s^3\,ds = 2h_v^4$ with $h_v = \sqrt{2}\big((1+\eta_v) - (1+\eta_v)^{-1}\big) \le 2\sqrt{2}\,\eta_v$. Each facet of Q has six vertices, so

$$\operatorname{vol}(V(Y) \setminus K(Y)) \le \sum_v 128\,\eta_v^4 \le 128 \sum_v \sum_{u \ni v} \eta_u^4 = 768 \sum_i \eta_i^4 .$$

Together. $\operatorname{vol}(V(Y)\cap K(Y)) \geq 8+\sum_i g(\eta_i)$ with $g(\eta)=\frac{1}{3}(1-(1-\eta)^4)-768\eta^4$. The derivative of $g(\eta)/\eta=\frac{1}{3}(4-6\eta+4\eta^2-\eta^3)-768\eta^3$ is at most $-2+\frac{8}{3}\eta<0$, and at $\eta=0.09855$ its value is $0.4137\ldots$; so $g(\eta_i)\geq\frac{2}{5}\eta_i=\frac{1}{5}\delta_i$ for every i. ∎

Corollary 7.41 (Centres on the rays of a root system)**.** *If exactly* 24 *centres other than* c *lie within* $\sqrt{6}$ *of* c*, all on the rays from* c *through the normalised roots of a copy of* R*, then* $\operatorname{vol}(V_c)\geq 8$*, with equality only when they are at distance* 2 *and* V_c *is the* 24*-cell.*

Proof. If $T(Y)>8$, then $\operatorname{vol}(V_c)\geq T(Y)>8$ by Lemma 7.6. Otherwise $\operatorname{vol}(V_c)\geq \operatorname{vol}(V(Y)\cap K(Y))\geq 8+\frac{1}{5}\sum_i\delta_i$ by Proposition 7.4 and Theorem 7.40, and equality forces every $\delta_i=0$, twenty-four contacts, and the 24-cell by Lemma 7.2. ∎

The inequality is far from tight: along the ray of one centre the part of $V(Y)$ outside the hull stays below 10^{-6} up to $\delta=0.197$, against $768\eta^4\approx 7\cdot 10^{-2}$ there. What stops the argument at the push-outs is the gain: Lemma 7.16 needs the normals to be the roots, and with the directions tilted the contact cell itself can fall to 8 (Proposition 7.19), so a proof off this family has to take the gain from the push-outs that the tilts force.

Off the push-outs, the pair terms still say how far out the centres of (G) can be. The counts below are those the case certificates of the proof of Theorem 7.32 give when they are run with twenty-four centres in place of thirty, as Proposition 7.34 does for twenty-five to thirty.

Proposition 7.42 (Where the twenty-four centres lie)**.** *Let* Y *be a packing set of exactly* 24 *centres with* $2\leq|y|<\sqrt{6}$ *and* $T(Y)\leq 8$*. Then at most* 13 *centres of* Y *lie beyond* 2.0161*, at most four beyond* 2.05*, at most two beyond* 2.1*, and at most one beyond* 2.15*.*

Proof. For a radius ρ and a count c, take the tree of the proof of Theorem 7.32 with $M=24$ and the single split $N(\rho)\leq c-1$ against $N(\rho)\geq c$. For $(\rho,c)=(2.0161,11)$, $(2.05,20)$, $(2.1,22)$ and $(2.15,23)$ the first case carries a certificate of degree 12 in the angle and 4 in the distance, and the largest value it gives over the count vectors of that case is at most 3.10018, 3.09836, 3.07883 and 3.09251 respectively, below $\frac{9}{8}\pi^2-8=3.10330$. So $T(Y)>8$ in the first case, and $N(\rho)\geq c$ whenever $T(Y)\leq 8$. Each certificate is checked as there: positivity by an exact LDL^{T} factorisation, the inequality $K\leq\Pi$ on the admissible pairs by a branch and bound in ball arithmetic,[41] the bounds on the bins, and the largest value over the count vectors by exact dynamic programming. ∎

The caps alone give much less. Since $\sum_y S(|y|)\geq U(Y)\geq\frac{9}{8}\pi^2-8$ when $T(Y)\leq 8$, and S decreases, k centres can lie beyond ρ only if $(24-k)S(2)+kS(\rho)\geq\frac{9}{8}\pi^2-8$. That allows 19 centres beyond 2.0161, six beyond 2.05, three beyond 2.1 and two beyond 2.15, and one beyond every radius below $\sqrt{6}$ (Figure 42); the pair terms remove six, two, one and one. Beyond 2.05 they cannot remove more at this degree: with the next count, 21 at 2.05, 23 at 2.1, or 24 at 2.2 or 2.25, the optimum of the same programme already lies above the level, at 3.1151, 3.1248, 3.1422 and 3.1306, so certificates of this kind cannot exclude a single centre far out. That is still far from the neighbourhood of Theorem 7.9, where $\sum_i\delta_i\leq 4\cdot 10^{-3}$, and it does not constrain the directions.

[41] At the antipodal pairs, $u=-1$, the caps are disjoint and Π vanishes, so the kernel must be nonpositive there. A kernel computed from too few samples along $u=-1$ can be slightly positive at such pairs and then fails the branch and bound, which is why the programme refines its samples along that edge as well; for 2.0161 and 2.05 it also keeps a margin of $3\cdot 10^{-5}$ and $5\cdot 10^{-5}$, rather than $2\cdot 10^{-5}$, wherever the caps are disjoint.

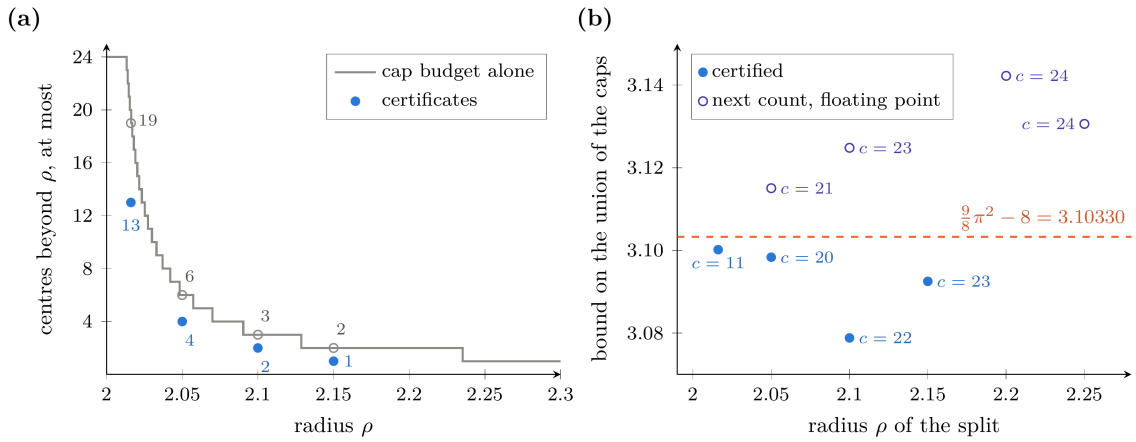


FIGURE 42. Proposition 7.42. (a) For twenty-four centres with $T(Y) \leq 8$, the number that can lie beyond a radius ρ: the cap budget alone (step curve, open marks) against the counts that the case certificates prove (filled marks), 13 beyond 2.0161, 4 beyond 2.05, 2 beyond 2.1 and 1 beyond 2.15. (b) For the split $N(\rho) \leq c-1$ against $N(\rho) \geq c$, the bound on the union of the caps in the first case: proved by a certificate (filled marks) and, for the next count, the optimum of the programme of the same degree (open marks), which is not a bound, against the level $\frac{9}{8}\pi^2 - 8$ that $T(Y) > 8$ needs.

Remark 7.43 (What (G) needs). Statement (G) is sharp at the root system and holds near it: Theorem 7.9 proves it whenever $\sum_i \delta_i \leq 4 \cdot 10^{-3}$ and $d(W) \leq \frac{1}{48}$, and Theorem 7.40 proves it for every push-out of the root system in the region, whatever its size. The region $T(Y) \leq 8$ in which it is asked is larger than the first, but not by much. Along the root system pushed out evenly the pair terms reach 8 at $\sum_i \delta_i = 0.155$, and along the ray of one centre at $\delta = 0.197$; every local minimum of T that a descent from random starts reaches with 24 centres is the root system itself; and configurations far from it, such as the packing with $\mathrm{vol}(V(Y) \cap K(Y)) = 4.750$ of Remark 7.22, have T far above 8. Inside the region the structure of the volume is known to second order. The push-outs raise it at the rate $\frac{2}{3}$ (Theorem 7.9), and on the first-order packing cone the Hessian loses at most $(\sum_i \delta_i)^2$ (Proposition 7.14), so that the second-order model $\frac{2}{3}S - \frac{1}{2}S^2$ is positive up to $S = \frac{4}{3}$, far beyond 0.197. The tilts alone never lower the volume at second order (Proposition 7.18), and along the closed curves of Proposition 7.19 they leave it at exactly 8; but those curves break the packing constraint at first order, the largest inner product rising to $4C^2/(4C^2+3) > \frac{1}{2}$, so a packing can follow them only by moving centres out, which the first-order term rewards at the rate $\frac{2}{3}$. A proof of (G) has to capture this mechanism; the searches of Remark 7.22 find no configuration below 8.

A proof needs two quantitative statements. The first is a localisation: if $|Y| = 24$ and $T(Y) \leq 8$, then Y lies within an explicit distance of a copy of $\sqrt{2}\,\mathsf{R}$. For contacts this is the uniqueness theorem of [LLM24]; its robust form is proved only at slack $2 \cdot 10^{-26}$ (Theorem 5.75), and Proposition 7.21 shows that no form of it that looks at the directions alone can pass slack 0.017, so the localisation has to use the distances, through T itself. Proposition 7.42 is the part of it that the pair terms give: it bounds how many centres lie far from c, but not where their directions point. The second is the volume bound in that neighbourhood, which needs the third-order remainder and the part of $V(Y)$ outside the

inversion hull under control (Remark 7.15); Theorem 7.9 provides it up to $\sum_i \delta_i = 4 \cdot 10^{-3}$, a factor of about forty short of the region. Together, (G) and the remaining counts of (C) would prove Conjecture 1.3 and with it the bound $\pi^2/16$ for the density of every packing of $\mathbb{R}^4$, which is open; the kissing problem in dimension four [Mus08] and the uniqueness of its solution [LLM24] each needed a new semidefinite method, and the computations of this section indicate that the remaining step does too.

8. A density bound from the cells alone

Statements (G) and (C) are needed for the value $\pi^2/16$, but not for a bound on the density. By Lemma 7.6 every cell has volume at least $T(Y) = \frac{9}{8}\pi^2 - U(Y)$, where Y is the set of centres within $\sqrt{6}$ of its centre. If $|Y| \le 23$ or $|Y| \ge 31$, the cell has volume above 8 (Theorems 7.8 and 7.25); for the seven counts in between, the certificates of Section 7.4, run with a level in place of $\frac{9}{8}\pi^2 - 8$, bound $U(Y)$ from above. Lemma 3.1 turns the least of the resulting lower bounds for the cells into a bound for the density of every packing.

Proposition 8.1 (The union of caps from twenty-four to thirty centres)**.** *Let $24 \le M \le 30$, and let Y be a packing set of exactly M centres with $2 \le |y| < \sqrt{6}$. Then $U(Y) < L_M$, with L_M as in Table 7. In particular $U(Y) < 3.3291$ for every such set.*

Proof. For $M = 24$, if $T(Y) > 8$ then $U(Y) < \frac{9}{8}\pi^2 - 8$. Otherwise the counts of Proposition 7.42 hold: at least 11 centres lie within 2.0161, 20 within 2.05, 22 within 2.1 and 23 within 2.15. With the types a, b, f of the distance ranges $[2, 2.05]$, $(2.05, 2.444]$ and $(2.444, \sqrt{6})$ and the bins with ends 2.0161, 2.05, 2.1, 2.15, 2.2 and 2.444, the derivation of (7.9) bounds $U(Y)$. A certificate of degree 6 in (4.2) gives at most 3.25388 over the 376 count vectors of the case, at $(24, 0, 0, 0, 0, 0, 0)$, where all twenty-four centres lie within 2.0161. It is made exact and checked as in the proof of Theorem 7.35, with margins of $2 \cdot 10^{-5}$ for the pairs and $2 \cdot 10^{-6}$ for the triples and the bins, except that the matrix of (b) is rounded to dyadic rationals as a whole: its negative eigenvalues are set to zero, 2^{-30} is added to its diagonal, and an exact LDL^{T} factorisation shows it positive semidefinite. Its six inequalities on pairs close 3 361 boxes, the bins take 285 intervals, and its seven inequalities on triples close 104 028 boxes; at most one centre lies beyond 2.15, so no triple has two centres of type f. With the pair terms dropped one has only $U(Y) \le \sum_y S(|y|) \le 24\, S(2) = 3.335124\ldots$, since S is decreasing. For $M = 26$, if $T(Y) > 8$ then $U(Y) < \frac{9}{8}\pi^2 - 8$. Otherwise the counts of Proposition 7.34 and Lemma 7.31 hold: at least 15 centres lie within 2.05, 20 within 2.1, 22 within 2.15, 23 within 2.2 and 24 within 2.444, and at most 24 within 2.0161. With the types a, b, f of the distance ranges $[2, 2.05]$, $(2.05, 2.444]$ and $(2.444, \sqrt{6})$ and the bins with ends 2.0161, 2.05, 2.1, 2.15, 2.2 and 2.444, the derivation of (7.9) bounds $U(Y)$, and a certificate of degree 8 in (4.2), made exact and checked as in the proof of Theorem 7.35 with margins of $2 \cdot 10^{-5}$ for the pairs and $2 \cdot 10^{-6}$ for the triples and the bins, gives at most 3.32906 over the 17 845 count vectors of the case, at $(15, 0, 9, 0, 0, 0, 2)$. Its six inequalities on pairs close 9 996 boxes, the bins take 6 323 intervals, and its nine inequalities on triples close 7 512 629 boxes. The kernel on pairs alone gives 3.3501 here, with the split $N(2.03) \le 25$ against $N(2.03) = 26$, whose second case is empty by Theorem 5.79. For the other counts, $25 \le M \le 30$ and $M \ne 26$, take in the proof of Theorem 7.32 the tree with M centres in place of thirty and no split, so that the only case is $N(2.0161) \le 24$. Each such case carries a certificate of degree 12 in the angle and 4 in the distance, with a multiplier $s \ge 0$, and

$$U(Y) \le \sum_b n_b m_b + \tfrac{t}{2} \le \max_n \sum_b n_b (m_b + s S(e_b)) + \tfrac{t}{2},$$

TABLE 7. The levels of Proposition 8.1: for M centres within $\sqrt{6}$, an upper bound L_M for the union of caps $U(Y)$, and the lower bound $\frac{9}{8}\pi^2 - L_M$ for the volume of the cell that it gives. Every entry is rounded in the safe direction. For $M = 24$ and $M = 26$ the bound is the value of the kernel on typed triples, and the others are values of two-point certificates.

M	24	25	26	27	28	29	30
L_M	3.2539	3.3274	3.3291	3.3090	3.2587	3.2080	3.1538
$\frac{9}{8}\pi^2 - L_M$	7.8494	7.7759	7.7742	7.7943	7.8446	7.8953	7.9495

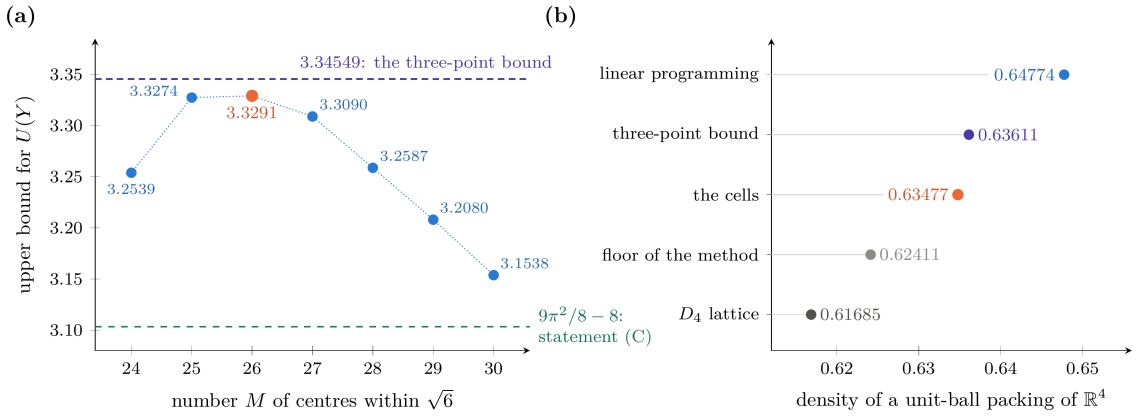


FIGURE 43. The density bound of Theorem 8.2. (a) The levels L_M of Proposition 8.1 against the level $\frac{9}{8}\pi^2 - 8$ that statement (C) asks for (green) and the level 3.34549 below which every count would have to fall for the bound to pass the three-point bound (violet); all of them do, the largest being L_{26} (orange). (b) The resulting bound (orange) with the others for $\mathbb{R}^4$: the density $\pi^2/16$ of D_4, the floor of Remark 8.3, the three-point bound [CdLS22], and the linear programming bound [CE03].

the first inequality by the chain of inequalities of that proof, which uses no assumption on $U(Y)$, the second because $s \geq 0$ and $S \geq 0$, and the maximum over the integer count vectors n of the case, found exactly by dynamic programming. The right side, evaluated in interval arithmetic, is the entry of Table 7. Each certificate is checked as those of Theorem 7.32. ■

Theorem 8.2 (Density from the cells)**.** *Every packing of unit balls in* $\mathbb{R}^4$ *has density at most*

$$\frac{\pi^2/2}{\frac{9}{8}\pi^2 - 3.3291} = 0.634766\ldots.$$

Proof. Enlarge the packing to a saturated one, which does not lower its density, and let c be a centre and Y the set of other centres within $\sqrt{6}$ of it. If $|Y| \leq 23$, then $\mathrm{vol}(V_c) > 8$ by Theorem 7.8. If $|Y| \geq 31$, then $\mathrm{vol}(V_c) \geq T(Y) > 8$ by Theorem 7.25. Otherwise $24 \leq |Y| \leq 30$, and $\mathrm{vol}(V_c) \geq T(Y) = \frac{9}{8}\pi^2 - U(Y) > \frac{9}{8}\pi^2 - 3.3291$ by Lemma 7.6 and Proposition 8.1. Every cell therefore has volume at least $v = \frac{9}{8}\pi^2 - 3.3291 = 7.7742\ldots$, and Lemma 3.1 bounds the density by $\mathrm{vol}(B^4)/v$, with $\mathrm{vol}(B^4) = \pi^2/2$. ■

Remark 8.3 (Where the bound stands). The bound lies below the three-point bound 0.63611 of Cohn, de Laat and Salmon [CdLS22], the best published bound in dimension four before it, and below the linear programming bound 0.64774 of Cohn and Elkies [CE03].[42] It uses no Fourier analysis and no global object, only the volume of each cell inside the ball of radius $\sqrt{3/2}$ about its centre, and its proof is local in the sense of Section 9.

The bound is limited in two ways. The first is the level at $M = 26$, 3.3291, with the level 3.3274 of the pairs alone at $M = 25$ just below it (Figure 43(a)). The kernel on typed triples brings the level at $M = 24$ from $24\,S(2) = 3.33512$, the value with the pair terms dropped, to 3.2539, against 3.19636 at the root system, and the level at $M = 26$ from 3.3501 to 3.3291. At $M = 26$ the optimum of its programme in floating point is 3.2456 on the sampled constraints; the certified level is higher because the exact check bounds each inequality on triples over the whole domain, where it exceeds its values at the samples. The pairs alone could not have brought the level at $M = 26$ below 3.34549. With at most twenty-five centres within 2.03, as Theorem 5.79 allows, the optimum of the programme on pairs at $M = 26$ puts twenty-four centres within 2.0161 and a twenty-fifth between 2.0161 and 2.03, a configuration that the pair terms cannot exclude. Without it the optimum falls to 3.34471, the larger of its values when at most twenty-three centres lie within 2.0161 (3.34336) and when twenty-four lie there and no other within 2.03 (3.34471), and a level there would already have carried the bound to 0.63605.[43] A form of Theorem 5.75 with $\kappa = 0.008$, placing the twenty-four directions within the tolerance of Corollary 7.28, would exclude it, since that corollary then leaves no further centre within $\sqrt{6}$; the deep holes of a root system of 2-vectors are at inner product $1/\sqrt{2}$ with its nearest directions, so a centre in one of them is at distance at least $4/\sqrt{2} = 2\sqrt{2}$.[44] The second is a floor that no level can pass: at the root system $U(Y) = 24\,S(2) - 96\,\Pi(1, 1, \frac{1}{2}) = 3.19636\ldots$ and $T(Y) = 7.90694\ldots < 8$, so a bound on T alone gives at best $(\pi^2/2)/7.90694 = 0.62411$. The rest of the way to $\pi^2/16$ is the volume that the cell of the root system has outside $B(\sqrt{3/2})$, which is the content of statement (G).

9. Comparison with other strategies

9.1. **Local against global.** The Cohn–Elkies method [CE03], and its sharpened form in dimensions 8 and 24 [Via17, CKMRV17], is global and Fourier-analytic: one auxiliary function certifies optimality over all configurations at once. The argument here is local. It bounds one Voronoi cell at a time, and it reduces the bound for a cell either to an extremal problem for one fixed polytope (Section 6) or to an inequality between the distances and pair angles of the neighbours inside a ball in which no three caps meet, which a semidefinite kernel certifies (Sections 5.11 and 7.1). In dimension four the global route is closed: by the result of Li [Li25] recalled in Section 1.4, no auxiliary function brings the linear programming bound down to $\pi^2/16$, and the three-point refinement of [CdLS22] stays above it as well.

[42] Cohn and Elkies state their bounds for the centre density, the density divided by $\mathrm{vol}(B^n)$; their value 0.13126 in dimension four is a density of $0.13126\,\pi^2/2 = 0.64774$.

[43] These three values are optima of the programme in floating point, on sampled constraints; only the certified levels of Table 7 enter a proof.

[44] The three-point programme of Theorem 5.78 extended by a twenty-fifth point of its own type, at inner product at most t_2 with the other twenty-four and with the twenty-four at most $t_1 = 0.508$ apart, would exclude the configuration if it excluded $t_2 = a(2.0161, 2.03) = 0.51135$, the value for a centre within 2.03. At degree 10 it does not: the margin by which that programme proves $N(2.0161) \le 24$ is used up before t_2 reaches 0.5101. At degree 12 the programme in floating point excludes $t_2 = 0.5101$ but not $t_2 = 0.5163$.

9.2. **Linear programming bounds in dimension four.** The functions that settle dimensions 8 and 24 are not found by search. They are pinned down by an interpolation formula, which works because in exactly those two dimensions the relevant space of modular forms is small enough, and rigid enough under the large automorphism group of the lattice, that the Fourier eigenfunction condition together with finitely many prescribed zeros determines the function; a contour integral then makes the sign conditions checkable. Nothing of that rigidity is present for D_4. Cohn and Elkies [CE03] already observed that the near-exactness of their bound appeared to belong to dimensions 1, 2, 8 and 24 alone, and the lower bound of Li recalled in Section 1.4 turns the observation into a theorem in dimension four: no admissible function brings the bound down to $\pi^2/16$. The refinements that do apply in low dimensions, the coupling to spherical-code bounds of [CZ14] and the three-point bound of [CdLS22], stop short as well. A method that bounds the Voronoi cell one configuration at a time, at the cost of a case analysis and of classifying the equality case by hand, is therefore not a second choice in this dimension.

9.3. **What each method asks of the reader.** A linear or semidefinite programming bound, whether of Cohn–Elkies type or the relaxation over contact graphs that Musin [Mus18] proposes for the twenty-four-cell conjecture, certifies positivity of a high-dimensional object found by a numerical solver. Such certificates can be converted to exact rational form [Par03, Las01, CdLL24], and the ones used here are: every certificate of this paper is a finite list of rational matrices whose positivity is checked by an exact factorisation, together with an inequality on a compact domain checked by branch and bound in interval arithmetic (Section 11). The magic-function constructions are closed form once written down, but checking that a given function has the stated decay, sign pattern and Fourier eigenvalue takes complex-analytic and modular machinery. The formal proof of the Kepler conjecture [Hal17] lies at the far end: every step is machine-checked. The arguments of Sections 2 and 6 need no solver at all, and the finite parts of the others are checked in Lean (Section 11.2).

9.4. **Other dimensions.** The lattices D_5, E_6 and E_7, with their Voronoi cells [CS88, Cox73], are the next cases for the same argument. The truncation to a ball in which no three caps meet, and the labelled kernels that bound the union of the caps, carry over in form to every dimension, and the second variation at the lattice takes the same shape in terms of the adjacency of the relevant root system [Hum90]. Two things change with rank. The number of combinatorial regions grows with the Weyl group, and the problem of several deviations becomes harder rather than easier, since the subsets of active directions grow combinatorially while the failure of the second-order method at the dense configurations (Appendix C.1.4) has no reason to disappear. The cap argument of Section 6 needs a fixed polytope containing the relevant body, an enclosure that is tight in the critical direction, and a cap formula for the enclosure; whether these exist for E_6 or E_7 is not investigated here.

10. Conclusion

10.1. **What is proved.** The twenty-four-cell conjecture holds at every centre whose neighbours within $2\sqrt{2}$ all touch it (Theorem 1.1). The proof divides by the number of contacts: at most 22 by the covering estimate (Corollary 5.17), 23 by the pair-angle inequality and its three-point certificate (Theorems 5.40 and 5.72), and 24 by the classification of twenty-four-point kissing configurations (Theorem 5.25), proved here from the positive kernel of [LLM24], every property of which that the proof uses is verified (Proposition 5.29). With

one deviating direction the bound has an elementary proof, through the extremal cap of a cross-polytope (Theorems 1.6 and 6.6).

The conjecture holds also at every centre with at most twenty-three other centres within $\sqrt{6}$ (Theorem 7.8) or with at least twenty-eight (Theorems 7.25, 7.32 and 7.35 and Propositions 7.37 and 7.38); at every centre whose twenty-four neighbours within $\sqrt{6}$ lie on the rays of a root system (Corollary 7.41), whose contacts contain enough of a root system (Corollary 7.5), or whose neighbours within $\sqrt{6}$ are within $\varepsilon_0 = 4 \cdot 10^{-26}$ of contact (Theorem 7.9); and, by Theorem 1.4, every periodic or saturated packing whose centres are all of these kinds has density at most $\pi^2/16$. With no assumption on the centres, the same lower bounds for the cells give 0.63477 for the density of every packing of unit balls in $\mathbb{R}^4$ (Theorem 8.2), below the three-point bound 0.63611 of [CdLS22]. The paper also proves that the kissing number of $\mathbb{R}^4$ is stable under a slack of 0.008 (Theorem 5.78), that no twenty-six points fit at a slack of 0.01468 (Theorem 5.79), that twenty-four centres near a root system leave no room for another within $\sqrt{6}$ (Corollary 7.28), that the root system and the deletions of one root are infinitesimally rigid (Proposition 5.48 and Corollary 5.49), that a saturated configuration of twenty-three contacts shares at most twenty directions with any copy of R (Corollary 5.57), and it computes the second order of the cell volume at the root system exactly (Section 7.3).

10.2. **The contact cases.** The proof for twenty-three contacts does not classify the configurations, and it does not need saturation. It measures the cell inside the ball of radius $\sqrt{3/2}$, where by Lemma 5.67 no three contact caps meet, so that the volume there is an exact function of the pair angles, $A_* + \sum_{i<j} \omega(\gamma_{ij})$ with $A_* = 7.907144\ldots$ (Proposition 5.68). All that is needed is the inequality $\sum_{i<j} \omega(\gamma_{ij}) \geq 8 - A_* = 0.092855\ldots$ for every 23 contact directions, which the deletion of a root satisfies with 0.034 to spare (Theorem 5.69), and that inequality is proved by the three-point method of Bachoc and Vallentin [BV08], the first level of the hierarchy whose second level settles twenty-four points in Theorem 5.30. Lemma 5.71 turns a feasible point of a semidefinite programme in one polynomial f and nine positive semidefinite matrices F_k into a lower bound for the pair sum, and Theorem 5.72 exhibits one of degree 8 with bound 0.0929, above the target by 0.0000444. The relaxation on pair angles alone reaches 79.5 per cent of the target (Remark 5.45); the triples supply the rest, and the polynomial part of the certificate is almost inert, so the three-point matrices carry the argument (Remark 5.73). A relaxation succeeds here, where no classification of the codes is available, because the inequality is strict at the deletion, whereas the deletion lies on the boundary of the classification (Remarks 5.61 and 5.70).

The configuration that the inequality excludes is described as well. Its facets would have to average more than $\frac{32}{23}$, more than any facet volume at twenty-four contacts, so it is not a perturbation of the root configuration; the deletion of a root is infinitesimally rigid, with an exact integer stress on its 88 tight pairs (Proposition 5.48), so it is not a deformation of the boundary configuration either; no contact configuration of 23 directions has a symmetry group transitive on them, all 528 rotation types of order 23 missing the contact condition, the closest by 0.074 in an inner product (Proposition 5.46); and it shares at most twenty directions with any copy of R (Corollary 5.57). Remark 5.19 explains why the missing volume cannot be collected one spherical Voronoi cell at a time, which sends the argument to the pair angles, and Proposition 5.41 makes the inequality concrete: sixty-five pairs at 60°, or the weighted equivalent, where the deletion has eighty-eight, against a ceiling of one hundred and fifteen.

The case of twenty-four contacts, Theorem 5.25, rests on a certificate computed by others, the positive kernel of [LLM24], and the proof from it is given in full here: the bound

of the second level of the hierarchy and its equality case are Lemma 5.27, the positivity of the kernel is Lemma 5.28, the properties of the rational point that these lemmas use are Proposition 5.29, checked in all seven steps of the procedure (Section 5.5.3), and the passage from the four inner products to the root system is Lemma 5.26. That the root system is isolated among contact configurations of 24 directions is also proved, with no certificate (Corollary 5.49). Corollary 5.35 reduces the volume statement for 24 contacts to the inequality $E(W) \geq 8 - 32\pi + 10\pi^2$ between pair and triple angles, through a third-order inclusion and exclusion inside the ball of radius $\sqrt{2}$ that is exact at the root configuration (Proposition 5.34). A certificate for that inequality must be sharp, and its three-point relaxation is not (Remark 5.36); nor is the three-point cardinality bound, at 24.047 even at degree 20 [Lei25]. No route to the statement for twenty-four contacts that stays at the three-point level is complete.

10.3. **Several deviations by elementary means.** Theorem 1.8 follows from the classification of Theorem 5.25 (Corollary 5.31). Elementary means establish the following about it, independently of any certificate. The second-order method gives positivity at every small configuration examined, but degenerates at dense ones: Numerical observations C.4, C.5 and C.7 find, at two configurations, near-null subspaces of the joint Hessian of dimensions 4 and 5 on which the second derivative vanishes to a working precision of up to 45 digits, and Numerical observations C.6 and C.8 sample them in 66 directions and find the quartic coefficient positive in every one. Proposition C.10 gives an exact symmetry of order 48 of the first configuration, Numerical observation C.11 identifies the representation it carries on the near-null subspace, and Proposition C.12 proves by character theory that the quartics invariant under it form a space of dimension 5 instead of 35. A fit of those five coefficients admits a sum-of-squares representation with a small margin (Numerical observation C.13); the fit is numerical, so this is not a proof. Along three swap paths, at $m = 2, 3, 6$, the inequality is proved over the whole angular range of the unconstrained defect (Appendix C.6).

The cap argument of Section 6 does not extend to several simultaneous deviations, for a structural reason. With one deviating direction the other twenty-three half-spaces are fixed, so the body Q is fixed and every parameter sits in the cutting half-space. With m deviating directions the body moves, and the quantity to be bounded compares two overlapping unions of m pieces, whose volumes are of first order in the tilts while their difference is of second order (Remarks 5.2 and 5.3). Splitting the comparison into one term per deviating root and settling each by Theorem 6.8 fails, because the piece E_j of an m-direction configuration is not the piece that the single-deviation theorem controls: the pyramid Pyr_j meets the hyperplane of an adjacent root in a facet, and exchanging that constraint for its tilted version changes $\mathrm{vol}(E_j)$ by an amount comparable to the whole defect. A fixed body is nonetheless available. Dropping the m deviating constraints leaves a polytope Q_D that is bounded for every packing-valid D with $m \leq 5$ and for most larger ones (Proposition 5.5), and the problem becomes one extremal question again, with all parameters in the cutting half-spaces. What remains hard is (5.4) itself: the volume of a union of m caps of Q_D cannot be replaced by a sum, and for the dense configurations Q_D is unbounded.

Written in terms of the contact directions alone, that inequality is the polar-volume inequality $\mathrm{vol}(\mathrm{conv}(W)^\circ) \geq 8$ (Proposition 5.8), equivalently $\mathrm{area}(\partial V_c) \geq 32$ (Lemma 5.10), and in that form it is the part of the conjecture in which two or more contact directions leave the root system at once. Section 5.3 proves that a worst configuration exists and may be taken maximal (Corollary 5.9), and it rules out four approaches by their behaviour at D_4, where a sharp argument has no room to lose anything: term-by-term reduction

(Remark 5.3), any bound built facet by facet from the pairwise contact condition, which delivers $8\pi/(3\sqrt{3}) = 4.8368\ldots$ and at best $96\sqrt{2} - 128 = 7.7645\ldots$ (Remark 5.12), and the convexity and Mahler-type bounds for the polar volume (Remark 5.13). Three more go the same way in Section 5.4: nothing resting on the areas of the spherical Voronoi cells beats $7.7989\ldots$ (Proposition 5.18), the shape that is left over cannot be collected one cell at a time in either the Voronoi or the Delaunay decomposition (Remark 5.19), and the measures of the contact caps and their pairwise intersections, used in full, return $7.997885\ldots$ at twenty-three contacts (Proposition 5.23).

10.4. **What is open.** What is not proved is Conjecture 1.3: a centre with between 24 and 27 other centres within $\sqrt{6}$, one of them at a distance between $2 + \varepsilon_0$ and $\sqrt{6}$. By Proposition 7.23 it follows from statement (G) for twenty-four centres and statement (C) for twenty-five to twenty-seven, and together these would give $\Delta_4 = \pi^2/16$ (Corollary 3.2). Statement (C) is proved from twenty-nine centres on (Theorems 7.25, 7.32 and 7.35) and at twenty-eight (Propositions 7.37 and 7.38). Statement (G) is proved along the push-outs of the root system and near it (Theorems 7.9 and 7.40), and its second order is nonnegative on the packing cone (Propositions 7.14 and 7.18).

Both statements need a localisation that the methods of this paper do not supply: that twenty-four centres with $T \le 8$ lie near a copy of $\sqrt{2}\,\mathsf{R}$. A localisation through the directions alone cannot pass slack 0.01698, where a second twenty-four-point code appears (Proposition 7.21); a margin on pairs at the second level of the hierarchy gives out at slack about $5 \cdot 10^{-4}$, because the root system itself uses it up (Proposition 7.10); and kernels on pairs, even labelled by distance, do not exclude twenty-five centres at any degree tried (Remark 7.27). The localisation therefore has to use the distances through T, at the level of triples or higher.

10.5. **Outlook.** Two routes are visible. The first is a labelled certificate at the second level of the hierarchy, a kernel on subsets of at most two centres that carries their distances from c, for the statement that twenty-four centres with $T \le 8$ lie near a root system; combined with the volume bound near the root system, which Theorem 7.9 provides in a smaller neighbourhood and Section 7.3 describes to second order, it would prove (G) and, through Corollary 7.28, the case of twenty-four close centres in (C). The unlabelled version of that programme, which settles twenty-four contacts, is already large [LLM24], and the labelled one is larger. The second route continues the kernel that settles thirty and twenty-nine centres, the labelled two-point kernel together with the three-point kernel on directions typed by distance (Lemma 4.4), downward through the counts. At twenty-eight centres it settles the case of at most fourteen centres within 2.0161, fifteen with at most three beyond 2.35, and sixteen with none beyond 2.35 (Proposition 7.37), and, with convex combinations of certificates of degree 10 found for single numbers of centres within 2.0161, all the rest, the last vector with a count of how many close directions can be nearly opposite a far one or at nearly the largest inner product with it (Proposition 7.38). On the whole case at twenty-eight, with three types, the value that the solver returns for the programme on its sample points is 3.107 at $d = 8$, above $\frac{9}{8}\pi^2 - 8 = 3.10330$, and 3.095 at $d = 10$, below it; at twenty-nine the step from $d = 8$ to $d = 10$ lowers it from 3.064 to 3.012. Whether a certificate of degree 10 exists for the whole case is not settled: after five rounds of refinement the solutions still violate the inequalities on triples between the samples by more than 10^{-4}, and with the 3276 triples of 28 centres a proof needs about $2 \cdot 10^{-6}$; the combinations, checked vector by vector, do not need one. At twenty-seven the values at $d = 8$, 10 and 12 are 3.157, 3.123 and 3.113, and at twenty-six the value at $d = 8$ is 3.246, all above the

level.[45] For the density alone much less is needed. The same kernel, run with a level in place of $\frac{9}{8}\pi^2 - 8$, already carries Theorem 8.2 below the three-point bound through its level at twenty-six centres, and its levels at twenty-four, twenty-five and twenty-seven centres would carry it further, towards the floor of Remark 8.3.

A third route avoids (G) and (C) as statements about single cells. Lemma 3.3 needs only a transfer of volume between neighbouring centres, antisymmetric and bounded, after which every centre holds at least 8. This is the shape of Hales's proof of the Kepler conjecture, where the score of a decomposition star replaces the volume of a Voronoi cell [Hal05]. In dimension four the transfer can vanish at every centre covered by Theorems 1.1 and 1.2; what has to be shown is that at a centre of Conjecture 1.3, with 24 to 28 neighbours within $\sqrt{6}$ and one of them not in contact, the neighbours whose cells exceed 8 can supply the deficit. Whether a transfer of this kind exists is open, and it is a weaker statement than the twenty-four-cell conjecture, since it can hold even if some cell has volume below 8.

For several deviations among contacts, where the classification already gives the bound, two elementary routes remain open. A lower bound for $\mathrm{vol}_3(F_w)$ that reads the rest of the configuration, not only the pairwise condition at w, and is summable over w, would give the bound through (5.6) and a second proof of Theorem 5.40; Remark 5.12 shows that the purely local constant falls short by an explicit amount, not that no such bound exists. And an exact derivation of the quartic tensor at a dense configuration, in place of the fitted coefficients of Appendix C.4.6, would turn Numerical observation C.13 into a proof at that configuration; the exact invariant space of dimension 5 is in hand, so what that route needs is a derivation, not a search.

11. Computer verification

Every computation on which a proof in this paper depends is of one of two kinds, and the kind is named where the computation is used. The first is exact arithmetic, in $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$ or $\mathbb{Z}[\phi]$, with no rounding at any step. It covers the vertex enumeration of Q and the box certificate of Section 6, the rigidity and root-meet arguments of Sections 5.7 and 5.8, the enumeration of Proposition 5.63, the identities of Propositions 7.19 and 7.21, the positivity of every certificate matrix by an exact LDL^{T} factorisation [GVL13, §4.1], and the value of every certified bound. The second is interval or ball arithmetic with outward rounding [MKC09, Ch. 5], [Joh17]: every computed quantity is an interval that contains the true value, and a sign is decided only when the interval excludes zero. It covers the inequalities that the certificates of Theorems 5.72, 5.78, 5.79, 7.8, 7.25, 7.32 and 7.35 and Propositions 7.34, 7.37, 7.38, 7.42 and 8.1 must satisfy on their domains, which are decided by branch and bound, and the constants of Theorems 7.9 and 7.40, Lemma 7.31, and Corollary 7.28.

Floating-point arithmetic enters in two places only. It is used to find certificates, which are then rounded to rationals and checked as above, so that a solver's tolerance never reaches a proof. And it produces the values quoted to show where a method stops, such as the optima of semidefinite programmes in Sections 7.2 and 7.4; these are marked where they occur, and no proof uses them. The numerical study of the dense configurations in Appendices C.3 and C.4 is of a third kind, a high-precision evaluation at a stated working

[45] These are the objective values that the solver returned for the programmes in floating point, on sampled constraints, and they only show where the method stands. The solver flagged its solutions at $d = 8$ as inaccurate and stopped at $d = 10$ and $d = 12$ with numerical errors, as it did at twenty-eight, where solutions of that kind still gave the certificates of Proposition 7.38, so the values are approximate. An optimum on samples is never larger than that of the full programme, so if these values are close to the optima, no certificate of that form exists; they do not prove it.

precision between 35 and 45 digits, cross-checked at a second precision; it too is reported as evidence and enters no proof.

11.1. **Independent arithmetic.** The certificate of [LLM24] is verified from the deposited data [LLM24d] in code that shares nothing with its authors', with the exact rational arithmetic of FLINT and the ball arithmetic of Arb [Joh17]. The semidefinite programmes behind Theorems 5.72, 5.78, 5.79, 7.25, 7.32 and 7.35 and Propositions 7.37 and 7.38 are posed with [DB16] and solved by the interior-point method of [GC26]; what enters a proof is the rounding of the solution to rationals, checked as above. The box certificates are regenerated by the rule of Appendix D, so that every quoted box count is exact and reproducible. The second-level programmes of Section 7.2 are solved by the solver of [LdL24] at a working precision of 128 bits; no certificate is extracted from them, and their values are quoted only to show where that method stops.

11.2. **The formal part.** Twenty parts of the paper are also verified in the Lean 4 proof assistant [MU21], with no external library; each is a finite computation or a polynomial identity, and Table 8 lists them. Identities are proved over every commutative ring by Lean's ring normaliser, so that they hold verbatim over $\mathbb{R}$. Finite computations are settled either by the kernel itself or by compiled evaluation; the second trusts the compiler as well as the kernel, and the axiom report of each theorem says which was used. No proof in the formal part is left incomplete. What is formalised is the arithmetic. That a certificate turns a programme's inequality into a theorem, that a form is the Hessian of the volume, or that a cap has the stated closed form is proved in the text and not formalised, because the geometry of polytopes and cap measures in $\mathbb{R}^4$ has no counterpart in the libraries on which a formalisation could be built.

TABLE 8. The formally verified parts of the paper.

Result	What Lean checks
Proposition 5.48, Corollary 5.49	The 24 roots, their inner products, the 96 and 88 tight pairs, the positivity of the stress and the equilibrium relation (5.33) at every direction; the neighbour sum of the full root system.
Propositions 5.55 and 5.56	The supports of the roots, the couples of complementary index pairs, the 48 exceptional ordered triples of supports, the 96 triples of roots pairwise at 60°, and the Hadamard symmetry used in the proof.
Proposition 5.63	The 120 vertices of the 600-cell in $\mathbb{Z}[\phi]$, their inner products, the twenty-five inscribed 24-cells, and the exhaustive search: no independent set of 25, five of 24 and 115 of 23.
Theorem 5.72	The exact half: nonnegativity of $f_1, \dots, f_8$, positive definiteness of the nine F_k by exact LDL^{T}, and the bound; and the branch and bound of (5.48) on the admissible domain in exact dyadic arithmetic (421 881 boxes).
Theorem 7.8	Regions I, II$_s$ and II$_f$ of the proof (419 913, 39 551 and 9 545 boxes), with ω, ω', A_*, the cap $S(d)$, the shares and the Gamma envelopes computed inside Lean in outward-rounded 256-bit interval arithmetic.
Proposition 5.29	The zeros of the two-point polynomial σ_2 on $[-1, \frac{1}{2}]$, with their multiplicities, by a Sturm sequence.

Result	What Lean checks
Lemma 5.26	Every positive definite Gram matrix of order at most 4 with 2 on the diagonal and $-1, 0, 1$ off it, and the count of its norm-2 vectors; 24 occurs only for D_4.
Section 7.1	The identities of Lemmas 7.3 and 7.6 and Proposition 7.4, the identity for $a(d_1, d_2)$, the counts 21 and 231, and the finite content of Corollary 7.5.
Section 7.3	The form H built from the root system, the certificate of Proposition 7.14 by exact LDL^{T}, the push-out block, the tilt block of Proposition 7.18, and the expansions of Lemma 7.16.
Theorem 7.9	The arithmetic of ε_0: the quotient of σ_2, its bounds on $[-1, \frac{1}{2}]$, the constants of Lemma 5.76 and the bracket β.
Theorem 5.78	Both certificates from their entries alone: positivity, the bound, and both branch and bounds (441 intervals and 181 869 boxes at degree 8; 2 957 intervals at degree 10).
Theorem 5.79	The certificate of degree 10 from its entries alone: positivity, the value of the bound, the inequality at $\lvert C\rvert = 26$, and both branch and bounds.
Theorem 7.25	The certificate from its entries alone: $A_1, \dots, A_{12}$ and the bordered matrix of (b) positive semidefinite by exact LDL^{T}, with t computed exactly; the pair inequality (a) by branch and bound in exact dyadic arithmetic, with Π bounded below in outward-rounded 256-bit interval arithmetic (191 740 boxes closed and 180 310 set aside); the bracket (c) on 356 intervals; and $31m + \frac{t}{2} < \frac{9}{8}\pi^2 - 8$.
Lemma 5.50 and Theorem 5.51	The annihilating polynomials of $4N$ and $2K$, the ranks over $\mathbb{Q}$, the diagonal $\frac{11}{16}$, the ratio $\sqrt{96/11}$ and the arithmetic of the radius $2/\sqrt{397}$.
Section 6	The 25 vertices of Q, $\operatorname{vol} Q = \frac{25}{3}$, Lemma 6.3, the fourth derivative of Lemma 6.5, the three ranges of the case $\lvert S\rvert = 1$ and the 303 boxes of the independent check of the case $\lvert S\rvert \geq 2$ (Remark 6.7).
Proposition 7.19	The integrals J_1 and J_2 in closed form in $\mathbb{Q}(\sqrt{3})[c, s]/(c^2 + s^2 - 1)$, the identity $3J_1^2 - 4\sqrt{3}J_2 = 32$, the conditions on the slices, the Gram matrix at $\theta = 0$ and the perfect square.
Proposition 7.21	The 24 rational points of S^3, their slack $\frac{849}{50000}$ and the inner product at distance $\frac{57}{250}$ from $-1, -\frac{1}{2}, 0, \frac{1}{2}, 1$.
Corollary 7.28	The quartic of part (ii), its double zero at $1/\sqrt{2}$, its values at τ and at 1, and the budget $3\beta + \frac{3}{2}\gamma = 6h^2$; the 5-design identities of the roots and the deep hole; τ, h and $6h^2$ at $\rho = 2/\sqrt{1 - 0.016}$, the signs for every $\rho \in [2, \sqrt{6})$, and the 200 intervals of part (iii).
Proposition 7.29	The Gegenbauer coefficients of p, the bound on the remainder R of the identity for $-p$, the sign of p on $[-1, 0.614039]$ again by 29 intervals, $0.614039 > \tau$, the bound $(24p_0)^2 > 0.26785 \sum_k p_k^2/c_k$, and $S_1 = \cdots = S_5 = 0$ for the roots.

Result	What Lean checks
Proposition C.10	The 384 signed coordinate permutations, all distinct and each mapping the roots onto themselves; exactly 48 of them fix A_{18}, those fixing one coordinate with its sign; and two orthogonal maps carrying the roots onto themselves and A_{18} onto the configurations grown from roots 5 and 12.

11.3. **What is printed and what is archived.** Every exact claim that a proof uses is stated in the text: as a displayed fraction or polynomial, as a table entry, or as the rule that produces it. A reader with a computer algebra system can re-derive every coefficient list of the arcs from Definition C.1 and the boundary curves of Theorems B.7 and B.9, and the roots are numbered as in Appendix E. The cap inequality of Section 6 is proved by hand in full, and the subdivision rule of Remark 6.7 regenerates the 303 boxes that check it independently. The objects that are not printed are the matrices of the semidefinite certificates and the lists of boxes of the branch and bounds; they are in the archive described below, together with the programmes that check them and the Lean development.

Data and code availability

The supplementary package, with the certificates, the programmes that check them, the logs of every run and the Lean development of Section 11.2, is archived on Zenodo [BBMB26]. The certificate verified in Section 5.5.3 is the data set of [LLM24d], deposited by its authors under the MIT licence; it is included, with its checksum and licence notice, so that the proof of Theorem 5.30 can be rerun from the archive alone. The programmes of the package, apart from the deposit of [LLM24d], were written by Deep Bhattacharjee. They are in Python, with NumPy, SciPy, SymPy and mpmath, with python-flint for the ball arithmetic of Arb, and with CVXPY and Clarabel for the semidefinite programmes; in Julia, with ClusteredLowRankSolver, LasserreSphericalCodes and Nemo for the second level of the Lasserre hierarchy and the certificate of [LLM24d]; in C, for the enumeration of the independent sets in the graph of the 600-cell and for an integral over $O(4)$ in the re-verification of that certificate; and in shell scripts that run the checks in sequence. The formalisation of Section 11.2 is written in Lean 4 alone, with no external library. Deep Bhattacharjee carried out every computation and every check of the paper, in C, Python, Julia and Lean 4, with the assistance of Claude Code, the command-line tool of Anthropic in which their AI model Claude writes programmes and runs them in a shell. Part of the code, including the re-verification of the computations, the programmes behind Sections 5.13, 7, 7.1, 7.3 and 7.4 and part of the Lean development, was developed with that assistance.

Competing interests

None.

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None.

Appendix A. Index of notation

Table 9 is the complete list of symbols the paper uses for the quantities it introduces; nothing in the argument depends on a convention fixed elsewhere.

TABLE 9. Index of notation.

Symbol	Meaning
R	the D_4 root system, $\{\pm e_i \pm e_j : 1 \le i < j \le 4\}$, $\lvert\mathsf{R}\rvert = 24$
Λ	the scaled D_4 lattice $\sqrt{2}\,D_4$, nearest-neighbour distance 2
$W(D_4)$	the Weyl group of D_4, of order 192
$\mathcal{V}$	the reference cell $\{z : \langle z, \alpha\rangle \le \sqrt{2}\ \forall \alpha \in \mathsf{R}\}$, a regular 24-cell of volume 8
V_c, V_c^{act}	the Voronoi cell of a centre c, and the cell cut out by the centres within $2\sqrt{2}$ alone
W	a contact configuration: unit vectors pairwise at least 60° apart (Definition 5.7)
u_i, u_0	the normalised roots $\alpha_i/\sqrt{2}$, and the root from which a direction deviates
φ, η	the tilt angle of a deviating direction and its deviation direction, a unit vector orthogonal to u_0
$\Delta(\varphi, \eta)$	the volume defect $\mathrm{vol}(V_c) - 8$ of a single deviation (Section 6)
E_j, C_j	the pieces of the free-volume identity (Theorem 5.1)
Q, Q_D	the polytope cut out by the 23 undeviated roots, of volume $\frac{25}{3}$, and by the roots outside D
B	the cross-polytope $\{\lvert x_0\rvert + \lvert x_1\rvert + \lvert x_2\rvert + \lvert x_3\rvert \le 2\}$ enclosing Q (Lemma 6.3)
$B(R)$	the closed ball of radius R about the centre of the cell, the origin of its coordinates
$\mathrm{Cap}_K(w)$	the cap $\{x \in K : \langle x, w\rangle \ge 1\}$ of a body K
$\omega(u)$	the pair weight: the overlap in $B(\sqrt{3/2})$ of the caps of two contacts at inner product u (Proposition 5.68)
A_*	$\frac{9}{8}\pi^2 - 23S(2) = 7.907144\ldots$, the volume of $B(\sqrt{3/2})$ less 23 contact caps
P, B	the polynomial and the bound of the three-point certificate (Lemma 5.71 and Theorem 5.72)
Φ, Ψ	the two distance criteria (Proposition 7.1 and Corollary 7.7)
$S(d)$, $s(d)$	the cap of $B(\sqrt{3/2})$ cut off by a centre at distance d, and $S(2) - S(d)$
H_z, $V(Y)$	the half-space $\{x : \lvert x\rvert \le \lvert x - z\rvert\}$ of a centre z, and the intersection of the H_y, $y \in Y$
$K(Y)$	the inversion hull: the convex hull of 0 and the points $4y/\lvert y\rvert^2$, $y \in Y$ (Proposition 7.4)
$U(Y)$, $T(Y)$	the volume of the union of the caps of Y in $B(\sqrt{3/2})$, and $\frac{9}{8}\pi^2 - U(Y)$ (Section 7.4)
$\Pi(h_1, h_2, u)$	the volume of the intersection of two caps (Lemma 7.24)
$a(d, d')$	the largest inner product of the directions of two centres at distances d, d' from c that the packing allows
$N(\rho)$	the number of centres within ρ of c (Theorem 7.32)

continued on the next page

Table 9 (continued)

Symbol	Meaning
ε_0	the constant $4 \cdot 10^{-26}$ of Theorem 7.9
(G), (C)	the two statements of Proposition 7.23

Appendix B. The single-deviation defect region by region

The identities of Section 5 hold for any number of deviating directions. This appendix returns to the case of one deviating direction, where the cell can be followed exactly, region by region, as the deviation varies. Nothing in it is needed for Theorem 1.6, which Section 6 proves by an argument of another kind; what it gives is the defect itself, as an explicit function, on the boundary of a fundamental domain.

B.1. The deviation domain and its fundamental triangle.

B.1.1. *The two remaining parameter spaces.* Away from the root-aligned case of Lemma 2.8, the free-volume identity of Theorem 5.1 has two parameter spaces of different structure.

- *Several directions at once* ($m \geq 2$): a choice of m directions on S^3, each with its own deviation in the 3-dimensional tangent space at its root. Appendix C studies this case to second order at the root system and along three exact paths.
- *One direction* ($m = 1$): a tilt φ and a unit deviation direction η orthogonal to the undeviated root, a point of a 2-sphere, which the symmetries of R reduce to a triangle (Appendix B.1.2). Appendix B.2 and the three subsections after it treat the boundary of that triangle.[46]

B.1.2. *The fundamental domain of deviation directions.* Fix a root $\alpha_0 \in \mathsf{R}$ and let $u_0 = \alpha_0/\sqrt{2}$ be the corresponding contact direction. The tangent space orthogonal to u_0 is 3-dimensional, and a deviation direction is any unit vector η in it; the deviating cap direction is then $u_1(\varphi, \eta) = \cos\varphi\, u_0 + \sin\varphi\, \eta$.

Choose coordinates in which $u_0 = (1,1,0,0)/\sqrt{2}$, and put

$$v_1 = \tfrac{1}{\sqrt{2}}(1,-1,0,0), \qquad w_1 = \tfrac{1}{\sqrt{2}}(0,0,1,1), \qquad v_2 = \tfrac{1}{\sqrt{2}}(0,0,1,-1),$$

an orthonormal basis of $u_0^\perp$ consisting of normalised roots.

Lemma B.1 (The fundamental triangle)**.** *The stabiliser of u_0 in $W(D_4)$ has order 8 and acts on $u_0^\perp$ as the group of sign changes of the three coordinates in the basis v_1, w_1, v_2. The spherical triangle with vertices v_1, w_1, v_2 is therefore a fundamental domain for its action on the unit sphere of $u_0^\perp$. Two further orthogonal maps, which fix u_0 and map R onto itself but do not lie in $W(D_4)$, map the triangle onto itself: $S = \mathrm{diag}(1,1,1,-1)$, which fixes v_1 and exchanges w_1 and v_2, and the matrix H of Lemma B.4, which fixes v_2 and exchanges v_1 and w_1.*

Proof. $W(D_4)$, the group of permutations and even sign changes of the coordinates, has order 192 and acts transitively on the 24 roots, so the stabiliser of a root has order 8. The transposition of the first two coordinates negates v_1; the transposition of the last two negates v_2; and the product of that transposition with the sign change of the last two coordinates

[46] The two cases differ in kind. With one deviating direction the twenty-three other half-spaces are fixed and every parameter sits in one cutting half-space; with several the body that is cut moves with the parameters (Proposition 5.5).

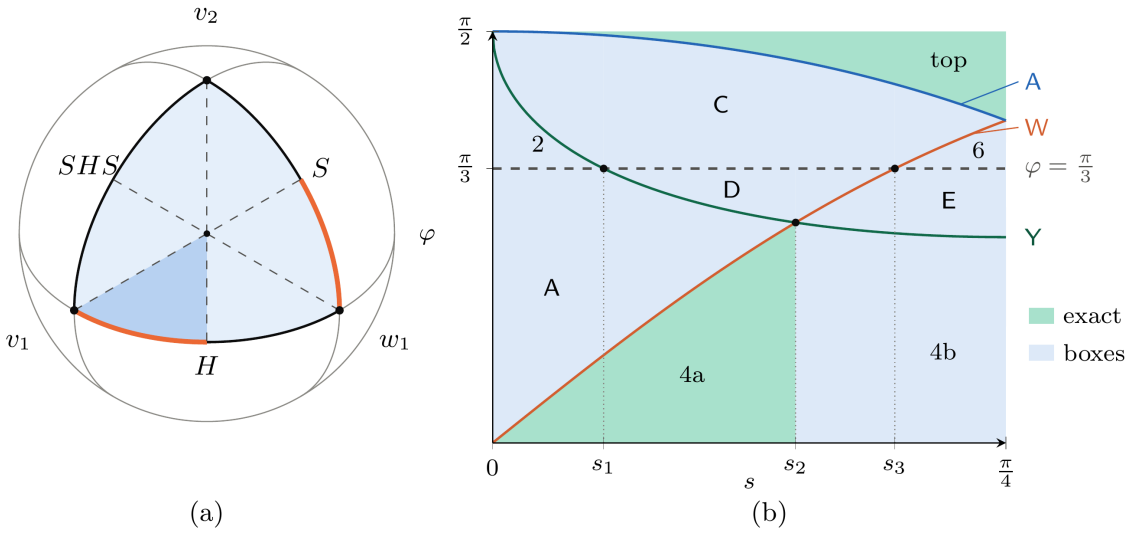

FIGURE 44. (a) The fundamental triangle of Lemma B.1 on the sphere of deviation directions, with vertices v_1, w_1, v_2. The stabiliser of u_0 in $W(D_4)$ carries it to the eight octants. The reflections S and H, through the dashed great circles, generate the group S_3 of Lemma B.4, which permutes the vertices and the edges; the six small triangles cut out by the three dashed circles are the images of one of them. (b) The first arc in the plane of tilt φ and arc position s, to scale: the breakpoint curves of Theorem B.7 and the eight regions they bound, shaded by the certificate of Theorem B.8 that each carries. By Propositions B.3 and B.5 the same picture holds on the other two edges. The interior of the triangle is settled by Theorem 6.8.

negates w_1. Each fixes u_0 and the other two basis vectors, so they generate the whole stabiliser, and its images of the triangle are the eight octants, which tile the sphere. The statements about S and H are direct evaluations. S changes one sign and H is not a signed permutation, so neither lies in $W(D_4)$; both lie in the full symmetry group $W(F_4)$ of the 24-cell, in which $W(D_4)$ has index 6. ■

Every deviation direction is therefore equivalent to one in this triangle. Its edges are the arc from v_1 to w_1, the arc from w_1 to v_2 and the arc from v_1 to v_2; the first two are parametrised as

$$\eta_{v_1 w_1}(s) = \cos s\, v_1 + \sin s\, w_1, \qquad \eta_{w_1 v_2}(s) = \cos s\, w_1 + \sin s\, v_2, \qquad s \in [0, \tfrac{\pi}{4}].$$

The range $s \in [0, \pi/4]$, half of each quarter circle, suffices because H reflects the first arc about its midpoint and S reflects the second about its own. Appendix B.2 through Appendix B.5 treat the two arcs in full; that is where the breakpoints and the exact defect polynomials are found. The interior is not reached this way. A convexity argument carrying the boundary inward would suffice, and Appendix B.1.6 shows that the natural one fails. The interior is settled by the argument of Section 6, which does not use the triangle. Figure 44 shows the triangle and which argument covers which part of it.

The question for Appendix B.2 through Appendix B.5 is the sign of $\Delta(\varphi, s)$ for $0 < \varphi < \pi/2$ and $s \in [0, \pi/4]$ on both arcs. Corollary B.6 settles it outside thin explicit margins, and Theorem 6.8 settles it everywhere.

B.1.3. *Where the worst direction sits.* Lemma B.1 says where every deviation direction can be moved; it does not say where in the triangle the defect is smallest. A direct search answers that, and the answer decides which edges the certificates have to reach.

Minimising $\Delta(\varphi, \eta)$ numerically over the entire transverse 2-sphere, at each of many sampled tilts, and comparing the result against direct evaluation along three symmetry-distinguished one-parameter families, shows the following. The three families are the six root directions orthogonal to u_0, the two coordinate-axis directions orthogonal to u_0 (the vertex v_1 family), and the half-integer-sign family exemplified by $e_* = (-1, 1, -1, 1)/2$ (the vertex w_1 family). One of the three matches the global minimum for almost the whole range of tilts: the coordinate family on $\varphi \in (0, 0.16] \cup [1.26, \pi/2)$, the root family on $\varphi \in [0.18, 1.09]$. The exception is the window $\varphi \in (1.09, 1.26)$, where the minimiser lies strictly below all three.

A finer search in that window locates it. At ten tilts spanning and bracketing the window, from $\varphi = 1.05$ to $\varphi = 1.30$, a Nelder–Mead search with 30 random restarts per tilt over the full 2-sphere returns a minimiser one of whose coordinates (a, b, c) relative to the frame (v_1, w_1, v_2) vanishes to at least five decimal places, values of order 10^{-5} against coordinates of order 1. To the resolution of the search, the minimiser is therefore on one of the three great circles $a = 0$, $b = 0$, $c = 0$ through pairs of $\{v_1, w_1, v_2\}$, and never at an interior point with all three coordinates bounded away from zero. For the five tilts $\varphi \in \{1.15, 1.18, 1.20, 1.22, 1.24\}$ it sits on the circle $b = 0$ through v_1 and v_2, which is the third edge of the triangle rather than either of the two arcs; at the window's ends it sits on $a = 0$ or $c = 0$.

This is a numerical finding, and the theorem of Section 6 does not depend on it. It shows that the certificates of Appendix B.2 through Appendix B.5 have to reach the third edge as well as the two arcs. The next subsection gives the exact symmetry that carries the first arc onto the third edge.

B.1.4. *The third edge, closed by an exact symmetry.* The edge from v_1 to v_2 needs no computation of its own: an isometry carries it onto the first arc.

Lemma B.2 (The third edge is the image of the first arc)**.** *Let $S = \mathrm{diag}(1, 1, 1, -1)$ in the coordinates of Lemma B.1. Then*

$$Su_0 = u_0, \qquad Sv_1 = v_1, \qquad Sv_2 = w_1, \qquad S^2 = \mathrm{Id}.$$

Moreover S permutes the 24 roots of R bijectively (a root $\pm e_i \pm e_j$ maps to another root of the same index pair, with the sign at index 4 flipped when $i = 4$ or $j = 4$ and left unchanged otherwise), so S is an automorphism of the root system fixing u_0.

Proof. Immediate: S negates only the fourth coordinate, and u_0, v_1 both have a zero fourth coordinate, while v_2 and w_1 differ exactly in the sign of that coordinate. $S^2 = \mathrm{diag}(1, 1, 1, 1) = \mathrm{Id}$ since $(-1)^2 = 1$. A root $\pm e_i \pm e_j$ goes to $\pm e_i \pm e_j$ with at most the sign of e_4 changed, which is again a root. ∎

Proposition B.3 (The v_1-v_2 edge needs no separate certificate)**.** *For every φ in the packing-valid range and every s,*

$$\Delta\big(\varphi, \eta_{v_1 v_2}(s)\big) \;=\; \Delta\big(\varphi, \eta_{v_1 w_1}(s)\big),$$

where $\eta_{v_1 v_2}(s) = \cos s\, v_1 + \sin s\, v_2$ is the third edge and $\eta_{v_1 w_1}$ is the first arc's own parametrisation (Lemma B.1). Consequently, by Theorems B.7 and B.8, $\Delta > 0$ on the v_1-v_2 edge wherever Theorem B.8 certifies the first arc: on two of the eight regions with no excluded margin, and on the other six outside a margin of less than one percent of the region's area.

Proof. Since S fixes u_0 linearly, S carries the deviated direction $u_1(\varphi,\eta) = \cos\varphi\, u_0 + \sin\varphi\, \eta$ to $u_1(\varphi, S\eta)$ for every φ, η. Since S is also an automorphism of R fixing u_0 (Lemma B.2), S carries the entire packing configuration at deviation η (the 23 undeviated roots, permuted among themselves, together with the one deviating root $u_1(\varphi,\eta)$) to a congruent configuration at deviation $S\eta$: the two Voronoi cells are related by the isometry S, hence have identical volume, hence identical defect. Applying this with $\eta = \eta_{v_1v_2}(s)$ and using $Sv_1 = v_1$, $Sv_2 = w_1$ from Lemma B.2 (so, by linearity, $S\eta_{v_1v_2}(s) = \cos s\, v_1 + \sin s\, w_1 = \eta_{v_1w_1}(s)$ exactly, for every s) gives the displayed identity. ∎

The third edge, where Appendix B.1.3 found the worst direction for part of the transition window, therefore carries the first arc's certificate. The isometry is S, the reflection that also halves the second arc in Appendix B.4.

B.1.5. *The full symmetry group of the fundamental triangle, and its consequence for the second arc.* Together with H, the symmetry S generates a group large enough to settle the second arc as well.

Lemma B.4 (The fundamental triangle's full symmetry group)**.** *The group generated by* $S = \mathrm{diag}(1,1,1,-1)$ *and the Hadamard matrix* $H = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$ *has order exactly* 6*, every element fixes* u_0*, and every element is an automorphism of* R*. The induced action on* $\{v_1, w_1, v_2\}$ *realises all six permutations of this three-element set, so the group is isomorphic to the symmetric group* S_3*.*

Proof. H is symmetric and orthogonal, so $H^2 = \mathrm{Id}$; it fixes u_0 and v_2 and exchanges v_1 and w_1, and it maps R onto itself (for instance $H(e_1+e_2) = e_1+e_2$, $H(e_1-e_2) = e_3+e_4$ and $H(e_1+e_3) = e_1+e_3$, and the remaining roots follow in the same way). S fixes u_0 and v_1 and exchanges w_1 and v_2. Two transpositions of $\{v_1, w_1, v_2\}$ generate its full permutation group, so the induced action is onto the six permutations; it is faithful, because an element fixing u_0, v_1, w_1, v_2 is the identity. In particular $M := SH$ has order 3 and acts as the cycle $v_1 \mapsto v_2 \mapsto w_1 \mapsto v_1$.[47] ∎

Proposition B.5 (The second arc is the image of the first)**.** *For every* φ *in the packing-valid range and every* s*,*

$$\Delta\big(\varphi, \eta_{w_1v_2}(s)\big) \;=\; \Delta\big(\varphi, \eta_{v_1w_1}(s)\big).$$

Proof. By Lemma B.4, $M = SH$ fixes u_0 and is an automorphism of R, so (exactly as in the proof of Proposition B.3) it carries the packing configuration at any deviation η to a congruent one at deviation $M\eta$, giving $\Delta(\varphi,\eta) = \Delta(\varphi, M\eta)$ for every φ, η. Since $Mw_1 = v_1$ and $Mv_2 = w_1$ (Lemma B.4), linearity gives $M\eta_{w_1v_2}(s) = \cos s\, v_1 + \sin s\, w_1 = \eta_{v_1w_1}(s)$ exactly, for every s. Substituting $\eta = \eta_{w_1v_2}(s)$ gives the displayed identity. ∎

Both identities were also checked numerically, independently of the algebra above, by evaluating the volume-defect function directly (half-space intersection and convex-hull volume). A scan of active-facet patterns over $s \in \{0.02, \ldots, 0.78\}$, covering every combinatorial subpart of the second arc including the B-bounded band, finds the breakpoints of the two arcs coinciding to grid resolution at every sampled s and the defect values agreeing to floating-point tolerance, including at the crossing of curve B. That crossing is produced by vertex b of Theorem B.9, which has no counterpart among the vertices Theorem B.7 names on the first arc. By Proposition B.5 the curve it produces is nonetheless reached on the first arc,

[47] The six matrices, their action on the 24 roots and on $\{v_1, w_1, v_2\}$, were also checked one by one in exact arithmetic over $\mathbb{Q}(\sqrt{2})$.

under M, by a differently labelled vertex, so the separate derivations of the two arcs arrive at the same function through different combinatorial routes.

Corollary B.6 (Positivity on the boundary)**.** *On both arcs $\eta_{v_1w_1}$ and $\eta_{w_1v_2}$, $s \in [0, \pi/4]$, and on the third edge, $\Delta(\varphi, s) > 0$ for $0 < \varphi < \pi/2$ outside the excluded margins of Theorem B.8: everywhere on two of the eight regions, and on the other six outside a margin of less than one percent of the region's area.*

Proof. By Propositions B.3 and B.5, the second arc and the third edge carry the same function of (φ, s) as the first arc, on which Theorems B.7 and B.8 give the stated sign and margins. ∎

The boundary of the triangle therefore carries the first arc's certificates with the first arc's margins. The direct computation on the second arc in Appendix B.5 is not needed for this; it gives the defect region by region there, by a route that shares nothing with Proposition B.5, and so checks that identity independently.

The symmetry argument stops at the boundary. No isometry preserving R and fixing u_0 moves a generic interior point onto an edge, so the interior of the triangle is not reached this way. It is reached in Section 6 by an argument that abandons the triangle altogether.

B.1.6. *The interior of the fundamental triangle.* With the boundary settled, the question is whether a structural argument carries the interior along without a two-parameter closed-form derivation. The obvious candidate fails; this subsection records why, and what direct computation shows about the structure of the interior.

A candidate reduction, tested and falsified.

> *Candidate argument (false).* If $\Delta(\varphi, \cdot)$ were concave, for every fixed φ, along every geodesic arc of the transverse 2-sphere lying inside the closed fundamental triangle, then for any interior point Q, picking a geodesic through Q with both endpoints on the triangle's boundary would give $\Delta(\varphi, Q) \geq \min(\Delta(\varphi, \text{endpoint}_1), \Delta(\varphi, \text{endpoint}_2)) \geq 0$, the last inequality by Corollary B.6, which covers all three edges. This would settle the whole triangle with no further derivation.

This argument is false. Along geodesics from each vertex to random points on the opposite edge, and along chords between random interior points, each sampled at 25 values of φ, the discrete second differences are positive at several points, most of them with φ near $\pi/2$; there the defect is locally convex, not concave. The effect does not come from the grid. At one such point, halving the step eight times, from $h = 0.08$ to $h \approx 3 \times 10^{-4}$, gives values of $d^2(h)/h^2$ that converge to $+0.3056$ and agree to four significant figures over the last five halvings. Discretisation error would shrink faster than h^2 and have no stable limit, and no combinatorial breakpoint lies nearby, since the local vertex count stays 37 in a window around the point; so the second derivative there is positive. Hence $\Delta(\varphi, \cdot)$ is not concave along every interior geodesic, and the interior cannot be reduced to the boundary in this way at any φ of the domain: the effect occurs throughout and is only largest near $\varphi = \pi/2$. This rules out every reduction of the interior to the boundary by concavity along geodesics. The interior's own combinatorial structure. Direct evaluation of the number of active vertices of the local Voronoi region, scanned over a fine grid of the closed triangle at a fixed $\varphi = 1.45$ (in the region-top band, where both boundary arcs have the simple 32-vertex type throughout), shows that the boundary's combinatorial simplicity is special to the boundary: it does *not* extend to the interior. A grid of 861 points finds the 32-vertex type only along a band adjacent to the edges (25% of the sampled points), a 34-vertex type in a second band

just inside that (17%), a 37-vertex type filling most of the remaining interior (53%, the plurality type), and a further 40-vertex type occupying a smaller region near the triangle's centre (5%). The interior is therefore combinatorially richer than either boundary arc: it is not one region but at least four; the exact curves separating them, and a volume formula on any one of them, are not derived here; the map above establishes that such curves exist and locates them to the resolution of its grid.

The partition respects the S_3 symmetry of Lemma B.4. The map is invariant under the reflection fixing v_1 and exchanging w_1, v_2, the symmetry S of Lemma B.2; since S is an isometry of the whole configuration, it maps each combinatorial type onto the type of the congruent point. By the same argument, the full 6-element group of Lemma B.4 acts on the interior as well as on the boundary: every one of its elements is an automorphism of R fixing u_0, hence carries any interior deviation direction to a congruent configuration at the image direction, with identical defect and identical combinatorial type. Hence it suffices to derive the combinatorial map and the volume formula on a single fundamental domain of the triangle under this S_3 action (one-sixth of the triangle, e.g. the wedge from v_1 to the midpoint of arc v_1w_1 to the triangle's centre) and extend by the group, instead of treating the interior as one undifferentiated two-parameter domain six times the necessary size.

These two findings close the regional approach to the interior. The natural structural reduction is false, and the combinatorial structure of the interior is strictly richer than that of the boundary, so a region-by-region derivation would need at least four new volume formulas before any certificate. Section 6 uses an argument that does not use the triangle.

B.2. **Breakpoint classification on the first arc.**

B.2.1. *Setup.* Fix the arc $\eta_{v_1w_1}(s) = \cos s\, v_1 + \sin s\, w_1$, $s \in [0, \pi/4]$, of Lemma B.1, and write $u_1(\varphi, s) = \cos\varphi\, u_0 + \sin\varphi\, \eta_{v_1w_1}(s)$ for the deviating cap direction. As φ increases from 0 at fixed s, the active-facet pattern of the local Voronoi region changes at finitely many exact values of φ, the arc's *breakpoints*. At a breakpoint the deviating hyperplane $\langle x, u_1\rangle = 1$ passes through a vertex of the polytope Q of the twenty-three undeviated roots (Lemma 6.1), so that vertex enters or leaves the cell.

Theorem B.7 (Complete breakpoint classification, first arc)**.** *Exactly four exact curves account for every active-facet-set transition of $u_1(\varphi, s)$ across the entire domain $s \in [0, \pi/4]$:*

(1) the universal curve *$\varphi = \pi/3$, from the apex $2u_0$ of Q, which lies on eight facets, independent of s;*
(2) a curve W from the sign vertex $w = (1,1,1,1)/\sqrt{2}$, satisfying $\sin s = \tan(\varphi/2)$;
(3) a curve A, the image of W under the symmetry H of Lemma B.1, from the axis vertex $a = \sqrt{2}\, e_1$, satisfying $\cos s = \tan(\varphi/2)$;
(4) a curve Y from the sign vertex $y = (1,-1,1,1)/\sqrt{2}$, satisfying $\sin\varphi\,(\sin s + \cos s) = 1$.

Each equation is the condition $\langle x, u_1(\varphi, s)\rangle = 1$ at the named vertex x, which lies on $\langle x, u_0\rangle = 1$ or, for y, on $\langle x, u_0\rangle = 0$. For w, for instance, $\langle w, u_0\rangle = \langle w, w_1\rangle = 1$ and $\langle w, v_1\rangle = 0$, so the condition reads $\cos\varphi + \sin\varphi \sin s = 1$, that is $\sin s = \tan(\varphi/2)$.[48] That no further curve exists is a numerical statement: a dense scan, over 150 values of s with a refined bisection in φ at each, finds at most four breakpoints at every sampled s, each on one of the four curves to a tolerance far below the separation of the curves.

[48]The apex: the eight roots β with $\langle \beta, \alpha_0\rangle = 1$ satisfy $\langle 2u_0, \beta/\sqrt{2}\rangle = 1$, so $2u_0$ lies on their eight facets, and $\langle 2u_0, u_1\rangle = 2\cos\varphi$ equals 1 exactly at $\varphi = \pi/3$.

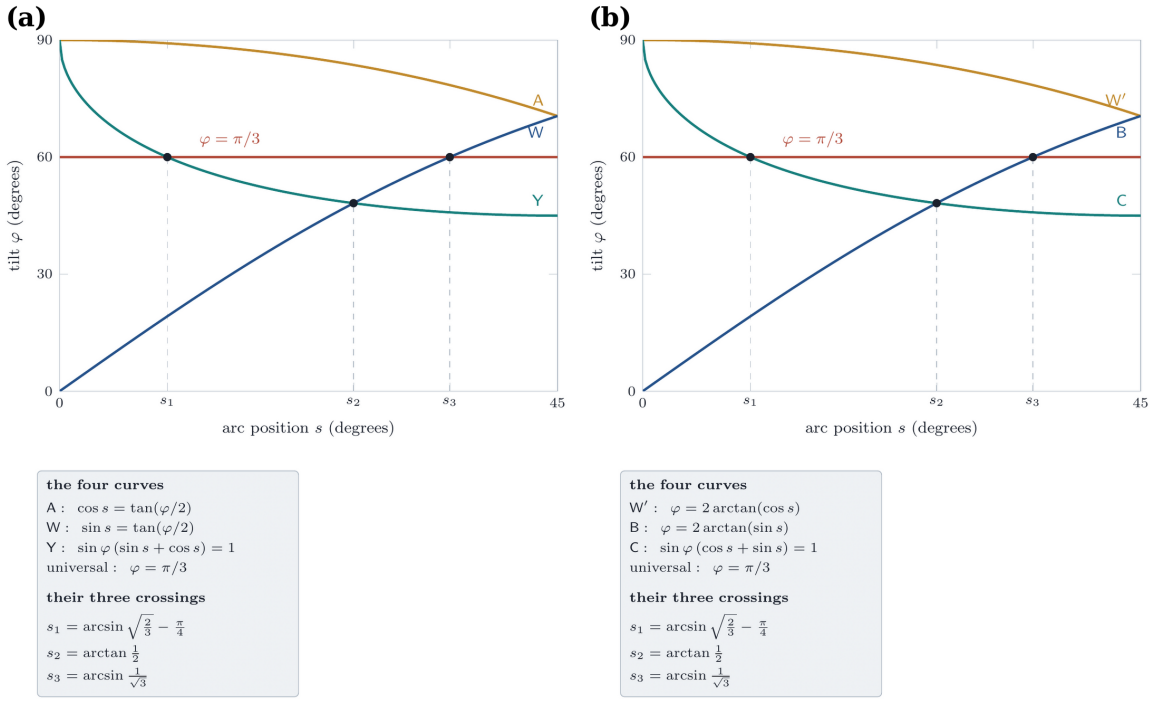

FIGURE 45. (a) The four breakpoint curves of the first arc, drawn to scale from the closed forms of Theorem B.7: the universal curve $\varphi = \pi/3$ and the curves A, W, Y. They meet in exactly three points, at the abscissae s_1, s_2, s_3 listed at the right, and the eight combinatorial regions of Appendix B.3.2 are the cells of the resulting arrangement. The curves A and W meet only at $s = \pi/4$, where the reflection symmetry of the arc identifies them. (b) The four breakpoint curves of the second arc, drawn to scale from the closed forms of Theorem B.9. Comparing with Figure 45(a) shows what Proposition B.5 later proves: the three equations here are exactly those of A, W and Y on the first arc, and the arrangement, crossings included, is the same one, even though the curves come from different vertices of $\mathcal{V}$. The band immediately below region top splits into three sub-parts at the crossings s_1, s_2, s_3.

B.2.2. *The eight combinatorial regions.* A point of the domain lies above or below each of the four curves, and the four signs determine the combinatorial type of the cell. Eight sign patterns occur, so the domain has eight regions; the vertex patterns at representative points of each confirm that each is one combinatorial type, and a dense grid scan finds no further boundary. We name them region top, the band above A; region 4, the band below both W and Y; and regions 2, 6, A, C, D, E between them, described in Appendix B.3, which certifies the sign on each. The names are labels only.

B.3. **Certificates on the first arc.**

B.3.1. *Exact remapping and box covering.* Each of the eight regions of Theorem B.7 is certified by one of two exact methods, according to whether its boundary curve is rational in Weierstrass coordinates.

Exact zero-margin certificates. When a region's lower (or upper) boundary curve is rational in $\mathsf{u} = \tan(\varphi/2)$ and $\mathsf{v} = \tan(s/2)$, as are the universal curve and the curves W, A of

Theorem B.7, a further exact substitution maps the curved region onto the unit square, after which the volume-defect numerator and denominator become two explicit polynomials with coefficients in $\mathbb{Z}[\sqrt{2}]$ (or, where a crossover point of the region's own two bounding curves is used to rescale the domain, $\mathbb{Z}[\sqrt{5}]$). If, in the Bernstein basis over the unit square, every coefficient of the numerator has one sign and every coefficient of the denominator has one sign, then Δ has one sign on the whole open region, with *no excluded margin* (Appendix G works through a small instance).

Box-covering certificates. When the region's own boundary is not rational (the curve Y is of genus one, see region 4 below), the region is instead covered by many small rectangular boxes in the plain Weierstrass coordinates, with corners chosen numerically inside the domain, and the same exact Bernstein sign test is applied to each box. This leaves out a thin margin of known size along the boundary.

B.3.2. *The region count.* A grid scan of the reduced domain $s \in (0, \pi/4)$, $\varphi \in (0, \pi/2)$ at 150×200 resolution, with a band of 0.01 radians excluded around each of the four breakpoint curves, finds *eight* combinatorially distinct types.[49] The four curves cross exactly three times in $s \in (0, \pi/4)$: Y meets the universal line at $s_1 = \arcsin\sqrt{2/3} - \pi/4 = 0.169918\ldots$, the W-curve meets Y at $s_2 = \arctan(\frac{1}{2})$, and W meets the universal line at $s_3 = \arcsin(1/\sqrt{3})$. Region 4, bounded above first by W and then by Y, is a single combinatorial region: direct convex-hull volumes at representative points on both sides of the crossover agree with one symbolic volume formula to twelve decimal places. It is certified in two pieces (Appendix B.3.3) because only one of them has a rational boundary.

B.3.3. *The eight regions, one by one.*

Theorem B.8 (Certificates, first arc)**.** *Each of the eight regions of Appendix B.3.2 carries a positivity certificate for Δ. On region top and on the first piece of region 4 it is an exact Bernstein certificate on the whole region; on the other regions it is a box-covering certificate that excludes a margin of less than one percent of the area of the region. In each certificate the Bernstein coefficients of the numerator and of the denominator have one sign throughout, so Δ has one sign on the certified part.*

For each of the eight regions we give the number of vertices at a representative point (moving vertices, from the deviating cap, plus fixed vertices; the numbers are the same at several sample points across the region), its exact fixed-volume contribution (the boundary simplices not involving the deviating cap, which sum to an exact rational constant independent of (φ, s)), and its certificate method.

Region top. Bounded below by the Hadamard-image curve A ($\cos s = \tan(\varphi/2)$) and above by $\varphi = \pi/2$, for the *entire* range $s \in (0, \pi/4)$, the only region needing no internal crossover point, since its two bounding curves never meet inside the reduced domain. Both boundary curves are rational in (u, v), which gives the simplest remapping onto a unit square, and the Bernstein coefficients of numerator and denominator have one sign throughout: an exact certificate with no excluded margin.

Region 2. Bounded by $\pi/3 < \varphi <$ the Y-curve ($\sin\varphi(\sin s + \cos s) = 1$) for $s \in (0, s_1)$. A generic point has 29 vertices, 24 fixed and 5 moving; of the fixed ones, 23 are axis and sign vertices of the kinds named in Theorem B.7, and the last is the apex $2u_0$, which the

[49] The band is needed because grid points on or numerically near a curve can register a spurious extra type through near-degeneracy in the floating-point half-space intersection; a coarser scan at 60×90 without it reports nine.

deviating hyperplane no longer reaches once $\varphi > \pi/3$. The fixed-only boundary-simplex sum is exactly 29/4. Box-covering certificate.

Region 4 (two-piece). Bounded above, for all $s \in (0, \pi/4)$, by the lower envelope of the W- and Y-curves, crossing exactly at $s_2 = \arctan(\frac{1}{2})$ (equivalently $\varphi_2 = \arccos(\frac{2}{3})$, verified by exact symbolic reduction of both defining equations to 0 under this substitution): $\varphi_{\max}(s) = 2\arctan(\sin s)$ for $s < s_2$ (piece 4a) and $\varphi_{\max}(s) = \arcsin(1/(\sin s + \cos s))$ for $s > s_2$ (piece 4b). Piece 4a's own ceiling curve is itself rational, $\mathsf{u} = 2\mathsf{v}/(1+\mathsf{v}^2)$, so the substitution $\mathsf{u} = \mathsf{a} \cdot 2\mathsf{v}/(1+\mathsf{v}^2)$, $\mathsf{v} = (\sqrt{5}-2)\,\mathsf{b}$, which sends the crossover point to the corner $\mathsf{b} = 1$ since $\tan(s_2/2) = \sqrt{5}-2$, gives an exact remapping onto the unit square in (a, b) over $\mathbb{Z}[\sqrt{5}]$, with numerator and denominator of degree $(10, 18)$ and $(10, 20)$ respectively, an exact, zero-margin certificate, the second (after region top) of the two exact certificates on this arc. Piece 4b's ceiling is not rational: in (u, v) the equation of Y reads $2\mathsf{u}(1+2\mathsf{v}-\mathsf{v}^2) = (1+\mathsf{u}^2)(1+\mathsf{v}^2)$, a curve of genus one. The numerator and denominator in (u, v), degree $(10, 10)$ with plain rational coefficients, are certified by box-covering instead, using only numerically-chosen safe box corners against the (non-algebraic, numerically-located) ceiling.

Region 6. Bounded below by the universal line and above by the W-curve, for $s \in (s_3, \pi/4)$. A representative point has 36 vertices, 22 fixed and 14 moving. One moving vertex lies on five facets; its exact coordinates are taken from whichever four of them give the simplest expression, all choices giving the same point. Box-covering certificate.

Region A. Bounded below by W and above by the universal line and then, past s_1, by Y, for $s \in (0, s_2)$; it pinches to a point at s_2. Its ceiling changes curve inside the region while the combinatorial type stays the same: convex-hull volumes on the two sides of s_1 agree with one formula to twelve decimal places. A representative point has 32 vertices (23 fixed +9 moving); the fixed-only sum is exactly 65/12. Box-covering certificate.

Region D. A lens, bounded above by the universal line and below by Y and then, past s_2, by W, for $s \in (s_1, s_3)$. It pinches to a point at both ends, since Y meets the universal line at s_1 and W meets it at s_3. A representative point has 37 vertices, 22 fixed and 15 moving; the fixed-only sum is exactly 29/6. Box-covering certificate.

Region E. Bounded below by Y and above by W and then, past s_3, by the universal line, for $s \in (s_2, \pi/4)$, pinching to a point only at s_2. A representative point has 37 vertices, 21 fixed and 16 moving, the most moving vertices of any region; the fixed-only sum is exactly 17/4. The floor curve is the same non-rational Y-curve responsible for region 4b's elliptic obstruction. Box-covering certificate.

Region C. Bounded above by A and below by Y, then, past s_1, by the universal line, then, past s_3, by W. Its floor changes curve twice, and facet computations at representative points in all three ranges confirm that it is one combinatorial type. Unlike regions A, D and E it does not pinch at a crossing; it narrows only towards $s = \pi/4$, where W and A meet. A representative point has 34 vertices (23 fixed +11 moving); the fixed-only sum is exactly 11/2. Box-covering certificate.

B.3.4. *What the eight certificates establish.* Together, the eight certificates of Theorem B.8 give $\Delta > 0$ on the first arc outside the excluded margins recorded above, none more than a few tenths of one percent of its region's area and none on the two exact regions. The margins, and the rest of the sphere of directions, are settled by Theorem 6.8.

B.4. **Breakpoint classification on the second arc.**

B.4.1. *Setup.* The second boundary arc of Lemma B.1 is $\eta_{w_1 v_2}(s) = \cos s\, w_1 + \sin s\, v_2$, $s \in [0, \pi/4]$, reduced to this range by an exact symmetry S that exchanges w_1 and v_2 and fixes u_0 and v_1 (Lemma B.1). Its breakpoints are classified as on the first arc.

Theorem B.9 (Complete breakpoint classification, second arc)**.** *Exactly four exact curves account for every active-facet-set transition of* $u_1(\varphi, s) = \cos\varphi\, u_0 + \sin\varphi\, \eta_{w_1 v_2}(s)$ *across the entire domain* $s \in [0, \pi/4]$*:*

(1) the universal curve $\varphi = \pi/3$*, inherited unchanged from Theorem B.7 (it is proved once, for every choice of deviation direction, and restricts to this arc);*
(2) a curve W'*,* $\varphi = 2\arctan(\cos s)$*, from the same sign vertex* $w = (1,1,1,1)/\sqrt{2}$ *that gave curve* W *on the first arc, but with a* different *functional dependence on* s*, since the same vertex plays a structurally different role on the two arcs;*
(3) a curve B*,* $\varphi = 2\arctan(\sin s)$*, from the sign vertex* $b = (1,1,1,-1)/\sqrt{2}$*, a vertex that plays no role on the first arc;*
(4) a curve C*,* $\sin\varphi\,(\cos s + \sin s) = 1$*, from the one-axis vertex* $c = \sqrt{2}\, e_3$*, the same functional form as curve* Y *on the first arc, but arising here from a different vertex.*

The two vertices responsible for the first arc's remaining curves, $y = (1,-1,1,1)/\sqrt{2}$ *and* $a = \sqrt{2}\, e_1$*, give no breakpoint in the open interior of this arc.*

As on the first arc, each equation is the passage of the deviating hyperplane through the named vertex. A dense scan (150 values of s, a 900-point grid in φ refined by bisection) finds at most four breakpoints at each sampled s, each on one of the four curves to within 2×10^{-4}, far below the separation of the curves, and no other transition.

B.4.2. *The combinatorial atlas.* Sorting the four curves by φ-value locates three exact crossing points on $s \in (0, \pi/4)$:

$$\begin{aligned} s_1 &= 0.169918\ldots \quad (\mathsf{C} = \text{universal}),\\ s_2 &= \arctan(\tfrac{1}{2}) = 0.463648\ldots \quad (\mathsf{B} = \mathsf{C}),\\ s_3 &= \arcsin(\tfrac{1}{\sqrt{3}}) = 0.615480\ldots \quad (\mathsf{B} = \text{universal}), \end{aligned}$$

each found exactly and checked numerically to eleven significant figures. The equations of W', B and C are those of A, W and Y on the first arc, so the arrangement, drawn in Figure 45(b), is the same, with the same three crossings and the same eight regions; Proposition B.5 explains why.

B.4.3. *Region top.* The topmost region, bounded above by $\varphi = \pi/2$ and below by curve W', spans the whole range $s \in (0, \pi/4)$ with no internal splitting, and its lower boundary is rational in Weierstrass coordinates (of the form $\mathsf{u} = (1-\mathsf{v}^2)/(1+\mathsf{v}^2)$, as the curve A of the first arc). Appendix B.5 gives an exact zero-margin certificate for this region.

B.4.4. *The band immediately below region top.* The next region down, bounded above by W' and below, depending on s, by whichever of curves C, the universal curve, or B is locally uppermost, is one combinatorial region on each of the three s-sub-intervals cut by the crossings s_1, s_3, and its *lower* boundary curve changes between them:

$$\text{lower boundary} = \begin{cases} \mathsf{C} & s \in (0, s_1),\\ \text{universal } (\varphi = \pi/3) & s \in (s_1, s_3),\\ \mathsf{B} & s \in (s_3, \pi/4). \end{cases}$$

At eight sample points across the middle sub-interval $s \in (s_1, s_3)$ the cell has the same combinatorial pattern, 34 vertices and the same active facets, so the middle sub-part is one combinatorial region. Appendix B.5 gives a direct certificate on this middle sub-part, the widest of the three by s-range. The two remaining sub-parts are handled not by a

direct computation but by the identification of Proposition B.5, which carries the first arc's certificates onto the whole of this one.

B.5. **Certificates on the second arc.**

B.5.1. *Region top: an exact, zero-margin certificate.*

Theorem B.10 (Certificate, second arc, region top)**.** *On the region bounded above by $\varphi = \pi/2$ and below by curve W' (Theorem B.9), for the entire domain $s \in (0, \pi/4)$, the volume-defect function Δ has an exact zero-margin certificate in the Bernstein basis: after the change of variables used for region top of the first arc, the numerator has 286 Bernstein coefficients (degree $(12, 21)$) and the denominator 231 (degree $(10, 20)$), and all of them are negative, exactly in $\mathbb{Z}[\sqrt{2}]$.*

Since numerator and denominator have one sign throughout, their ratio is positive on the whole open region, with *no* excluded margin, as for the zero-margin certificates of the first arc. The cell has 32 vertices (8 moving under the deviating cap, 24 fixed) at a representative point and at five sample points across nearly the whole s-range; the exact fixed-volume contribution is $77/12$, and the closed-form moving volume is checked against a direct convex-hull volume computation to eleven significant figures at the representative point, before the symbolic change of variables. As an independent check, the defect evaluated numerically at 55 points spread over the whole (s, φ)-domain is positive at all of them, as the certificate requires.

B.5.2. *The band below region top.* For the middle sub-part of the band described in Appendix B.4.4, bounded above by W' and below by the universal curve $\varphi = \pi/3$, for s in the sub-interval $(s_1, s_3) = (0.169918, 0.615480)$, we do not use an exact zero-margin certificate as in Appendix B.5.1, because the lower boundary here is the constant curve $\varphi = \pi/3$: mapping between it and the curved upper boundary W' brings $\sqrt{3}$ into the arithmetic beside $\sqrt{2}$, and the exact sign test used here works over $\mathbb{Q}(\sqrt{2})$. The box covering of the harder regions of the first arc is used instead.

Theorem B.11 (Certificate, second arc, band below region top)**.** *On the middle sub-part described above, the volume-defect numerator and denominator, mapped to the coordinates (u, v) (degrees $(12, 12)$ and $(10, 11)$, with exact coefficients in $\mathbb{Z}[\sqrt{2}]$, each step of the change of variables compared with a 200-digit evaluation of the original trigonometric expression), have positive Bernstein coefficients, all 169 of N and all 169 of D, on each of the 40 boxes that cover the sub-interval. Hence $\Delta > 0$ on the covered region, which is the sub-part minus the excluded margin described below.*

The excluded margin consists of a strip of about 1.7% of the s-range of the sub-interval at each end, from the safety factor used in placing box corners near s_1 and s_3, and a strip of 0.1 to 0.2% along the numerically located floor ($\varphi = \pi/3$) and ceiling (W') of each box. It is wider than the margins of the first arc, which are below 0.2%, because the covering was run at a single moderate resolution; a finer covering would reduce it at the cost of running time. Positivity on the whole domain, margins included, is Theorem 6.8.

B.5.3. *The band's C-bounded sub-part.* The first of the two remaining sub-parts of the band described in Appendix B.4.4, bounded above by W' and below by curve C (Theorem B.9), for $s \in (0, s_1) = (0, 0.169918)$, has one combinatorial type (34 vertices, 11 moving under the deviating cap and 23 fixed, the same at eight sample points across the sub-interval). It is certified by the box covering of Theorem B.11, with the constant lower boundary $\varphi = \pi/3$

replaced by the curve C (evaluated numerically when the lower corner of each box is placed; box covering does not need the boundary to be rational).

Theorem B.12 (Certificate, second arc, band C-sub-part)**.** *On the sub-part described above, the exact fixed-volume contribution is* $17/3$*; the moving-volume numerator and denominator, mapped to plain coordinates* (u, v)*, have degree* $(12, 12)$ *and* $(10, 11)$ *respectively, as in Theorem B.11, with exact coefficients in* $\mathbb{Z}[\sqrt{2}]$*, each step of the change of variables compared with a* 200*-digit evaluation of the original trigonometric expression. On each of the* 40 *boxes that cover the sub-interval all* 169 *Bernstein coefficients of* N *and all* 169 *of* D *are positive. Hence* $\Delta > 0$ *on the covered region, which is the sub-part minus the excluded margin described below.*

The excluded margin is wider than that of Theorem B.11: about 4% of the s-range of the sub-interval at each end, since a wider safety factor was used near $s = 0$ and near s_1, where the numerical evaluation of curve C is closer to degenerate, and the same strip of 0.1 to 0.2% along the floor and ceiling of each box. A finer covering would reduce it.

B.5.4. *The band's* B*-bounded sub-part.* One sub-part of the band, bounded above by W' and below by curve B for $s \in (s_3, \pi/4)$, is not certified by this method. Its combinatorial data are as elsewhere: the representative point has 34 vertices, 11 moving and 23 fixed, the pattern is the same at eight sample points across the sub-interval, and the fixed volume contribution is 6. But for one of the 74 moving boundary simplices the exact simplification of the determinant over the relevant algebraic extension could not be completed, so there is no closed form for the volume and the direct computation gives no certificate on this sub-part.

No certificate is needed there. By Appendix B.1.5 the defect on the second arc, this sub-part included, is the defect on the first arc after a change of variables, so the certificates of Appendix B.3 cover it. The three direct certificates above check that identification independently on part of the domain.

B.5.5. *Coverage of the boundary.* Figure 44(b) shows the certificate carried by each region of the first arc. The computations of this subsection certify three regions of the second arc directly and do not reach the others. By Proposition B.5 they are not needed: the second arc and the third edge carry the certificates of the first arc, with its margins. The interior of the triangle is settled by a different argument in Section 6.

B.6. **What the single-deviation case contains.** The single-deviation problem is treated here in two ways: one proves the theorem, the other gives the values of the defect.

The regional route occupies Appendix B.2 through Appendix B.5. On the first boundary arc of the fundamental deviation triangle, the breakpoint classification is complete: eight combinatorial regions, each with its own defect polynomial in the variables (u, v), two certified exactly with no margin and six by box covering outside a margin of less than one percent of their area (Theorems B.7 and B.8). On the second arc three of the four units were certified directly and the fourth was left without a closed form, and none of this is needed: the group generated by S and H is isomorphic to S_3, fixes u_0, realises every permutation of $\{v_1, w_1, v_2\}$, and carries both the second arc and the third edge onto the first arc exactly as functions of (φ, s) (Lemma B.4 and Propositions B.3 and B.5). The whole boundary of the triangle therefore inherits the first arc's certificate together with its margins (Corollary B.6).

That route cannot reach the interior. The symmetry group does not move an interior point to an edge, the reduction by convexity is false (Appendix B.1.6), and the interior

carries at least four distinct combinatorial region types against the boundary's one, so a regional treatment there would need four new volume formulas before any certificate could be attempted.

The global route is Section 6. It discards the triangle, the arcs and the combinatorial types alike, and replaces them by one question about a fixed polytope, how much a half-space at unit distance from the centre can cut off. The answer is $\frac{1}{3}$, the volume of the pyramid that the undeviated configuration already cuts off, which is why the bound is sharp and why equality forces the D_4 configuration (Theorem 6.8). The argument does not depend on the tilt, the deviation direction or the combinatorial type of the cell.

Neither route contains the other. Theorem 6.8 gives the sign everywhere and nothing else; it does not produce the defect as a function, and it cannot say where along an arc the defect is small or how the breakpoints sit. The regional certificates give exactly that, region by region. Between the two lies Proposition 6.9, which comes from the same reformulation as the global route but replaces the cross-polytope by a cone: it gives an explicit positive lower bound on the defect, of size $\varphi^2/3$ near $\varphi = 0$, on the range $\varphi < 42.17598\ldots^\circ$, and gives no bound above it. For the theorem alone, Section 5 and Section 6 suffice; the shape of the defect function is in the earlier subsections of this appendix.

Appendix C. Several simultaneous deviations

This appendix studies a centre at which several contacts leave the root system at once. For packings that case does not occur: twenty-four contacts form a copy of the root system (Theorem 5.25), so Corollary 5.31 settles it, and no proof in the main text uses what follows. The computations here are of two kinds, and the kind is named where each occurs: exact computations, which prove what they state, and evaluations at a stated working precision, which are evidence and enter no proof (Section 11).

C.1. **The joint Hessian technique.**

C.1.1. *Simultaneous deviations.* Let $m \geq 2$ of the twenty-four directions of the root system be deviated at once, each in the 3-dimensional tangent space at its root, and the other $24-m$ held at their roots. The cell they cut is still a polytope circumscribed about the unit ball, and its volume less 8 is a function Δ_A of the $3m$ deviation parameters. Two moving directions can come within 60° of each other or of a fixed root, so these are configurations of directions, not of packings, and by Corollary 5.31 no packing with twenty-four contacts realises any of them except the root system itself. The question here is whether the bound $\mathrm{vol} \geq 8$ needs the packing condition at all near the root system.[50]

Definition C.1 (Joint Hessian)**.** Number the roots $0, \ldots, 23$ as in Appendix E. For a set $A = \{k_1, \ldots, k_m\}$ of roots, the *joint Hessian* $H_A \in \mathbb{R}^{3m\times 3m}$ is the Hessian at the origin of Δ_A as a function of the deviations $x_i \in \alpha_{k_i}^{\perp}$, each written in an orthonormal basis of $\alpha_{k_i}^{\perp}$. It has one 3×3 block for each deviating direction and a cross-block for each pair.

If H_A is positive definite, the defect increases along every small deviation of the directions in A. That is a second-order statement at the root system only: it excludes neither a later decrease nor a zero at a finite angle. The results of Appendix C.6 are of a different kind, exact statements that the defect is positive along three paths over their whole range. No theorem of this paper uses Appendices C.1 to C.4, and every result in them other than

[50] A positive answer for all m and all deviations would give the local bound, with no use of the classification, at every centre whose active directions are roots except for m of them: the cell of such a centre contains the polytope of the $24-m$ undeviated roots and the m deviated directions.

Propositions C.10 and C.12 is numerical: the Hessians are computed by finite differences of exactly computed cell volumes.

C.1.2. *Two directions and chains.*

Numerical observation C.2 (Two-direction case)**.** For $m = 2$ deviating directions with Gram value $\frac{1}{2}$ (an adjacent pair), the minimum eigenvalue of the joint Hessian is $\frac{1}{3}$, and for a single deviating direction considered alone, $H_A = \frac{2}{3} I_3$, in both cases to within 10^{-6} of finite-difference computations. The second value agrees with Proposition 6.9, which gives $\Delta \geq \varphi^2/3$ near $\varphi = 0$ for a single deviation.

C.1.3. *The case $m = 2$ from first principles.* The cross-term formula behind Numerical observation C.2 was obtained by fitting the numerically computed cross-Hessian to a small family of $W(D_4)$-covariant invariants, and the fit was checked to within 10^{-6} at several sample points and at two adjacent pairs. It is not derived from the vertex structure of $\mathcal{V}$. A derivation would begin as follows.

For an adjacent pair α, β (Gram value $\frac{1}{2}$), exact vertex enumeration over all $\binom{24}{4}$ facet quadruples of $\mathcal{V}$ shows that the two corresponding facets, each an octahedron with 6 vertices, share 3 vertices, so they meet along a triangular ridge. A simultaneous perturbation of both facet normals therefore moves $6 + 6 - 3 = 9$ vertices, and each of them is the solution of a 4×4 linear system, by Cramer's rule, once the facets that govern it under the joint perturbation are known.

That set has to be determined vertex by vertex. Each of the 9 vertices lies on 6 facets in the unperturbed cell, more than the generic 4, so perturbing one or two of those facets can make others redundant at the new position. At the shared vertex $(\sqrt{2}, 0, 0, 0)$, with active facets $\{\alpha, \beta_1, \ldots, \beta_5\}$, perturb α alone and complete the perturbed facet to a determining 4-set in each of the $\binom{5}{3} = 10$ possible ways: 4 completions are feasible and give the same new vertex (one previously active facet becomes redundant), 1 gives a singular 4×4 system, and 5 give points that violate another facet. So the vertex moves smoothly and does not split, but its governing facet set is found by a case analysis and not by a uniform rule. A first-principles proof of Numerical observation C.2 requires this analysis at all 9 vertices for a general perturbation direction, followed by a second-order expansion of the resulting volume formula. It is not carried out here, and Numerical observation C.2 rests on the numerical computation.

Numerical observation C.3 (Chains)**.** Let A be an induced path (a *chain*) of m directions with consecutive Gram values $\frac{1}{2}$ and no other adjacency. The Richardson-extrapolated minimum eigenvalue of H_A, computed from three step sizes, decreases from about 0.276 at $m = 3$ and levels off near 0.168 for $m = 6, 7, 8$; it is positive for every m from 3 to 8. Exhaustive search from every starting vertex shows that 8 is the largest length of an induced path in the 8-regular Gram-$\frac{1}{2}$ adjacency graph on R, in which each root is joined to the eight roots at 60° from it.

The plateau has a geometric source. At an interior chain vertex, the two tangent axes pointing toward its left and right neighbours (within its own 3-dimensional orthogonal complement) are not independent: their inner product is $-\frac{1}{3}$ or -1 at every interior vertex tested, so they are partially or fully antipodal. This constraint keeps the pairwise cross-terms from reaching their individual extremes simultaneously. It explains the plateau qualitatively; no bound is derived from it.

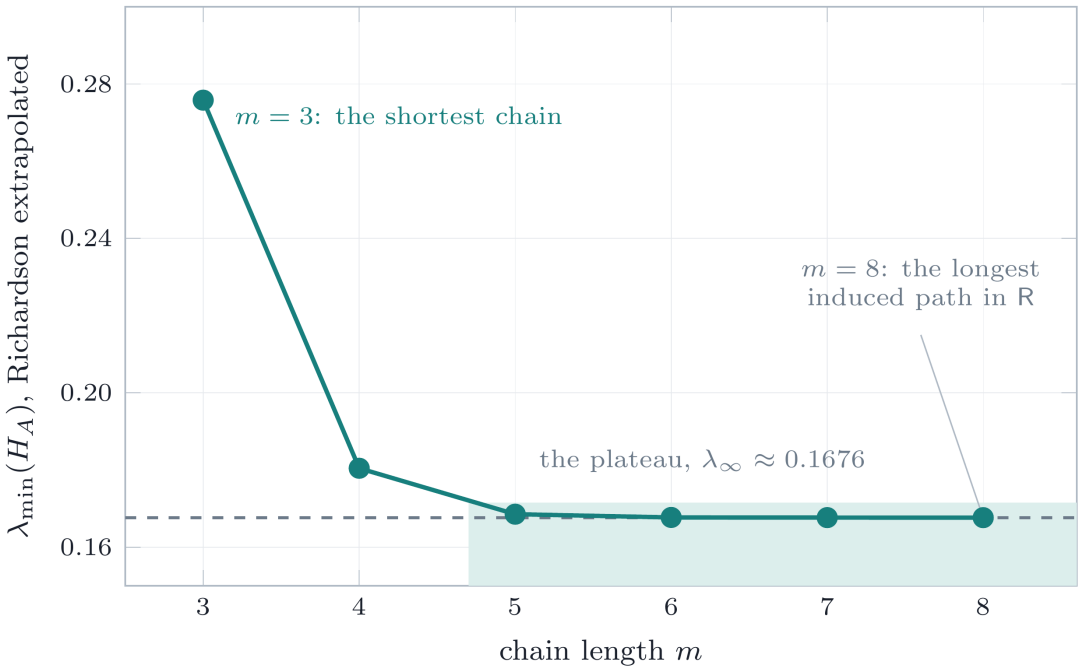

FIGURE 46. Richardson-extrapolated minimum eigenvalue of the joint Hessian H_A for chains of length $m = 3, \dots, 8$ (Numerical observation C.3; full values in Appendix E). The value falls from $m = 3$ to $m = 5$ and then levels off near 0.1676. It stays away from zero up to $m = 8$, the longest induced path in the contact graph of R, so numerically the second-order method gives positivity for every chain.

C.1.4. *Dense configurations.* Chains are not the worst case. A *star*, one root together with its eight Gram-$\frac{1}{2}$ neighbours ($m = 9$), gives a minimum eigenvalue whose Richardson extrapolation is $\frac{1}{12} \approx 0.0833$ to the accuracy of the computation, below the chain plateau. Growing the set greedily by local density, adding at each step a remaining root with the most neighbours already in the set, lowers the minimum eigenvalue further: about 0.083 at $m = 9$, 0.073 at $m = 12$ and 0.045 at $m = 15$, and from $m = 18$ on a value that double precision does not distinguish from zero (Figure 47). In this range a small positive value, zero and a small negative value cannot be told apart at machine precision. Appendices C.2 and C.3 examine one such configuration at higher precision.

C.2. **A degeneracy of the Hessian technique.**

C.2.1. *The limit of a second-order method.* The joint Hessian technique proves positivity to second order at the root system for a set A by showing that H_A is positive definite. Where H_A has a nontrivial kernel, the sign of the defect along the kernel is decided by terms of order four or higher, and no second-order computation can decide it. Appendices C.1.4 and C.3 indicate that such sets occur among the dense ones. We know of no earlier study of this degeneracy for the joint Hessian of D_4; the work of Musin [Mus08, Mus18] on this circle of problems concerns the kissing number and the 24-cell conjecture.

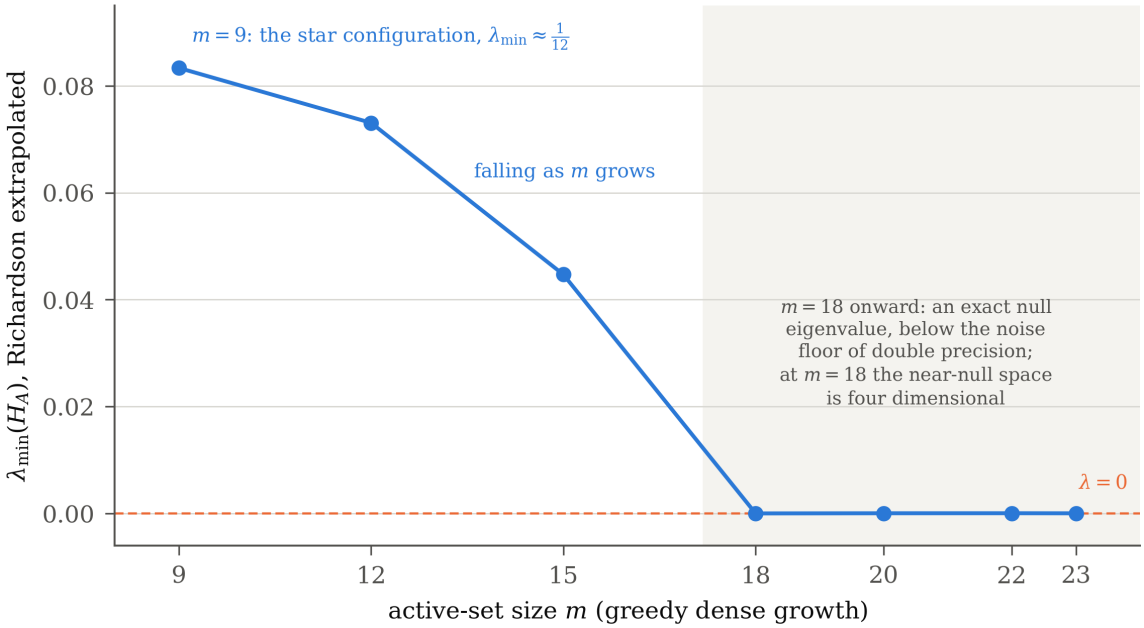

FIGURE 47. Richardson-extrapolated minimum eigenvalue of H_A under greedy dense growth, $m = 9$ to $m = 23$ (full values in Appendix E). Unlike the chains of Figure 46, the values do not level off above zero. From $m = 18$ on, shaded, the extrapolated value is below what double-precision finite differences resolve; Appendix C.3 examines this range in arbitrary-precision arithmetic, where the second derivative vanishes to the working precision.

C.2.2. *Precision.* In double precision the computed volumes carry an absolute error of order 10^{-10}, which a finite-difference second derivative with step h divides by h^2. At the step sizes the dense configurations require, this error exceeds the eigenvalues in question. Deciding between a small positive value, zero and a negative value therefore requires a higher working precision, not a smaller step, and Appendix C.3 uses one.

C.2.3. *Quartic positivity.* Where the second-order term vanishes on a subspace, the next test is the sign of a quartic form on it, and at A_{18} the subspace is 4-dimensional (Appendix C.3). For quadratic forms, Sylvester's criterion and an $LDL^\top$ decomposition settle the sign algorithmically. For quartic forms no such device exists in general. A theorem of Hilbert [Hil88] identifies the cases in which every non-negative real form is a sum of squares of forms of half the degree: forms in two variables, quadratic forms, and ternary quartics. In every other case, quartic forms in four variables included, there are non-negative forms that are not sums of squares; the Choi–Lam form used in Appendix C.4.3 is one. A non-negative quartic form on the near-null subspace at A_{18} therefore need not have a sum-of-squares certificate, and a proof there may require a case-specific argument, of the kind of the exact factorisations and Sturm sequences of Appendix C.6, in place of a Gram matrix with a positive-semidefinite square root.

C.3. **A singular dense configuration.**

C.3.1. *The configuration A_{18}.* The set reached by greedy dense growth from root 0 at $m = 18$ is

$$A_{18} = \{0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 16, 17, 20, 21, 22, 23\}.$$

It lies in the range of Appendix C.1.4 where double precision does not resolve the sign of the smallest eigenvalues of the joint Hessian. The computations of this subsection use an arbitrary-precision volume computation, with up to 45 decimal digits (Appendix C.3.2), independent of the double-precision one.

Numerical observation C.4 (Quartic degeneracy at A_{18})**.** At A_{18} the near-null subspace of the joint Hessian is 4-dimensional (Numerical observation C.5). Along every direction sampled in it (Appendix C.4), the central-difference estimate of the second derivative of the volume-defect function decreases by a factor close to 4 at each of three successive halvings of the step, as it does when the second derivative is zero, and the leading behaviour is $F(\sigma) = a_4\sigma^4 + O(\sigma^6)$ with $a_4 > 0$, where $F(\sigma)$ is the defect at the deviation σv for a unit vector v of the subspace.

The quartic test covers the whole near-null subspace. A test confined to the span of three of its eigenvectors would miss the fourth direction, which is the one with the smallest second derivative.

C.3.2. *The high-precision volume method.* The volume of a polytope containing the origin in its interior, as a Voronoi cell does, is computed by star triangulation from the origin:

(1) Every vertex is computed at full working precision as the solution of the 4×4 linear system of some four active facet equations $\langle n_k, x\rangle = 1$, and kept if it satisfies every facet inequality $\langle n_k, x\rangle \le 1$ up to a small tolerance.
(2) The vertices of each facet, which lie in a 3-dimensional affine hyperplane, are triangulated into tetrahedra. A double-precision projection to 3 local coordinates is used only to decide which vertices are adjacent; the triangulation is then applied to the high-precision coordinates.
(3) Each tetrahedron and the origin span a 4-simplex of volume $\frac{1}{4!}|\det M|$, where M is the 4×4 matrix of the tetrahedron's vertices; the sum over all tetrahedra of all facets is the volume.

The method reproduces the volume 8 of the 24-cell to within 10^{-49} at 50 digits, and at a generic small perturbation of an 18-element set it agrees with the double-precision convex-hull computation to within 1.1×10^{-9}, the precision of the latter.

C.3.3. *The dimension count.*

Numerical observation C.5 (Four-dimensional near-null space)**.** At A_{18}, the four smallest eigenvalues of the double-precision finite-difference joint Hessian at the step sizes $h = 0.02$, 0.01 and 0.005 each decrease by a factor close to 4 at every halving of h, as a zero eigenvalue does, while the fifth, about 0.075, stays constant. The near-null subspace is therefore 4-dimensional to the accuracy of the computation.

C.3.4. *The quartic behaviour.* For each tested direction v in the near-null subspace, $F(\pm\sigma v)$ is evaluated at $\sigma = 0.02$, 0.01 and 0.005, with 35 to 45 decimal digits. The central-difference estimate $(F(\sigma) + F(-\sigma))/\sigma^2$ of the second derivative decreases by a factor close to 4 at each halving of σ, and the quartic coefficient, estimated as $a_4 \approx (F(\sigma) + F(-\sigma))/(2\sigma^4)$ for small σ, is positive at every direction tested.

C.3.5. *Consequence.* If the second derivative vanishes on a 4-dimensional subspace at A_{18}, as these computations indicate, then no second-order argument, including the computations of Numerical observations C.2 and C.3 and the sweep of Appendix C.1.4, can establish Theorem 1.8 at A_{18}, and any argument there needs information of order four or higher. The volume-defect function is positive at every direction tested, through its positive quartic coefficient. This agrees with Theorem 1.8, which Corollary 5.31 proves by a different route. Appendix C.4 describes the sampling.

C.4. **Broadened directional sampling.**

C.4.1. *Sampling the full near-null subspace at A_{18}.* The near-null subspace at A_{18} is sampled along 36 directions: its four basis eigenvectors, the six normalised pairwise sums and the six normalised pairwise differences of these, and twenty directions drawn at random from the whole subspace (Gaussian coefficients, normalised). The sample spans all four dimensions of the subspace.

Numerical observation C.6 (Sampling at A_{18})**.** The quartic coefficient estimate $a_4 \approx (F(\sigma) + F(-\sigma))/(2\sigma^4)$ at $\sigma = 0.015$, with 35 digits, is positive along all 36 directions, with values from about 0.0088 to 0.314. Six directions spanning this range, re-evaluated at $\sigma = 0.0075$ with 40 digits, agree with the first estimate to within 1 to 5%, which confirms quartic scaling.

C.4.2. *A second configuration: $m = 20$.* Two further steps of the same greedy growth give

$$A_{20} = \{0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 19, 20, 21, 22, 23\} \supset A_{18}.$$

The same growth started from roots 5 and 12 gives two other 18-element configurations, which reproduce the double-precision spectrum of A_{18} to four decimal places. Both are images of A_{18} under orthogonal maps preserving R: of the 1152 such maps, 48 send A_{18} onto each of them, and 48 onto itself (an exact check). They are therefore congruent to A_{18} and are not examined further.

Numerical observation C.7 (Five-dimensional near-null space at A_{20})**.** At A_{20} the scaling test of Numerical observation C.5 finds a 5-dimensional near-null subspace: the five smallest eigenvalues decrease by a factor of 3 to 4 at each halving of h, while the sixth, about 0.035, stays constant.

Numerical observation C.8 (Sampling at A_{20})**.** Along 30 directions in this subspace (the five basis eigenvectors, their ten normalised pairwise sums, and fifteen random directions), the estimate at $\sigma = 0.015$ with 35 digits is positive in every case, with values from about 0.0076 to 0.314. Four directions spanning this range, re-evaluated at $\sigma = 0.0075$, agree to within 1 to 5%.

C.4.3. *A sum-of-squares test of a fitted quartic at A_{18}.* Sampling tests individual directions and leaves the sign of the quartic form between them open. For a test on the whole subspace, the 35 coefficients of a general quartic form $Q(x_0, x_1, x_2, x_3)$ on the near-null subspace at A_{18} were fitted by least squares to 166 estimates of a_4 (at $\sigma = 0.015$, 35 digits): the 16 basis, sum and difference directions of Numerical observation C.6 and 150 further random directions. The design matrix has full rank 35, the relative residual is 0.91%, and the largest pointwise residual is 0.0022, against values from 0.0064 to 0.314.

Write $Q(x) = m(x)^\top M\, m(x)$, with $m(x)$ the vector of the 10 monomials of degree 2 in 4 variables. The symmetric matrices M that represent Q form an affine space of dimension $55 - 35 = 20$, and Q is a sum of squares if this space contains a positive-semidefinite matrix.

The semidefinite program $\max t$ subject to $M - tI \succeq 0$ over this space is solved by an interior-point method [DB16, GC26]. On two test forms in the same variables it gives the expected answers: $t \approx 1$ for $(x_0^2 + x_1^2 + x_2^2 + x_3^2)^2$, and infeasibility for the Choi–Lam form $x_0^2x_1^2 + x_1^2x_2^2 + x_2^2x_0^2 + x_3^4 - 4x_0x_1x_2x_3$, which is non-negative by the inequality of arithmetic and geometric means and is not a sum of squares by the theorem of Choi and Lam [CL77].

Numerical observation C.9 (Sum of squares for the fitted quartic at A_{18})**.** For the 35 fitted coefficients, the semidefinite program is feasible with optimal value $t \approx 0.0066$, so the fitted quartic is a sum of squares with a small margin.

A sum-of-squares representation of the fitted form proves that the fitted form is non-negative on the whole subspace. It does not prove this for the true quartic form at A_{18}: the fitted form comes from numerical estimates with a relative residual of 0.91%, and the margin $t \approx 0.0066$ is of the same order as the fitting error.

C.4.4. *What the sampling establishes.* The two configurations give 66 sampled directions, in near-null subspaces of dimensions 4 and 5, and the quartic coefficient is positive along every one; no symmetry relating A_{18} to A_{20} was found. A proof that the quartic form is positive definite on either subspace would need the exact quartic tensor and a proof of its definiteness, and by Appendix C.2.3 the latter need not take the form of a sum-of-squares certificate. The many other dense configurations with m from 15 to 23 were not examined, and Theorem 1.8 concerns every packing-valid configuration of every size $m \geq 2$. The sampling is evidence at two configurations and proves nothing beyond them.

C.4.5. *An exact symmetry of A_{18}.* A_{18} was produced by greedy growth without regard to symmetry, but it has a large symmetry group.

Proposition C.10 (The stabiliser of A_{18})**.** *Within the group of signed coordinate permutations of $\mathbb{R}^4$ that map R to itself, of order $2^4 \cdot 4! = 384$, the subgroup fixing A_{18} setwise has order 48. It consists of the signed permutations that fix one specific coordinate axis together with its sign and act as an arbitrary signed permutation, a group of order $2^3 \cdot 3! = 48$, on the other three coordinates.*

Proof. Every element of the group of order 384 is a monomial matrix with entries $0, \pm 1$, so whether it maps R to itself and whether it fixes the index set A_{18} are finite integer checks. All 48 elements of the described subgroup fix A_{18}, and none of the other 336 elements does; both checks are part of the formal verification of Section 11.2. ∎

Each element of this group is an isometry of $\mathbb{R}^4$ that maps R and A_{18} to themselves, so it acts on the 54-dimensional joint tangent space of Definition C.1 by permuting the 18 active blocks, each with its induced 3×3 orthogonal action, as it permutes the active roots. For a sample of group elements g, $\mathrm{Rep}(g)^\top H_{A_{18}} \mathrm{Rep}(g)$ agrees with the finite-difference Hessian $H_{A_{18}}$ to within 4×10^{-6}, which is the accuracy the step $h = 0.02$ allows. The quartic coefficient estimate at a random direction v of the near-null subspace and at its images $\mathrm{Rep}(g)v$ under six non-identity elements agree to within 8×10^{-5}, again the accuracy of the step. The symmetry also accounts for the agreement among the symmetric sample directions of Numerical observation C.6: many of the 36 directions lie in one orbit of the group, so they are not independent samples.

Restricted to the numerically computed near-null subspace, the representation has trace 4 at two elements (the identity among them), 1 at sixteen, 0 at eighteen and -2 at twelve, and the space of 4×4 matrices commuting with all 48 restricted images has dimension 2. To the accuracy of the computation, the near-null subspace is therefore the sum of two non-isomorphic irreducible representations of the group.

C.4.6. *The invariant quartics at A_{18}.* The character table of the stabiliser, with its 10 conjugacy classes, is computed exactly from 10 irreducible characters built from the natural representations of the group and checked by the first orthogonality relation [Ser77, §2.3].

Numerical observation C.11 (The near-null representation)**.** The traces of the near-null representation at A_{18} agree, at each of the 10 conjugacy classes, with the character of $\varepsilon \oplus (\mathrm{std} \otimes \delta)$, where ε is the 1-dimensional character given by the sign of the permutation part, std is the defining 3-dimensional representation of the group, and δ is the determinant character. Of the two candidate characters with the same distribution of trace values, only this one matches class by class.

Write x_0 for a coordinate on the ε-line and $y = (y_0, y_1, y_2)$ for coordinates on the $\mathrm{std} \otimes \delta$ summand, so that $y \mapsto \delta(g)\, g(y)$. A quartic $Q(x_0, y)$ invariant under the group splits by its degree in x_0 into five parts. Since ε and δ have order 2, invariance at even degree in y is invariance under std, and at odd degree it is relative invariance under std with the character $\varepsilon\delta$, the product of the signs of a signed permutation.

Proposition C.12 (The invariant quartics)**.** *For the representation $\varepsilon \oplus (\mathrm{std} \otimes \delta)$ of the stabiliser of A_{18}, the space of invariant quartics $Q(x_0, y)$ has dimension 5, with basis*

$$x_0^4, \qquad x_0^2\, S_2(y), \qquad x_0\, P_3(y), \qquad S_4(y), \qquad S_{22}(y),$$

where $S_2 = y_0^2 + y_1^2 + y_2^2$, $S_4 = y_0^4 + y_1^4 + y_2^4$, $S_{22} = y_0^2 y_1^2 + y_1^2 y_2^2 + y_2^2 y_0^2$ and $P_3 = y_0 y_1 y_2$, the last being the unique relative invariant of degree 3 up to scale.

Proof. The multiplicity of the required character (triv at even degree, ε at odd degree) in $\mathrm{Sym}^b(\mathrm{std} \otimes \delta)$ for $b = 0, 1, 2, 3, 4$ follows from the power-sum formula applied to $p_k = \mathrm{tr}(M^k)$, $M = \delta(g)\, g$, at the 10 class representatives, in exact rational arithmetic. The multiplicities are $1, 0, 1, 1, 2$, with sum 5. Independently, the Reynolds operator (the exact average over the 48 group elements) applied to y_0^2, y_0^4 and $y_0^2 y_1^2$ gives S_2, S_4 and S_{22} up to scale, and applied with the character $\varepsilon\delta$ to y_0^3, $y_0^2 y_1$ and $y_0 y_1 y_2$ it gives 0, 0 and P_3. This matches the count. ∎

Since the quartic form at A_{18} is invariant under the exact symmetry of Proposition C.10, Numerical observation C.11 places it in the 5-dimensional span of Proposition C.12, and only its five coefficients $c_1, \ldots, c_5$ are unknown. They were fitted from seven directions chosen to separate them (the pure x_0-direction and six combinations of x_0 and y that separate the contributions of S_2, S_4, S_{22} and P_3), each evaluated with 40 digits. At $\sigma = 0.02$ the fit is

$$(c_1, \ldots, c_5) \approx (0.3119, -0.0516, 0.2307, 0.0273, 0.0492),$$

with residual below 1.5×10^{-4} at all seven directions. The fit at $\sigma = 0.01$ and the combined 14-point fit at both step sizes give

$$(0.3157, -0.0495, 0.2302, 0.0308, 0.0563) \quad \text{and} \quad (0.3138, -0.0505, 0.2304, 0.0290, 0.0527).$$

The three agree to within a few percent on every coefficient. In double precision the fits at the two step sizes differ by 100 to 700%, the precision limit of Appendix C.2; those fits are not used.

Numerical observation C.13 (Sums of squares on the reduced family)**.** The semidefinite program of Appendix C.4.3 is feasible for each of the three fits, with optimal margins $t \approx 0.0085$, 0.0131 and 0.0108, so each fitted quartic is a sum of squares.

The reduced fit determines five coefficients from targeted directions at two step sizes, with consistent results and larger, more stable margins than the 35-coefficient fit of Numerical observation C.9. The coefficients are still finite-difference estimates and the margins are still small, so this is evidence and not a proof. The elementary route to Theorem 1.8 at A_{18} stops here.

C.5. **What is known about several simultaneous deviations.**

C.5.1. *Proved and observed.* The second-order computations are numerical. For two adjacent directions and for chains up to the largest induced-path length of the contact graph, Numerical observations C.2 and C.3 find the joint Hessian positive definite. Appendix C.1.4 shows that chains are not the worst case: along dense configurations the minimum eigenvalue falls to the resolution of double precision. Numerical observations C.4, C.5 and C.7 examine two such configurations in high precision and find near-null subspaces of dimensions 4 and 5 on which the second derivative vanishes to the working precision, and Numerical observations C.6 and C.8 find the quartic coefficient positive along 66 directions spanning these subspaces.

Two statements about the first configuration are exact. Proposition C.10 gives a group of order 48 that stabilises it, and Proposition C.12 shows that for the representation identified numerically in Numerical observation C.11 the invariant quartics form a space of dimension 5, not the generic 35. Fitting the five coefficients from high-precision finite differences at two step sizes and testing the fitted forms for a sum-of-squares representation gives margins $t \approx 0.009$ to 0.013 across three fits (Numerical observation C.13). The coefficients are measured, not derived, so this is not a proof.

By a different method, exact and at finite angle, Appendix C.6 proves the defect positive along three swap paths, at $m = 2, 3, 6$, across their whole angular range.

By a third method, which uses no deviation structure, Theorem 5.16 bounds the cell volume from below by $\frac{\pi m}{3}\tan^3 r_m$ for a contact configuration of m directions, with $2r_m - \sin 2r_m = 2\pi/m$. That quantity exceeds 8 for every $m \le 22$, so the local bound holds with any number of directions deviating by any amount, at every centre with at most twenty-two contacts (Corollary 5.17). Only $m = 23$ and $m = 24$ are left, and at $m = 24$ the same estimate returns $7.7989\ldots$. Proposition 5.18 says that this is as far as the method goes: the covering bound is exactly the Jensen collapse of a per-cell bound, so it is the strongest conclusion the areas of the spherical Voronoi cells can support, and at the root configuration those areas are all equal.

C.5.2. *What the classification settles.* For packings the question does not arise. Corollary 5.31 proves Theorem 1.8 by the classification of Theorem 5.25: twenty-four contacts form a 24-point kissing configuration, and by Theorem 5.30 there is only one of those, so no direction deviates. The results of this appendix do not use the classification. They concern the stronger statement without the packing condition, show what elementary methods give on it, and rest on no external certificate. The next subsection states their limits.

C.5.3. *Not established by the methods of this appendix.* None of this proves that the quartic form is positive definite at either configuration, and none of it proves the defect positive for every set of deviations. That statement ranges over every set of deviating roots at every $m \ge 2$; the dense range of Appendix C.1.4 contributes two examined configurations out of a large family, and the rest of that family, with every other value of m, is not examined. The degeneracy of Appendix C.2 restricts what an elementary proof can look like: if the second derivative vanishes on the near-null subspaces, as the computations indicate, no argument

that stops at second order can decide the sign there, and a proof has to use the quartic term. The three swap paths do not reach that range either, being low in m and highly symmetric.

The symmetry of order 48 narrows the search without closing it. Together with Numerical observation C.11 it confines the quartic tensor to a 5-dimensional space instead of a 35-dimensional one, but the five numbers used are finite-difference measurements, not a symbolic derivation, and the feasibility margin is small. No analogous symmetry of the second configuration, or of any other dense configuration, has been sought.

The cap argument of Section 6 does not extend here. With one deviating direction the other twenty-three half-spaces are fixed, so there is a fixed body Q and every parameter of the problem sits in the cutting half-space. With m deviating directions the body moves with the parameters, and there is no fixed polytope to take caps of. What has to be bounded is then a comparison between two overlapping unions rather than a single cap, and Remark 5.3 records why the obvious way of splitting that comparison into m separate ones cannot work. Proposition 5.5 does supply the substitute for Q, namely the polytope Q_D obtained by dropping the m deviating constraints, and recasts the whole problem as one extremal inequality, (5.4), for the union of m caps of that fixed body. An argument that uses no classification has to prove that inequality.

C.5.4. *What the reformulations settle.* Section 5.3 adds two further exact restatements, the polar form $\mathrm{vol}(\mathrm{conv}(W)^\circ) \geq 8$ over contact configurations (Proposition 5.8) and the boundary form $\mathrm{area}(\partial V_c) \geq 32$ (Lemma 5.10). Neither proves anything by itself. They locate the multi-direction case precisely: by Proposition 5.8 the case this appendix treats is the part of the 24-cell conjecture in which two or more contact directions leave the root system at once, and nothing here settles it without the classification of Theorem 5.25, which settles it for packings.

They also rule out approaches. Four natural approaches are eliminated, each by an evaluation at the D_4 configuration itself, where an argument that is to produce the sharp constant 8 must lose nothing. Term-by-term reduction to Theorem 6.8 is eliminated by Remark 5.3. Any bound assembled facet by facet from the pairwise contact condition is eliminated by Remark 5.12: the bound such an argument delivers is $8\pi/(3\sqrt{3}) = 4.8368\ldots$, and the best one it could deliver is $96\sqrt{2}-128 = 7.7645\ldots$, since an admissible configuration of eight tight neighbours in square-antiprism position has a facet of volume $16\sqrt{2}-\frac{64}{3}$, below the octahedron's $\frac{4}{3}$. The convexity bound for the polar volume integral is eliminated because Jensen's inequality is an equality only at a ball, and at the root configuration it returns $7.7351\ldots$; the Mahler-type product bound is eliminated because it carries a factor $\frac{3}{2}$ of slack there. The last two are Remark 5.13.

A fifth approach fails as well. Proposition 5.18 shows that no argument resting on the areas of the spherical Voronoi cells can do better than $7.7989\ldots$, since the covering bound already saturates that information and the root configuration has cells of equal area. What is left over is their shape: at D_4 each cell is an eight-faceted spherical polytope of circumradius 45° against the $34.0986\ldots^\circ$ of the cap with the same measure, and the whole of the remaining 0.2010 is that difference. Proposition 5.23 adds the sharpest version of the same accounting: putting every pairwise cap overlap back exactly, which below r_m is not an estimate but an identity, gives $7.997885\ldots$ at $m = 23$, so even that stops short.

A sixth fails with it. That shape deficit cannot be collected one cell at a time, in either decomposition the configuration provides (Remark 5.19): the per-cell inequality (5.14) is sharp at every cell of the root configuration and fails at a direction carrying nine contacts at 60°, while the Delaunay decomposition does not localise the integrand, and the functional it would have to localise to falls 4.78% short at the regular simplex of edge 60°. What

makes the root configuration extremal is how its cells fit together, not the shape of any one of them.

What survives is a positive but soft statement, Corollary 5.9: a worst configuration exists and may be taken maximal, so the inequality is a question about maximal contact configurations and about an extremal point, not about a supremum approached but never reached. The difficulty lies in identifying that point, and at $|W| = 24$ it coincides with the uniqueness half of the kissing problem in dimension four, which the kernel of Theorem 5.30 supplies and Theorem 5.25 records.

C.6. **Swap configurations at finite angle.** Every result of Appendices C.1 to C.4 is of second order: a statement about the joint Hessian H_A, which controls the defect only infinitesimally close to the root system. This subsection proves the defect positive at finite angle along three highly symmetric paths of several deviating directions, each over its *whole* angular range. Along each path the moving directions are less than 60° apart, so the polytopes are not cells of packings; the results bear on the question of Appendix C.1, whether the bound needs the packing condition near the root system. Figure 48(a) shows the three functions.

C.6.1. *The swap path and its symmetry.* Fix an adjacent pair of roots $\alpha, \beta \in \mathsf{R}$ (Gram value $\frac{1}{2}$), and let the two directions deviate simultaneously along the *swap path*

$$u_1(\theta) = \cos\theta\,\alpha/\sqrt{2} + \sin\theta\,\beta^{\perp},$$
$$u_2(\theta) = \cos\theta\,\beta/\sqrt{2} + \sin\theta\,\alpha^{\perp},$$

where $\beta^{\perp}$ and $\alpha^{\perp}$ are the unit vectors along the components of β orthogonal to α and of α orthogonal to β, with the other 22 roots held fixed, for θ in $(0, \theta_*)$, $\theta_* = \pi/3$. Both directions move along the great circle through α and β, towards each other; at θ_* they have exchanged places and the configuration is the root system again.

Lemma C.14 (Swap-path symmetry)**.** *$u_2(\theta) = u_1(\theta_* - \theta)$ identically. Consequently $\theta \mapsto \theta_* - \theta$ is an involution of $(0, \theta_*)$ exchanging the roles of u_1, u_2 while leaving the (unlabelled) geometric configuration unchanged, and $F_{\mathrm{swap}}(\theta) = F_{\mathrm{swap}}(\theta_* - \theta)$ for the resulting volume-defect function F_{swap}.*

The involution has the single fixed point $\theta = \pi/6$, where the two moving directions coincide. Tracking every one of the 33 vertices of the cell by its set of facets, at 80 sample points, finds exactly one combinatorial transition on $(0, \theta_*)$, located at $\theta = \pi/6$ to five decimal places by bisection. Since the involution maps transitions to transitions, a single one must sit at its fixed point, so it is at $\pi/6$ exactly. Hence F_{swap} consists of two analytic pieces, on $(0, \pi/6]$ and $[\pi/6, \theta_*)$, exchanged by the involution, and positivity on one gives positivity on the other.

C.6.2. *The closed form for F_{swap}.* On $(0, \pi/6)$ the local region has 33 vertices, computed exactly by Cramer's rule over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, t)$ with $t = \tan(\theta/2)$, and its boundary triangulates into 157 signed simplices, the combinatorial type being read off at $\theta = 0.3$ and constant across the interval. The volume is the sum of the corresponding determinants.

Combining those 157 rational functions into a single quotient in one step is not practical. At least one of them carries a degree-24 denominator in t with coefficients in $\mathbb{Z}[\sqrt{3}]$, and the common denominator of the whole sum is large enough to defeat a direct cancellation. The sum is a rational function of t all the same, and grouping the terms into batches, cancelling

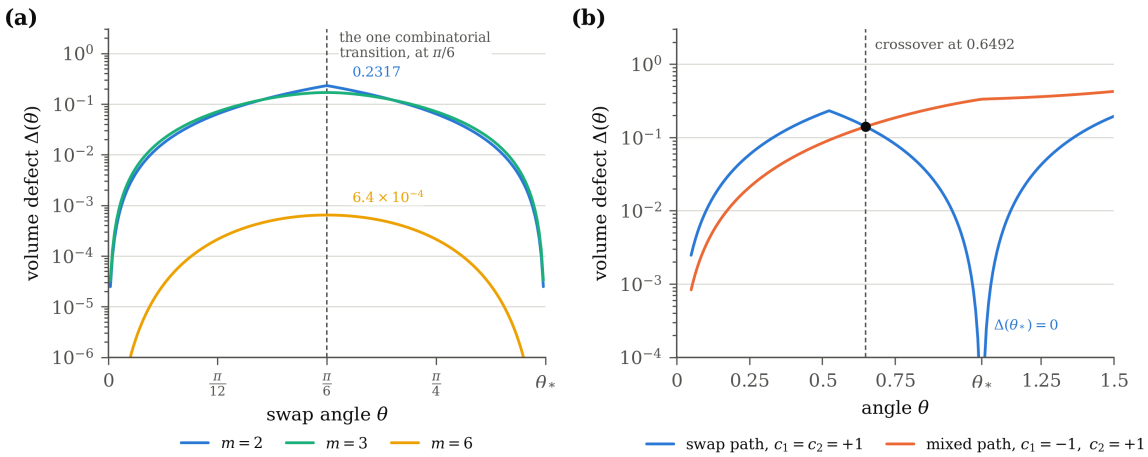

FIGURE 48. (a) The volume defect along the three swap paths of this subsection, over the whole range $(0, \theta_*)$, $\theta_* = \pi/3$. The vertical scale is logarithmic because the three sit nearly three orders of magnitude apart: the two-direction and three-direction paths peak at 0.2317 and 0.1692 at the midpoint, the six-direction path at 6.4×10^{-4}. All three are positive throughout and vanish at both ends, as Theorems C.15 to C.17 assert exactly. The dashed line is the single combinatorial transition, at $\theta = \pi/6$, visible as the corner in the $m = 2$ curve; the $m = 6$ path has none, and its curve is smooth there. The values plotted are enumerated cell volumes, computed by a route independent of the closed forms, and for $m = 6$ they agree with the closed form of Appendix C.6.6 to 9×10^{-9}. (b) The swap path against the mixed path, the two differing only in the sign c_1. The more adversarial of the two is the lower one, and which that is changes at $\theta = 0.6492$: below the crossover the mixed path is lower, at $\theta = 0.05$ by 0.000834 against 0.002474, and above it the swap path is, through θ_* and beyond. Second-order theory cannot tell the two apart, since the Hessian entry that distinguishes them enters only through its magnitude. The zero of the swap curve at θ_* is the swap point itself, where the configuration is again a root system.

within each batch and only then combining the batches reaches it. Carried out that way, the result is

$$F_{\text{swap}}(t) = \frac{\text{num}(t)}{\text{den}(t)}, \qquad t = \tan(\theta/2),$$

with num and den of degree 471 in t and coefficients in $\mathbb{Z}[\sqrt{3}]$. That is large, and it is exactly the form the next step needs.

Two independent checks accompany it. The 157-term sum, evaluated exactly at rational values of t, agrees with the closed form at every point tested, and both agree with a floating-point half-space intersection and convex-hull volume to between 6 and 8 significant figures.

C.6.3. *The norm trick, and the $m = 2$ result.* Exact real-root isolation works with rational coefficients, and these lie in $\mathbb{Z}[\sqrt{3}]$. Writing $\text{num}(t) = P(t) + \sqrt{3}\,Q(t)$ with $P, Q \in \mathbb{Z}[t]$, the *Galois conjugate* $\text{num}'(t) = P(t) - \sqrt{3}\,Q(t)$ is a second, real-valued (not complex-conjugate) function of the same real variable t, and the product

$$N(t) \;:=\; \text{num}(t)\,\text{num}'(t) \;=\; P(t)^2 - 3Q(t)^2$$

has purely integer coefficients: the $\sqrt{3}$ cancels exactly. The same construction applied to $\mathrm{den}(t) = R(t) + \sqrt{3}\,S(t)$ gives $D(t) := R(t)^2 - 3S(t)^2$, again integral. If $N(t)$ has no real root on the open interval in question, then neither $\mathrm{num}(t)$ nor $\mathrm{num}'(t)$ can vanish there (their product would vanish if either did), so $\mathrm{num}(t)$ itself is continuous and nowhere zero on a connected interval, hence of constant sign there, checkable at a single point. The same argument applies to den via $D(t)$.

Theorem C.15 (Finite-angle positivity, $m = 2$ swap)**.** *$F_{\mathrm{swap}}(\theta) > 0$ for every $\theta \in (0, \theta_*)$, $\theta_* = \pi/3$, the midpoint $\theta = \pi/6$ included.*

Proof. By Lemma C.14 it suffices to take $\theta \in (0, \pi/6]$, that is $0 < t \le \tan(\pi/12)$. $N(t)$ and $D(t)$ are even polynomials in t, so they are polynomials of degree 471 in $u = t^2$. Sturm sequences in rational arithmetic show that the one for N has a root of multiplicity exactly 2 at $u = 0$, which corresponds to the excluded endpoint $\theta = 0$, and no other root on a rational interval containing $(0, \tan^2(\pi/12)]$, and that the one for D has no root there. So num and den have constant signs on the piece, and exact evaluation at four points, one of them $\theta = \pi/6$, shows both negative. Hence $F_{\mathrm{swap}} = \mathrm{num}/\mathrm{den} > 0$ there. ∎

C.6.4. *The swap path is not the worst path at every angle.* Theorem C.15 concerns one path, and a direct check shows how far it is from the whole two-direction case. Write the deviations of the pair as $e_1 = c_1\beta^\perp$ and $e_2 = c_2\alpha^\perp$ with signs $c_1, c_2 \in \{\pm 1\}$; the swap path is $c_1 = c_2 = 1$. At $\theta = 0.05$ the smallest defect among the four sign choices is not on the swap path but on the *mixed* path $c_1 = -1$, $c_2 = 1$, with 0.000834 against 0.002474. The two paths have the same smallest eigenvalue of the joint Hessian: the entry that couples the two directions along the path has absolute value $\frac{1}{3}$ in both, and the eigenvalues of the 2×2 block it sits in do not depend on its sign. No second-order argument separates them. Scanning both paths over $\theta \in (0.05, 1.5)$ finds a single crossover, at $\theta = 0.6492$ (Figure 48(b)): below it the mixed path has the smaller defect, above it the swap path does, through θ_* and beyond. The worst pair of deviations is therefore not one fixed configuration over the whole range, as Appendix B.1.3 found for a single deviation.

Theorem C.15 is a statement about one path and is not weakened by this. What the comparison rules out is reading it as evidence about the worst pair of deviations at small θ, where the mixed path is lower. The values for the mixed path are floating-point findings; nothing about that path is proved here.

C.6.5. *The triangle swap, $m = 3$.* The next configuration is a real 3-clique: three pairwise Gram-$\frac{1}{2}$ roots rotated simultaneously in a cyclic swap, with the remaining 21 roots fixed. Cramer's rule and staged simplification, as for $m = 2$, give a closed form in two pieces, split at $\theta = \pi/6$.

The boundary triangulation of the second piece needs 176 exact 4×4 determinants over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, t)$. Every determinant in this subsection is computed by a division-free expansion and compared with the floating-point determinant of the same four points at a reference angle before it is used; all 176 agree.[51]

Theorem C.16 (Finite-angle positivity, $m = 3$ triangle swap)**.** *$F_{\mathrm{tri}}(\theta) > 0$ for every $\theta \in (0, \theta_*)$, $\theta_* = \pi/3$.*

[51] The comparison is not a formality. With fraction-free elimination one of these determinants came out as -5.7369 at $t = \tan(0.425)$, where the division-free expansion, Gaussian elimination over the field, a cofactor expansion and the floating-point value all give 0.30580.

Proof. On each piece the argument of Theorem C.15 applies to that piece's numerator and denominator. The two closed forms agree exactly at $\theta = \pi/6$, and $F_{\text{tri}}(\theta_*) = 0$, where the configuration is the root system again. ∎

C.6.6. *The hexagon swap, $m = 6$.* The third configuration is a cyclic rotation of six directions, a hexagon, which generalises the pairwise swap. Unlike the cases $m = 2$ and $m = 3$ it is one combinatorial piece on the whole interval $(0, \pi/3)$: the vertex-degree signature is constant throughout. The closed form is short:

$$F_{\text{hex}}(t) = \frac{8t^4\big(A(t) + \sqrt{3}\,B(t)\big)}{(t^2-3)^2(t^2+1)^2(3t^2-1)^2},$$
$$A(t) = 27t^8 - 108t^6 + 90t^4 + 100t^2 + 3,$$
$$B(t) = 48t^5 - 96t^3 - 16t.$$

Theorem C.17 (Finite-angle positivity, $m = 6$ hexagon swap)**.** *$F_{\text{hex}}(\theta) > 0$ for every $\theta \in (0, \pi/3)$.*

Proof. The denominator is a product of three squares, hence non-negative for every real t, vanishing only at $t = \pm\sqrt{3}, \pm 1/\sqrt{3}$; since the interval of interest is $0 < t < 1/\sqrt{3} = \tan(\pi/6)$ (i.e. $\theta \in (0, \pi/3)$), it is strictly positive throughout. The norm-trick quantity $N(t) = A(t)^2 - 3B(t)^2$ factors exactly as $(t^2-3)^2(3t^2-1)^6$, by direct expansion, again a product of squares, strictly positive on the same open interval. Since $A(0) = 3 > 0$ and Sturm sequences on two rational ranges whose union contains $[0, 1/\sqrt{3}]$ find no root of A, $A(t) > 0$ on the whole interval. Combined with $A^2 > 3B^2$, this gives $A > \sqrt{3}|B| \geq -\sqrt{3}B$, hence $A + \sqrt{3}B > 0$. Since $t^4 > 0$ for $t \neq 0$ and the denominator is strictly positive, $F_{\text{hex}}(t) > 0$ throughout $(0, 1/\sqrt{3})$. ∎

At the endpoint $t = 1/\sqrt{3}$ numerator and denominator both vanish through the factor $3t^2 - 1$, and the limit is $F_{\text{hex}}(\theta_*) = 0$, since there the configuration is the root system again. A 50-digit evaluation at twelve points of the interval confirms the sign, also near the endpoint, where double precision loses the small positive value to cancellation.

C.6.7. *Scope.* Theorems C.15 to C.17 are exact and hold over the whole angular range of their paths, not only near $\theta = 0$, which the Hessians of Appendix C.1 cannot give. Each concerns one path, with one highly symmetric rotation pattern (pairwise, triangular or hexagonal) and every other root held fixed. They say nothing about $m = 4, 5, 7$ or $m \geq 8$, nothing about less symmetric configurations of two, three or six directions, and nothing about deviations off these paths. Carrying the finite-angle method to the dense sets of Appendices C.1.4 and C.3, where the second-order method degenerates, is the natural next step for an elementary treatment of several deviations.

Appendix D. Methods of computation

This appendix states the two procedures that the certificates of the paper share, and then how each certificate is found and checked.

The box-covering certificate. Let R be a box, let F be a function on R whose sign is to be certified, and let $\beta(R')$ be, for every sub-box R', an outward-rounded enclosure of the range of F on R' in the sense of [MKC09, Ch. 5]. The certificate is produced by the following procedure. A list of boxes starts as $\{R\}$. A box R' is removed from the list; if $\beta(R')$ has the required sign, R' is recorded as certified; otherwise, if the longest side of R' is below a tolerance ε fixed before the run, the procedure stops and reports failure; otherwise

R' is bisected along its longest side and both halves are put on the list. The procedure ends when the list is empty.

A successful run returns finitely many boxes whose union is R, with disjoint interiors, and this is checked on the box boundaries in exact rational arithmetic rather than assumed from the construction; so every quoted box count is exact. There is one way to fail, a box of width below ε on which the sign is still undecided, and then no partial certificate is reported. In the independent check of Theorem 6.6 (Remark 6.7) the enclosure is the corner value supplied by Lemma 6.5; in Theorems 5.72, 5.78 and 7.8 it is a second-order Taylor form with rigorous tables of ω, ω' and ω''; in Theorems 7.25, 7.32 and 7.35 and Propositions 7.37 and 7.38 it is the largest Bernstein coefficient [Far12] of the kernel on the box less the corner value of Π in ball arithmetic.

Rounding a semidefinite solution. A semidefinite programme is solved in floating point with its inequalities imposed at finitely many sample points. After each solution the inequality is tested at a much larger set of further points, weighted towards the edges and corners of the domain, and the worst are added to the sample until none is violated. The matrices are then rounded to dyadic rationals, and from that point on everything is exact: positivity by LDL^{T}, the value of the bound, and the inequality on the whole domain by the procedure above. A margin is kept in the sampled inequality, so that the rounded kernel still satisfies it between the sample points.

The certificates. *The cap inequality.* The chain of Section 6 is exact from end to end and needs no computer: the vertex enumeration of Q, with 25 vertices and $\operatorname{vol} Q = \frac{25}{3}$, Lemma 6.3, the monotonicity of Lemma 6.5, the three polynomial inequalities of the case $|S| = 1$, and the five vertex values of the case $|S| \geq 2$, the largest $\frac{2}{9}$. The 303 boxes of Remark 6.7, whose largest corner value is $0.99755\ldots$, check the case $|S| \geq 2$ independently.

The arcs. On each region of Appendices B.3 and B.5 the numerator and the denominator of Δ are written in the Bernstein basis over the box or the remapped square, with coefficients in $\mathbb{Z}[\sqrt{2}]$ or $\mathbb{Z}[\sqrt{5}]$, and the sign of every coefficient is decided exactly; by the variation-diminishing property of the basis [Far12] that finite check is a proof of the sign on the box (Appendix G works through a small instance).

The pair inequality. The polynomial P of Theorem 5.72 is expanded exactly from the certificate entries, multiplied by 315 so that every coefficient is an integer over a power of two, and bounded on the ordered admissible domain over 421 881 boxes. The labelled certificate of Theorem 7.8 is checked by the same procedure on the regions of its proof.

The cited certificate. The verification of Proposition 5.29 reads the deposited data [LLM24d], checks their format and the positive definiteness of the 127 blocks, rebuilds the zonal matrices without holding the expansion of the integrand at once, checks the four polynomial identities and the value 24 of the bound, and isolates the zeros of σ_2 by a Sturm sequence.

The cardinality certificates. The certificates of Theorem 5.78 are found by a sampled programme, rounded, and checked: positivity by LDL^{T}, the bound, and both branch and bounds of the three-point condition in outward-rounded interval arithmetic.

The second order at the root system. The form H of Proposition 7.13 is assembled from the integral model of the 24-cell over the triangles of the tight pairs; the certificate of Proposition 7.14 is the dual of a Shor relaxation, averaged over the symmetry group, made exact, and checked by exact LDL^{T}.

The explicit neighbourhood. The constant ε_0 of Theorem 7.9 is computed from the deposited sums of squares in ball arithmetic, and the bracket β of its proof in exact rational arithmetic on a grid of step 10^{-5}.

The counts. The kernel of Theorem 7.25 is found as above, the inequality being tested after each solution at two million further admissible triples, weighted towards the edges $|y| = 2$ and $|y'| = \sqrt{6}$ and the boundary $u = a(|y|, |y'|)$. The Bernstein subdivision of the kernel has its floating-point error bounded a priori: each midpoint step is an average, so it adds at most one unit in the last place of the largest coefficient. The certificates of Theorem 7.32 are found and checked in the same way, one for each case of the tree, with a margin in the pair inequality tried at $2 \cdot 10^{-5}$ and lowered until the bound of the case stays below $\frac{9}{8}\pi^2 - 8$ by $2 \cdot 10^{-4}$. The certificate of the fifth case, with the kernel on triples, is found from sampled constraints refined in rounds: each round adds the points at which the previous solution violates its inequalities most and keeps, of each kind of pair and of triple, the 5000 pairs and 2500 triples that come closest to violation. The tree is checked from its splits alone: each child adds one bound on one $N(\rho)$, and its other bound on $N(\rho)$ is no tighter than what the bounds of the parent, together with $N(2.0161) \le 24$ and $N(d_{\max}) = 30$, imply. The largest value of $\sum_b n_b(m_b + sS(e_b))$ over the count vectors of a case is found by dynamic programming over the bins in exact rational arithmetic, with the values $S(e_b)$ rounded up from ball arithmetic. The single splits of Propositions 7.34 and 7.42, and the certificates of the levels of Proposition 8.1, are found and checked in the same way. So are the certificates of Theorem 7.35 and Proposition 7.37, except that the pairs of the starting grid are kept in every round: without them the pruning to the pairs nearest to violation can leave a region of pairs unsampled for a round, and the next solution then violates its inequality there. The certificate of Proposition 7.38 is a convex combination of certificates of degree 8 and 10 found in this way for the cases at twenty-eight and twenty-six centres or for parts of them, a certificate of degree 8 counting as one of degree 10 whose blocks of degrees 9 and 10 vanish. The negative eigenvalues of each solution are set to zero first, so that every matrix of the combination is positive semidefinite before the rounding. The weights minimise, by linear programming, the largest value of the right side of (7.9) over the count vectors, with the constant of each inequality taken as its largest value on a sample of pairs and triples; the sample is enlarged in four rounds by local maximisation of the combined inequalities from their largest sampled values. The certificates of degree 10 for exactly seventeen, eighteen, nineteen, twenty and twenty-one centres within 2.0161 were found one after the other, in one round each, each from the samples of the one before, which carry its refinements; for twenty-two, five rounds of this kind were run, and the certificates of four of them enter the second combination of Proposition 7.38. For twenty-three, further rounds were run from the samples of the last round for twenty-two; the certificates of the first, third and fourth of them, with 0.90 of the weight on the last two, enter the third combination, which settles twenty-four. The fourth combination, for twenty-three with at most three centres beyond 2.35, puts 0.62 of its weight on two rounds for twenty-two and 0.35 on the second to fifth rounds for twenty-three. The combinations for the single vectors $(23, 0, 1, 0, 0, 0, 4)$, $(23, 0, 0, 1, 0, 0, 4)$, $(23, 0, 0, 0, 1, 0, 4)$ and $(23, 0, 0, 0, 0, 1, 4)$ draw on the same certificates and on a first round for twenty-three close centres with four beyond 2.35; the weights of each minimise the bound at its vector alone, with the ranges of the types narrowed to the bins that the vector occupies, so that the sampled pairs and triples are those that occur there. For $(23, 0, 1, 0, 0, 0, 4)$ and $(23, 0, 0, 0, 0, 1, 4)$ the local maximisation starts from 400 points of each kind, among them the largest values that a dense search found for an earlier combination, since with fewer starts it missed a peak of the inequality

for one centre of type a and two of type f on the face where the two f directions are at their largest inner product $\frac{2}{3}$. For $(23, 0, 0, 0, 0, 1, 4)$ the local maximisation that sets the constant of each inequality on triples also starts from every local maximum of a grid of 121^3 points on its domain, since the inequality for two close centres and one far centre has a narrow interior peak that the other starts miss, and the two further constants of the count in the proof of Proposition 7.38 are set in the same way on their parts of the domain.

Holes. The labelled programmes of Remark 7.30 have two classes of directions, n of slack 0.008 and one further direction. They carry a Schoenberg kernel of each degree up to d with a 2×2 matrix over the classes, and three-point kernels of Bachoc–Vallentin type with one matrix for each class of the pole. Their inequalities are imposed at sampled pairs and triples and refined at the worst of 20 001 further pairs and of several tens of thousands of further triples. Only values that stay positive are quoted, and sampling can only lower a value.

The swap paths. An exact symbolic determinant is accepted only when two independent algorithms agree on it. In the derivation behind Appendix C.6.5 fraction-free elimination returned a wrong value for one symbolic matrix without warning, and the method of Berkowitz and direct cofactor expansion, which agreed with each other, exposed it. The three functions of Appendix C.6 are also recomputed from the cell itself, by enumerating the local region of the deviated configuration and subtracting 8 at 211 angles across $(0, \theta_*)$, on a route that shares no step with the closed forms. For the hexagon, whose closed form is printed in Appendix C.6.6, the two agree to $9 \cdot 10^{-9}$ across the interval, and the same route reproduces the comparison of Appendix C.6.4: 0.000834 against 0.002474 at $\theta = 0.05$, and the crossover at $\theta = 0.6492$.

The fitted quartic. The Gram-matrix test of Appendix C.4.3 is a floating-point semidefinite programme of the kind of [Par03, Las01], run on a least-squares fit; it was first run on two quartics of known status, a perfect square and the form of Choi and Lam, and returned the right answer on each. Its output is evidence about the fitted form only.

Software. The exact and ball arithmetic is that of FLINT and Arb [Joh17], with [Meu17] on a second, independent path for the exact rational computations; convex hulls are computed by [BDH96], and the numerical linear algebra uses [Har20, Vir20]. The formal part uses Lean 4 [MU21].

Appendix E. Numerical tables

The roots are numbered $0, \dots, 23$ throughout as follows: the pairs $i < j$ of coordinates are taken in lexicographic order, and within each pair the signs (ϵ_i, ϵ_j) in the order $(+,+)$, $(+,-)$, $(-,+)$, $(-,-)$; root number $4k + l$ is $\epsilon_i e_i + \epsilon_j e_j$ for the k-th pair and the l-th sign pattern. Thus roots 0 to 3 are $\pm e_1 \pm e_2$, roots 4 to 7 are $\pm e_1 \pm e_3$, and roots 20 to 23 are $\pm e_3 \pm e_4$. The tables below collect the numerical values behind Numerical observation C.3 and Appendices C.1.4 and C.4. They are floating-point or high-precision values, as stated with each, and no proof uses them.

Chains. Table 10 lists the Richardson-extrapolated least eigenvalue of H_A for chains of length 3 to 8, with the three values it is extrapolated from.

The values level off near 0.1676 instead of falling further past length 5 or 6, and 8 is the largest length of an induced path in the adjacency graph of R at 60°, by exhaustive search from every vertex. Numerical observation C.3 states only what these values support: a floor near 0.168, strictly positive.

TABLE 10. Least eigenvalue of H_A on chains (double precision).

Length m	Extrapolated $\lambda_{\min}(H_A)$	Values at $h = 0.04,\ 0.02,\ 0.01$
3	0.275801	0.276304, 0.275929, 0.275833
4	0.180435	0.181187, 0.180632, 0.180486
5	0.168532	0.169316, 0.168736, 0.168585
6	0.167689	0.168474, 0.167894, 0.167742
7	0.167650	0.168424, 0.167847, 0.167700
8	0.167637	0.168411, 0.167833, 0.167687

Dense configurations. Table 11 lists the active sets of the greedy growth of Appendix C.1.4, which at each step adds the root with the most active neighbours, together with the extrapolated least eigenvalue.

TABLE 11. Least eigenvalue of H_A along the greedy growth (double precision).

m	Active roots	Extrapolated $\lambda_{\min}(H_A)$
9 (star)	a root and its 8 neighbours at 60°	0.083336
9 (greedy)	$0, 4, 5, 8, 9, 12, 13, 16, 17$	0.083342
12	$0, 4, 5, 8, 9, 12, 13, 16, 17, 21, 22, 23$	0.073070
15	$0, 2, 4, 5, 8, 9, 11, 12, 13, 16, 17, 20, 21, 22, 23$	0.044694
18	$0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 16, 17, 20, 21, 22, 23$	-0.000013
20	$0, 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 19, 20, 21, 22, 23$	0.000017
22	22 of the 24 roots	0.000023
23	23 of the 24 roots	0.000008

The least eigenvalue falls strictly through $m = 15$. From $m = 18$ on the extrapolated value is below what a double-precision finite difference can resolve, an absolute volume precision of order 10^{-10} divided by h^2, and the entries are at that noise floor.[52] This is why Appendices C.3 and C.4 work in high precision.

Sampling at the singular configurations. At A_{18} the near-null space of H_A has dimension 4 to the accuracy of the computation: under halving of the step, the eigenvalues 0 to 3 fall by a factor of about 4, as zero eigenvalues do, while eigenvalue 4 ≈ 0.075 does not change. The quartic coefficient a_4 of Δ was sampled along 36 directions of that space, the 4 basis vectors, the 12 normalised sums and differences of two of them and 20 random unit combinations, at $\sigma = 0.015$ with 35 digits. At A_{20}, whose near-null space has dimension 5, 30 directions were sampled in the same way. Table 12 gives the results.

The last row compares, on six directions at A_{18} and four at A_{20}, the estimate at half the step with the first one; a ratio near 1 confirms the quartic scaling instead of assuming it. The configuration A_{20} comes from a different greedy path than the row $m = 20$ of Table 11. The evidence covers the whole near-null space at two dense configurations, with structured and random directions, and only those two.

[52] The value at the star, 0.083336, is within $3 \cdot 10^{-6}$ of $\frac{1}{12}$, the value that the symmetry of the star suggests; no exact derivation of it is attempted here.

TABLE 12. The quartic coefficient a_4 along the near-null spaces (high precision).

	A_{18}	A_{20}
Directions tested	36	30
Least a_4	0.008779	0.007628
Largest a_4	0.313708	0.313731
Directions with $a_4 < 0$	0	0
Ratio at $\sigma = 0.0075$ to $\sigma = 0.015$	0.954 to 0.991	0.952 to 0.991

APPENDIX F. EXPLICIT COORDINATES OF THE REFERENCE CELL

Lemmas 2.3 and 2.5 are proved in Section 2 by an exact argument. The coordinates are recorded here with a direct numerical check, for a reader who wants to compute with the polytope.

The 8 axis vertices are $\pm\sqrt{2}\,e_i$, $1 \le i \le 4$, and the 16 sign vertices are $\frac{1}{\sqrt{2}}(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4)$ with $\epsilon_i \in \{\pm 1\}$; all 24 have norm $\sqrt{2}$. A convex-hull computation in floating point on these 24 points alone, without the half-space description of Definition 2.2, finds all 24 extreme, returns the volume 7.999999999999984, which is 8 to fourteen digits, and splits the boundary into 96 simplices: the 24 facets of the 24-cell are octahedra, and each is cut into 4 tetrahedra. Maximising over the vertex list, the support function $h(u) = \max_{z \in \mathcal{V}} \langle z, u \rangle$ equals 1 at each of the 24 unit root directions and $\sqrt{2}$ at each of the 24 unit vertex directions, as Lemma 2.5 states.

APPENDIX G. A MODEL OF THE BERNSTEIN CERTIFICATE

The certificates of Appendices B.3 and B.5 decide the sign of a polynomial on a box by the signs of its coefficients in the Bernstein basis. Their polynomials have degree up to 21 in one of the two variables, too large to show the mechanism, so a small instance is worked here.

On $[0, 1]$ the Bernstein basis of degree n consists of the polynomials $B_{k,n}(t) = \binom{n}{k} t^k (1 - t)^{n-k}$, $0 \le k \le n$, which are nonnegative on $[0, 1]$ and sum to 1. Take

$$p(t) = t^2 - t^3 + \tfrac{1}{4} t^4,$$

which, like the defect along an arc, vanishes to second order at the left end: $p(0) = p'(0) = 0$ and $p''(0) = 2$. Matching the coefficients of t^j gives

$$p = \sum_{k=0}^{4} b_k B_{k,4}, \qquad (b_0, \ldots, b_4) = (0,\, 0,\, \tfrac{1}{6},\, \tfrac{1}{4},\, \tfrac{1}{4}).$$

Every coefficient is nonnegative, so $p \ge 0$ on $[0, 1]$, term by term. Since $B_{k,4}(t) > 0$ for $0 < t < 1$ and $b_2 > 0$, in fact $p > 0$ there, and p vanishes only at $t = 0$. Five rational numbers decide the sign, with no calculus and no root-finding.

The test is sufficient and not necessary. The polynomial $q(t) = (2t - 1)^2$ is nonnegative, but its coefficients of degree 2 are $(1, -1, 1)$, and raising the degree to 6 gives

$$(1,\, \tfrac{1}{3},\, -\tfrac{1}{15},\, -\tfrac{1}{5},\, -\tfrac{1}{15},\, \tfrac{1}{3},\, 1),$$

still with three negative entries. No degree helps: all basis functions are positive at $t = \frac{1}{2}$, so nonnegative coefficients and $q(\frac{1}{2}) = 0$ would force every coefficient, and with it q, to

vanish. A zero in the interior of the domain defeats the test, while a zero at an end of the domain, as for p, does not. The certificates of the arcs are run on regions whose closures meet the zero set of the defect only on their boundary, and the regions with an irrational boundary are covered by boxes that keep a thin margin from it (Appendix B.3).

Every number in this appendix is an exact rational, obtained by exact linear algebra: the change to the Bernstein basis of a given degree is a fixed invertible linear map, and the elevation of degree is another. No floating-point step enters, as none enters the certificates of the arcs.

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